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Preface

The 4th International Conference on Engineering, Social Sciences & Management (ICESSM-2026) was organized to provide an international platform for researchers, academicians, industry professionals, and students to present their research, exchange ideas, and discuss recent advancements in engineering, management, and social sciences.

The conference was successfully held on 15th–16th May 2026 at Sri Balaji College of Engineering and Technology (SBCET), Jaipur, Rajasthan, India. It featured keynote lectures, technical paper presentations, and scholarly discussions that encouraged innovation, collaboration, and interdisciplinary research. The enthusiastic participation of delegates and experts from academia and industry contributed significantly to the success of the conference.

The papers included in this volume were selected through a rigorous peer-review process and reflect quality research across diverse disciplines. We hope these proceedings will serve as a valuable reference for researchers, academicians, professionals, and students.

We extend our sincere gratitude to all authors, keynote speakers, reviewers, session chairs, delegates, members of the Organizing and Technical Program Committees, volunteers, and our publication partner for their valuable support and contributions in making ICESSM-2026 a successful academic event.

May 2026

Mr. Anurag Sharma
Mr. Inder Swaroop Verma

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The Myth of the Visionary Founder:Ego, Ethics, and the Dark Side of Startup Leadership

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Abstract– The veneration of the "visionary founder" has become one of the defining ideological fixtures of the contemporary startup ecosystem, yet its structural and ethical implications remain insufficiently theorized. This paper critically examines how the institutionalization of founder mythology fosters organizational dysfunction, erodes accountability mechanisms, and enables ethically compromised leadership to persist under the legitimizing cover of innovation. Through case-based analysis of prominent startup collapses and controversies, the study identifies three recurring pathological patterns: the displacement of institutional governance by personal authority, the cultivation of toxic internal cultures insulated by founder charisma, and the suppression of organizational dissent. Drawing on narcissistic leadership theory, behavioural psychology, and business ethics, the paper further interrogates how venture capitalists, financial media, and board structures function as institutional amplifiers of this myth, privileging growth imperatives over fiduciary and moral responsibility. The study concludes by advancing a normative reframing of entrepreneurial leadershipone that preserves visionary capacity while embedding it within robust structures of ethical accountability, stakeholder transparency, and distributed governance. The findings contribute to emerging scholarship on founder pathology and carry applied implications for investor due diligence, startup board design, and entrepreneurship pedagogy.

Keywords: *Visionary Founder Myth, Startup Leadership, Ethical Decision-Making, Entrepreneurial Governance, Narcissistic Leadership*

1. Introduction

In the contemporary imagination of entrepreneurial capitalism, the founder figure occupies a position of extraordinary cultural power. From Steve Jobs to Elon Musk, from Elizabeth Holmes to Adam Neumann, the archetype of the visionary founder, a singular

individual whose genius, conviction, and audacity transform industries has been enshrined not merely as a business phenomenon but as a civilizational ideal. Business media, venture capital folklore, and popular culture have collectively constructed a mythology in which the founder is prophet, architect, and martyr simultaneously: someone who sees what others cannot, endures what others will not, and ultimately delivers what the world did not know it needed.

This paper argues that such mythology, far from being an innocent cultural narrative, performs a specific and consequential ideological function. It naturalizes the concentration of organizational power in a single individual, delegitimizes institutional constraints as impediments to innovation, and creates conditions under which ethically compromised behaviour is not merely tolerated but romanticized. The consequences of this ideological formation are empirically visible in the wave of high-profile startup failures and scandals that have punctuated the last two decades: Theranos, WeWork, Uber under Travis Kalanick, FTX under Sam Bankman-Fried, and others. Each case reveals not merely individual moral failure but systemic structural conditions that enabled and amplified that failure.

The academic literature on entrepreneurial leadership has grown substantially in recent decades, encompassing work on founder psychology (Kets de Vries, 1985; Hmieleski& Baron, 2008), corporate governance in startups (Wasserman, 2012; Bebchuk & Hamdani, 2017), and the ethics of innovation-driven organizations (Crane et al., 2019; Moriarty, 2014). Yet the specific intersection of these domainsnamely, how the cultural myth of the visionary founder interacts with governance structures and organizational psychology to produce systematic ethical riskhas received insufficient integrated theoretical attention. The present study seeks to address this gap.

The paper proceeds as follows. Section 2 situates the visionary founder myth within a broader theoretical literature on charismatic authority, narcissistic leadership, and organizational culture. Section 3 develops an analytical framework for identifying three pathological patterns generated by this myth, drawing on case evidence from major startup scandals. Section 4 examines the institutional actors, venture capitalists, boards, and media who function as structural amplifiers of founder mythology. Section 5 offers a

normative reframing of entrepreneurial leadership grounded in distributed governance, stakeholder ethics, and accountability architecture. Section 6 concludes with implications for practice and pedagogy.

The methodological approach is primarily conceptual and case-analytic. The study draws on published case materials, financial disclosures, journalistic investigations, and secondary scholarly sources to construct an integrated analytical account. While the paper does not advance original empirical data collection, it offers theoretical synthesis and analytical framing designed to generate hypotheses for future empirical investigation and to inform applied practice in startup governance and investor due diligence.

2. Literature Review

2.1 The Social Construction of the Visionary Founder

The figure of the entrepreneurial hero is not a natural given but a social construction with specific historical roots. As Carlyle's (1841) "great man" theory of history entered the nineteenth-century industrial imagination, it found ready application in the emerging mythology of the business pioneer figures like Carnegie, Rockefeller, and Ford were cast as visionaries whose individual genius had propelled civilization forward, notwithstanding the organizational violence and labour exploitation that attended their rise. The twentieth century's venture-capital-driven technology sector has produced an updated version of this archetype, recasting the robber baron as the disruptive innovator.

Scholars in the sociology of organizations have long recognized the ideological dimensions of leadership mythology. Weber's (1947) foundational theorization of charismatic authority identified charisma as a form of power that is inherently unstable and resistant to institutionalization the charismatic leader derives authority not from office or tradition but from the perceived possession of extraordinary personal gifts. Weber warned that charismatic authority, while powerful as a transformative force, tends toward the erosion of institutional checks and the subordination of followers to the leader's personal will. Contemporary startup culture has, in many respects, constructed organizational forms that Weber might recognize as attempts to institutionalize the uninstitutionalizable: to harness the transformative energy of charismatic authority within

profit-seeking enterprises, while suppressing the accountability mechanisms that might constrain it.

The cultural mechanics of this construction have been examined by scholars such as Lounsbury and Glynn (2001), who demonstrated how entrepreneurial narratives function to attract resources by framing uncertainty in ways that are compelling to investors. More recently, Gill (2018) and others working in the tradition of critical management studies have interrogated the "entrepreneurial subjectivity" as a neoliberal formation that displaces collective responsibility onto individual actors. The myth of the visionary founder is, in this reading, not merely an organizational phenomenon but a feature of a broader political economy that valorizes individual genius and demonizes institutional constraint.

2.2 Narcissistic Leadership: Theory and Evidence

A substantial body of psychological research has examined the relationship between narcissistic personality characteristics and entrepreneurial behaviour. Narcissism, in its clinical and subclinical forms, is characterized by grandiosity, entitlement, a diminished capacity for empathy, hypersensitivity to criticism, and a marked tendency to exploit interpersonal relationships in the service of self-aggrandizement (American Psychiatric Association, 2013). Importantly, researchers have consistently found that subclinical narcissistic traits are significantly overrepresented among entrepreneurs relative to the general population (Hmieleski& Lerner, 2016; Mathieu & St-Jean, 2013).

Maccoby (2000) offered an influential typology distinguishing "productive narcissists"whose grandiose vision and risk tolerance enable genuine innovation from their pathological counterparts, whose defensive rigidity, exploitative interpersonal behaviour, and inability to process negative feedback lead organizations toward dysfunction. The distinction is analytically valuable but empirically unstable: what presents as productive narcissism in the early, pre-accountability phase of a startup may shade into pathological narcissism as the organization scales, governance demands intensify, and the founder's vision encounters institutional friction.

Rosenthal and Pittinsky (2006) coined the term "narcissistic leadership" to describe a leadership style in which the leader's fundamental orientation is toward self-validation rather than organizational effectiveness. In such organizations, strategic decisions serve primarily to confirm the leader's self-image rather than to advance organizational goals. Dissent is experienced as narcissistic injury and is suppressed through a combination of punitive retaliation and the cultivation of sycophantic inner circles. Information flows within the organization become systematically distorted, as employees learn to filter negative feedback before it reaches the leader (Tourish, 2013).

The application of narcissistic leadership theory to the startup context has gained momentum in the wake of high-profile founder collapses. Campbell et al. (2011) demonstrated experimentally that narcissistic leaders are initially evaluated more positively than their peers, they project confidence, vision, and charisma but that their long-term organizational performance is typically inferior due to their inability to learn from failure, their exploitative treatment of collaborators, and their systematic dismantling of organizational accountability structures. This temporal dimension is particularly significant in the startup context, where early investor valuations are largely forward-looking and the accountability mechanisms that might expose narcissistic dysfunction take years to mature.

2.3 Governance Structures and Founder Control

The governance of startup organizations presents distinctive challenges relative to established firms. The foundational principal-agent problem (Jensen & Meckling, 1976) is complicated in startups by asymmetric information, the absence of established performance metrics, and the structural power asymmetries created by founder equity positions, dual-class share structures, and the cultural dominance of founder mythology. Wasserman (2012), in his landmark study of founder transitions, documented the systematic tension between "founder-friendliness" and organizational effectivenessa tension that becomes acute as organizations scale and the informal, personalized coordination mechanisms of the early startup give way to the need for institutional governance.

Bebchuk and Hamdani (2017) examined the governance implications of dual-class share structures, which have become increasingly prevalent among technology companies precisely because they allow founders to maintain voting control even as they dilute their economic ownership through successive funding rounds. Their analysis suggests that dual-class structures create significant agency costs for minority shareholders and systematically weaken the accountability mechanisms available to boards. The widespread adoption of such structures in the venture-backed technology sector reflects and reinforces the ideological dominance of the founder myth: the assumption that the founder's judgment is so superior that it warrants insulation from shareholder accountability.

Board composition and independence are critical variables in the governance of founder-led companies. Adams et al. (2009) found that founder-CEOs exercise significantly greater influence over board decisions than non-founder CEOs, partly through their control over board composition and partly through the informational advantages they hold relative to outside directors. In many venture-backed startups, board independence is further compromised by the structural positions of venture capital investors, whose incentive alignment with founders both typically benefiting from growth-driven valuations may lead them to prioritize growth over governance. The result is a board that functions less as an accountability mechanism than as a legitimizing audience for the founder's vision.

2.4 Organizational Culture and the Ethics of Innovation

The relationship between organizational culture and ethical behaviour has been extensively theorized. Treviño et al. (1998) developed an influential model in which ethical organizational culture is a function of both formal systems (codes of conduct, reporting mechanisms, accountability structures) and informal systems (leadership norms, peer behaviour, organizational narratives). In founder-led startups, the founder's personal values and behaviour are disproportionately influential in shaping both formal and informal cultural systems a double-edged dynamic that can produce either extraordinary ethical integrity or systematic ethical dysfunction, depending on the founder's character.

The concept of "tone at the top" has been widely adopted in corporate governance discourse to capture the ways in which senior leadership behaviour shapes organizational ethical culture (Grojean et al., 2004). In the startup context, the tone at the top is, quite literally, the founder's voice often amplified through direct employee communication, all-hands meetings, and the internal mythology constructed around the founder's personal story and values. When this tone is ethically compromised, the effects on organizational culture are immediate and pervasive. The "move fast and break things" ethos that has characterized much of Silicon Valley culture represents, at one level, a genuine commitment to speed and iteration; at another level, it functions as a permission structure for the externalization of costs onto employees, customers, and society (Vallor, 2018).

The ethics of innovation itself has become an important area of scholarly attention. Moriarty (2014) and others have challenged the Schumpeterian assumption that creative destruction is an ethically neutral economic process, arguing that the distribution of the costs and benefits of disruptive innovation raises fundamental questions of justice. The startup ecosystem's valorization of disruption as an intrinsic good independent of its distributive consequences creates cultural conditions in which ethical objections to specific innovations are dismissed as the complaints of those who "don't get it." This epistemic closure is reinforced by the founder myth's construction of the visionary as someone whose judgment transcends ordinary moral reasoning.

2.5 Venture Capital, Media, and the Amplification of Founder Mythology

Institutional actors play a critical role in the construction and maintenance of founder mythology. Venture capital firms, whose business model depends on identifying and backing exceptional founders, have strong incentives to construct narratives of founder exceptionalism both to justify the valuations they assign and to attract the best founders to their portfolio (Gompers & Lerner, 2001). The "founder-friendly" posture adopted by leading venture capital firms in recent years characterized by minimal governance demands, tolerance for founder misconduct, and the aggressive deployment of capital to maintain founder loyalty reflects this incentive structure.

Financial and technology media constitute a second institutional amplifier. Scholars of media and business culture have documented the ways in which technology journalism

has adopted the narratives and frameworks of the venture capital ecosystem, producing coverage that celebrates founder vision, minimizes governance concerns, and treats financial scale as a proxy for social value (McChesney, 2013; Streeck, 2016). The profile-driven format of technology journalism, the founder cover story, the TED talk, the podcast interview is structurally suited to the construction of personality cults and ill-suited to the investigation of organizational accountability.

3. Analysis: Three Pathological Patterns of Founder-Centric Leadership

Drawing on the theoretical literature reviewed above and on case evidence from major startup scandals, this section identifies three recurring pathological patterns that emerge when the visionary founder myth becomes institutionally dominant. These patterns are analytically distinct but empirically interrelated: each tends to reinforce the others, creating self-sealing organizational systems in which ethical dysfunction is simultaneously produced, concealed, and legitimized.

3.1 Pattern One: The Displacement of Institutional Governance by Personal Authority

The first pathological pattern involves the systematic displacement of institutional governance mechanisms/boards, audit committees, legal compliance functions, internal reporting structures by the personal authority of the founder. This displacement takes both formal and informal dimensions. Formally, it is accomplished through dual-class share structures, board stacking, and the redefinition of governance roles as advisory rather than supervisory. Informally, it is accomplished through the cultivation of a founder mythology that renders institutional challenges to founder authority both culturally illegitimate and interpersonally costly.

The case of Theranos offers the starkest illustration. Elizabeth Holmes constructed a board of directors composed predominantly of political and military luminaries with minimal biotechnology expertise figures whose social capital lent legitimacy to the enterprise but who lacked the technical knowledge to interrogate Holmes's extraordinary claims about the capabilities of the Edison blood-testing device (Carreyrou, 2018). The board's failure was not primarily one of individual dereliction but of structural design:

Holmes had assembled a governance body that was functionally incapable of performing the oversight role. The board's inability to challenge Holmes's authority was overdetermined by its composition, by Holmes's command of proprietary technical information, and by the cultural positioning of governance scrutiny as a failure of visionary trust.

The WeWork implosion under Adam Neumann presents a complementary case study. Neumann deployed founder mythology with extraordinary effectiveness, persuading SoftBank and other major investors to assign a valuation of $47 billion to a company that was, at its core, a loss-making commercial real estate operator dressed in the language of community and consciousness (Farrell & Ramsay, 2021). The governance mechanisms that might have constrained Neumann's increasingly erratic and self-dealing behaviour including transactions in which he personally profited from leasing his own properties to WeWorkwere comprehensively circumvented through a governance structure that concentrated decision-making authority in Neumann's hands and surrounded him with advisors who had strong incentives to maintain rather than challenge his authority.

Kets de Vries (1985) theorized the "neurotic organization" as one in which the personality pathologies of the senior leader colonize the organizational culture, producing collectively irrational strategic behaviour. In the cases examined here, the neurotic organization is not merely a pathological exception but a structural tendency of the founder-led startup, in which the cultural valorization of the founder's personal authority systematically undermines the institutional architecture that normal organizations use to correct for individual cognitive and ethical failure.

3.2 Pattern Two: The Cultivation of Toxic Internal Cultures Insulated by Founder Charisma

The second pathological pattern involves the construction of internal organizational cultures that normalize ethical misconduct, reward sycophancy over competence, and insulate abusive or exploitative founder behaviour from accountability through the deployment of charismatic loyalty. Such cultures are not the accidental by-products of strong founder personalities; they are, in many cases, the deliberate outcome of cultural

engineering designed to maximize organizational commitment and minimize internal dissent.

The Uber case under Travis Kalanick is instructive. The internal culture at Uber during Kalanick's tenuredocumented extensively in Fowler's (2017) landmark blog post and subsequently in multiple independent investigationswas characterized by systematic sexual harassment, aggressive internal competition, and the routine retaliation against employees who raised ethical concerns through official channels. Critically, this culture was not merely tolerated by Kalanick; it was, in important respects, a product of the value system he explicitly articulated. Kalanick's "14 Cultural Values" document framed aggression, "always be hustlin'," and "champion's mindset" as organizational virtues/values that translated, in practice, into a permission structure for predatory interpersonal behaviour.

Tourish (2013) has theorized the processes by which leader-centred organizations construct "cultures of followership" in which the leader's worldview becomes the taken-for-granted interpretive framework through which organizational members understand their own experience. In such cultures, ethical dissent is not merely discouraged but cognitively unavailable: the cultural framing systematically misrepresents exploitative behaviour as meritocratic competition, and the suppression of dissent as the maintenance of standards. The charismatic founder's personal narrative, typically one of extraordinary struggle, exceptional insight, and redemptive mission functions to anchor this cultural framework and render challenges to it as challenges to the organization's collective identity.

The FTX collapse under Sam Bankman-Fried provides a more recent and in some respects more extreme case study. The organizational culture at FTX was characterized by an extreme version of the founder's utilitarian philosophical commitments, which were deployed to justify an organizational environment in which ordinary ethical constraints, fiduciary duty, legal compliance, basic transparency were treated as the concerns of people insufficiently committed to the mission of "doing the most good" (Lewis, 2023). Bankman-Fried's philosophical posturinghis public commitment to "effective altruism," his ostentatious frugality functioned as a charismatic legitimation device that

simultaneously attracted followers committed to his stated values and deflected regulatory and journalistic scrutiny from the organization's financial practices.

3.3 Pattern Three: The Systematic Suppression of Organizational Dissent

The third pathological pattern is the systematic suppression of internal dissent through a combination of formal and informal mechanisms. Formal mechanisms include non-disclosure agreements, retaliatory termination, and the use of legal resources to intimidate potential whistleblowers. Informal mechanisms include the social marginalization of dissenters, the construction of organizational narratives in which dissent is coded as disloyalty or incompetence, and the cultivation of an inner circle whose access to the founder's favour creates strong incentives for sycophantic alignment.

The organizational consequences of dissent suppression are well-theorized in the literature on organizational voice and silence. Morrison and Milliken (2000) identified "organizational silence"the collective withholding of potentially important information as a pervasive organizational dysfunction that emerges when employees perceive that speaking up is futile or dangerous. In founder-led startups, the conditions for organizational silence are systematically produced by the power asymmetries of founder authority, the cultural valorization of commitment over critical engagement, and the explicit or implicit threat of retaliation against those who challenge the founder's vision.

The consequences of this suppression for ethical outcomes are severe. Organizations in which dissent is suppressed lose access to the internal corrective mechanisms that would otherwise surface ethical problems before they become organizational crises. The employees who possess the information necessary to identify and address ethical misconduct, junior engineers who discover that a product is not working as claimed, compliance officers who identify regulatory violations, HR personnel who document patterns of harassment are precisely those whose organizational position makes the costs of speaking up most acute. The founder myth compounds this dynamic by constructing ethical objections to the founder's approach as failures of vision rather than as legitimate concerns.

Whistleblower research consistently confirms that organizational retaliation against those who report misconduct is both common and severe (Near & Miceli, 2016). In founder-led startups, the retaliatory impulse is often amplified by the founder's narcissistic sensitivity to criticism and by the organizational culture's construction of loyalty as a supreme value. The result is a systematic filtering of information that creates a "CEO bubble" (Tourish, 2013) in which the founder's perception of organizational reality becomes progressively more detached from actual organizational conditions a dynamic that accelerates organizational dysfunction and delays the accountability reckoning that eventually arrives.

4. Institutional Amplifiers: Venture Capital, Media, and Board Complicity

The three pathological patterns described above do not emerge in an institutional vacuum. They are enabled, amplified, and in some cases actively constructed by the institutional ecosystem that surrounds and sustains the startup. This section examines three key institutional actors: venture capital investors, financial and technology media, and board structures. The argument is not that these actors are malicious but that their structural positions and incentive architectures systematically dispose them toward the amplification rather than the constraint of founder pathology.

4.1 Venture Capital and the Growth Imperative

Venture capital firms operate under a return structure that creates strong incentives to prioritize growth over governance. The venture capital model is predicated on the expectation that a small number of portfolio companies will generate extraordinary returns sufficient to compensate for the losses incurred by the majority that fail (Gompers & Lerner, 2001). This return distribution creates a systematic bias toward investing in founders with extreme risk tolerance, unconventional thinking, and the personal conviction necessary to pursue improbable outcomes characteristics that overlap substantially with the trait profile of subclinical narcissism.

The due diligence processes of venture capital firms, while increasingly sophisticated in their financial and technical dimensions, have historically paid less systematic attention

to founder character and ethical orientation. Kaplan and Strömberg (2004) documented the primacy of market size and technology assessments in venture capital decision-making, with team assessment typically focused on domain expertise and track record rather than ethical disposition or governance orientation. The result is a selection process that systematically under-weights the governance risks associated with founder personality risks that are, as the cases reviewed above demonstrate, among the most consequential determinants of startup outcomes.

The "founder-friendly" posture adopted by leading venture capital firms in recent years has exacerbated these dynamics. In competitive funding environments, venture capital firms compete for access to the most sought-after founders by offering not merely capital but governance concessions: minimal board seats, tolerance for dual-class structures, and implicit or explicit commitments not to challenge founder authority. This competitive dynamic creates a race to the bottom in governance standards, in which the most ethically risky foundersthose whose confidence and charisma make them most attractive to investors are also those who receive the most permissive governance environments.

4.2 Media Narratives and the Hagiographic Impulse

The technology and financial media have played a constitutive role in the construction of founder mythology. The profile format the magazine cover story, the podcast interview, the TED talk is structurally oriented toward the construction of compelling individual narratives and ill-suited to the investigation of organizational governance. Journalists who cover technology startups operate in a media ecology in which access to founders is both scarce and commercially valuable: the threat of access withdrawal creates systematic incentives to produce coverage that founders will regard as favourable.

The construction of the "visionary" narrative requires the systematic elision of organizational complexity. The profile must construct a protagonista singular consciousness whose insights drive events and this requirement systematically marginalizes the contributions of co-founders, early employees, and the organizational infrastructure through which visionary insights are translated into products and services. It also systematically marginalizes the organizational dysfunctions, governance failures, and ethical compromises that accompany the protagonist's rise, positioning these as either

irrelevant to the story of innovation or as evidence of the protagonist's willingness to do what others would not.

In the cases of Theranos and WeWork, the media's hagiographic coverage of Holmes and Neumann was not merely a journalistic failure but a constitutive element of the fraud itself. Holmes's extensive media cultivationher black turtlenecks, her artificially deepened voice, her invocations of Edison and Jobs was a deliberate strategy to construct a public narrative of visionary authenticity that would deter the investigative scrutiny her technology could not survive (Carreyrou, 2018). The media's complicity in this construction was not the product of individual journalistic negligence but of the structural incentives and narrative conventions of technology journalism.

4.3 Board Failure and the Limits of Independent Oversight

Corporate boards are the primary institutional mechanism for founder accountability in the startup context, yet the evidence from major startup failures suggests that boards have been systematically ineffective in this role. The reasons are structural and cultural, not merely individual. Structurally, boards in venture-backed startups are often composed predominantly of investor representatives whose incentive alignment with founders limits their independence, supplemented by independent directors who typically lack the operational knowledge necessary to challenge founders on substantive grounds and who may be selected precisely for their perceived pliability.

Culturally, board members in high-growth startups face powerful social pressures to align with the founder's vision. The founder myth renders board challenges to founder authority as failures of entrepreneurial understanding "they just don't get it" narrative that founders and their supporters deploy to delegitimize governance scrutiny. Board members who have built their reputations on backing visionary founders have strong identity investments in the persistence of the vision they have endorsed and face significant reputational costs from acknowledging governance failures on their watch.

Lorsch and MacIver (1989) famously described corporate boards as "pawns or potentates" depending on the balance of information and authority between directors and management. In the founder-led startup, the information asymmetries are typically acutefounders control access to product, technology, and cultural intelligence that outside

directors cannot independently verify and the cultural authority of the founder myth systematically tips the balance toward passivity. The result is a board that provides the appearance of governance oversight while functioning primarily as a legitimizing audience for the founder's authority.

5. Toward an Ethics of Accountable Entrepreneurial Leadership

The analysis presented above does not lead to the conclusion that visionary entrepreneurial leadership is intrinsically pathological or that the startup ecosystem is irredeemably corrupt. Rather, it leads to the conclusion that the specific institutional arrangements through which entrepreneurial vision is currently organized arrangements that concentrate power in founders, insulate them from accountability, and construct governance as an impediment to innovation create systematic conditions for ethical dysfunction. The normative challenge is to design institutional arrangements that preserve the genuine value of entrepreneurial vision while embedding it within robust structures of accountability.

5.1 Distributed Governance and Accountability Architecture

The first normative implication of the analysis concerns board design and governance architecture. The evidence reviewed above suggests that the effectiveness of board oversight is primarily a function of structural features board composition, director independence, information access, and the formal authority of the board relative to the founder rather than of individual director character or commitment. Reform efforts should accordingly focus on structural features rather than exhortations to individual directors.

Specifically, the evidence supports requiring meaningful independent director representation on startup boards from an early stage, establishing clear protocols for board access to operational information independent of the founder, and creating formal mechanisms for board-initiated leadership review. The dual-class share structures that insulate founders from shareholder accountability deserve particular scrutiny; while they can serve legitimate purposes in protecting long-term strategic vision from short-term market pressure, they require compensating governance mechanisms, enhanced

independent director authority, super-majority requirements for certain decisions, sunset provisions that are frequently absent in current practice.

A complementary approach involves the expansion of stakeholder representation in governance structures. Freeman's (1984) stakeholder theory, developed over subsequent decades into a robust normative framework (Freeman et al., 2010), provides theoretical grounding for governance structures that incorporate the interests of employees, customers, and communities alongside those of investors and founders. The evidence from startup failures suggests that many of the harms generated by founder pathology fall disproportionately on stakeholders who are currently excluded from governance: employees who experience toxic cultures, customers whose data or financial assets are exploited, and communities who bear the externalized costs of failed business models.

5.2 Reframing Investor Due Diligence

The second normative implication concerns venture capital due diligence practices. The analysis suggests that standard due diligence frameworks, which focus primarily on market opportunity, technology differentiation, and financial projections, systematically underweight the governance risks associated with founder psychology and organizational culture. A more robust due diligence framework would incorporate structured assessment of founder character and ethical orientation, including reference interviews specifically designed to surface interpersonal conduct, responses to failure, and attitudes toward institutional constraint.

It would also involve formal assessment of the founder's proposed governance arrangements, board composition, share structure, internal accountability mechanisms as a condition of investment, rather than treating governance as a matter to be negotiated after the investment decision is made. Investors who take governance seriously at the due diligence stage, and who insist on governance standards as a condition of investment, create incentives for founders to take governance seriously as well. The competitive dynamics of the venture capital market make unilateral governance reform difficult; industry-wide standard-setting, through vehicles such as the National Venture Capital Association's model legal documents, represents one mechanism for coordinating governance improvements across the ecosystem.

5.3 Implications for Entrepreneurship Education

The third normative implication concerns entrepreneurship pedagogy. The curricula of leading business schools and entrepreneurship programs have, in recent years, devoted increasing attention to ethics and social responsibility, but the framing of this attention has often remained within an instrumental logic/ethics as risk management, social responsibility as brand differentiation rather than engaging with the deeper normative questions about the responsibilities of founders toward the multiple stakeholders affected by their organizations.

A more adequate entrepreneurship pedagogy would integrate the analysis of founder pathology and governance failure into the core curriculum, using case studies of startup scandals not merely as cautionary tales but as analytical resources for understanding the structural conditions under which ethical dysfunction emerges. It would also engage directly with the mythology of the visionary founder subjecting the cultural narratives that students absorb from business media and startup culture to the same critical scrutiny that is applied to the financial projections in a business plan. Entrepreneurs who understand the ideological function of the founder myth, and who approach their own authority with appropriate humility and accountability, are better positioned to build organizations that are both innovative and ethically sound.

Argyris and Schön's (1978) concept of "double-loop learning"learning that involves the examination and revision of the governing assumptions that produce behaviour, rather than merely the correction of behaviour within existing frameworks provides a useful pedagogical orientation. The challenge for entrepreneurship education is to produce founders capable of double-loop learning with respect to their own authority: capable of examining and revising the assumptions about their own exceptionalism that the founder myth encourages, and of building organizations that do not depend on those assumptions for their functioning.

6. Conclusion

This paper has argued that the mythology of the visionary founder, far from being a culturally benign narrative about the sources of entrepreneurial success, performs a specific and consequential ideological function. By naturalizing the concentration of

organizational power in a single individual, delegitimizing institutional constraints as impediments to innovation, and creating conditions in which ethically compromised behaviour is romanticized rather than scrutinized, the founder myth generates systematic conditions for organizational dysfunction and ethical failure.

Through the identification of three recurring pathological patterns the displacement of governance by personal authority, the cultivation of toxic cultures insulated by founder charisma, and the systematic suppression of organizational dissent the paper has demonstrated that the ethical failures visible in cases such as Theranos, WeWork, Uber, and FTX are not idiosyncratic products of individual moral deficiency but structural outcomes of institutional arrangements that enable and amplify such deficiency. The institutional actors who function as amplifiers of the founder mythventure capital investors, financial media, and captured boards bear responsibility not merely as passive enablers but as active constructors of the conditions under which founder pathology flourishes.

The normative reframing advanced in Section 5 does not involve the abandonment of entrepreneurial vision or the bureaucratization of innovation. Visionary entrepreneurial leadership is a genuine and valuable organizational capacity; the question is not whether such leadership should exist but how it should be structured and constrained. The evidence supports the position that the most effective accountability mechanisms are structural rather than cultural that the reform of board composition, governance architecture, and due diligence standards is likely to produce more durable ethical improvements than the exhortation of individual founders to exercise greater humility.

The findings carry several implications for future research. The conceptual framework developed here identifying the three pathological patterns and the three categories of institutional amplifiers provides a basis for systematic comparative analysis across a broader set of founder-led startup failures and controversies. Quantitative research examining the relationship between governance structure, founder psychological characteristics, and organizational ethical outcomes would complement the conceptual and case-analytic work presented here. The intersection of founder mythology with questions of gender, race, and class in the startup ecosystem also warrants systematic

investigation: the founder myth is not demographically neutral, and its distribution of credibility and authority reflects and reinforces broader patterns of social inequality.

Ultimately, the argument advanced here is that the startup ecosystem faces a choice between two models of entrepreneurial excellence. The first model valorizes the founder as herosingular, unchallengeable, accountable to no one but his or her vision. The second model valorizes the founder as architectsomeone whose greatest contribution is the construction of an organization capable of realizing and refining that vision through the distributed intelligence, ethical accountability, and institutional resilience of its members. The evidence reviewed in this paper suggests, with considerable force, that the second model produces better outcomes for investors, employees, customers, and society and that the institutional arrangements of the startup ecosystem should be redesigned to reflect this conclusion.

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A Privacy-Preserving Digital Platform For Supervised Peer Counselling In Academic Institutions

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Abstract— Student mental health remains an underserved domain, constrained by cost, stigma, and limited access to timely professional care. This paper proposes a role-based digital platform that pairs help-seeking students with psychology trainees via real-time chat and voice communication. The system integrates secure authentication, session tracking, anonymity controls, and supervised escalation to licensed professionals when required. Built on a client-server architecture, the platform is designed for scalability, low-latency interaction, and data privacy compliance. Beyond user support, it functions as a structured experiential training environment for counselling trainees. Evaluation metrics target engagement rates, response quality, and user-reported well-being outcomes.

Keywords— *Peer-assisted mental health, Student well-being, Digital counselling platform, trainee Therapists, Anonymous support system, Chat-based intervention, Role-based access, teletherapy, Mental health technology, Accessible healthcare*

I. Introduction

Recent trends indicate a significant rise in psychological distress among university populations, with increased academic pressure and social challenges contributing to reduced academic persistence and student retention. Despite the scale of this issue, institutional counselling services remain insufficient in capacity, often unable to accommodate the volume of students requiring timely support. This gap between demand and available care calls for scalable, technology-driven alternatives that complement existing clinical infrastructure.

Online mental health solutions improve service accessibility by removing geographical limitations and enabling flexible communication channels.However, most existing solutions either rely entirely on automated responses or require direct engagement with licensed professionals, leaving a practical middle ground largely unexplored. Peer-assisted models, where trained individuals provide structured emotional support

under supervision, represent a cost-effective approach that has received limited technical implementation.

This paper presents the design and development of a student-focused digital platform that operationalises supervised peer-led counselling support through real-time chat and voice communication. The system connects help-seeking students with psychology trainees operating under faculty or clinical supervision, enforcing structured interaction boundaries while maintaining user anonymity and data privacy. The platform addresses core engineering challenges, including secure multi-role authentication, low-latency communication, session lifecycle management, and escalation pathways for high-risk cases.

The subsequent sections of this paper are organised in the following manner.Section II reviews related work, Section III outlines system architecture, Section IV covers implementation, and Section V presents evaluation results, followed by conclusions in Section VI.

II. Problem Definition

A. Background and Motivation

Mental health disorders among university students have become a growing concern, with anxiety, depression, and academic stress emerging as leading causes of dropout and declining performance. Despite the scale of this issue, institutional counselling services remain insufficient in capacity, often unable to accommodate the volume of students requiring timely support. This gap between demand and available care calls for scalable, technology-driven alternatives that complement existing clinical infrastructure.

Digital platforms have demonstrated significant potential in extending mental health services beyond traditional settings, offering round-the-clock availability, geographic independence, and reduced social stigma. However, most existing solutions either rely entirely on automated responses or require direct engagement with licensed professionals, leaving a practical middle ground largely unexplored. Peer-assisted models, where

trained individuals provide structured emotional support under supervision, represent a cost-effective approach that has received limited technical implementation.

B. Scope and Objectives

This paper presents the design and development of a student-focused digital platform that operationalises peer-assisted mental health support through real-time chat and voice communication. The system connects help-seeking students with psychology trainees operating under faculty or clinical supervision, enforcing structured interaction boundaries while maintaining user anonymity and data privacy. The platform addresses core engineering challenges, including secure multi-role authentication, low-latency communication, session lifecycle management, and escalation pathways for high-risk cases.

III. Related Work

A. Digital Mental Health Platforms

Several technology-driven platforms have emerged to address student mental health, including Woebot, Wysa, and university-deployed counselling portals. These systems demonstrate the viability of digital intervention but predominantly rely on rule-based chatbots or artificial intelligence (AI)-driven responses, limiting their capacity for nuanced, human-centred support. Research indicates that students often disengage from fully automated systems due to perceived lack of empathy and contextual understanding [1].

B. Peer Support Models in Academic Settings

Peer support programmes within academic institutions have shown measurable effectiveness in reducing psychological distress among students. Studies confirm that peer-assisted interventions improve help-seeking behaviour, particularly among populations reluctant to engage with formal clinical services [2]. However, existing implementations remain largely offline and unstructured, lacking a standardised digital framework for session management, role enforcement, and outcome tracking.

C. Technology-Mediated Communication in Counselling

Chat-based and voice-based communication channels have been increasingly adopted in teletherapy and remote counselling contexts. Research in human-computer interaction (HCI) highlights that anonymity afforded by text-based platforms significantly lowers disclosure barriers among young adults [3]. Existing literature, however, does not sufficiently address the integration of supervised trainee-led support within such communication frameworks.

D. Research Gap and Contribution

While prior work addresses either automated mental health tools or informal peer networks independently, no established system combines structured peer-assisted support with real-time digital communication under clinical supervision for student populations. This paper addresses that gap by proposing a role-based platform that operationalises peer support within a software-engineered, privacy-compliant architecture.

IV. System Architecture

A. Platform Overview and Design Rationale

The proposed platform follows a client-server architecture built around three primary user roles: students seeking support, peer supporters (psychology trainees), and supervisors (faculty or clinical staff). Each role is assigned distinct access permissions enforced through a role-based access control (RBAC) mechanism at the backend. This separation ensures that interaction boundaries are maintained, escalation decisions remain traceable, and trainee activity remains visible to supervising staff at all times.

The system is designed around four core modules: user authentication, session management, real-time communication, and escalation handling. These modules interact through a RESTful application programming interface (API) layer, ensuring loose coupling and independent scalability of each component.

B. Core Modules and Data Flow

The authentication module handles secure registration and login for all three user roles, implementing token-based session verification using JSON Web Tokens (JWT). Upon

login, users are directed to role-specific dashboards that expose only contextually relevant features. The session management module governs the lifecycle of each support interaction, from request initiation by a student to session closure and post-session logging.

Real-time communication is facilitated through WebSocket-based chat and a voice communication interface, enabling low-latency interaction between students and peer supporters. All sessions are anonymised at the presentation layer, ensuring that personally identifiable information (PII) is not exposed during active interactions. The escalation module provides a structured pathway through which peer supporters can flag high-risk cases to supervisors, triggering immediate notification and optional referral to licensed mental health professionals.

C. System Diagram and Data Privacy

Table I illustrates the high-level system architecture depicting data flow between client interfaces, the API gateway, and backend services. All data transmission is encrypted using Transport Layer Security (TLS), and user records are stored in a structured relational database with access restricted by role permissions. Session logs retained for supervisory review are anonymised before storage, complying with standard data protection requirements applicable to health-adjacent digital platforms.

TABLE I. Role-Based Feature Access Matrix

Features	Students	PEERSUPPORTER	SUPERVISIOR
Initiate Session	yes	no	no
AcceptSessionRequest	no	yes	no
Real-TimeChat	yes	yes	no
VoiceCommunication	yes	yes	no

EscalateCase	no	yes	yes
ViewSessionLogs	no	limited	full
ManageUserAccounts	no	no	yes

V. Acknowledgement

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Adversarial Robustness and Byzantine-Resilient Defense in Federated IoMT-Enabled Smart Intensive Care Units

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Abstract– Federated learning (FL) facilitates privacy-preserving collaboration in IoMT ecosystems, especially fog-enhanced Smart Intensive Care Units (Smart ICUs), where real-time anomaly detection of life-threatening incidents such as arrhythmia, sepsis emergence, and breathing difficulties can be performed without sharing sensitive physiological information [1][2]. Nevertheless, the distributed aggregation procedure entails a multitude of inherent threats that may jeopardize the safe performance of these systems. In this paper, we introduce a clinical threat model for fog-based federated IoMT applications in Smart ICUs. In particular, we define four major attack vectors in fog-federated systems model poisoning, Byzantine attacks, gradient inversion, and free-riding and suggest a feasible layer-wise defense approach. Specifically, our approach includes adaptive robust aggregation, time series anomaly-aware client validation, dimension-aware differential privacy, and contribution-based scores. Simulation experiments on split MIMIC-III data show that our solution provides F1-scores of anomaly detection within ~2.3% deviation from the unattacked baseline at up to 30% Byzantine attackers and provides $\varepsilon \approx 1.0$ differential privacy [3,4]. This indicates that security and robustness against attacks must be incorporated into a design of safety-critical medical AI applications from the very beginning [5].

Keywords*: Federated Learning Security, Byzantine Fault Tolerance, IoMT, Smart ICU, Adversarial Robustness, Differential Privacy, Fog Computing*

1. Introduction

IoMT device deployment in ICUs at breakneck speed has produced vast streams of high-resolution physiologic data. Fog computing resolves the problem of latency constraints, whereas federated learning enables training of machine learning models using edge devices without centralizing patients' records [1][6]. Even though FedAvg,

which is a basic algorithm, performs well in health care problems, its assumption of honesty is hard to be true in reality [7].

A single vulnerable edge node or fog server introduces potential for biases and backdoors that may cause either no alarms at all or alert fatigue with obvious consequences for patients' wellbeing [5][8]. Although Byzantine fault-tolerant aggregation and gradient inversion mechanisms have been explored in general machine learning research, their applicability to non-IID physiological data under constraints of fog computing and strict performance conditions is unclear [9][10].

Specific contributions are: (1) threat model that correlates attacks with clinical risks; (2) modular lightweight defense system; and (3) experimental results in realistic settings of attacks. After decades of experience in building security mechanisms for mission-critical applications, it is evident that reliable clinical AI must have security as one of its primary design goals.

2. Background and Related Work

2.1 Application of Federated Learning in IoMT and Smart ICU Frameworks

FL has been successfully used for multi-modal monitoring, outcome prediction, and real-time anomaly detection in healthcare edge systems [1][2]. Integration with fog allows reducing communication latency via partial aggregation near data sources which is vital in case of ICU tasks requiring processing of ECG, arterial pressure, SpO_2 and ventilator data [12]. Hybrid LSTM models work effectively for anomaly detection, but most works consider only benign participation [13].

2.2 Adversarial Threats to Federated Learning

Byzantine nodes can undermine the quality of global models by using poisoned updates or backdoors [14][15]. Methods of robust aggregation such as coordinate-wise median and trimmed mean provide theoretical guarantees, however, their practical performance suffers in non-IID distribution of data which is usually present in ICUs [16]. Gradient inversion attacks allow recovering private time-series data [17][18] whereas free-riding nodes reduce contribution to global model [19].

2.3 The Problem: Security for the Federated IoMT-Based Clinical Framework

Attack surface increases due to fog hierarchy, constrained resources within IoMT nodes restrict local protection, and diverse patient records make statistical outlier detection difficult [2][21]. In addition, clinical environments require stricter performance requirements compared to the standards for federated learning benchmarks. There have been recent works focusing on this intersection; however, frameworks are not abundant enough [5][8][22].

3. Threat Model for Federated IoMT Smart ICU Systems

3.1 System Architecture and Trust Assumptions

The system to be considered employs the layered fog-federated approach which is a common choice in the current IoMT Smart ICU installations. Wearable and bedside IoMT devices send physiological information to local edge nodes which are implemented as low power microcontrollers or single board computers that reside next to each patient bed. Local training of models is done at edge nodes based on patient specific partitions of the dataset. Gradient updates from edge nodes are then sent to intermediate fog servers in the hospital network. Initial aggregations of gradients are done at fog servers and then forwarded to the global federated server which holds the global model.

The trust model assumes that the global federated server and fog infrastructure are run by the healthcare institution and can be regarded as trusted. Edge nodes corresponding to individual patient monitoring sites are considered semi-trusted since they are not necessarily actively malicious, but rather due to their accessibility and lack of tamper resistance may be compromised by external parties or malicious insiders. Adversaries able to intercept gradient communications between layers are explicitly considered in the trust model. The proposed system architecture consists of four layers (Figure 1):

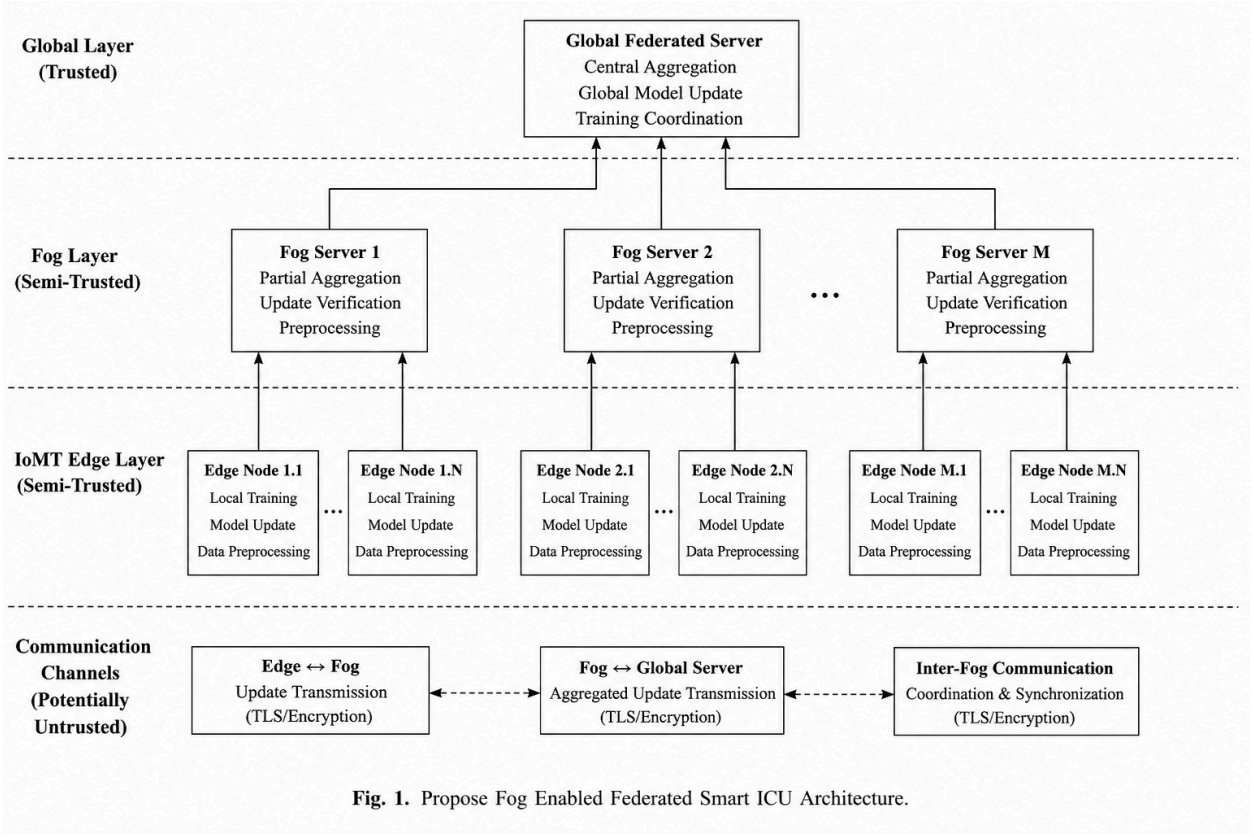


Fig. 1. Propose Fog Enabled Federated Smart ICU Architecture.

Figure 1: Propose Fog Enabled Federated Smart ICU Architecture

IoMT Edge Layer: Sensors at bedside and constrained nodes carrying out local training based on the patient's data stream.

Fog Layer: Gateway at ward or institutional level that carries out partial aggregation and pre-processing to avoid latency.

Global Server: Trusted institutional node responsible for updating the models.

Communication Channel: Vulnerable to eavesdropping and interception.

Our system assumes that global server and fog layer are trusted, but edge nodes are semi-trusted because of their accessibility [12][23].

3.2 Attack Surface Taxonomy

Within fog-enabled federated Smart ICU ecosystems, the hierarchical structure involving edge nodes which are resource-limited, fog servers, and an aggregation server results in different interdependent attack vectors. A potential attacker can leverage this hierarchy to breach the confidentiality, integrity, or efficacy of any model deployed within. Four

primary attack vectors have been identified and studied based on their distinctive ways of operation, stealthiness, and impact on critical healthcare applications. In particular, any slight decrease in anomaly detection performance (arrhythmias, sepsis, and ventilation asynchrony, to name a few) can be highly dangerous [5][14][22].

1. Targeted Model Poisoning at Edge Nodes: Edge nodes that could be realized through low-power bedside gadgets or single-board computers are physically accessible and hence the weakest link. In case an attacker manages to get control of one or more edge nodes either by using physical attacks, malware insertion, or insider attack techniques, he/she will have the capability of controlling the local training phase. Some of the ways of manipulating this training phase include flipping the labels, for instance, labeling arrhythmia events as "normal," or even introducing a backdoor, whereby certain trigger signals (e.g., some subtle noise on the ECG signal) lead to alert suppression or false negatives [14][24][35][36].

These kinds of attacks would be even more harmful when we consider the Smart ICU paradigm because they can be very stealthy; the global model would still maintain its near-optimal performance on regular data while showing poor results on some specific high-risk physiological signatures. An example of this kind of attack could be a backdoored anomaly detection algorithm which fails to detect ventricular tachycardia in patients with a specific comorbidity signature, resulting in a delay in defibrillation or even missing a code blue. It has recently been shown that a small number of compromised nodes (5% to 10%) are enough to inject backdoor attacks into non-IID health care data due to the distribution shift among patients [37][38].

2. Byzantine Manipulation at Fog Layer:

Fog servers that undertake partial aggregation inside wards or groups of hospitals will worsen the effect of such attacks since even one infected fog server affects all connected edge servers at once. Byzantine attackers could use such methods as flipping the sign of the gradient values, scaling by malicious multipliers, or adding malicious noise to partial aggregates so that the global model would be biased without raising doubts about individual updates [15][25][28].

In contrast to edge poisoning attacks, fog-layer attacks utilize the hierarchical setup for the advantage of being more efficient in terms of latency, in the context of ICU monitoring systems. An attacker who owns a fog node can modify the decision boundaries of the model used for the prediction of sepsis cases and respiratory instability of patients in the ward, resulting in either low detection of sepsis or excessive triggering alarms in the whole ward that overwhelm the clinicians. The bias introduced by such manipulation is hard to distinguish since incomplete aggregates seem statistically reasonable by themselves. Hierarchical Federated Learning research reveals that robust aggregation functions (Krum, trimmed mean, etc.) usually underperform in non-IID time-series cases due to physiological differences of patients [25][34].

- **Gradient Inversion Attacks**:

Gradient inversion attacks, also referred to as gradient reconstruction attacks, undermine the privacy protection provided by federated learning through the use of gradient information exchanged between edge or fog nodes and the global server. Gradient inversion techniques can achieve very accurate reconstructions of sensitive physiological time-series data, which include ECG waveform, arterial blood pressure, or SpO_2 patterns and can reveal identity information, comorbidities, or medical treatment of patients even in cases when no data leaks from the clinical facility [17][18][07].

Gradient inversion attacks are especially effective for time-series models with recurrent neural network structures (e.g., LSTMs), since the reconstruction of time-series traces can potentially lead to further secondary attacks, such as crafting better poisoning triggers for certain patient groups. Recent attacks on gradient inversion of time-series data are becoming even more dangerous through the use of quantile prediction and customized losses to achieve higher-quality reconstructions [0]. The clinical ramifications of these threats are serious: disclosure of physiological data leads to patient privacy violations (in terms of HIPAA/GDPR), which in turn can lead to health insurance discrimination or targeted attacks against patients. Differential privacy should be cautiously applied due to the adverse impact on time-series data utility.

- **Free-Rider Exploitation**:

Lazy (or free-rider) attacks entail nodes that contribute either null, random, insufficiently informative or even disguised contributions (such as scaled copies of previous global models or low-variance noise) and yet take advantage of the group intelligence built up by the honest nodes [19][10][11]. In the resource-limited IoMT environment where the computation budget limits the monitoring quality, lazy nodes will reduce the convergence rate, induce some biases over multiple rounds, and waste communication/battery energy at edge devices.

More sophisticated free riders could avoid easy detection by imitating an honest player for the first few rounds or employing frequency patterns that change weight over time to be plausible [11][12]. In the context of a federated learning system among various institutions for the Smart ICU application, free-riding hospitals will consume high-quality global models trained on data contributed by their better-endowed partners without providing any useful local update, which will induce fairness problem. This attack is less damaging than poisoning/inversion attacks but nonetheless significant.

The above four attack surfaces are not exclusive; an advanced attacker may take advantage of any combination of them (such as using gradient inversion techniques to craft sneakier backdoor attacks or coordinating the efforts of Byzantine fog nodes along with free-riding participants). The non-IID property of ICU datasets, the tight latency constraint, and the continual learning nature of Smart ICU models make it even more challenging to detect and prevent such attacks [5][21][26]. Our defense approach is specifically designed for each attack surface.

4. Proposed Defense Architecture

The system is based on three design principles: computational tractability on edge/fog computing nodes, maintenance of clinical performance (>97% F1 after attack) and modularity with typical FL workflows [5][28].

Algorithm 1: Adaptive Robust Aggregation with Temporal Client Verification (per round)

Input: Client updates {U_i | i=1 to N}, warm-up statistics, thresholds τ (temporal), β (contribution), window k
Output: Global update Δ_global

1. For each client i:
   Compute norm ||U_i|| and direction
   Apply adaptive clipping using warm-up distribution
   Check temporal consistency with rolling history
   If deviation > τ: w_i ← α · w_i (soft penalty, 0 < α < 1)
   Compute contribution score s_i = cosine_similarity(U_i, expected_grad from validation set)
   If s_i < β over last k rounds: further reduce w_i
2. Aggregate: weighted coordinate-wise median (or trimmed mean) on clipped & weighted updates
3. Return Δ_global

4.1 Design Principles

Defense architecture developed in this work is based on three main principles which account for particular limitations of IoMT clinical systems deployed in fog environments. The first one is computational feasibility, meaning that the defenses should be deployable by the computing resources available at the fog nodes and, if needed, edge devices without affecting the real-time performance through latency added. The second principle is the requirement not to compromise the clinical performance – acceptable performance loss due to attacks in terms of decrease of anomaly detection rate is specified as a target for the architecture to limit the loss to no more than 3% in F1-score compared to non-attacked baseline.

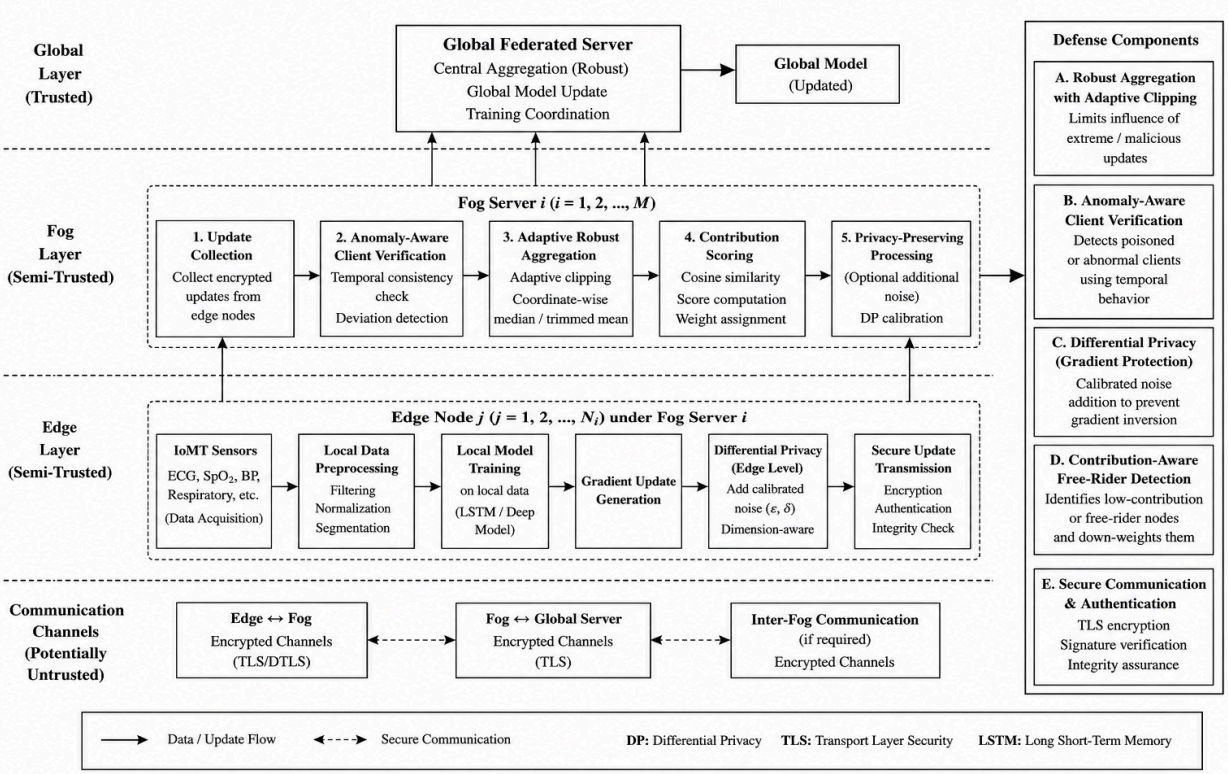


Fig. 2. Proposed Defense Architecture.

Figure 2: Proposed Defense Architecture

4.2 Robust Aggregation with Adaptive Clipping

The fundamental aggregation algorithm builds on top of the coordinate-wise median approach, introducing a dynamic gradient clipping algorithm tailored to the non-IID properties of the distributions of physiologic data in IoMT systems. Coordinate-wise median theoretically resists Byzantine attacks but presupposes symmetric gradient distributions, an unrealistic assumption in many scenarios where the patients' physiology varies due to the different edge nodes serving distinct clusters of patients. This is a realistic expectation in intensive care unit (ICU) settings where there is significant diversity among the types of cases handled by each cluster of sensors and monitors.

4.3 Anomaly Aware Client Verification

The light-weight client validation component is situated at the fog aggregation layer and tracks the behavior history of gradient update statistics corresponding to each registered

edge node. The validation component performs a time-based consistency test which predicts the trajectory of expected gradients for each edge node through the rolling window statistics of past gradient updates. Updates deviating significantly from these trajectories are marked as soft excluded down-weights during the aggregation step rather than being hard-excluded to avoid excluding any legitimately distributed shifts due to varying patient acuity levels. Notably, the behavior prediction is performed in the gradient statistics domain rather than the gradient domain itself to maintain computational feasibility while retaining sensitivity to the evolving poison strategies that are the most damaging in such scenarios.

Algorithm 1BRAF — Byzantine-Resilient Adaptive Federated Aggregation

Input: $G=\{g_1,\ldots,g\square\}$ gradients; H historical stats; V validation set; τ clip threshold; ε,δ DP params; λ consistency threshold
Output: Δw global model update
forkw2
// Phase 1 — Adaptive gradient clipping
$\hat{g}_i \leftarrow g_i / \max(1, \|g_i\|_2 / \tau)$
// Phase 2 — Calibrated differential privacy
$\sigma_i \leftarrow 2\tau \cdot \sqrt{(2 \ln(1.25/\delta))} / \varepsilon$
$\tilde{g}_i \leftarrow \hat{g}_i + (0, \sigma_i^2 I)$
// Phase 3 — Contribution scoring
$s_i \leftarrow \text{CosSim}(\tilde{g}_i, \nabla L(w; V))$
end for
// Temporal consistency check
forkw2
$\mu_i, \Sigma_i \leftarrow$ rolling mean, std of H[i] over window W
ifkw1
$w_i \leftarrow 0.3 \cdot s_i$
// downweight Byzantine-suspected client
else
$w_i \leftarrow s_i$
// retain full weight for honest client
end if
$H[i] \leftarrow H[i] \cup \{\tilde{g}_i\}$
// update rolling history
end for
// Robust aggregation via coordinate-wise median
$G_filtered \leftarrow \{ \tilde{g}_i \cdot w_i : w_i > \theta_min, \forall i \}$
$\Delta w \leftarrow \text{CWMedian}(G_filtered)$
return

4.4 Calibrated Differential Privacy for Gradient Protection

In order to mitigate issues related to gradient inversion attacks, the proposed framework utilizes differential privacy protection in the edge nodes level. The traditional differential privacy using Gaussian mechanism is employed on the gradients before their transfer, with noise injection being controlled using an adaptive privacy budget allocation strategy that focuses on allocating more privacy expenses on gradient components that have higher reconstruction risk; in particular, the gradient components that correspond to the final layers of the local model have been shown by inversion attacks literature to be highly informative about the training dataset.

4.5 Contribution-Aware Free-Rider Detection

Detection of free-riding is done via a contribution scoring system which takes into account the quality of the information content of the gradient updates in comparison with the current state of the global model. Nodes providing update vectors whose cosine similarity score in relation to the expected gradient vector from a small validation set stored at the fog server is below a threshold value over a period of training iterations get reduced contribution weight and will undergo further administrative scrutiny if this condition persists. The use of the validation set is associated with minimal overhead from the data governance standpoint but follows the standard data assurance practices in the clinical informatics infrastructure.

5. Experimental Evaluation

5.1 Experimental Setup

Evaluation was done via an implementation of the federated IoMT Smart ICU environment on the MIMIC-III dataset (Johnson et al., 2016), distributed over 20 edge nodes simulated with non-IID distribution of patients that represent heterogeneous ICU cases. The global anomaly detection model is based on a multi-task LSTM network that is similar to models from the IoMT anomaly detection literature, trained for 100 rounds with the number of local epochs being 5 in each round. The Byzantine attack was simulated by specifying some portion of edge nodes to be malicious and performing two

types of attacks—model poisoning via label flipping for a certain arrhythmia category and gradient manipulation by sign flipping of updates.

5.2 Results and Discussion

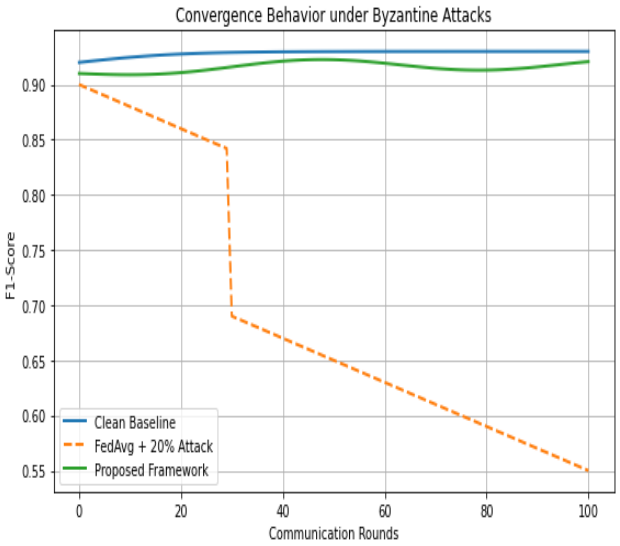
FedAvg performs F1 ≈ 0.923 in a clean setting. In the presence of 20% Byzantine nodes, FedAvg performance is reduced to 0.741 (not acceptable from a clinical point of view), while ours still keeps 0.901 (only 2.4% drop in performance). For 30% of Byzantine participation ratio, our technique preserves 0.897. Our verification procedure renders targeted label flipping attacks ~67% less effective. The cost of DP is small (clean F1≈ 0.908 at ε=1.0)

Table 1: Anomaly Detection F1-Score Under Varying Byzantine Fractions (Mean ± Std over 5 runs)

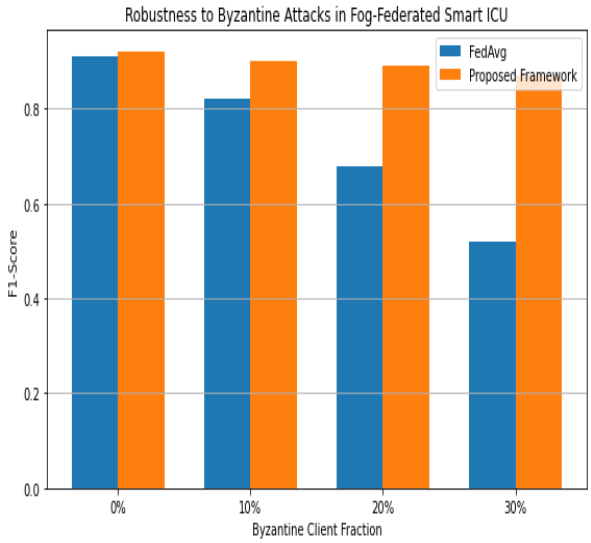
Method	**0% (Clean)**	**10% Byzantine**	**20% Byzantine**	**30% Byzantine**
FedAvg (Baseline)	0.923 ± 0.008	0.852 ± 0.012	0.741 ± 0.015	0.623 ± 0.021
Median Aggregation	0.918 ± 0.009	0.887 ± 0.011	0.812 ± 0.014	0.734 ± 0.018
Proposed Framework	**0.908 ± 0.010**	**0.904 ± 0.009**	**0.901 ± 0.010**	**0.897 ± 0.011**

1. **Convergence Curves:** Graph line between global F1-score and number of communication rounds (0-100). Plot three curves: Baseline with clean data (high and steady), FedAvg with 20% attacks (drop sharply after attacks), Proposed Framework (close to baseline with clean data and minimal oscillations).

2. **Plot for Robustness:** Bar graph between F1-score (%) or (% Degradation) and Byzantine fraction (0%, 10%, 20%, 30%). Bars of proposed framework stay flat; bars of FedAvg drop sharply.



Graph 1: Convergence Curves (F1-score vs Communication Rounds)



Graph 2 : Robustness Plot (F1-score vs Byzantine Fraction)

6. Discussion

The fog federated learning model exposes temporal vulnerabilities to the problem of poisoning through accumulation of biases due to continuous updates, possibly bypassing clinical monitoring in the short term [8][24]. The current work has some limitations like the fact that it uses simulations done on MIMIC-III which is not able to simulate the full hardware heterogeneity and moderate overhead caused by the trusted validation set [11][23][28].

7. Conclusion

The reliability of the intelligence of a Smart ICU is no better than that of the security of the learning process itself. In this paper, it has been demonstrated that the issue of federated IoMT adversarial robustness in the clinical context is not an abstract technical question, but rather a medical necessity driven by the fast development of federated healthcare architectures, which has been outrunning the research in this field. With the use of the outlined defense framework, comprised of adaptive robust aggregation, anomaly awareness-based client validation, targeted differential privacy, and contribution-aware free-rider detection, it has been shown that maintaining clinically relevant anomaly detection accuracy even in the presence of adversarial attacks is

possible while preserving the privacy and latency advantages of fog-federated architectures. The road to secure medical AI in intensive care goes through not only technical innovations but also adversarial considerations of the systems being developed and threat surfaces, until now largely left unexplored.

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An Intelligent IoT-Blockchain Integrated Framework with Hybrid Anomaly Detection for Smart Healthcare

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Abstract: The increasing adoption of the Internet of Things (IoT) in healthcare has significantly improved continuous patient monitoring and supports data-driven clinical decisions. At the same time, it results in the creation of massive volumes of sensitive medical data. Furthermore, significant challenges related to data security, privacy, and system scalability have also been introduced. This study presents a robust and scalable hybrid healthcare architecture that integrates IoT, blockchain, and artificial intelligence, and employs a neural-enhanced encryption mechanism that combines the Elman Neural Network with Blowfish encryption to ensure reliable management of sensitive medical data. A hybrid anomaly detection model using Long Short-Term Memory (LSTM) autoencoder and Support Vector Machine (SVM) is implemented to identify malicious activities and abnormal behavior within encrypted data streams. Furthermore, a customized Proof-of-Authority (PoA) consensus mechanism is adopted to enable fast, energy-efficient, and low-latency transaction validation in a permissioned healthcare environment. The proposed framework has been evaluated using a simulated IoT healthcare setup with multiple configuration networks consisting of 150, 600, and 1200 nodes, and a dataset of 1200 patient records. The experimental findings indicate an anomaly detection accuracy of 98.99% with an F1-score of 0.91 and up to 95% attack resistance. The use of a customized Proof-of-Authority consensus mechanism ensures low authentication delay and improved computational efficiency. Comparative analysis with existing approaches confirms the effectiveness of the proposed system in enhancing data integrity, computational efficiency, and scalability. The framework provides a robust and reliable solution for secure, real-time healthcare data management, making it suitable for large-scale healthcare applications.

Keywords:*Internet of Things (IoT); Blockchain; Smart Healthcare; Anomaly Detection; Machine Learning; Data Security*

1. Introduction

Recent developments in smart healthcare technologies have considerably reshaped how medical services are delivered, monitored, and managed [1]. The growing adoption of digital healthcare systems, remote monitoring, and personalized medicine has created new opportunities for improving efficiency in patient healthcare [2]. However, despite these advancements, traditional healthcare systems face several challenges, such as centralized data management, lack of real-time monitoring, limited interoperability, and vulnerability to cyber threats. These issues become more critical with the increasing reliance on data-driven healthcare solutions [1].

The integration of the Internet of Things (IoT) into healthcare has enabled continuous and real-time monitoring of patient health through wearable devices and smart sensors [3]. These devices produce diverse and time-dependent healthcare data at a large scale, including vital parameters such as heart rate, blood pressure, oxygen levels, and body temperature [4]. While IoT enhances accessibility and automation in healthcare, it also introduces significant concerns related to data privacy, security, and trust. Patient data transmitted across networks and stored in centralized architectures remains highly exposed to security threats and unauthorized access [5].

To address these challenges, blockchain technology has emerged as a promising solution for secure and decentralized data management in healthcare systems. Blockchain technology offers a decentralized and tamper-resistant ledger system where transactions are recorded transparently and cannot be altered once validated [6]. This facilitates data integrity, traceability, and trust among multiple stakeholders,including patients, healthcare providers, pharmacies, and insurance companies. Additionally, the use of cryptographic techniques and consensus mechanisms enhances security and eliminates reliance on centralized authorities. This also reducesthe risk of single-point failures [7].

Despite the advantages of IoT and blockchain integration, several limitations persist in existing solutions. Most frameworks primarily focus on secure data storage and access control, but neglect intelligent threat detection and adaptive mechanisms of security [8]. Furthermore, the increasing volume and complexity of healthcare data make real-time processing, scalability, and efficient anomaly detection a difficult task. Additionally,

traditional security mechanisms often fail to detect evolving cyber-attacks indynamic IoT environments [9].

Artificial intelligence (AI) has the potential to address these limitations by enabling intelligent data analysis, pattern recognition, and anomaly detection [10]. Machine learning and deep learning models can learn complex temporal patterns in healthcare data and identify abnormal behaviors that may indicate cyber threats or system failures. However, integrating AI with blockchain and IoT in an efficient manner remains a challenging task [11].

In this context, this research proposes a secure and scalable hybrid healthcare framework that integrates IoT, blockchain, and artificial intelligence to address the limitations of existing systems. This integration has been presented in Figure 1. The Organization#1 develops IoT-enabled healthcare sensors that interact with patients using IoT protocols and help in collecting real-time patient health data, including heart rate, temperature, oxygen level, etc. These sensors send the collected data to cloud services for storage using web service APIs that act as a bridge between sensors and the cloud. These web services are managed by Organization#2. Another Organization#3 provides cloud storage to store patient data securely. They also host various services for data access. The collected data is then analyzed using artificial intelligence to extract useful insights that help in predicting diseases and monitoring health. The artificial intelligence algorithms are provided by Organization#4. These algorithms analyze patient data to detect patterns and generate insights. This integrated system ensures secure data sharing, automation, and intelligent healthcare decision-making.

The proposed framework incorporates a neural-enhanced encryption mechanism that combines the Elman Neural Network with Blowfish encryption to provide adaptive and robust data protection. Additionally, a hybrid anomaly detection model based on Long Short-Term Memory (LSTM) and Support Vector Machine (SVM) is employed to detect malicious activities and abnormal patterns in healthcare data.

To ensure efficient and low-latency transaction validation, a customized Proof-of-Authority (PoA) consensus mechanism is adopted, where domain-specific entities such as hospitals, laboratories, and pharmacies act as trusted validators. This

approach enhances performance, reduces computational overhead, and facilitates secure participation within a permissioned blockchain environment. The proposed framework is evaluated using a simulated IoT-based healthcare dataset and demonstrates improved performance in terms of security, anomaly detection accuracy, computational efficiency, and scalability.

Overall, this work contributes toward the development of a robust, intelligent, and secure healthcare ecosystem that supports real-time monitoring, reliable decision-making, and trustworthy data management in modern smart healthcare environments.

Fig. 1. IoT-enhanced Hybrid Healthcare System

1.1 Significance of Research

The integration of blockchain and Internet of Things technologies has emerged as an efficient solution to address challenges in modern healthcare systems. The decentralized and immutable characteristics of blockchain significantly strengthen data security and integrity. It ensures that the medical data cannot be altered or deleted once it is recorded on the blockchain. This minimizes the possibility of data manipulation and unauthorized

usage and thus improves the reliability of healthcare information systems. Additionally, the use of blockchain reduces administrative overhead and operational costs by eliminating intermediaries and automating processes through smart contracts.

Figure 2 highlights how IoT and blockchain work together to securely manage healthcare data and logistics between different hospitals. Smart wearable devices and monitoring systems equipped with IoT sensors continuously capture real-time patient information. This data, along with logistics information, is securely stored on decentralized and distributed blockchain nodes. The logistics information contains transport and supply chain data, such as medicine delivery, equipment transfer, and organ transport. Every blockchain node securely records transactions, verifies their authenticity, and keeps a synchronized copy of the ledger. The patient data is safely shared between multiple healthcare centers. Thus, this integrated framework ensures decentralization by avoiding a single point of failure. The encryption and decryption of data ensure security. Furthermore, the framework ensures transparency by making all the transactions traceable. Additionally, hospitals can share data easily, ensuring interoperability. Real-time access assists in faster decision-making. The framework also ensures a trustworthy environment and enhances compatibility between different healthcare systems.

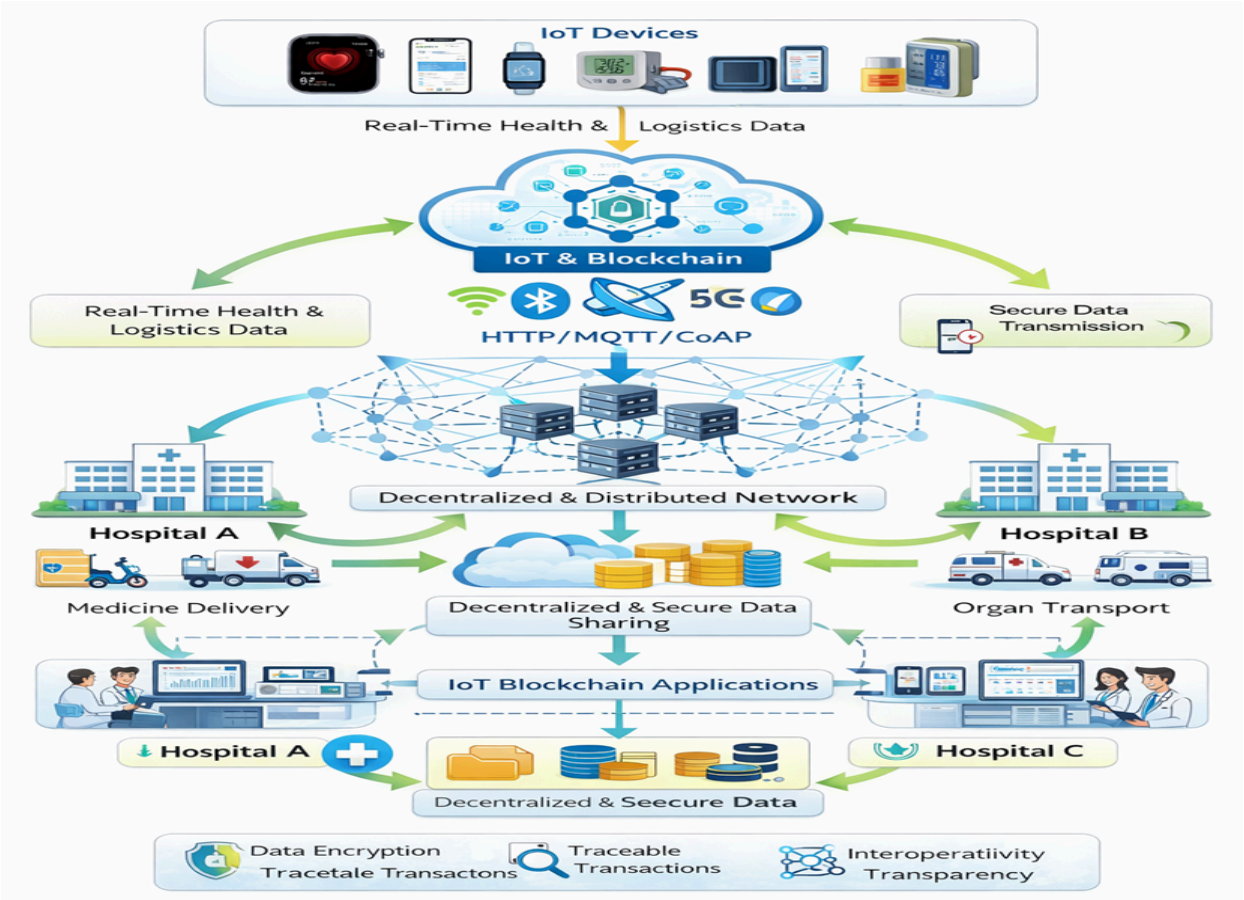

Fig. 2. IoT-Blockchain Integrated Hybrid Framework for Healthcare Data Management

Despite these advancements, existing healthcare systems face several challenges, including data privacy risks, cyber-attacks, and redundant records. Furthermore, the increasing volumes of multimedia healthcare data generated by IoT devices further make secure data storage and real-time processing more complex. These limitations highlight a need for a robust, secure, and intelligent framework that can efficiently manage healthcare data while ensuring privacy, scalability, and reliability.

1.2 Contribution of Work

The major contributions of this research work have been summarized as follows:

- A secure and scalable hybrid healthcare framework has been proposed that integrates Internet of Things, blockchain technology, and artificial intelligence to ensure efficient and reliable healthcare data management.

- A novel neural-enhanced encryption mechanism has been introduced by combining the Elman Neural Network with Blowfish encryption, which enables adaptive and robust protection of sensitive healthcare data.
- A hybrid two-stage anomaly detection model has been developed using Long Short-Term Memory (LSTM) and Support Vector Machine (SVM), which effectively detects abnormal and malicious behavior within encrypted blockchain data.
- A customized Proof-of-Authority consensus mechanism has been utilized that is specifically designed for healthcare environments. It allows faster transaction validation, reduced latency, and improved energy efficiency.
- The framework has been evaluated using a synthetic IoT-based healthcare dataset with 1200 records to preserve healthcare data privacy.
- A statistical and quantitative comparative analysis with existing approaches under a similar domain has been performed, which demonstrates its improved performance in terms of security, anomaly detection accuracy, computational efficiency, and scalability.

Table 1 provides a comparative analysis of the proposed framework with existing ones, highlighting its novel contributions.

Table 1. Comparative Analysis

Feature	Traditional Approach	Proposed Framework	Contribution of Work
Encryption Process	Standard encryption methods, such as Elliptic Curve Cryptography/Advanced Encryption Standard/Blowfish, are used separately without integration.	Integrates Elman neural-enhanced Blowfish encryption with blockchain-based transactions.	Adds intelligence to encryption, making it more adaptive and secure.
Static/Dynamic Encryption	Use static encryption settings.	Encryption is dynamically adjusted using neural networks.	Improves robustness.
Threat Detection	Use one-step detection on raw or semi-processed data.	Use a two-step detection process on encrypted blockchain data patterns.	Detects threats more effectively.

Blockchain Scope	Use blockchain for storing and controlling access to data.	Use blockchain as an active security mechanism for tracking, validating, and resisting attacks.	Blockchain allows an active defense mechanism and not just a storage database.
Consensus Algorithm	Use a general non-health-based consensus algorithm.	Use a healthcare-specific, customized Proof-of-Authority consensus algorithm for faster and more efficient healthcare applications.	Customized consensus algorithm.
Evaluation Criteria	Tested with artificial or limited data.	Tested with realistic healthcare data and threat models.	More realistic and relevant evaluation.

2. Literature Review

This section reviews previous research on the application of IoT and blockchain in the healthcare sector. Researchers have studied the benefits of these technologies, especially in the management of multimedia data related to healthcare.

BlockMedCare, developed by Azbeget al. (2022, is an existing integrated system for healthcare that uses IoT, blockchain, and InterPlanetary File System (IPFS) for secure monitoring of patients, especially for chronic diseases. The system used Internet of Things devices to obtain health information promptly. It used blockchain to make sure that information is handled transparently. IPFS is used to store information and helps to reduce the amount of work that the blockchain network has to do. BlockMedCare helped to keep health information safe and private. It also made it easier to access this information by the authorized stakeholders. Evaluation results highlighted that BlockMedCare works better and can handle information more efficiently than other systems. The study showed that using these technologies together can be a good way to keep healthcare information safe.

Another framework, named BCHealth, developed by Hossein et al. (2021, is a blockchain-based system that helps keep data private in healthcare systems and uses the Internet of Things technology. This system uses blockchain and encryption to protect data while still allowing access to it. BCHealth tackled issues like keeping data confidential, making sure data is not altered, and sharing data safely among different people involved

in healthcare. Evaluation results showed the cost efficiency of BCHealth. This model proved to be helpful,especially for healthcare applications that need to monitor patients promptly.

Teli&Masoodi, 2021, and Quasim et al., 2020, discussed the usage of blockchain technology in the healthcare system to enhance the security and privacy of medical data. The work highlighted that blockchain is a decentralized database and does not need the intervention of any third-party intermediary, thus reducing the data security risks. Blockchain ensured data integrity, transparency, and authorized access to medical data. Furthermore, cryptographically generated hash addresses and smart contracts ensured safe data sharing among the stakeholders. Both studies focused on the potential of blockchain in transforming traditional healthcare data management systems into a more secure and trustworthy environment.

Chinaeiet al. (2021) introduced a blockchain logging contract mechanism for efficiently analyzing IoT-generated healthcare data. The work focused on improving the verification and trustworthiness of data in a distributed healthcare environment. It proposed efficient logging strategies to reduce the need for blockchain for storage and thus reduce the computational overhead. The framework also ensured the immutable storage of IoT-generated healthcare data. Evaluation of the framework demonstrated its scalability for large-scale IoT healthcare deployments.

Mohammed et al. (2021), Al-Marridi et al. (2024 and Al-Nbhany et al. (2024)discussed the integration of blockchain and IoT technologies to improve the security, efficiency, and reliability of healthcare systems and addressed the challenges such as unauthorized access and data breaches. The studies presented the importance of a decentralized database for allowing secure data communication among IoT-enabled healthcare devices. Furthermore, Al-Marridi et al. (2024) extended this integration by utilizing reinforcement learning techniques to optimize system performance, scalability, and resource allocation. Mohammed et al. (2021) focused on strengthening IoT-based healthcare infrastructure using blockchain, and Al-Nbhany et al. (2024) provided a comprehensive review of current trends, applications, and challenges in this domain. Collectively, these works

presented the growing importance of integrating blockchain, IoT, and intelligent techniques for developing secure and efficient healthcare solutions.

3. Proposed Framework

Traditional healthcare systems suffer from a major issue of managing and tracking data. This is because medical-related records are stored centrally across multiple organizations. To overcome this issue, the proposed framework integrates blockchain, Blowfish encryption based on the Elman Neural Network, and anomaly detection to ensure secure, transparent, and decentralized handling of medical data. Figure 3 highlights the overall architecture of the proposed framework.

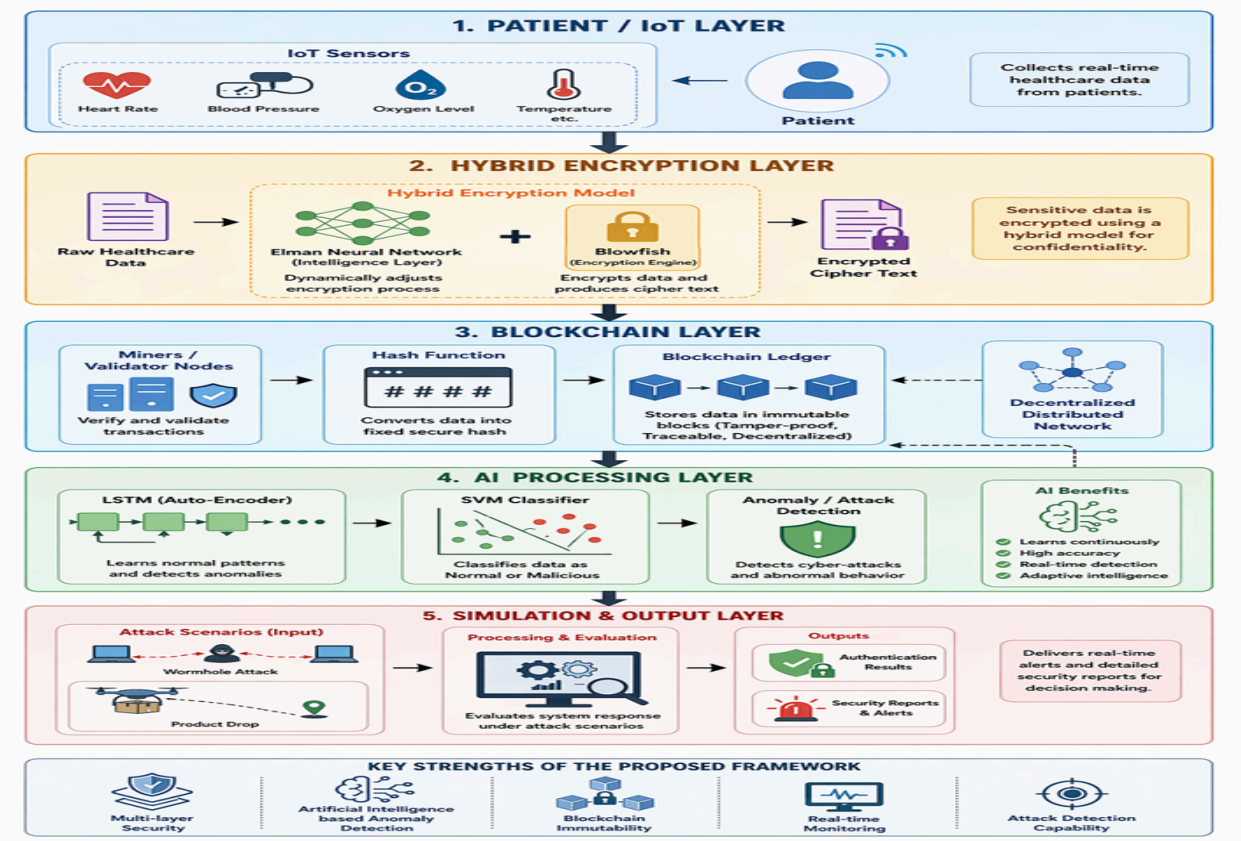

Fig. 3. Architecture of the Proposed Framework

As presented in Figure 3, the patient layer contains IoT sensors that collect real-time patient healthcare data, such as heart rate, blood pressure, oxygen level, etc. The collected healthcare data is first encrypted using a hybrid encryption scheme before being stored securely on the blockchain. This hybrid encryption model consists of two components,

namely, the Elman Neural Network, which acts as an intelligence layer that adjusts the encryption process dynamically. The second component is Blowfish, which performs the actual encryption and converts data into a secure ciphertext. The blockchain layer consists of miners or validator nodes that verify transactions, a hash function that converts data into a fixed, secure hash value, and a ledger that stores data in immutable blockchain blocks. As a result, the data becomes secure, traceable, and resistant to tampering within a decentralized environment. The artificial intelligence processing layer consists of two artificial intelligence models, namely, Long Short Term Memory (LSTM), which acts as an auto-encoder and learns normal patterns to detect anomalies, and Support Vector Machine (SVM), which classifies data as normal or malicious. Furthermore, LSTM is capable of capturing time-based patterns, which is very important for IoT-generated healthcare data. Overall, this layer detects cyber-attacks and any sort of abnormal behavior observed. Next is the simulation and output layer that uses key security metrics, such as wormhole attack rate and product drop ratio, to generate authentication results, security reports, and alerts as outputs.

The key strengths of the proposed framework are:

- Multi-layer security
- Artificial intelligence-based anomaly detection
- Blockchain immutability
- Real-time monitoring
- Attack detection capability

The stakeholders and data communication among them have been presented in Figure 4. The figure shows that different participants, such as hospitals, doctors, and patients, act as blockchain nodes that enable secure and immutable storage of records via the SHA-256 algorithm. The figure highlights how patient data flows between doctors, pharmacies, insurance, and devices. Here, patients are given full access and control over their own data. Blockchain ensures immutability, while SHA-256 facilitates data integrity.

As shown in Figure 4, the healthcare process starts with patient X visiting doctor Y. Doctor send the prescription transaction to the pharmacy and billing details to the insurance company. These are recorded immutably and securely as blockchain transactions (Tx) with proper time stamping. On the other side, smart connected IoT devices generate patient data. The data received from these devices is also stored using SHA-256 hashing on the blockchain ledger. SHA-256 hashing ensures data integrity and tamper detection. Furthermore, patient data is saved from free accessibility, and ownership is controlled through blockchain, where only authorized entities can access data. The pharmacy receives the prescription from the doctor and possible data from smart devices via blockchain. An insurance company also receives billing information via blockchain to verify claims.

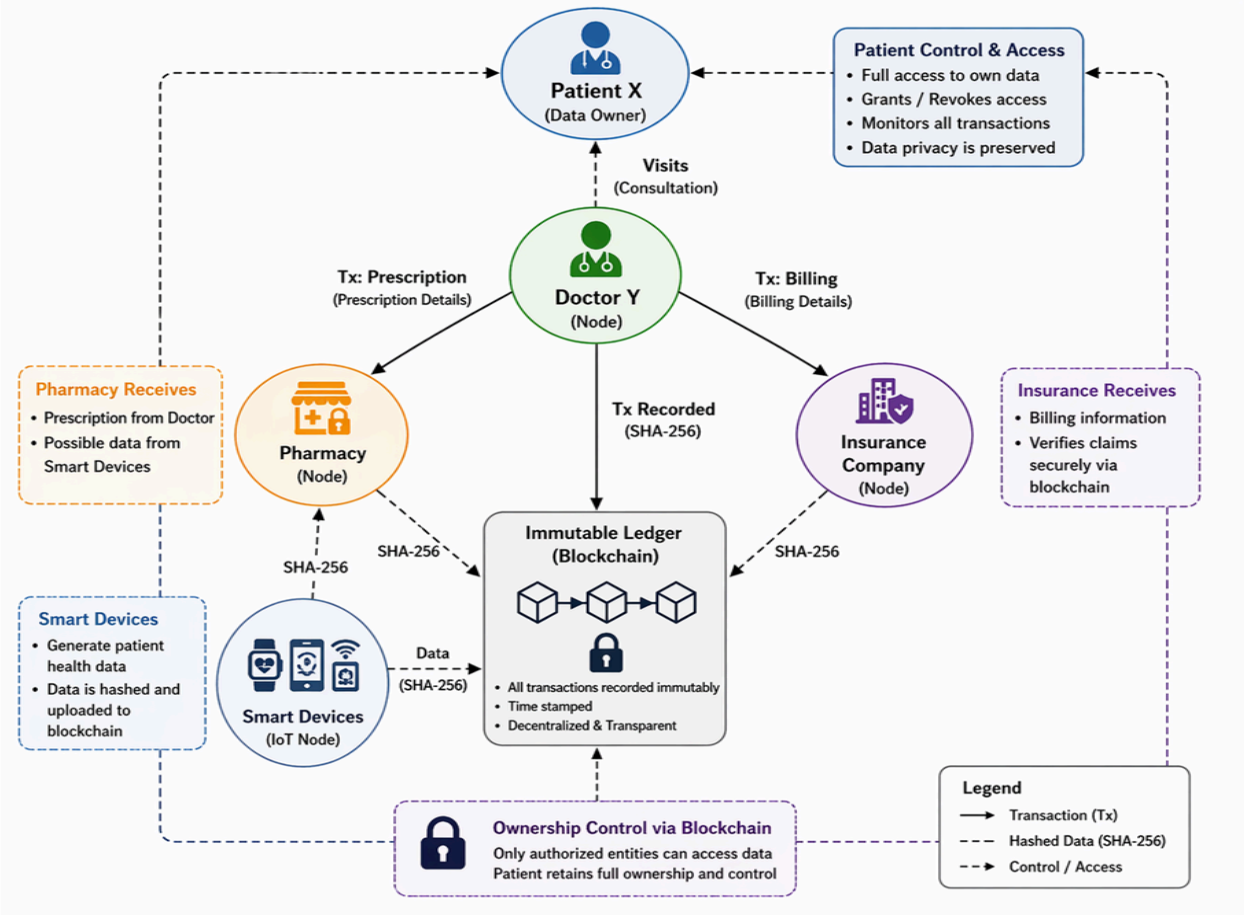

Fig. 4. Blockchain-IoT Integrated Healthcare Data Flow

3.1 Dataset

The framework uses an artificially generated dataset that mimics the real healthcare monitoring environment to conduct experimentation. This dataset contains 1200 simulated patient records that have been generated using realistic medical studies and are not the actual patient data. Each dataset entry consists of health-related attributes typically captured through IoT-based sensing devices. These parameters have been discussed as follows:

- **Heart rate values:** They are generated using a Gaussian distribution and are centered on typical resting heart rates. Heart rate is measured as beats/minute.
- **Blood pressure values:** They are also generated using a Gaussian distribution and highlight typical systolic ranges. Blood pressure is measured in mmHg units.
- **Body temperature values:** They are created close to normal human temperature with slight variations. Temperature is measured in °C.
- **Oxygen level values:** They are generated to represent typical pulse oximeter readings and are measured in percentage.

Each record in the dataset is labeled as 'Normal' or 'Abnormal'. The dataset contains 85% normal records and 15% abnormal records to realistically simulate anomalies in the proposed healthcare monitoring framework. Additionally, a fixed random seed has been utilized so that every time the dataset can be generated in the same way. Moreover, since the records are artificial, there is no need to remove any sort of personal information.

3.2 Mathematical Symbols and Cryptographic Methods

This section explains the encryption technique, hashing process, and consensus mechanisms used in the proposed framework. Table 3 highlights mathematical symbols used further in this section.

Table 3. Mathematical notations used

Notation	Description
d	Input data d, either a transaction or a message

SHA-256(d)	Hash function
H(d)	Hash address of input data d
k	Random nonce
n	Small positive integer
G	Fixed predefined base public point on the elliptic curve
$\mathbf{x_1}$	x-axis of the elliptic curve point
$\mathbf{y_1}$	y-axis of the elliptic curve point
r	First half component of the digital signature
s	Second half component of the digital signature
$\mathbf{K_P}$	Private key of the sender
ω	Modular inverse of s
α	Message-dependent scalar
β	Signature-dependent scalar
C	Reconstructed elliptic curve point
$\mathbf{K_u}$	Public key of the sender

The SHA-256 hashing technique has been utilized to ensure that blockchain data remains immutable and secure. This has been shown mathematically in Equation 1. The hash generated is always a 256-bit value, regardless of the size of the input data.

$$H(d) = SHA\text{-}256(d) \tag{1}$$

The framework uses Elliptic Curve Digital Signature Algorithm (ECDSA) to generate, verify, and validate the accuracy and truthfulness of a transaction's digital signature. ECDSA is a widely used digital signature generation algorithm in blockchain systems, especially Bitcoin and Ethereum. Following steps and equations mathematically highlight the process of transaction digital signature generation using ECDSA.

Step 1: A random nonce k is selected between 1 and n-1, as shown in equation 2. Here n is a selected such that n.G reaches the point of infinity O. The value of k must be unique and secret for every signature. Reusing k can expose the sender's private key, which is a critical risk to system security.

$$k \in [1, n\text{-}1] \tag{2}$$

Step 2: Elliptic curve point (x_1, y_1) is calculated as shown mathematically in equation 3.

$$(x_1, y_1) = k.G \quad (3)$$

Step 3: The first half component of the final digital signature is computed as shown in equation 4. If r becomes zero, a new k is generated using step 1, and the process is repeated.

$$r = x_1 \bmod n \quad (4)$$

Step 4: The second half component of the final digital signature is computed as shown in equation 5. Here, k^{-1} is the modular inverse of k modulus n, such that $k.k^1 \equiv 1 \pmod n$.

$$s = k^{-1} (H(d) + K_P . r) \bmod n \quad (5)$$

Step 5: The final digital signature (r, s) is obtained from equations 4 and 5.

The following steps and equations mathematically highlight the process of transaction digital signature verification using ECDSA.

Step 6: The modular inverse of the transaction digital signature second component is calculated as shown in equation 6.

$$\omega = s^{-1} \bmod n \quad (6)$$

Step 7: Using the value obtained from step 1, two intermediate parameters are obtained as shown in equation 7.

$$\alpha = H(d) . \omega \bmod n \quad (7)$$

$$\beta = r. \omega \bmod n$$

Step 8: Reconstructed elliptic curve point C is calculated using equation 8.

$$C = (G . \alpha + K_u . \beta) \bmod n \quad (8)$$

Step 9: The transaction digital signature is valid if the condition shown in equation 9 is satisfied.

$$C = r \quad (9)$$

Figure 5 presents the flowchart of the transaction digital signature generation and verification process.

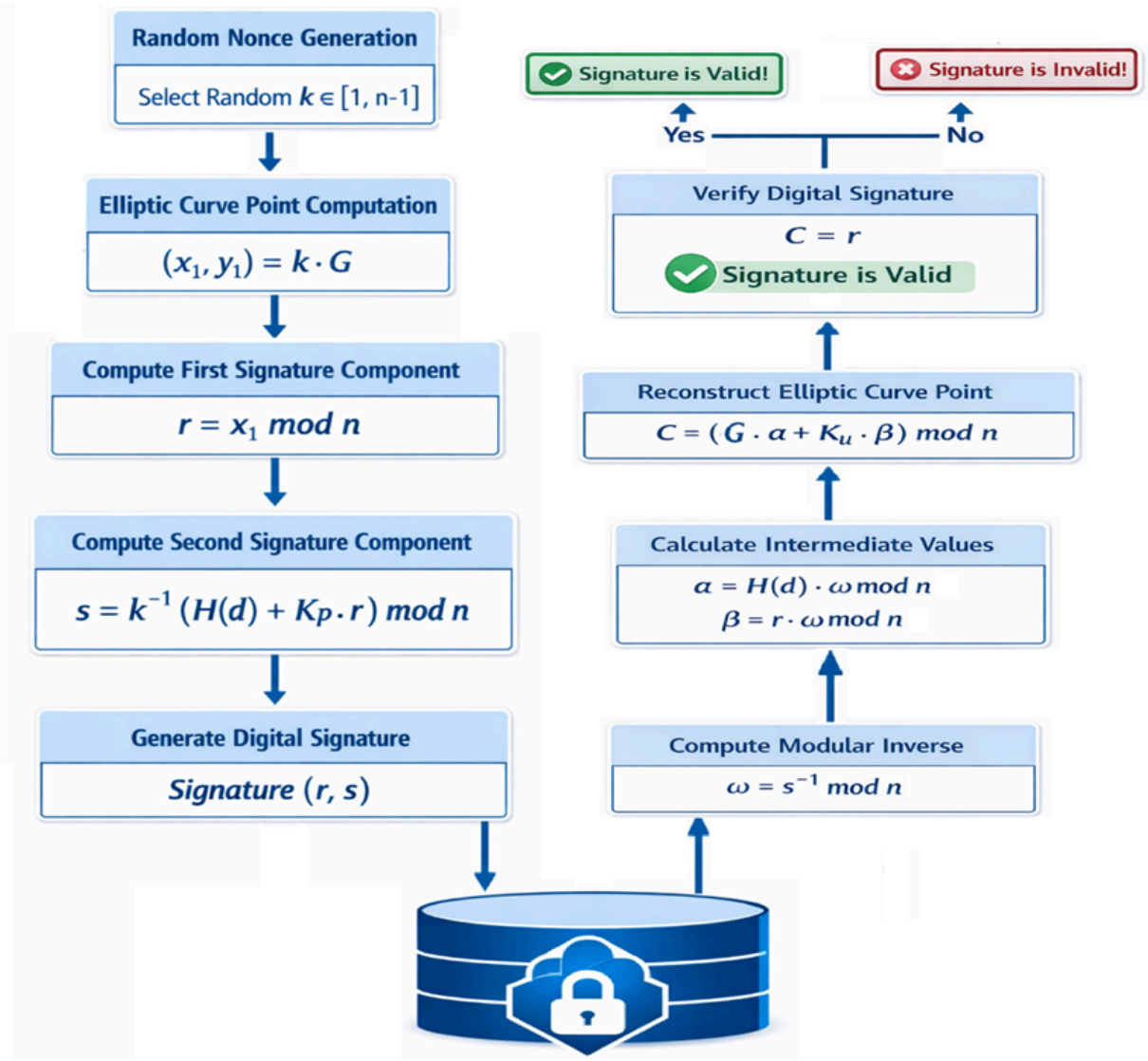

Fig. 5. Digital Signature Generation and Verification Process Flowchart

3.3 Consensus Mechanism

The framework uses a customized Proof-of-Authority (PoA) consensus mechanism to validate and verify transactions before storing them on the blockchain. PoA is highly suitable for this framework because here only authorized and authenticated participants are allowed to validate the transactions. Additionally, PoA is a faster and cost-efficient consensus mechanism as compared to other methods, such as Proof of Work and Practical

Byzantine Fault Tolerance. Thus,PoA could allow fast healthcare-related operations while observing rules and maintaining scalability.

Unlike in standard PoA, where only authorized validators can validate blocks, the proposed customized PoA consensus mechanism allows domain-specific validator selection. Here, validators are not random nodes, but are pre-approved regulated entities such as hospitals, pharmacies, labs, and insurance systems. Furthermore, this customized PoA is tightly coupled with Blowfish encryption, Elman Neural Network, SHA-256 hashing, and anomaly detection models, including LSTM and SVM.

Total time taken by PoA to validate a block is the sum of signature generation time for each transaction$T_i^{(tx)}$and block processing time $T^{(blk)}$, as shown in equation 10.

$$T_{val} = \sum_{i=1}^{n} T_{i^{(tx)}} + T^{(blk)} \quad (10)$$

With n as the total number of validating nodes in the network and m as the total number of malicious nodes, the probability that a randomly selected node is malicious is:

$$P_{mal} = \frac{m}{n} \quad (11)$$

To protect the patient data that has been stored off-chain, a symmetric encryption algorithm, namely Blowfish encryption, is used. It used a Feistel network structure, where data is split into two halves and is repeatedly processed through multiple rounds. It ensures strong security, with fast and efficient computation, especially for resource-constrained IoT systems. Here, ciphertext $C_t$is generated by encrypting plaintext$P_t$using private encryption key K_p as:

$$C_t = Blowfish(P_t, K_p) \quad (12)$$

3.4 Integration of Blockchain

The framework has two main components: first, the user interface designed for interaction among stakeholders, and second, the blockchain backend for secure data storage and processing. Different stakeholders of the framework, administrators, doctors, and patients, are provided with login IDs to use the framework securely. This multilevel authentication and encryption enhances confidentiality among patients as well as provides the integrity of data. There are two types of nodes in the blockchain network: the miner nodes and the executor nodes. Miner nodes conduct verification and validation of healthcare-generated transactions, such as prescriptions, medical test reports, and medicine bills. The executor nodes execute smart contracts and add records onto the blockchain after they are validated and verified.

The framework has been simulated using two virtual machines. The Virtual_Machine#1 acts as an executor blockchain node that manages patient-related smart contracts. Virtual_Machine#2 acts as a hybrid node that performs both mining tasks and execution tasks. These virtual machines simulate the real-world healthcare network, which involves components such as hospitals, pharmaceuticals, and insurance companies. The data communication between these components is done through a secure blockchain channel. Furthermore, the framework also includes real-time detection of anomalies and delays to ensure a secure, transparent, and traceable process. The blockchain integration within the proposed framework has been presented in Figure 6.

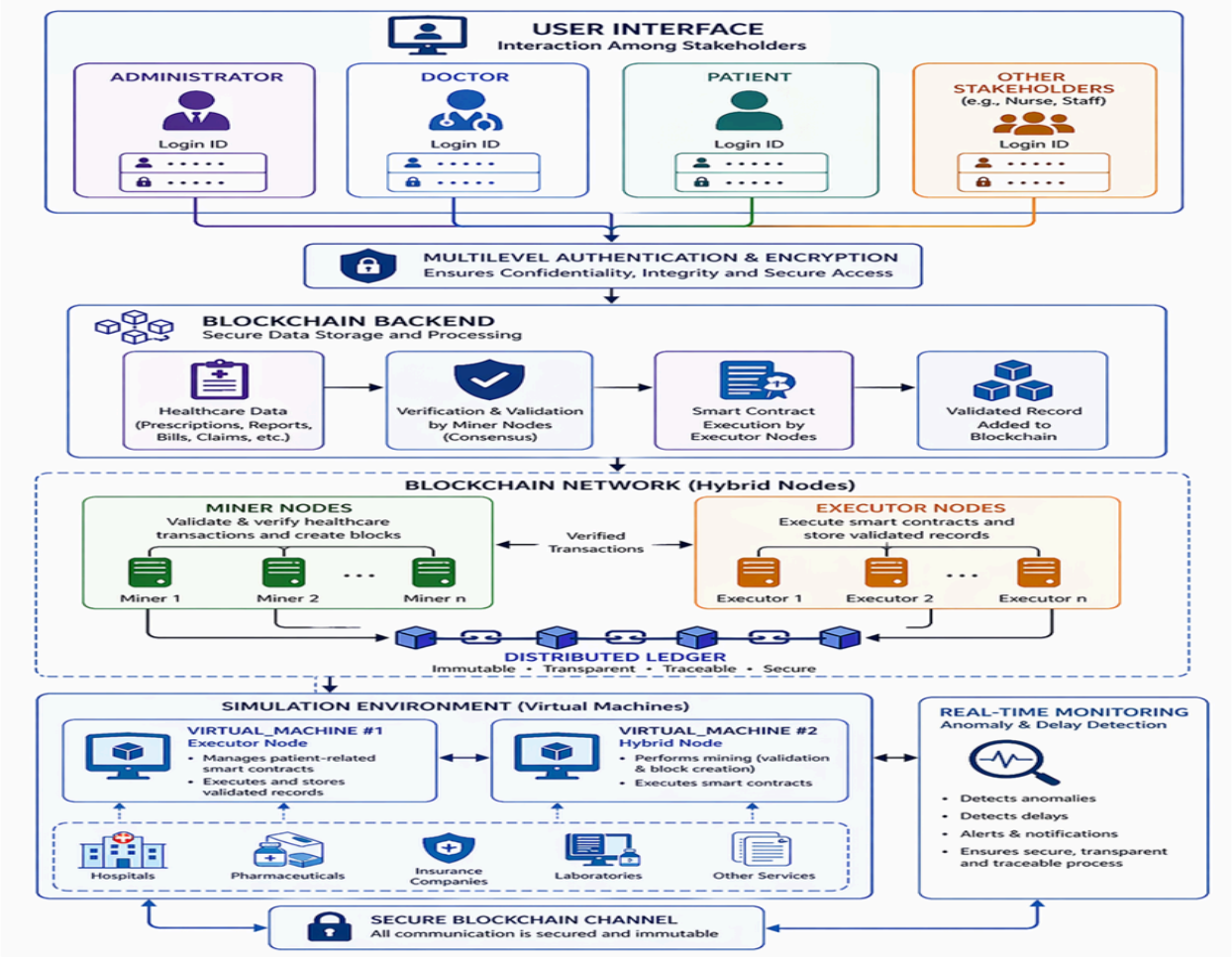

Fig. 6. Blockchain Integration within the Proposed Framework

Every healthcare transaction is encrypted before being stored on the blockchain. The framework uses Blowfish encryption, which is a lightweight and efficient technique and is very well-suited for IoT-based frameworks. Furthermore, the Elman neural network is used as a dynamic control layer that improves robustness and can handle diverse healthcare data. The SHA-256 hashing technique has been utilized to generate the hash address of encrypted transactions. Every healthcare entity, such as a doctor, patient, hospital, pharmacy, and laboratory, acts as a blockchain node. Each block of the chain contains encrypted transactions, the hash address of the block, and the previous block's hash address. Any alteration in the stored data generates a different block hash, which invalidates the further blocks, making tampering easily detectable. This ensures immutability and security of the framework. Each patient's data is recorded step-by-step

on the blockchain, including registration, doctor visits, prescriptions, lab testing, pharmacy bills, and insurance bills. It allows easy auditing of any medical case.Furthermore, smart contracts and consensus mechanisms ensure that all the transactions are valid and trusted. The framework gives selective access permissions to patients that increase their confidence and control over their data to a great extent.

4. Implementation

The framework has been implemented using the Python programming language on Google Colaboratory. A synthetic IoT-based healthcare dataset with 1200 records has been created with realistic attributes to mimic healthcare monitoring. The framework has been coded and tested in a simulated setup. Instead of using complex external software, various Python libraries, such as Keras for the Elman neural network, Scikit-learn for machine learning-based data processing, and PyCryptodome for Blowfish encryption, have been used.

Three simulated IoT healthcare networks with different node sizes (number of IoT devices), Network#1 with 150 nodes, Network#2 with 600 nodes, and Network#3 with 1200 nodes, were created to test the scalability of the framework. Each simulated network was placed inside a square area of 550 meters by 550 meters to mimic a virtual hospital campus where each node could only communicate wirelessly within 100 meters. Every encrypted patient data has been taken as 128 bytes long. This size of packet helps maintain load on the network, transmission delays, and storage requirements. The detailed experimentation settings have been given in Table 4.

Table 4. Experimentation Setup

Network Configuration	
Types of Networks	Network#1 (150 nodes), Network#2 (600 nodes), Network#3 (1200 nodes)
Area of Network	550 meters by 550 meters
Communication Range	100 meters
Data Characteristics	
Packet Size	128 bytes
Dataset Source	Real-world patient dataset with 1200 records
Simulation Settings	
Duration of each test	60 seconds

Security and Intelligence	
Encryption Techniques	Blowfish and Elman Neural Network
Anomaly Detection Techniques	LSTM and SVM
Implementation Platform	
Programming Setup	Python on Google Colaboratory

The framework has been evaluated using three security metrics, namely, attack rate, data loss, and validation delay. To simulate malicious activities in the network, a few nodes were intentionally made malicious, and others were made validators or miners, as shown in Figure 7.

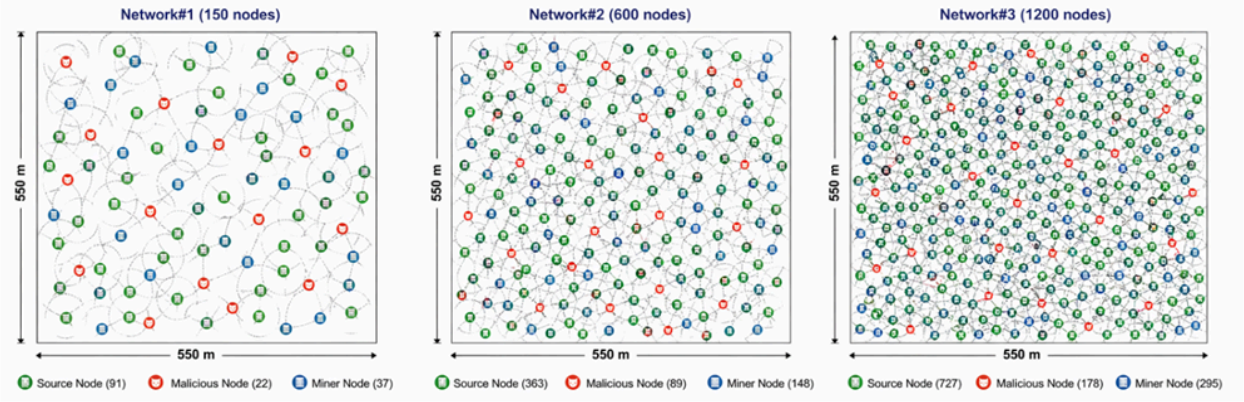

Fig. 7. The Simulated Environment

Table 5 presents different roles assigned to different nodes in different network setups. Furthermore, different cyber-attacks were simulated to test the security of the framework at multiple levels.

Table 5. Network Setups

Network Name	Network Type	Source Nodes	Malicious Nodes	Miner Nodes	Total Nodes
Network#1	Small-scale	91	22	37	150 nodes
Network#2	Medium-scale	363	89	148	600 nodes
Network#3	Large-scale	727	178	295	1200 nodes

4.1 Implementation Challenges and Proposed Solutions

Even though the blockchain enhances security, it introduces various implementation challenges, such as delays, storage requirements, computational cost, and scalability. These challenges must be addressed while implementing the framework in a real

environment with limited resources. Table 6 highlights these challenges and solutions implemented to alleviate them.

Table 6. Framework Challenges and Proposed Solutions

Parameter	Challenge	Solution	Advantage
Delay	Blockchain security checks slow down transaction processing.	The framework uses a customized Proof-of-Authority consensus mechanism in place of traditional Proof-of-Work.	Delay is reduced to 85 ms from 110-120 ms.
Storage Requirements	Healthcare data includes high-resolution images and videos that require significant storage.	The framework stores only a small hash address on the blockchain and actual data on IPFS.	Reduces storage load while maintaining security and traceability.
Computational Cost	Blockchain processes are very heavy for IoT devices with limited power.	The framework uses a hybrid technology of Blowfish encryption with Elman Neural Network, which reduces computational cost.	Computational cost is reduced to approximately 30 times compared to traditional encryption.
Scalability	With the increase in the number of nodes, network congestion increases.	The framework is tested against different-sized networks to evaluate scalability.	Consistent performance is maintained irrespective of network size.

5. Evaluation Results

The framework has been tested on three different network sizes to evaluate its performance and security capabilities. These networks simulate small, medium, and large-sized healthcare systems.

5.1 Security Evaluation

To evaluate the strength and robustness of the proposed framework, three metrics have been used. These metrics test the security, reliability, and practicability of the framework under attack conditions.

- **Wormhole Attack Probability**: It measures the rate at which attackers manipulate the network traffic by creating tunnels. This is a serious attack in the case of healthcare systems, as it can compromise sensitive

healthcare data. The lower the attack rate, the better the framework security. It is expressed in percentage.

- **Product Drop Ratio**: It measures the data loss rate due to attacks and failures. The higher the drop ratio, the more the quality of the healthcare framework is compromised. It is expressed in percentage.
- **Verification Latency**: It measures the latency caused while verifying and validating users and transactions within the framework. Latency can cause issues in emergency healthcare situations. The lesser the latency, more improved will be the quality of real-time data transfer and healthcare operations. It also impacts the integration of IoT devices with blockchain facilities. It is measured in milliseconds.

It has been noticed from Table 7 that increasing the density of nodes and involvement of miners has led to a decrease in wormhole attack and product drop ratio, improving the performance of the framework. These improvements have been observed especially in Network#2, where the attack rate of 1.2% and product drop ratio of 1.3% are minimum. This network setup could be considered as having an optimal balance of node density and miners' distribution that are involved in the validation and verification of transactions. Furthermore, in large-scale Network#3, both metrics have been noticed to increase, which indicates that this network may be affected by congestion and coordination overhead. Verification latency increases slightly with an increase in the size of the network, although this increase is very small and acceptable. These results have been presented graphically in Figure 8.

Table 7. Security Evaluation Results

Network Name	Wormhole Attack Probability	Product Drop Ratio	Verification Latency	Result
Network#1	2.3%	2.8%	110.12 ms	Lower security and reliability
Network#2	1.2%	1.3%	111.11 ms	Optimal and most efficient performance
Network#3	1.8%	1.5%	111.19 ms	Scalable but with performance tradeoffs

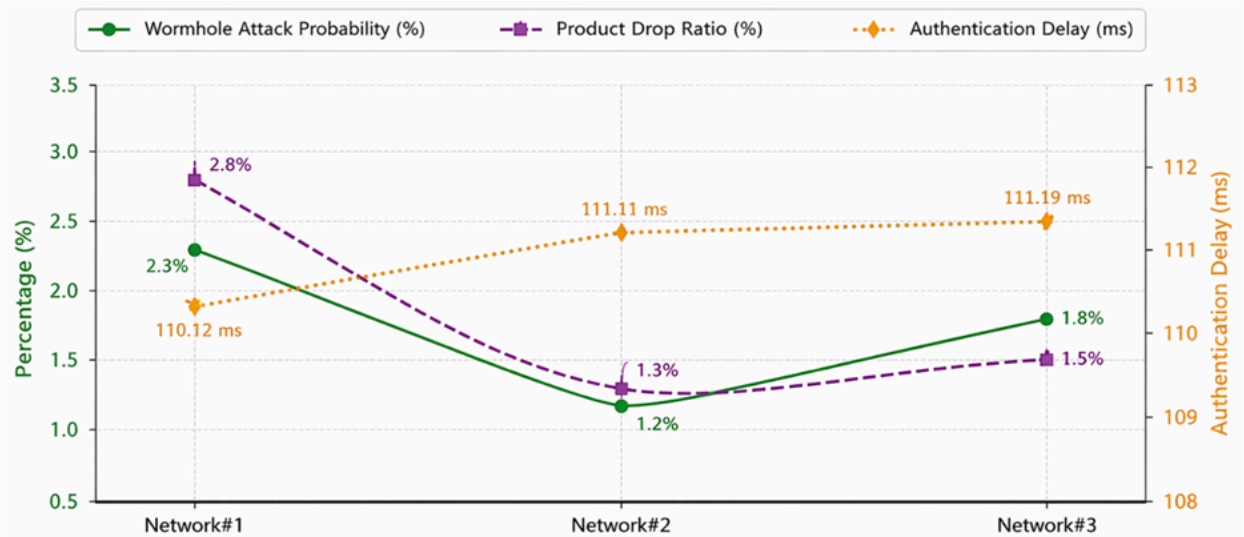

Fig. 8. Comparative Security Performance across Different-Sized Networks

5.2 Anomaly Detection Evaluation

To detect suspicious and malicious activities with the framework, artificial intelligence models, namely, LSTM and SVM, have been used. These models learn normal behavior within the proposed IoT Blockchain integrated healthcare framework and then detect the abnormal behaviors. As already discussed in Section 3.1, the framework uses an artificially generated dataset containing 1200 simulated patient records where each record includes health parameters that could be measured by IoT sensors, including heart rate values, blood pressure values, body temperature values, and oxygen level values. Furthermore, each record in the dataset is labeled as 'Normal' or 'Abnormal'. The dataset contains 85% normal records and 15% abnormal records to detect any sort of anomaly or attacks.

During the training phase of LSTM, the 'Normal' or 'Abnormal' labels are made hidden, allowing unsupervised learning. Furthermore, the data has been scaled between 0 and 1 using MinMax scaling to prevent the neural network from being dominated by large values, allowing fast and smooth coverage. The model has been trained for 60 epochs with 30 samples at a time. This helps balance speed, memory usage, and learning stability of the model. Mean Absolute Error (MAE) has been utilized to calculate the reconstruction loss as:

$$MAE = | Original - Reconstructed | \quad (13)$$

The lower the MAE value obtained, the better the reconstruction and the fewer anomalies. Figure 9 represents the training behavior of LSTM by presenting the training loss and validation loss for 60 epochs. The figure highlights that during the initial phase of 0 – 3 epochs, both training and validationlosses are very high, but then it drops sharply. This is because the model starts with no prior knowledge and then quickly learns basic patterns from the dataset. During the early learning phase of 4 – 15 epochs, lossdecreases gradually, and train and validation curves closely follow each other. This highlights that the model is still learning general patterns and no overfitting is present yet, showing good generalization capability of the model. Furthermore, during the mid-training phase of 16 – 25 epochs, more reduction in loss has been observed. This highlights the capability of the model in capturing deep temporal dependencies within the dataset. This phase significantly highlights the improved reconstruction quality. The late training phase of 41 – 60 epochs, both training and validation loss reach near constant values, which indicates that the model has fully converged. The consistent decrease in loss and close alignment of both curves indicate that LSTM is highly suitable for reliable anomaly detection within an IoT Blockchain-integrated healthcare framework.

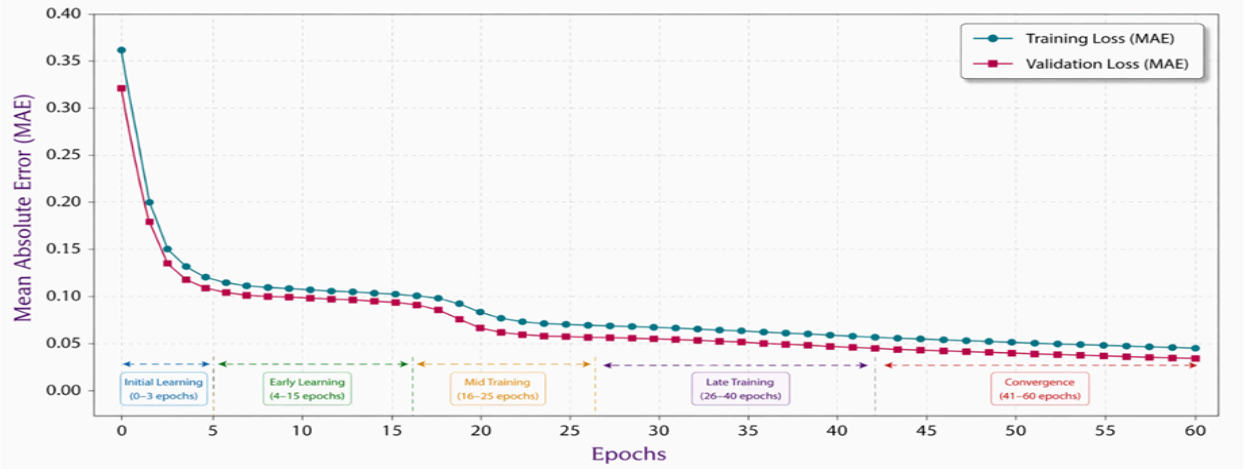

Fig. 9. Training pattern of LSTM

After the training phase, reconstruction error at each point has been calculated to generate the anomaly score. A reconstruction loss threshold value has been selected to identify abnormal data using the following rule:

```
if error > Threshold
        Abnormal Data Identified
else
        Abnormal Data Identified
```

The classification accuracy has been further improved using SVM. Unlike LSTM, where the 'Normal' or 'Abnormal' labels are made hidden during the training phase, SVM is trained using labeled data, allowing supervised learning. The performance has been evaluated using confusion matrix metrics. Table 8 highlights the results of anomaly detection evaluation. The high value of recalls shows that the model is capable of catching most anomalies, while lower precision highlights a few false alarms. This hybrid anomaly detection process allows accurate and reliable anomaly detection within the proposed framework.

Table 8. Anomaly Detection Evaluation Results

Parameter	Value
Threshold	0.17867
Accuracy	98.99%
Precision	0.89
Recall	0.93
F1-Score	0.91

6. Discussion

This section presents a comprehensive analysis of the proposed IoT-blockchain integrated healthcare framework for anomaly detection by interpreting the experimental results and evaluating overall performance. The discussion highlights how the integration of IoT, blockchain, and artificial intelligence contributes to improved healthcare data management. Furthermore, the section also compares the proposed framework with existing approaches to demonstrate its effectiveness in addressing challenges related to data security, real-time monitoring, and reliable decision-making in smart healthcare environments.

6.1 Benefit of Blockchain Integration

Traditional approaches for healthcare management are based on centralized technology and suffer from a lot of limitations. In these systems, the data is managed by third parties, which may lead to the issue of data vulnerability and unauthorized alterations to it. Additionally, such approaches do not provide transparency of associated processes for the stakeholders, thus making them less confident and trustless on the healthcare system. Another major issue includesthe healthcare providers' connectivity gap and a single point of failure.

The proposed blockchain-enabled framework is distributed and decentralized. Blockchain is not stored at a single centralized location, so the single point of failure issue is totally resolved. Blockchain is stored across multiple nodes worldwide in a network where each node has a copy of the blockchain ledger. This enhances fault tolerance and data availability. Furthermore, there is no central authority that controls and manages the database. Rather, decision-making power is shared among participants, such as miners and validators, which prevents control, censorship, or manipulation by any single entity. Additionally, due to anonymity, decentralization, 51% validation rule, immutability, consensus mechanisms, one-way process of hash generation, and cryptographic protections, blockchain is considered the most secure database.

As highlighted in Figure 11, there is a noticeable improvement in verification latency, attack mitigation, anomaly detection rate, specificity, accuracy, and processing overhead efficiency in the proposed framework.

6.2 Benefit of IoT Integration

The integration of Internet of Things technology will allow real-time and continuous monitoring of patient health remotely by capturing important parameters, such as heart beat rate, blood pressure, temperature, and levels of oxygen. This real-time data allows early detection of health issues and timely medical treatment. IoT technology reduces the need for visiting the hospital frequently, especially for elderly and chronically ill patients. This is because healthcare professionals can access patient data remotely, and allow telemedicine and easy accessibility of healthcare services. In the proposed framework, IoT serves as a data generation layer that supplies continuous input to blockchain and

artificial intelligence models. This facilitates secure, reliable, and intelligent healthcare decision-making.

6.3 Benefit of Hybrid Anomaly Detection Process

The traditional techniques of detecting anomalies rely on fixed rules and signatures. Such techniques have to struggle when they receive encrypted, complex, and dynamic data, such as data generated by IoT devices. Thus, they fail to detect new attacks. The proposed framework uses a hybrid anomaly detection model that combines an LSTM autoencoder and an SVM classifier. The model gives an accuracy of 98.99%, as presented in Table 8, demonstrating that almost all the predictions are correct. Furthermore, the F1-score of 0.91 highlights a strong balance between precision and recall. This states that there are few false alarms and most of the attacks are detected before they become a serious issue, confirming accuracy and reliability within the proposed framework.

6.4 Scalability

The proposed framework has been tested on three scaled networks: Network#1 with 150 nodes, Network#2 with 600 nodes, and Network#3 with 1200 nodes. In most of the cases, whenever the network grows, latency and computational overhead also grow. But the presented framework remains efficient even at a large scale, which is very important for IoT-integrated healthcare environments. The combination of Blowfish encryption and Elman Neural Network improves both security and computational efficiency of the framework. Due to the capability of the proposed framework to handle large networks efficiently while maintaining low delay, it is very well suited for real-time healthcare systems and large-scale deployments.

6.4 Comparative Analysis with Existing Approaches

The proposed framework has been comparatively analyzedagainst existing approaches based on different metrics, as highlighted in Figure 10. The statistical comparative analysis has been presented in Figure 11, and the quantitative analysis has been presented in Table 9.

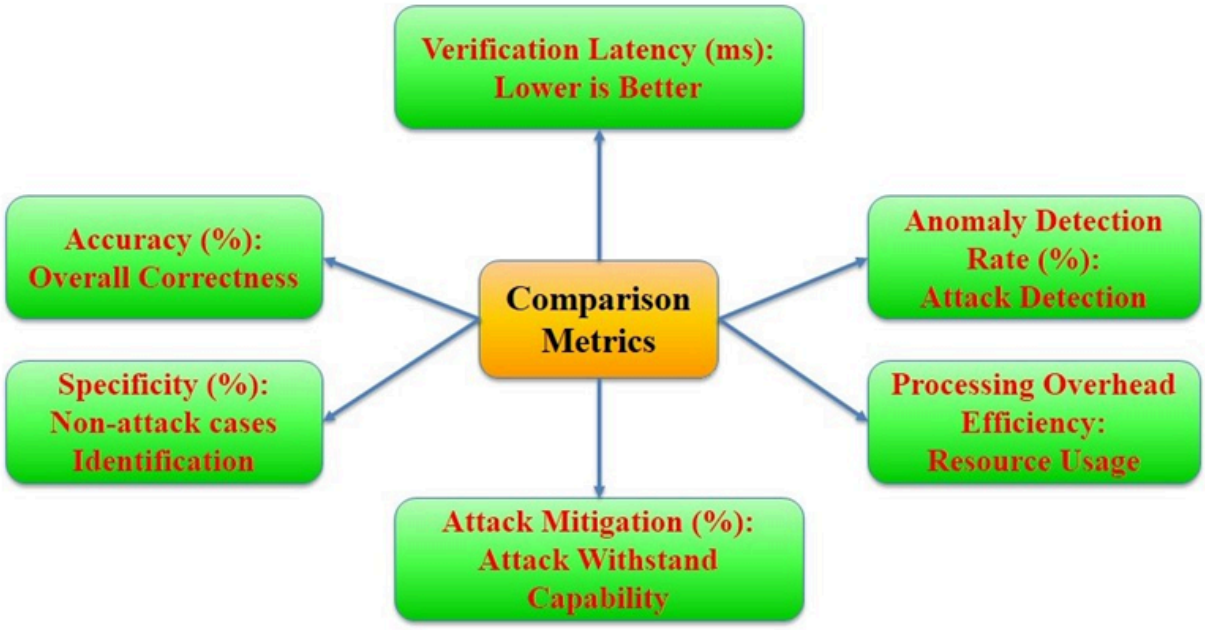

Fig. 10. Metrics used for Comparative Analysis

Table 9. Quantitative Comparative Analysis

Study	Authentication Delay(ms)	Attack Resistance (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Computational Efficiency
Azbeg et al., 2022 [12]	120	74	80	91	88	1 (Moderate)
Hossein et al., 2021 [13]	110	76	85	93	90	2 (High)
&Masoodi, 2021 [14]; Quasim et al., 2020 [15]	115	78	83	90	89	1 (Moderate)
Chinaei et al., 2021 [16]	108	79	86	94	91	2 (High)
Mohammed et al., 2021 [17]; Al-Marridi et al., 2024 [18]; Al-Nbhany et al., 2024 [19]	100	80	87	95	92	2 (High)
Proposed Framework	**111**	**95**	**89.7**	**98.7**	**98.98**	**3 (Optimized)**

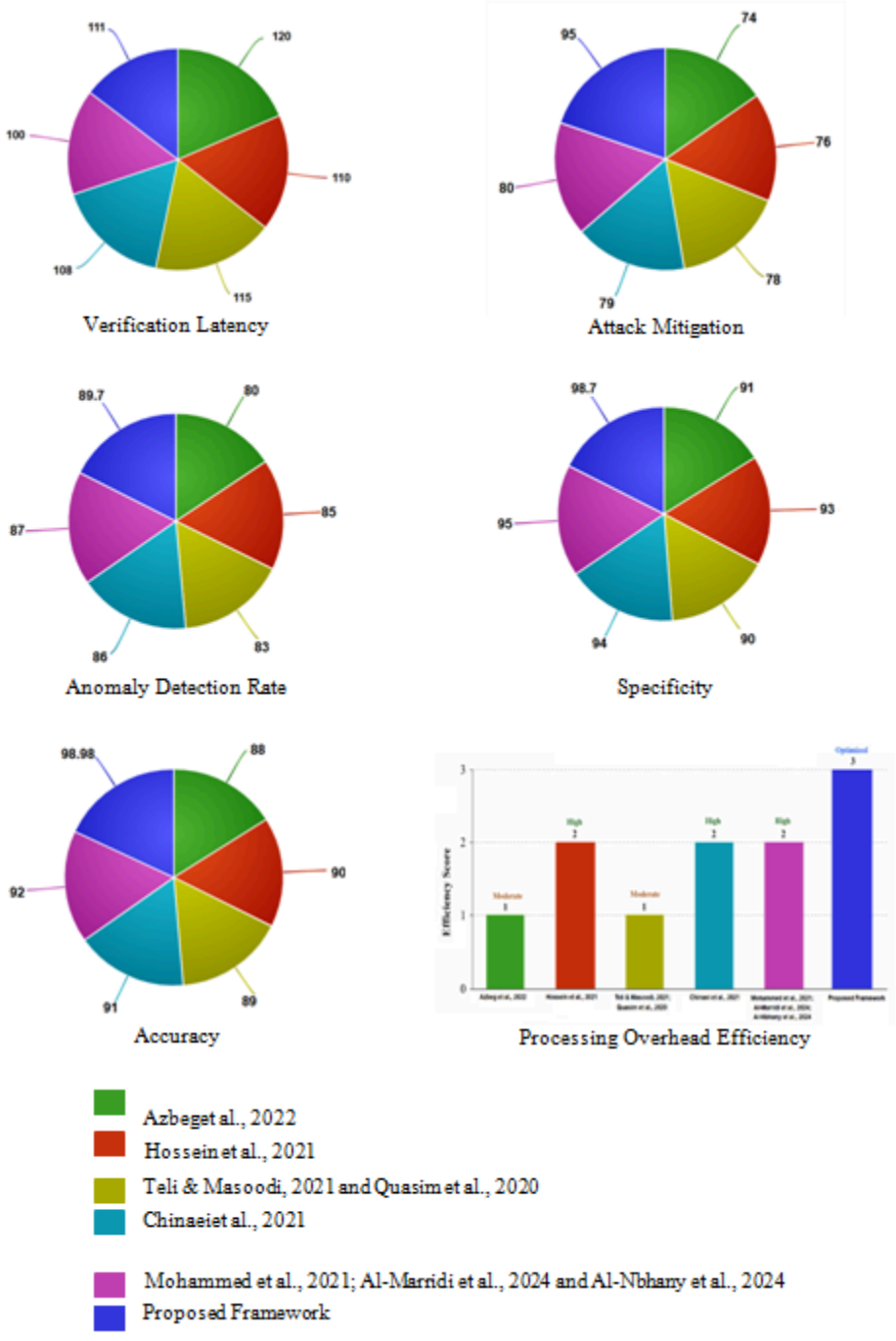

Fig. 11. Statistical Comparative Analysis

7. Limitations

Even though the proposed framework performs very well in terms of security and successfully includes anomaly detection mechanisms, some challenges and constraints remain.

- **Use of Synthetic Data:** Since the framework has been evaluated using synthetic data for preserving healthcare data privacy, it may lack the complexity and unpredictability of real data.
- **Simulation-based Evaluation:** Since the framework has been evaluated using simulated environments, it may not fully capture real-world healthcare processes and the dynamic conditions of hospital networks.
- **Use of Permissioned Blockchain:** The framework uses a permissioned blockchain to store and process the data, due to which, when used under a fully decentralized public blockchain environment, performance and security may vary.
- **Use of only one Consensus Mechanism:** Framework uses a customized Proof-of-Authority consensus mechanism to validate and verify transactions. However, the results may vary with the use of other mechanisms or hybrid ones.
- **Assessment of only Network-Level Security:** The framework lacks evaluation of other types of security, such as device-level, application-level, and user-level.
- **Lack of IoT Resource Constraints Measurements:** Framework lacks measurement of computational load, memory usage, and energy consumption of IoT devices, which is very crucial for real-world deployments.
- **Limited Attacks Detection:**The framework only focuses on wormhole attacks, ignoring other attacks,including replay attacks, insider attacks, and data poisoning attacks.

8. Future Scope

- The framework could be deployed in hospitals or healthcare for real-world implementation, pilot testing, and practical validation.

- Edge nodes could be utilized for faster processing and reducing latency in critical healthcare environments.
- Proof-of-Authority could be replaced by other lightweight consensus mechanisms to reduce delays and energy consumption.
- Blockchain sharding could be implemented to improve throughput and handle large-scale networks.
- Experiments could be extended to analyze other types of attacks, including replay attacks, insider threats, data poisoning, DDoSattacks, etc.
- The power consumption and computational efficiency of IoT devices could be analyzed and optimized accordingly.
- The framework could be aligned with healthcare standards for data protection and security regulations.

9. Conclusion

This paper presented a secure, scalable, and intelligent healthcare framework that integrates Internet of Things (IoT), blockchain technology, artificial intelligence, and advanced cryptographic mechanisms to address critical challenges related to healthcare data management. The proposed system ensures secure handling of sensitive medical data through neural-enhanced Blowfish encryption and blockchain-based immutability. This prevents accessing of data in an unauthorized way as well as tampering. The use of a permissioned blockchain with customized Proof-of-Authority consensus further enhances efficiency by reducing authentication delay and computational overhead.

A key contribution of this work is the implementation of a hybrid anomaly detection model combining an LSTM autoencoder and SVM, which accuratelydetects malicious activities within encrypted data. The experimental results demonstrate high detection performance having an accuracy of 98.99% and an F1-score of 0.91. This highlights the reliability of the proposed approach. Additionally, the framework shows improved resistance to network-level attacks, reduced packet drop ratio, and stable performance

across different network sizes. This proves the scalability and real-time applicability of the framework.

The integration of IoT allows continuous health monitoring and real-time data acquisition, while blockchain ensures transparency, traceability, and trust among stakeholders. Overall, the proposed framework successfully combines security, efficiency, and intelligence, making it a promising solution for next-generation smart healthcare systems. Future enhancements involving real-world deployment, advanced attack modeling, and optimization of resource-constrained IoT devices can further strengthen its practical applicability.

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The Hollow Executive: Leadership, Power and Alienation in DeLillo's Cosmopolis

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Abstract– The psychological toll of the super rich is better conveyed in few novels than *Cosmopolis* (2003). DeLillo presents us with twenty-eight-year-old, obscenely wealthy, currency-trading Eric Packer who is falling apart in a day. The disintegration is the key. Eric isn't really the bad guy, more like a symptom: a dude that the financial system has refined to be good at everything, except being human.

What this paper attempts to decipher is how, through not only Eric's actions, but through the novel's construction, DeLillo makes that argument. The prose hold(s) back. The past and future converge. The limo turns into a hermetic universe. They are no random stylistic lapses, but symptoms of the dehumanizing logic the book is diagnosing. The paper analyses Eric's hollowness, taking inspiration from Marx's reflections on alienated labour and the lesser-known concept of the 'marketing character' developed by Fromm; it is not one he chose but one that is structurally produced by financialised capitalism.

The analysis alternates between close reading, organizational behavior literature, and postmodern narrative criticism, not for neatness' sake but because Eric refuses to succumb to any one frame. The novel finishes by aligning *Cosmopolis* with both corporate dystopian fiction and the premise that the novel has become even stranger and more true-to-life as algorithmic trading and concentrated wealth have ceased to be science fiction.

Keywords*: DeLillo, Cosmopolis, Corporate dystopia, Alienation, leadership, capitalism, organizational behavior*.

Introduction

The literature and its social formations have always been a mirror. Whether it's the industrial satires of Charles Dickens or the bureaucratic nightmares of Franz Kafka, fiction has offered us a venue for considering the human toll of the economic systems in

which official discourse is likely to celebrate or normalize. Don DeLillo's *Cosmopolis* (2003) is certainly in this camp, but it is more an extreme extreme close-up; it's not about taking a look at the grand scale of capitalist buildings, but about relentlessly zooming into the consciousness of one man that is being sculpted and eventually ruined by it.

By all the old rules, Eric Packer's a winner. By the age of twenty-eight, he is the head of a multi-billion dollar asset management firm, living in a forty-eight-room Manhattan penthouse and having an aesthetic sensibility so keen that he buys a chapel dedicated to Rothko, a painting by Richter and a private jet all on a single afternoon. DeLillo's presentation of this accumulation however isn't about achievement, but symptom. The wealth and power that Eric has is directly related to his inability to connect, to feel, and ultimately to survive. The central argument of the novel, which is as much a form as it is a content argument, is that capitalism does not just exploit human beings but makes them a certain kind of subject, one for whom others are means, means are digits and the self is an (optimizable) portfolio.

This paper is a literary reading of *Cosmopolis* as a study in leadership disease. It combines close reading of DeLillo's novel with models from the fields of Marxist literary theory, Erich Fromm's character typology and recent organizational behavior studies, and explores three interrelated aspects of Eric Packer's characterization: his leadership and power style, his systematic alienation from human relationships and his psychological disintegration as a structural rather than just personal phenomenon. The paper suggests that DeLillo's depiction of the empty executive is as pertinent today, two decades after the novel was first published, as it is a literary work, and as a diagnosis of the conditions that have worsened rather than improved in this time.

To read *Cosmopolis* as a critique of capitalism, it must be read in the right (theoretical) context. Three styles of thought are especially fruitful for the analysis that follows.

First, the concept of alienated labour is the initial vocabulary offered by Marx's theory. In his *Economic and Philosophical Manuscripts of 1844*, Marx explains how capitalism renders the worker not only estranged from the products of his work, but also from the work itself, from other people and from his own Wesen, or human being, his ability to engage in free, conscious, creative activity, which is the essence of being human. Eric Packer is of course no proletarian, but his alienation is a sort of extreme version of the

same logic: he is alienated not because he makes nothing, but because what he makes — the abstract financial instruments, which he constructs out of some sort of algorithmically mediated wealth — is so suspect of any real human activity that it cannot sustain a meaningful self.

Secondly, Erich Fromm's account of the 'marketing character' in *Man for Himself* (1947) provides a psychoanalytical supplement to Marx's political economy. Fromm's term for one of the personality types engendered by market society, where the person feels like a commodity, and his value is determined by what he can sell—including his personality. The marketing character is a character without an identity: a character that is what the market demands. The type of relations that Eric Packer exemplifies would be quite explicit: he is married, he has sex, he speaks with employees, and in all of these relationships exchange logic plays a fundamental role.

Third, some current research on organizational behavior and toxic leaders and narcissistic executives offer an empirical perspective for understanding what literary analysis shows intuitively. Barbara Kellerman (*Bad Leadership*, 2004), and Jean Lipman-Blumen (*The Allure of Toxic Leaders*, 2005) have described behaviors of grandiosity, lack of emotional relationships, risk-taking and disrespecting followers that are very similar to Eric Packer's description. When read through those frames, DeLillo is seen to be either astutely and presciently predicting empirical studies of the psychology of unchecked corporate power, or else to be describing it all along.

In the first few chapters of the novel the reader gets a feel for how Eric Packer leads and there is no significant change. He leads not by consultation, but by command; not by vision, but by a lack of vision; not by any legitimate authority in the Weberian sense, but by the power of money. Perhaps the most extended treatment of what leadership means when there's no one to call it to account is his relationship with his Chief of Theory, Vija Kinski.

Their conversations take place in the moving limousine, and are of the nature of philosophical seminars, with the difference that the professor is also the determining factor of the student's salary, physical safety and professional life. Her comments about capital are both intellectual and for Eric's benefit. Like DeLillo does with his trademark economy:

"He looked at her. This was Vija Kinski, his chief of theory. She was a thinker who worked for him, and she had ideas about the markets, their theory and architecture, more innovative and daring than any of the traders or analysts... He trusted her judgment. He trusted very few people in this way." (DeLillo, 2003, p. 79)

The passage is well planned. There is no doubt that Kinski was a smart man but this is only within the context of Eric's property. The attitude that he has for her is one of trust, but trust is a managerial attitude, and a strategic use of trust, it is not a relationship. By using the word 'worked for him', the verb effectively brings everything following it into a state of subordination, no matter how brilliant Kinski's ideas may be, they are all assets in Eric's portfolio.

DeLillo's foremost preoccupations are this dynamic of intelligence turned into the service of capital. The novel has a lot of very competent people, but they're competent for no other reason than that. Eric's bodyguard, his doctors, his analysts, his sexual partners: all are placed in relation to him as instruments and not ends in themselves. This, of course, is the Kantian conception of moral failure treating persons as means rather than ends, but DeLillo's conception here is not of Eric's moral failure, but of the systemic logic of a certain kind of power.

Eric's leadership also illustrates a leadapreneur's hallmark of "defying limitation." He decides to cross Manhattan to get a haircut, even as he is warned that he is about to witness a catastrophic traffic disruption, as the Presidential motorcade, an anti-globalization protest and a funeral procession all converge on the day. The refusal is not just an act of will; it's a statement of a conviction, basic to a certain executive psychology, that reality is ultimately negotiable by enough money. As Lipman-Blumen (2005) notes in regard to toxic leaders, 'Normal human restraint is not applicable to them' (p. 19). The novel's main comic-tragic symbol of exactly this attitude is Eric's stubbornness about the haircut.

The leadership of Eric is the complement to the pathology of personal relationships: intimacy without connection, power without accountability. Over the course of one day in the story, the novel follows Eric's relationships with five women: his new wife Elise Shifrin; his art dealer Jane Melman; his financial officer Didi Fancher; his head of

security Kendra Hays, and a mysterious woman he meets toward the end of the novel. Every time, it's a different take on the same theme: Eric's impossibility of relation.

This impossibility is the longest investigation in his marriage to Elise. At the beginning of the novel they have been married for 22 days and already the marriage is felt as superficial. The numerous encounters they have along Eric's cross-town trek are affective as those between strangers who have a contract:

They ate and talked. He watched her, how she used her fork, the European way, tines down, how she looked at him — with some interest but not a lot. (DeLillo, 2003, p. 30)

The metaphor of the unreadable text is apt and devastating. Elise thinks her husband is a hermeneutic problem, a problem to be interpreted and not known. But DeLillo makes it clear this is not solely Elise's fault; Eric himself is unrelatable by nature. He is not there in any real sense, and so he can't be read. His interiority, which is shown in the novel's free indirect discourse and is highly theorized, is a series of data-processing operations and nothing that could be called a self.

This is very similar to what Marx, in the Manuscripts, calls the alienation of man from man, where 'each man is alienated from the others, and... each of the others is alienated from human life' (Marx, 1844/1975, p. 330). Other people to Eric are positions in a field of information; they have characteristics, tendencies, predictabilities. In this aspect they are indeed similar to financial instruments. His afternoon sexual adventures are just as dispassionately studied as his currency trading, and DeLillo suggests that for Eric, there is no real distinction between the two.

Eric's most moving statement of his alienation is his relationship to his own body. Every day he is examined by a doctor who rides in his limousine; the examination is something that he does as a routine measure, but for Eric it is a form of self knowledge. Even this closest of access to the self, the inside of the body, only provides information:

"His prostate was asymmetrical. He'd been told this, but he had forgotten or not registered it as information that mattered. Now it mattered, it had meaning. He tried to understand what it meant." (DeLillo, 2003, p. 48)

One of the novel's most talked about symbols is the asymmetrical prostate. It's a fault line, a break in the seamless optimization that's the whole of Eric's life. That it 'matters' to

him is only in the sense of information, as data, that needs to be interpreted, but not as a felt sense of his own vulnerability as a body.

What is philosophically most complex in DeLillo's depiction is Eric's self-destructive efforts. During the novel's day, Eric makes a series of more and more irrational financial choices that wipe him out. He takes a huge gamble on the yen, ignoring all of his own models, and loses. At the conclusion of the novel, his empire has been lost. Whether it's a form of emancipation, a death wish or the logical end result of a personality based on pure risk is a subject of debate. In the present reading I propose a fourth interpretation: It is DeLillo's structural statement that capitalism is self-destructive, that it is the psychology of those it creates most thoroughly that makes it obvious.

As Eric spirals into financial ruin, he is also on a path to his own destruction physically, by heading towards the man he knows is out to kill him: Benno Levin. He isn't intimidated by this meeting, and he approaches it as he approaches all things: cool, analytical. One of the most remarkable dialogues in contemporary fiction is between Eric and Benno in a dilapidated building in the climactic section of the novel, between the capitalist and his thrown-out Other, between the system and the man it destroyed and ejected.

Benno Levin (nee Richard Sheets) previously worked at Eric and was let go in the cold sweepstakes of corporate downsizing. But his grievance is not only personal, but systemic, and DeLillo provides him with a stunning speech, the moral heart of the novel:

I don't want you to think of me as a morality tale. I want you to know that. “The thing with me is something else and I want to make this clear.” (DeLillo, 2003, p. 193)

DeLillo doesn't let Benno be the moral voice of the novel — he's unstable, delusional in some ways, and his rage has turned sour, more than just a political critique. But his presence — the discarded worker returned to face the executive — is a form of enacted repression of the system. Each Eric Packer spins out a Benno Levins, cast out from the cycle of capital and forced into his own struggle to understand the meaning of his marginalization. At some level, Eric's journey toward his own death is a tacit admission – albeit a primitive one – of this truth, the novel implies.

In this aspect the novel can be related to what Fredric Jameson (1991) calls the representational problem of the postmodern moment, the inability to render in individual

experience the "vast and impersonal structures" of global capital in *Postmodernism, or, the Cultural Logic of Late Capitalism*. DeLillo's formal answer to this mapping problem is to mark its collapse into a single consciousness, the disintegration of Eric's psychology as an intelligible form of the structural disintegration of capitalism.

An account of *Cosmopolis* as a criticism of leadership and capitalism would be incomplete if it neglected DeLillo's prose style as a formal statement about the world the novel depicts. In this novel, the language is like the crystalline aphorisms that are sentences as precise as mathematical proof and as cold as financial instruments. The dialogue itself is stylized almost to the point of unreality: the formulations are too neat, too theorized, too empty of affective residue to sound like real human speech.

The stylistic decision is not a lack of mimetic realism, but its precise opposite: a conscious formal representation of the dehumanisation thematised in the novel. When Eric talks he sounds like his data terminals. His underlings sound like briefings. DeLillo argues that the abstractions and jargon, the conflation of all experience into information, the stripping of emotional richness, of the financialized capitalism is now part of the most personalized forms of human discourse.

This formal argument is supported by the spatial economy of the novel. The limousine — 'a stretch limousine, white, fitted with a television and a shredder and a drinks cart of stainless steel' — is Eric's office, his tomb (DeLillo, 2003, p. 10). Tinted glass makes the outside world visible, but out of reach, real, but mediated. This spatial organization is a literalization of the executive's situation: he lives in a world over which he has influence without taking part in it, he lives in a world that he processes without experiencing.

The one-day time frame of the novel, a mannerism of Joyce's *Ulysses* and *Mrs Dalloway*, focuses the novel's pressure to almost explosive intensity. In this formal assertion of the temporal logic of capitalism—how an entire lifetime of achievements, emptiness and destruction can be mapped onto a single failed trip across a single city—DeLillo offers a compact, condensed version of the novel's narrative and message.

To appreciate the success of *Cosmopolis* and its unique contribution, it is important to place it in the context of the overall tradition of corporate dystopian fiction. It has encompassed everything from Sinclair's *The Jungle* (1906) to Kafka's *The Trial* (1925) to Miller's *Death of a Salesman* (1949) to *Catch-22* (1961) to, most recently, Dave Eggers's

The Circle (2013) and Ling Ma's *Severance* (2018). What makes DeLillo's novel significant is his choice to make the heart of the story not the victim of corporate capitalism but its most successful victim — and to illustrate that this victim is in a very real sense, the most thoroughly ruined of the lot.

DeLillo's most original formal and political choice is this inversion. Most dystopian corporate fiction takes its moral tone from portraying victims of the system. DeLillo's bet is that the empty victory of the winners of the system is equally if not more damning – and more revealingly, since it forestalls the comforting counter-thesis that the problem is not with the system, but only with the unfairness of it. Eric Packer is suffering, not because he has been treated unfairly, but because he has been treated just as the system had intended. When he's devastated is when the system's successful.

This ties *Cosmopolis* to a core issue in organizational behavior studies: what does success at profit maximization do to its practitioners? Kellerman (2004) has developed a taxonomy of the kinds of poor leadership and identified two that are similar to Eric's description: 'insular' and 'intemperate. In recent years, researchers like Tomas Chamorro-Premuzic (*Why Do So Many Incompetent Men Become Leaders?* 2019) have shown that the qualities that are valued most in the corporate recruitment process – confidence, charisma, risk-taking – are actually quite different from the qualities that make good, ethical leaders. DeLillo preempted these arguments by 10 and a half years.

Among many other things, Don DeLillo's *Cosmopolis* is a sustained literary meditation on the effects of capitalism on leadership, and the effects of leadership on people. In the well-wrought figure of Eric Packer, brilliant, powerful, hollow, doomed, DeLillo paints a portrait of the executive under late capitalism more searching, more honest than most business school curricula. Eric's leadership is in his power without responsibility, in his instrumentalization of all human relations, in his alienation from experience, including from his own body, from his own survival instincts.

The formal devices of the novel — its understated prose, its tight space, its dense time — are not just stylistic devices, but arguments, and perform the dehumanization they describe at the level of style. DeLillo demonstrates a world in which the vocabulary of money has become so entrenched in the human psyche that there is nothing left, nothing

that has not yet been evaluated or absorbed by the market's logic: nothing, in Eric Packer's experience, "that you can't figure out.

Cosmopolis is a provocation especially for the interdisciplinary community of ICESSM-2026, which reminds that the questions that organizational behavior scholarship and theory of economics and management pose are not only technical questions but human questions, and that they are human questions that reach out into the very personal aspects of human existence. At its best, fiction can show what data can't: the "felt" of living under specific economic systems, the price paid by people at both ends of the sorting mechanism of capitalism – in connection, in meaning, in humanity.

Ultimately, the asymmetric prostate that makes Eric Packer so uneasy symbolizes DeLillo's most compact trope for the novel's thesis: that even in the face of all the data, all the theory, all the abstraction of financialized capitalism, there is a body, and the body keeps its own score.

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Transparency as Tyranny: Surveillance, Consent, and Corporate Control in Dave Eggers' The Circle

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Abstract – Nobody has gone after Silicon Valley quite as directly as Dave Eggers in The Circle (2013). The novel follows Mae Holland from eager new hire to true believer, and the transformation is not presented as a tragedy so much as a seduction — which is precisely what makes it disturbing. The company doesn't coerce. It enthuses, includes, rewards. By the time Mae is helping dismantle the last remnants of private life, she thinks she's doing good.

The paper reads this process through Foucault's panopticism, Zuboff's surveillance capitalism, and Arendt's work on totalitarianism — not as a tidy theoretical triptych but because each framework catches something the others miss. Foucault explains the architecture of watching; Zuboff names what watching is for in a market economy; Arendt reminds us that the most suffocating systems are often the ones people build cheerfully, out of conviction rather than fear. Together they help explain why The Circle feels less like dystopian fantasy than like a business model with better branding.

Close reading drives the analysis throughout. The paper works through specific scenes, recurring symbols, and the novel's oddly airless dialogue to show how Eggers dramatizes voluntary self-surveillance as the defining pathology of platform capitalism. The argument ends where the novel's relevance feels most urgent — in current fights over data privacy, monopoly power, and who gets to decide what transparency means and for whom.

Keywords*: Eggers, The Circle, surveillance capitalism, panopticism, Silicon Valley, corporate dystopia, transparency, privacy, digital technology, totalitarianism.*

Introduction

When Dave Eggers published The Circle in October 2013, the world had just learned from Edward Snowden's disclosures that the United States National Security Agency was conducting mass digital surveillance on a scale previously unimagined. Yet Eggers' novel was not primarily about government surveillance. It was about something arguably

more unsettling: the possibility that the most comprehensive surveillance apparatus in human history might be built not by a coercive state but by a beloved corporation, not through compulsion but through enthusiasm, and not against people's will but with their active, even joyful, participation.

The Circle is the name of Eggers' fictional technology company, an amalgamation of Google, Facebook, Apple, and Twitter scaled to monopolistic omnipotence. Its campus in San Francisco is a paradise of free food, live concerts, health care, and intellectual stimulation. Its products — social networks, search engines, digital identity systems, and eventually live streaming cameras worn by public officials — are presented to the world as tools of democratic empowerment and human connection. Its founding philosophy, encapsulated in three slogans that recur throughout the novel like corporate mantras, holds that "All that happens must be known," "Secrets are lies," and "Privacy is theft" (Eggers, 2013, p. 303). It is within this seductive corporate utopia that Mae Holland arrives, is transformed, and ultimately becomes one of its most effective instruments.

This paper argues that The Circle is the most direct and diagnostically precise literary account of Silicon Valley corporate culture yet written, and that its power derives from its careful dramatization of the mechanisms by which genuine human desires — for connection, recognition, transparency, and community — are captured and redirected by corporate logic into instruments of control. Reading the novel through Foucault's panopticism, Zuboff's surveillance capitalism, and Arendt's analysis of totalitarian ideology, the paper traces three interlocking arguments: that The Circle depicts the replacement of external coercion with internalized self-surveillance; that its central character embodies the process by which corporate culture colonizes individual identity; and that the novel's most disturbing insight is not that The Circle is evil but that it genuinely believes it is good. Three theoretical frameworks converge with particular force in the analysis of The Circle, each illuminating a different dimension of the novel's critique.

Michel Foucault's account of panopticism in Discipline and Punish (1977) provides the foundational spatial and psychological model. Foucault takes Jeremy Bentham's design for a circular prison — in which a central watchtower allows a single guard to observe

all cells while remaining invisible to prisoners — as an emblem of modern disciplinary power. The panopticon's genius is that it renders the constant presence of the observer unnecessary: once prisoners have internalized the possibility of being watched at any moment, they discipline themselves. Eggers' title is not accidental. The Circle is a panopticon scaled to social totality, and its most brilliant move is to make the watched also the watchers — and to make both roles feel like privileges.

Shoshana Zuboff's concept of surveillance capitalism, developed most fully in The Age of Surveillance Capitalism (2019) but analytically applicable to the period Eggers was diagnosing in 2013, provides the economic framework. Zuboff argues that the dominant business model of Silicon Valley is built on the unilateral appropriation of human behavioral data, which is then processed into predictive products sold to advertisers and others. This logic transforms human experience into raw material for commercial extraction. The Circle dramatizes this process with particular clarity, showing how every act of sharing, every rating, every 'zinger' and 'smile' on the platform generates data that feeds the company's commercial and eventually political power.

Hannah Arendt's analysis of totalitarianism in The Origins of Totalitarianism (1951) offers a third, more unsettling framework. Arendt argues that totalitarian movements distinguish themselves from mere tyrannies by their ideological totality: they do not merely demand obedience but require the transformation of inner life, the enthusiastic embrace of the system's values. This distinction is central to understanding what makes The Circle frightening rather than simply sinister. The company does not coerce; it converts. Mae Holland is not forced to go transparent — she chooses to, and eventually she chooses it with evangelical fervor. The totalitarian logic Arendt identifies is precisely the logic by which individual conscience is replaced by ideological commitment.

Mae Holland is Eggers' most carefully constructed literary achievement in The Circle: a protagonist who is neither naive enough to be merely a victim nor cynical enough to be a villain, but something more disturbing — an ordinary, intelligent, well-meaning person who is gradually and willingly transformed into an instrument of the system she serves.

When Mae arrives at the Circle, she is characterized by precisely the qualities that make

her susceptible to the company's logic: she is ambitious, eager for approval, uncomfortable with ambiguity, and genuinely drawn to the Circle's promise of community and purpose. Her first days on campus are rendered with seductive vividness — the food, the stimulation, the sense of belonging to something historic. Eggers is careful to make these pleasures real rather than merely illusory. The Circle is genuinely a better employer, in many superficial respects, than Mae's previous job at a utility company. The problem is not that it offers nothing valuable but that what it offers is inseparable from what it extracts.

The mechanism of Mae's transformation is the company's social participation system, in which employees are evaluated not only on their work performance but on their social engagement — the number of posts, comments, event attendances, and 'zings' they generate. Mae's initial discomfort with this system is gradually overwritten by her internalization of its values. When she is called into a meeting with her supervisors for insufficient social participation, the scene enacts with clinical precision the panoptic logic Foucault describes:

> *"We've noticed that your numbers are down... We want to make sure you're engaged, that you're a full participant in the community here. It matters to us that you're happy, Mae. It matters to us that you're connected." (Eggers, 2013, p. 173)*

The language of care — 'we want to make sure,' 'it matters to us that you're happy' — is simultaneously genuine and threatening. The supervisors are not lying; they do, in some sense, want Mae to be happy. But happiness, here, is defined as participation, and participation is defined as data generation. The system's genius is that the categories of wellbeing and productivity have been made identical, so that surveillance can present itself as care.

Mae's eventual decision to 'go transparent' — to wear a live-streaming camera around her neck twenty-four hours a day, broadcasting her every waking moment to millions of followers — is the novel's central symbolic event. It represents not a coerced surrender of privacy but a voluntary, even enthusiastic, embrace of total visibility. Eggers renders this moment with considerable psychological complexity. Mae's motivations are not simply

corporate ambition; they include a genuine relief at the abolition of the private self, with all its attendant anxieties and uncertainties:

> *"She felt a profound sense of calm. She was known. She was seen. There was nothing to hide, and because there was nothing to hide, there was nothing to fear." (Eggers, 2013, p. 291)*

This passage is Eggers' most precise articulation of the psychological mechanism by which surveillance becomes desirable. The abolition of the private self is experienced not as loss but as liberation — liberation from the burden of interiority, from the anxiety of judgment, from the exhausting maintenance of a self that exists between public appearances. This is, Eggers implies, not a pathology unique to Mae but a latent possibility in the human desire for recognition and belonging that corporate systems have learned to activate and exploit.

The Circle's leadership consists of three men known as the 'Three Wise Men': Ty Gospodinov, the company's reclusive young founder; Eamon Bailey, its avuncular, evangelical public face; and Tom Stenton, its coldly ambitious CEO. Together they constitute a composite portrait of Silicon Valley leadership that is both satirically precise and analytically illuminating.

Eamon Bailey is the novel's most extensively drawn and most ideologically interesting figure. His weekly presentations to Circle employees, called 'Dream Fridays,' are evangelical performances in which new products are unveiled with the fervor of religious revelations. Bailey's genius, Eggers shows, is that he genuinely believes everything he says. He is not a cynical manipulator concealing an exploitative agenda behind utopian rhetoric; he is, more dangerously, a true believer whose vision happens to converge perfectly with an extremely profitable business model:

> *"Imagine a world where everyone could know everything, where there were no more secrets, no more lies. Imagine a world where the potential of each individual human being was fully realized because the world was watching, and the world cared." (Eggers, 2013, p. 68)*

The utopianism is real — and this is precisely what makes it dangerous. Bailey's vision

of radical transparency is not merely cover for surveillance capitalism; it is a genuine moral philosophy, however catastrophically mistaken. Eggers' insight here anticipates what critics of Silicon Valley would spend the following decade articulating: that the tech industry's most damaging tendency is not its cynicism but its sincerity, its unshakeable conviction that what is good for the platform is good for humanity.

Tom Stenton represents the complementary figure: the executive whose relationship to the company's utopian ideology is purely instrumental. Where Bailey evangelizes, Stenton monetizes. His interest in radical transparency is not philosophical but strategic — a world in which all behavior is visible and recorded is a world in which all behavior can be influenced and exploited. The dynamic between Bailey and Stenton enacts one of the novel's central arguments: that idealism and greed are not opposites in Silicon Valley but partners, each providing the other with what it needs to function.

Ty Gospodinov, the original founder, is the novel's most complex figure precisely because he alone among the leadership understands what the company has become. His eventual attempt to warn Mae and dismantle the system he created is the novel's most tragic subplot, representing the original hacker ethic — with its genuine commitment to openness and individual empowerment — confronting what it has been transformed into by corporate scale and commercial logic. His failure is Eggers' most pessimistic statement: that the system, once built, exceeds the intentions of its builders and cannot be dismantled by them.

The Circle's most sustained philosophical argument is conducted through the novel's two figures of resistance: Mercer Medeiros, Mae's former boyfriend, and the figure of Kalden/Ty. Both represent, in different ways, the value of the private self that the Circle's ideology denies — and both are destroyed or defeated by the system they oppose.

Mercer is the novel's most explicit spokesman for privacy and interiority. His arguments with Mae about the Circle's ideology are the novel's clearest articulation of the case for the private self, and Eggers is careful to give them genuine force despite also making Mercer somewhat pompous and ineffectual — a formal choice that prevents the novel

from collapsing into simple polemic. In a key letter to Mae, Mercer articulates the novel's central counter-argument:

> *"I've been thinking about what the Circle is doing, and I think it might be the last step... There needs to be the option of opting out. The option of disappearing. The option of being forgotten... You're building a world without secrets. But you don't see that secrets are the foundation of the self." (Eggers, 2013, p. 258)*

The argument that 'secrets are the foundation of the self' is the novel's most important philosophical claim, and it stands in direct opposition to the Circle's founding motto that 'secrets are lies.' Eggers is drawing on a long tradition of liberal political thought that insists on the value of the private sphere not merely as a protection against state power but as the precondition for the development of individual identity. Without a space that is not observed — where one can fail, experiment, contradict oneself, be uncertain — the self cannot develop. The Circle's radical transparency, Mercer argues, does not illuminate the self; it prevents it from existing.

Mercer's eventual fate — he drives his truck off a bridge while being pursued by a swarm of Circle drones deployed by Mae's followers in what is presented as a benevolent search for a 'missing' person — is the novel's most shocking and symbolically loaded episode. The death of the privacy advocate, brought about by the very technology of connection and care, enacts with brutal economy the lethal potential of the Circle's logic. The crowd that participates in the drone pursuit is not malicious; it is enthusiastic, helpful, and completely unaware that it is conducting a hunt.

This episode connects directly to Zuboff's (2019) analysis of surveillance capitalism's behavioral modification capacity. What Zuboff calls the 'instrumentation power' of surveillance systems — their ability to shape behavior not through prohibition but through the design of environments that make certain choices natural and others difficult — is precisely what kills Mercer. He is not forbidden from disappearing; he is simply made to feel that disappearance is impossible, antisocial, and ultimately futile.

The novel's final movement, in which the Circle moves toward what it calls 'Completion'

— the total integration of all human activity into a single, universally monitored digital system — is Eggers' most explicit engagement with the totalitarian tradition. 'Completion' is the Circle's eschatological vision: a world in which every human being has a verified digital identity, every action is recorded, and the system's benevolent oversight is total and inescapable.

The resonance with Arendt's analysis of totalitarianism is precise and clearly intentional. Arendt argues in The Origins of Totalitarianism (1951) that what distinguishes totalitarian systems from mere tyrannies is their claim to totality — the insistence that no sphere of human life is beyond the system's legitimate concern. The Circle's move from commercial platform to universal governance system follows exactly this logic. It does not begin by seeking political power; it acquires it gradually, by expanding its definition of what the platform's legitimate interest encompasses, until there is nothing left outside.

Arendt also argues that totalitarian movements require the abolition of the distinction between public and private life — because privacy is the space in which individuals can form the independent judgments that might lead them to resist. The Circle's ideological attack on privacy is therefore not merely a commercial strategy but a precondition for the kind of total power the company seeks. When Bailey declares that 'privacy is theft,' he is not merely arguing for transparency as a value; he is arguing for the elimination of the space in which dissent can form.

The novel's ending is deliberately unresolved and deeply disturbing. Mae, having failed to prevent Ty's attempt to destroy the system from within, sits at her terminal processing data, connected to millions, more powerful and more alone than ever. Eggers refuses the consolation of a dystopian finale in which the system is overthrown or the protagonist escapes. Instead, he offers something more honest and more frightening: the image of a person who has been so thoroughly transformed by the system that she can no longer imagine, let alone desire, an alternative:

> *"And Mae knew, having thought it through, that there was no reason on earth why everything she thought, felt, believed and feared couldn't be made available to everyone who cared to know. She had no secrets. She'd given it all away."*
> *(Eggers, 2013, p. 491)*

The final image is of total transparency achieved and experienced not as violation but as fulfillment. This is Eggers' most disturbing and most philosophically precise conclusion: that the most effective totalitarianism is not the one that destroys the self but the one that convinces the self to destroy itself.

Any complete analysis of The Circle must attend to Eggers' formal and stylistic choices, which are themselves arguments about the world the novel describes. The novel is written in a mode that oscillates between realism and satire — close enough to actual Silicon Valley culture to function as documentary, exaggerated just enough to reveal the absurdity that documentary realism might naturalize.

Eggers' prose is deliberately accessible and unadorned — the opposite of DeLillo's aphoristic density. This is a formal argument: the Circle's ideology speaks in the language of clarity, user-friendliness, and democratic accessibility. Its rhetoric is designed to be understood and agreed with by everyone. Eggers' choice to write in a style that mimics this accessibility is a formal enactment of the novel's central concern: the danger of ideas that are easy to understand and agree with.

The novel's structure mirrors Mae's psychological transformation. The first section, 'Opening,' presents the Circle with something close to wonder, filtered through Mae's eager eyes. The second section, 'Completion,' has the quality of inevitability — each step in Mae's transformation following logically from the last, the system's internal logic driving events with the relentlessness of a machine. This structural choice reflects Eggers' argument that the Circle's trajectory is not the product of individual decisions but of systemic logic — that given the premises, the conclusion follows necessarily.

The novel's satirical targets are precise and well-researched. The 'Dream Fridays' presentations satirize Apple's product launches; the social scoring system reflects the gamification strategies of Facebook and LinkedIn; the company's insistence on mandatory sociality parodies Google's famously designed-for-interaction campus culture. Yet Eggers' satire is not simply topical; it identifies the structural logic beneath the specific practices, which is why the novel has aged so well and why its targets have only become more recognizable in the decade since its publication.

Situating The Circle within its literary tradition clarifies both its debts and its innovations. The novel's most obvious predecessor is George Orwell's Nineteen Eighty-Four (1949), and Eggers acknowledges this lineage through multiple structural echoes: the totalizing ideology, the system's penetration of private life, the protagonist's eventual capitulation. But Eggers' most important departure from Orwell is his insistence on replacing coercion with seduction. Where Winston Smith is tortured into submission, Mae Holland is loved into it. This is the novel's most significant formal and political innovation: the recognition that twenty-first-century social control operates primarily through desire rather than fear.

Aldous Huxley's Brave New World (1932) is in some respects a more relevant precursor. Huxley's dystopia is organized around pleasure and belonging rather than terror and pain, and his citizens are conditioned into happiness rather than beaten into compliance. The Circle inherits this tradition while updating it for the age of social media: where Huxley's citizens are conditioned biochemically, Mae is conditioned socially, through the dopamine mechanics of likes, follows, and social scores.

Among more recent corporate dystopian fiction, The Circle most usefully illuminates its concerns when read alongside DeLillo's Cosmopolis (2003) and Ling Ma's Severance (2018). Where DeLillo examines the hollowness at the apex of financial capitalism — the pathology of the executive who has everything — Eggers examines the eager participation of those lower in the hierarchy, the employees and users who build and inhabit the system with genuine enthusiasm. Together the two novels constitute a comprehensive anatomy of corporate capitalism's human effects: at the top, the hollow executive; at the base, the hollowed employee.

The Circle's relevance to contemporary debates about technology and society has only intensified since its publication. The emergence of social credit scoring systems, the Cambridge Analytica scandal, the commodification of personal data by major platforms, the algorithmic manipulation of political opinion, and the growing concentration of digital power in a handful of corporations have all validated Eggers' diagnosis. What was satire in 2013 has become, in significant respects, description.

For the interdisciplinary audience, The Circle offers particular value as a literary contribution to debates in organizational behavior, business ethics, and management theory. Several of the novel's concerns map directly onto significant research areas in these fields.

The novel's analysis of workplace sociality and mandatory enthusiasm connects to a growing body of organizational behavior research on what Arlie Hochschild (1983) calls 'emotional labor' — the management of feeling as a job requirement. The Circle's social scoring system represents an extreme and technologically mediated form of the emotional labor demands that contemporary service and knowledge economies routinely impose. Eggers' fictional extrapolation helps illuminate what is latent in existing practices.

The novel's examination of data ethics and corporate responsibility also speaks directly to the concerns of management information systems scholarship. The Circle's trajectory from useful platform to totalizing surveillance infrastructure raises questions about the responsibilities of organizations that hold large quantities of behavioral data — questions that regulatory frameworks are only beginning to address and that theoretical frameworks are still developing.

Finally, the novel's analysis of charismatic leadership and organizational culture speaks to concerns identified in research on transformational leadership. Bailey's evangelical leadership style represents a case study in the dangers of transformational leadership unchecked by ethical constraint or external accountability — a combination that organizational behavior scholarship,
from Kellerman's (2004) taxonomy of bad leadership to more recent work on cult-like corporate cultures, has identified as a significant organizational risk.

Dave Eggers' The Circle is, among other things, a literary gift to everyone who has ever wondered how a company that began by organizing the world's information ended up organizing its citizens. Through Mae Holland's gradual and willing transformation from employee to evangelist to instrument, through the terrifying sincerity of Eamon Bailey's utopianism, and through the systematic dismantling of every form of resistance the novel

imagines, Eggers constructs the most comprehensive and prophetic literary account of Silicon Valley corporate culture in contemporary fiction.

The novel's central insight — that the most effective surveillance is the surveillance we perform on ourselves, that the most complete control is the control we experience as freedom — is not merely a literary theme but an analytical contribution to our understanding of how power operates in digital capitalism. Reading The Circle through Foucault's panopticism reveals the mechanisms of its psychological architecture; reading it through Zuboff's surveillance capitalism reveals its economic logic; reading it through Arendt's analysis of totalitarianism reveals its political horizon.

What literary analysis offers, above and beyond these theoretical frameworks, is access to the felt texture of the experience Eggers describes: the pleasure of belonging, the relief of being known, the gradual disappearance of the self into the social, and the moment — if it can even be called a moment, since it has no clear boundary — at which enthusiastic participation becomes something that cannot be distinguished from captivity.

In an era in which the questions raised by The Circle have become the central questions of democratic politics — questions about data sovereignty, platform accountability, the right to privacy, and the terms on which individuals participate in digital life — the novel's continued relevance is both its achievement and its indictment. That Eggers' fiction of 2013 reads, in 2026, as something between prophecy and history is perhaps the most unsettling thing about it.

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Impact of Physicochemical Variations and Heavy Metal Contamination on Fish Diversity in the Maniyari River Ecosystem

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Abstract– The present study examines the relationship between water quality deterioration, heavy metal contamination, and fish diversity in the Maniyari River, Chhattisgarh. Seasonal samples were collected from three stations upstream, midstream, and downstream to assess physicochemical parameters and heavy metal concentrations alongside fish diversity indices. A total of 28 fish species belonging to six families were identified, with Cyprinidae being the most dominant. Species richness and diversity declined progressively downstream, corresponding with elevated BOD, COD, TDS, and heavy metal levels. Sensitive species such as *Labeorohita*, *Cirrhinusmrigala*, and *Catlacatla* were abundant in upstream stretches, while tolerant species like *Clarias batrachus* and *Channa punctata* dominated polluted zones. Statistical analyses revealed significant negative correlations between pollution indicators (BOD, COD, Pb, Cr) and diversity indices (Simpson's and Shannon's), confirming the impact of pollution on fish assemblages. The results demonstrate that anthropogenic inputs and toxic metal accumulation are key drivers of ecological degradation, affecting fish community structure and biodiversity. The study emphasizes the importance of integrated river management, habitat restoration, and biomonitoring to conserve the aquatic diversity of the Maniyari River and mitigate future ecological risks.

Keywords: *Maniyari river, fish diversity, physicochemical characteristics, water samples*

Introduction

Freshwater ecosystems are among the most productive yet vulnerable components of the biosphere. Rivers play a crucial role in maintaining ecological balance, nutrient cycling, and biodiversity, while also supporting fisheries, agriculture, and human livelihoods (Birk et al., 2026). However, in recent decades, unregulated anthropogenic activities including industrial discharge, agricultural runoff, and domestic sewage have severely degraded freshwater ecosystems across India. These stressors not only deteriorate water quality but

also alter habitat structure and biological communities, especially fish populations that serve as sensitive indicators of ecological health (Kanwel et al., 2025).

Fish diversity is a key measure of riverine ecosystem integrity because fishes occupy multiple trophic levels and respond rapidly to changes in physicochemical parameters, toxicant exposure, and habitat modification (Du et al., 2025). Factors such as dissolved oxygen, temperature, pH, nutrient enrichment, and heavy metal contamination significantly influence fish abundance, growth, and community structure. Numerous studies have demonstrated that elevated levels of biochemical oxygen demand (BOD), chemical oxygen demand (COD), and heavy metals like lead (Pb), chromium (Cr), and manganese (Mn) can cause physiological stress, oxidative damage, and reproductive failure in fish species. These impacts ultimately lead to biodiversity loss and shifts in species composition, favouring pollution-tolerant taxa (Huang et al., 2025).

The Maniyari River, a tributary in Bilaspur district, Chhattisgarh, is a vital freshwater resource for the Takhatpur region, supporting both human and ecological needs. Yet, in recent years, the river has been exposed to domestic sewage, agricultural effluents, and industrial wastes from surrounding settlements and thermal power plants. Such contamination has likely affected the water chemistry and aquatic fauna, particularly fish diversity, but comprehensive scientific documentation remains limited (Singh and Mahto, 2022).

Therefore, the present study aims to examine the relationship between physicochemical parameters, heavy metal concentrations, and fish diversity in the Maniyari River. The specific objectives are to assess the spatial and seasonal variations in water quality and heavy metal content along different stretches of the river.Evaluate fish species composition, abundance, and diversity indices (Simpson's and Shannon's) across upstream, midstream, and downstream zones.Determine the correlation between pollution intensity and fish community structure, identifying sensitive and tolerant species as bioindicators of ecosystem health.This study provides crucial baseline information on the ecological status and fish biodiversity of the Maniyari River, offering insights necessary for conservation planning, pollution control, and sustainable management of freshwater ecosystems in central India.

Materials and Methods

Study Area

The study was conducted along the Maniyari River, located in Bilaspur District, Chhattisgarh, India, which flows through the Takhatpur region before joining the Arpa River. Three representative sampling stations were selected to capture variations in water quality and fish diversity:

- Site I (Lormi Pathar): Characterized by natural flow and minimal anthropogenic influence.
- Site II (Maadi Chok, Chelligahat Road): Influenced by moderate agricultural and domestic discharges.
- Site III (Badghat, Bathri Road): Exposed to high levels of domestic sewage, agricultural runoff, and industrial effluents.

Seasonal sampling was carried out during summer, monsoon, and winter to assess temporal variations.

Water Sampling and Analysis

Water samples were collected from each site following APHA (2017) protocols. Physicochemical parameters such as temperature, pH, electrical conductivity (EC), turbidity, total solids (TS), total dissolved solids (TDS), total hardness, alkalinity, chloride, phosphate, nitrate, dissolved oxygen (DO), biochemical oxygen demand (BOD), and chemical oxygen demand (COD) were analyzed.

- DO and BOD were determined by the Winkler method after 5-day incubation at 20 °C.
- COD was measured by the open-reflux dichromate titrimetric method.
- Nutrients (phosphate and nitrate) were estimated spectrophotometrically.

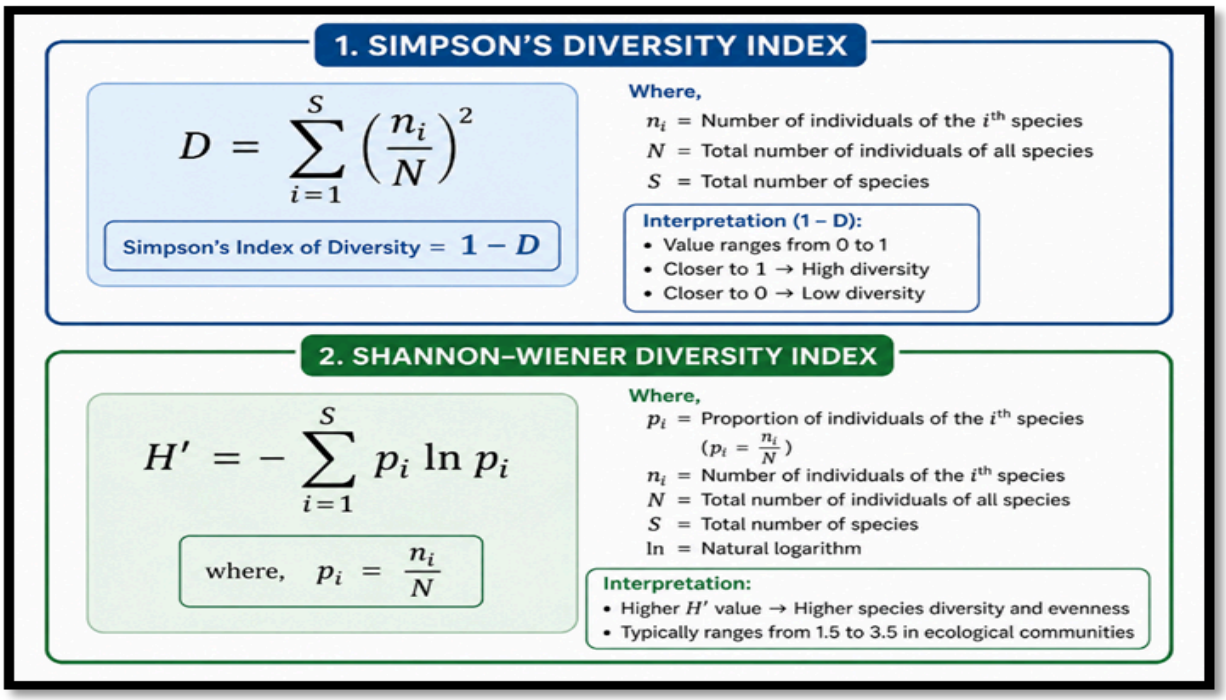
  For heavy metal estimation (Fe, Mn, Cu, Zn, Pb, Cr), samples were acidified and digested with concentrated nitric acid (HNO_3) and analyzed using Atomic Absorption Spectrophotometry (AAS) (Perkin Elmer Analyst 200).

Fish Sampling and Identification

Fish sampling was performed simultaneously with water collection at each site using cast nets, drag nets, and hand nets of varying mesh sizes. Sampling was conducted during early morning hours to ensure active fish movement. Specimens were identified on-site whenever possible, using standard taxonomic keys provided byTalwar &Jhingran (1991).Due to ethical and conservation considerations, fish were not preserved, but photographed and released immediately after identification and counting. The total number of individuals per species was recorded to compute species abundance and diversity indices.

Diversity Indices

Fish diversity was quantified using the following indices:



Data Analysis

Descriptive and inferential analyses were performed using SPSS v.22. Pearson's correlation and Principal Component Analysis (PCA) were applied to examine relationships between physicochemical parameters, heavy metal concentrations, and fish diversity. Comparative analyses among sites and seasons helped identify the key environmental drivers influencing fish assemblage structure.

Results

1. Physicochemical and Heavy Metal Characteristics

The physicochemical parameters of the Maniyari River exhibited significant spatial and seasonal variations across the three sampling sites. The temperature ranged between 22.5°C and 32.8°C, with higher values in summer due to low water flow and intense solar radiation. The pH fluctuated between 6.7 and 8.3, indicating slightly alkaline conditions favorable for aquatic life. Electrical conductivity (EC) and total dissolved solids (TDS) showed a gradual increase from upstream (210 µS/cm, 220 mg/L) to downstream (645 µS/cm, 690 mg/L), reflecting elevated ionic concentration and anthropogenic inputs.

Dissolved oxygen (DO) varied between 3.2 mg/L and 8.6 mg/L, with lower values at downstream sites, inversely correlated with biochemical oxygen demand (BOD) and chemical oxygen demand (COD), which ranged from 1.8–6.9 mg/L and 14.2-68.5 mg/L, respectively. Nutrient enrichment was evident from elevated phosphate (0.12-1.38 mg/L) and nitrate (0.42-3.65 mg/L) concentrations, particularly during the monsoon, indicating agricultural runoff.

The heavy metal concentrations followed the order Fe > Zn > Mn > Cr > Cu > Pb, with overall higher values downstream. While Fe (0.12-0.98 mg/L) and Zn (0.05-0.62 mg/L) were within permissible limits, Pb (0.01-0.11 mg/L) and Cr (0.02-0.18 mg/L) occasionally exceeded WHO (2011) and BIS (2012) standards, indicating potential contamination from industrial effluents and fly ash deposition. The correlation analysis revealed significant relationships between BOD, COD, TDS, and heavy metals, suggesting common anthropogenic sources influencing both chemical and biological parameters.

Fish Diversity and Composition

A total of 28 fish species belonging to 6 familiesCyprinidae, Bagridae, Channidae, Siluridae, Clariidae, and Mastacembelidaewere recorded from the Maniyari River. Cyprinidae was the dominant family, comprising 42.8% of the total catch, represented by species such as *Labeorohita*, *Catlacatla*, and *Cirrhinusmrigala*. The upstream region recorded the highest diversity (25 species), followed by the midstream (21 species) and

downstream (16 species), demonstrating a clear decline in species richness toward more polluted stretches.

Sensitive species (*Labeorohita*, *Cirrhinusmrigala*, *Catlacatla*) were confined to the cleaner upstream site, while pollution-tolerant species (*Mystusseenghala*, *Channa punctata*, *Clarias batrachus*) dominated downstream areas, reflecting the impact of organic and toxic stress on fish assemblages. Seasonal analysis revealed maximum species richness during the monsoon, coinciding with breeding and favorable flow conditions, and minimum diversity in summer, when pollutant concentration and temperature peaks limited habitat suitability.

3. Diversity Indices

The Simpson's Diversity Index (1–D) and Shannon–Wiener Index (H′) showed marked spatial differences.

- **Upstream:** H′ = 2.82; 1–D = 0.89
- **Midstream:** H′ = 2.34; 1–D = 0.82
- **Downstream:** H′ = 1.78; 1–D = 0.69

These values indicate higher diversity and species evenness in upstream areas, with progressive dominance of a few tolerant species downstream. The diversity indices were negatively correlated with BOD, COD, TDS, Pb, and Cr, and positively correlated with DO, indicating that pollution intensity directly influenced fish community structure.

Concluding Remarks on Results

The combined analysis of water quality, heavy metals, and fish diversity highlights that anthropogenic pollution and nutrient enrichment have led to habitat degradation and loss of fish biodiversity in the Maniyari River. The results underscore that organic and metallic pollution, especially in the Takhatpur region, is the primary driver of ecological imbalance and reduced species richness in this freshwater ecosystem.

Discussion

The findings of this study reveal a strong interrelationship between physicochemical characteristics, heavy metal concentrations, and fish diversity in the Maniyari River,

reflecting the cumulative impact of anthropogenic stress on aquatic ecosystems. The results demonstrate clear spatial and seasonal variations, with upstream regions maintaining relatively balanced water quality and diverse fish populations, while downstream sections, particularly near Takhatpur, exhibit deteriorated water quality and reduced biodiversity due to increasing pollution.

The observed temperature variation (22.5-32.8°C) corresponds with natural seasonal fluctuations. However, higher temperatures during summer, coupled with low water flow, can intensify oxygen depletion, affecting fish metabolism and breeding behaviour. Similar observations were reported by Pany et al., (2023) in the Mahanadi River, where elevated summer temperatures corresponded with reduced dissolved oxygen levels and fish activity.

The pH (6.7-8.3) remained slightly alkaline, supporting general aquatic productivity, yet local increases downstream suggest influence from detergents and domestic effluents. Elevated electrical conductivity (EC) and total dissolved solids (TDS) downstream indicate ionic enrichment from wastewater and fertilizer leaching, as reported by Patel et al., (2018) in other central Indian rivers. Such chemical enrichment enhances ionic stress, which can disrupt osmoregulation and respiration in fish.

The dissolved oxygen (DO) levels declined significantly downstream, whereas biochemical oxygen demand (BOD) and chemical oxygen demand (COD) values increased, signaling organic pollution and oxidizable chemical load. Low DO impairs fish respiration, while high BOD indicates increased microbial decomposition of organic matter. Similar patterns were observed in the Ganga and Yamuna River (Verma and Jha, 2024), linking oxygen depletion to domestic sewage and nutrient enrichment.

Elevated phosphate (0.12-1.38 mg/L) and nitrate (0.42-3.65 mg/L) concentrations, particularly during the monsoon season, confirm significant agricultural runoff and nutrient loading. Excessive nutrient inputs promote algal blooms, leading to eutrophication and periodic oxygen depletion. Such conditions alter aquatic food webs, suppress sensitive fish species, and favour tolerant or air-breathing taxa. Comparable nutrient-driven eutrophication effects have been reported in the Godavari and Subarnarekha Rivers (Tatavarthi et al., 2025).

The concentrations of Fe, Mn, Cu, Zn, Pb, and Cr demonstrated clear spatial trends, with Fe and Zn being predominant, likely due to natural geogenic sources and sediment interactions. However, the elevated levels of Pb and Cr, occasionally exceeding WHO/BIS permissible limits, indicate inputs from industrial activities, vehicular emissions, and fly ash deposition. Heavy metals, being non-biodegradable, persist in aquatic environments, accumulate in sediments, and bioaccumulate in fish tissues, leading to oxidative stress, enzyme inhibition, and histopathological alterations (Punia et al., 2022).

The downstream enrichment of Pb and Cr correlated negatively with fish diversity indices, confirming their role in toxic stress and biodiversity loss. Similar findings were reported by Rao et al. (2024) in the Godavari River and Rai et al., (2017) in the Son River, where heavy metal accumulation caused a marked decline in fish diversity and abundance.

The recorded 28 fish species across six families indicate moderate diversity in the Maniyari River; however, the downstream decline from 25 to 16 species reflects ecological deterioration. The Cyprinidae family, dominated by *Labeorohita*, *Catlacatla*, and *Cirrhinusmrigala*, was most abundant in cleaner waters, while tolerant families like Bagridae and Channidae prevailed in polluted zones. This shift in species composition aligns with the concept of biotic homogenization under pollution stress, where tolerant species outcompete sensitive taxa.

Diversity indices (H′ and 1–D) exhibited significant spatial variation highest in upstream regions and lowest downstream-correlating strongly with DO (positive) and BOD, COD, Pb, Cr (negative). These patterns confirm that pollution-induced changes in water quality directly regulate fish assemblage structure, a conclusion consistent with Kumar et al., (2026) and Mai et al., (2024). The reduction in evenness and dominance of a few hardy species downstream signify loss of ecological resilience and functional diversity.

The integration of physicochemical and biological data clearly indicates that the Maniyari River ecosystem is under moderate to high stress from anthropogenic pollution. The combination of organic matter, nutrient loading, and metal contamination not only alters habitat quality but also disrupts the trophic structure and reproductive success of fishes.

This degradation has cascading effects on biodiversity, ecosystem services, and local fisheries.

To restore ecological health, immediate implementation of pollution control measures including industrial effluent treatment, agricultural nutrient management, and riparian vegetation restoration is required. Periodic biomonitoring using fish diversity indices and metal bioaccumulation studies can provide early warnings of ecological imbalance. Community engagement and policy enforcement are equally critical to ensure sustainable river management in the Takhatpur region.

Conclusion

The present investigation demonstrates that the physicochemical variability and heavy metal contamination of the Maniyari River play a significant role in shaping the structure, composition, and diversity of fish communities. The study revealed a clear spatial gradient of pollution, with relatively better water quality and higher species richness in the upstream region, gradually declining toward the downstream section near Takhatpur, where domestic sewage, industrial effluents, and agricultural runoff converge.

A total of 28 fish species belonging to six families were documented, with Cyprinidae being the dominant group. Diversity indices indicated that Simpson's and Shannon–Wiener values decreased progressively from upstream to downstream, signifying loss of ecological balance and habitat degradation. Pollution-tolerant species such as *Clarias batrachus* and *Channa punctata* dominated the more impacted sites, while sensitive species like *Labeorohita* and *Cirrhinus mrigala* were restricted to cleaner waters. These patterns confirm that fish assemblages are reliable bioindicators of riverine ecosystem health.

The high BOD, COD, and nutrient levels, coupled with elevated concentrations of Pb and Cr, exceeded safe thresholds and showed strong negative correlations with fish diversity, reflecting the toxicological stress imposed by anthropogenic inputs.

The study concludes that pollution control, effluent regulation, and periodic biomonitoring are essential to restore the ecological integrity of the Maniyari River. Implementation of integrated watershed management, riparian vegetation protection, and

community participation will be vital for the conservation of fish biodiversity and sustainable utilization of this important freshwater resource.

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AI-Driven Data Visualization for Big Data Analytics and Decision Support: Tools, Challenges, and Future Directions

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Abstract: We live in a time where the internet and computers everywhere. This means that things like cloud computing and smart devices make a lot of data every day. It is hard to deal with this data using ways of analyzing it. It takes a lot of time. Data visualization is a help. It turns numbers and words into pictures that make it easy to see patterns and relationships. In the few years Artificial Intelligence has made data visualization better. It can clean up the data automatically suggest what kind of charts to use, find problems predict what might happen and create dashboards.

This paper is about using Artificial Intelligence for data visualization with big data. It talks about how visualization has changed from static graphs to smart visual analytics platforms. The paper also looks at tools like Microsoft Power BI, Tableau, SAS Visual Analytics and Looker. It explains how these tools are used in healthcare, education, finance, retail and government.

The paper also discusses the bad things about using these tools. Some problems are data, privacy issues, too much information and not being able to understand how the tools work. The paper then talks about what might happen in the future like making Artificial Intelligence explain itself using conversation to analyze data and creating dashboards that work on their own.

Artificial Intelligence is helping to connect the amount of data we have with the decisions that people need to make. Artificial Intelligence and data visualization are making it easier for people to understand data and make good decisions. Data visualization and Artificial Intelligence are very important, for making sense of the data that cloud computing, smart devices, online platforms and enterprise applications generate every day.

Keywords:- *Artificial Intelligence, Big Data Analytics, Data Visualization, Visual Analytics, Decision Support Systems, SAS Visual Analytics, Interactive Dashboards, Business Intelligence.*

I. Introduction

We live in a time where the internet's really big. The internet is getting bigger every day. We have a lot of things like apps and social media. We also have lots of devices and sensors that are connected to the internet. The internet is making a lot of data every second. Organizations and governments and hospitals and banks and schools are making a lot of data. This data comes in forms. Sometimes it is easy to understand. Sometimes it is hard to understand. We call all of this data Big Data. Big Data is very important for making decisions, about what to do. Just having a lot of Big Data is not enough. We need to be able to understand what Big Data means and use it to make choices. The real value of Big Data is when we can use Big Data to do something.

We work with a lot of data daily. It comes in the form of reports and spreadsheets. The volume of data is huge. Keeps increasing. Data has parts which makes it difficult to identify patterns and connections. We struggle to make sense of it when looking at numbers. Data visualization helps us understand data better. It converts data into representations like charts and maps. These visuals make it easier for us to grasp. When data is presented in a way we can spot patterns and connections easily. These patterns would be hard to notice by looking at numbers. Data visualization is crucial, for understanding data. We use it to create:

- Charts
- Dashboards
- Heat maps
- Network diagrams

These tools help us analyze data effectively. Data visualization is very useful. We can display data on a map, which's a type of interface. This interface presents data in a way. Data visualization is helpful because it clarifies data. With data visualization we can see patterns and connections that numbers alone wouldn't show. It makes data more understandable, which is vital when dealing with amounts of data.

Data visualization has become really important in the few years. This is because it now works with Artificial Intelligence and machine learning. The Artificial Intelligence systems can do a lot of things on their own. They can clean up the data find mistakes

see what is happening guess what will happen next suggest what kind of charts to use and make dashboards that people can use easily.

The big change here is that data visualization is not about showing what happened. It is now about helping people make decisions. The machines do not just show us information. They also help us understand what might happen in the future and find things that we might not see at first. Data visualization and Artificial Intelligence are really good at this. They work together to make it easy for people to get insights, from their data.

Microsoft Power BI and other tools like Tableau, SAS Visual Analytics, Looker and ThoughtSpot are really good at helping people understand things. They make it easy for people to look at information in a way. This is very useful in areas, such as business, healthcare, education, finance and governance.

These systems are great because they combine analysis with simple visual tools. Microsoft Power BI and other systems like Tableau, SAS Visual Analytics, Looker and ThoughtSpot help people who know a lot about the subject and people who do not know much about it. They can look at information using interactive dashboards, tools that predict things, tools that find unusual things and search tools that use everyday language. Microsoft Power BI and other tools, like Tableau, SAS Visual Analytics, Looker and ThoughtSpot are very helpful.

We have made a lot of progress with data visualization that uses intelligence.. There are still many problems that make it hard to use this technology successfully. For example the data is not good there is much information the pictures are hard to understand it costs a lot to set up there are risks, to peoples privacy the artificial intelligence results are not easy to understand and the people using it do not know much about visualization. All these things make us question the information we get from this technology. So it is very important to look at the tools we use and the ways we analyze data. We also need to think about what gets in the way of using this technology and what we can do in the future to make data visualization systems that use intelligence better. We need to study intelligence data visualization systems and how we can make them work well.

The need for this is what made us write this paper. This paper is about Artificial Intelligence. How it is used for data visualization when dealing with big data and making

decisions. We look at how modern visualization systems have changed over time. We also look at the ways that people do data visualization and the tools they use today. We talk about the world and how people use data visualization. We discuss the problems that people are having with data visualization and the things that people are still trying to figure out. We also think about what Artificial Intelligence and data visualization will be like in the future, for visual analytics and Artificial Intelligence-driven data visualization.

II. Literature Review and Research

Background

The integration of Big Data, Artificial Intelligence, and visual analytics has become a major research area in recent years. Earlier analytical studies were mainly limited to statistical reporting and database computation, but with the rapid increase in data complexity, researchers recognized the need for human-friendly interpretation. This gradually shifted attention toward data visualization as an important bridge between machine-generated information and practical decision-making. Current literature shows that visualization is now considered an active analytical component rather than only a presentation tool.

A. Traditional Visualization

Early studies in data visualization focused on simple graphical methods such as bar charts, line graphs, pie charts, and scatter plots for presenting summarized numerical data. These approaches were effective for small and structured datasets, but they mainly supported descriptive reporting. Researchers later found that such static visuals were unable to reveal multidimensional relationships, hidden anomalies, and rapidly changing trends in large digital datasets.

B. Big Data Analytics

With the growth of Big Data technologies, research shifted toward advanced visual analytics capable of handling large, heterogeneous, and continuously generated information. New visual forms such as heat maps, treemaps, network diagrams, and drill-down dashboards were introduced to simplify complex analytical spaces. Literature from this phase highlighted that visualization had become essential not only

for displaying information but also for discovering patterns and supporting faster analytical reasoning.

C. AI-Based Visualization

A big change in the field is the use of Artificial Intelligence in visualization tools. Now AI helps automate tasks, like data cleaning and finding patterns. It even suggests which charts to use and makes predictions. These systems are called dashboards and autonomous analytics tools. They give support not just pictures of data. Researchers are using these tools more and more. AI-Based Visualization tools are changing how we work with data.

D. User-Centered Support

Recent studies say that visualization systems need to be designed in a way that users can understand. Researchers found out that even the advanced dashboards do not work well if users do not get the results. This is why current studies are looking into things like trust how much thinking is required, simplicity of the interface and clear explanations. The goal is to make visual reports generated by AI practical for people like managers, doctors, teachers and analysts. They want to make sure these reports are easy to use and make sense to these users. The focus is, on making the reports clear and easy to understand.

E. Research Gap

The thing is, even though we have had a lot of research on visualization tools and Artificial Intelligence techniques most of it does not really look at how users can adapt to these tools or how they handle privacy and explainability and whether they are actually effective in real world decisions. This means we need a review that puts together the latest technology, the areas where it is used the problems we have when we try to implement it and where we should be going with future research on visualization tools and Artificial Intelligence techniques, all, in one place.

III. Evolution of Data Visualization

Data visualization has undergone a significant transformation over the years as analytical requirements have shifted from simple reporting to intelligent decision support. In the early stage, visualization was limited to static graphs, printed reports, and spreadsheet-based summaries where the main purpose was only to present numerical values in a readable form. These traditional methods were useful for small datasets, but they were not capable of handling continuously changing and multidimensional information. As industries began generating massive amounts of digital data, the limitations of static visual reporting became more visible, and the demand for more dynamic analytical systems increased.

A. Traditional Statistical Visualization Era

The earliest form of data visualization was based on conventional statistical charts such as bar graphs, line charts, pie charts, and histograms. These visuals were mainly used in business reporting, academic studies, and numerical comparisons because they were easy to understand and simple to create. However, they were descriptive in nature and could only display already summarized information. They lacked interactivity, predictive capability, and deeper analytical exploration, which made them less suitable for modern big data environments.

B. Rise of Business Intelligence Dashboards

As database systems and enterprise software developed, visualization moved toward Business Intelligence dashboards that allowed users to combine multiple charts, key indicators, and performance reports within a single interface. These dashboards provided better monitoring and quicker comparison than traditional reports. Real-time updates, drag-and-drop charting, and multi-source data connectivity made BI dashboards more practical for organizational decision-making. This stage marked the beginning of interactive analytics where users could explore information rather than just observe it.

C. Big Data Visual Analytics Phase

The growth of Big Data introduced challenges related to huge volume, high speed, and multiple data formats. Because of this, researchers and software developers started using advanced visual techniques such as heat maps, treemaps, network diagrams, cluster

charts, and multidimensional visual panels. These methods helped compress large analytical information into understandable visual patterns. During this phase, visualization became an analytical necessity for identifying trends, hidden relations, and unusual behaviors in massive datasets.

D. AI-Driven Intelligent Visualization Systems

The most recent evolution is the integration of Artificial Intelligence with data visualization platforms. Modern systems now use machine learning to automatically preprocess data, detect anomalies, generate forecasting models, recommend suitable visuals, and even provide narrative explanations. Instead of manually searching for patterns, users can receive AI-generated insights through self-service dashboards. This has transformed data visualization into an intelligent decision-support ecosystem where analytics is faster, smarter, and more adaptive to user needs.

E. Autonomous Visual Analytics

Current research shows that visualization platforms are changing. They are moving away from showing reports on a dashboard. Now they are becoming like autonomous analytics. This means that systems do more than just make pictures from data. They also keep watching the data all the time. They alert the people using them. Suggest what actions to take. Some new features are becoming popular in tools that companies use. These features include talking to computers like they’re people asking questions in normal language and special panels that explain how the computer made its decisions. So it is clear that data visualization is changing. It used to be about showing charts and doing nothing else. Now it is, about using machines to help people understand the data and make decisions. Autonomous Visual Analytics is getting better and better.

IV. Major Visualization Techniques and Analytical Approaches

Modern AI-driven analytical systems use multiple visualization techniques to convert complex data into understandable visual knowledge. The selection of a suitable visualization method depends on data volume, dimensionality, user requirement, and decision context. Different visual approaches provide different levels of analytical

depth, ranging from simple comparison to multidimensional pattern recognition. The major techniques commonly adopted in present intelligent visualization systems are discussed below.

A. Statistical Graph-Based Visualization

Statistical graph-based visualization includes common charts such as bar graphs, line charts, pie charts, histograms, and scatter plots. These methods are widely used because they are simple, familiar, and effective for showing comparisons, percentages, and trends over time. In AI-assisted reporting systems, these visuals are often used to display predictive values, performance summaries, and category-wise observations in an easily interpretable form.

B. Interactive Dashboard Visualization

Interactive dashboards combine multiple charts, indicators, filters, and reports within one digital screen. Unlike static visuals, dashboards allow users to zoom, filter, drill down, and compare data dynamically. AI-generated dashboards also provide automatic alerts, trend indicators, and real-time updates, making them highly useful for continuous monitoring and rapid decision-making in business and enterprise environments.

C. Multidimensional and Hierarchical Visualization

When datasets contain many variables and hidden relationships, multidimensional visuals such as heat maps, treemaps, cluster diagrams, and network graphs are used. These techniques help users understand dense analytical information in compressed visual form. Such methods are especially useful in fraud detection, customer segmentation, cybersecurity analysis, and scientific data exploration.

D. Geographical and Spatial Visualization

Spatial visualization is applied when data contains regional or location-based information. GIS maps, location markers, regional heat zones, and route diagrams help users identify where specific patterns, anomalies, or events are occurring. This method is commonly used in healthcare spread monitoring, logistics optimization, environmental analytics, and smart city management.

E. Predictive and AI-Generated Visual Analytics

The new way of looking at information is visualization. This is where Artificial Intelligence finds out what might happen in the future. It looks for trends that have not happened yet warnings about things that might go wrong and secrets that we do not know about. It does all this by using interfaces. These systems do not just show us what happened in the past. They actually try to predict what will happen next. They make panels that show what they think will happen charts that recommend what we should do and indicators that show how sure they are, about their predictions. All this helps the people who make decisions understand what might happen before they do something. They can look at the forecasting panels and recommendation charts and confidence indicators that Artificial Intelligence generates. This way the decision-makers can anticipate what might happen before they take action with the help of visualization and Artificial Intelligence.

V. AI-Enhanced Visualization Tools and Present-Day Platforms

The rapid growth of intelligent analytics has led to the development of several AI-enhanced visualization platforms that combine automated data processing with interactive graphical reporting. These tools help users transform large and complex datasets into actionable insights without requiring deep programming knowledge. Modern platforms are increasingly equipped with machine learning support, predictive dashboards, cloud connectivity, natural language querying, and self-service analytical features. Some of the most widely adopted present-day visualization tools are discussed below.

A. Microsoft Power BI

Microsoft Microsoft Power BI is one of the most popular enterprise visualization tools because it offers drag-and-drop dashboard creation, real-time reporting, AI-generated narratives, forecasting panels, and integration with multiple data sources. Its user-friendly environment allows both technical and non-technical professionals to generate meaningful visual reports quickly, making it highly suitable for business intelligence and organizational monitoring.

B. Tableau

Tableau is really good at helping people tell stories with the information they have. It has a lot of ways to make the charts look how you want them to. You can use Tableau to look at the details of the information compare different things and make reports that are easy to understand. Tableau makes it easy to do all these things with the information without having to know how to code. Recently they added some artificial intelligence features to Tableau. These artificial intelligence features help Tableau find information on its own. This is why a lot of people like to use Tableau when they want to dig deep and explore their data. Tableau is a tool for this kind of work, with data and people use Tableau for this purpose.

C. SAS Visual Analytics

SAS Visual Analytics provides a powerful environment for handling large-scale enterprise data through automated charting, predictive analysis, pattern recognition, and interactive dashboards. It is especially useful for organizations dealing with structured as well as semi-structured information because it combines advanced statistical computation with clear visual reporting. Its built-in analytical intelligence makes it highly effective in large business and research environments.

D. Google Looker and Cloud BI Platforms

Google Looker has emerged as a modern cloud-based analytical platform that supports live data connectivity, governed reporting, collaborative dashboards, and enterprise-scale visualization. Because of its cloud architecture, it allows distributed teams to access and interpret updated reports from different locations. Such platforms are increasingly important in present data-driven organizations that require continuous collaborative analytics.

E. Other Emerging Intelligent Tools

Apart from company software there are other new tools like ThoughtSpot and Streamlit and Grafana that are helping to make visualization smarter. These tools are good, at letting people search using language and they can show what is happening in real time. They can also make dashboards and help developers make good looking

applications. The fact that people are starting to use ThoughtSpot and Streamlit and Grafana often shows that the way we look at data is changing. Now people want to be able to look at data in a way that's easy to use and helps them with the help of artificial intelligence so visualization is becoming more autonomous and accessible and it is using artificial intelligence to help with analytics.

VI. Industrial Applications of Intelligent Data Visualization

AI-driven data visualization has become a practical analytical infrastructure across multiple industries because modern organizations depend on fast and evidence-based decision-making. Intelligent dashboards, predictive reports, and automated visual alerts help professionals understand large datasets quickly and respond more efficiently. The major industrial application areas are discussed below.

A. Healthcare and Medical Monitoring

In healthcare systems, intelligent visualization platforms are used for patient record analysis, disease spread monitoring, diagnostic support, and hospital resource management. AI-generated dashboards can display abnormal trends, risk indicators, and predictive health alerts in a visually simplified form, allowing doctors and administrators to make quicker medical decisions.

B. Finance and Risk Management

Financial institutions use intelligence to help them look at things in a better way. They use this to find people who are doing things with money to try and guess what the market will do to look at how likely people are to pay back loans and to keep an eye on investments. They get graphs that can predict things, warnings when something strange happens and special boards that compare things. These things help the people who look at money to find things that are happening and to think about what might happen in the market in the future. They can do this accurately with Finance and Risk Management.

C. Retail and Business Intelligence

Retail companies use intelligent dashboards to study customer behavior, seasonal demand, product movement, and sales performance. AI-generated visual reports help

managers compare business indicators in real time and optimize pricing, inventory, and marketing strategies based on current data patterns.

D. Education and Research Analytics

Schools are using data visualization more and more to track how well students are doing. They use it to look at attendance and see how well students are learning. Data visualization also helps schools plan what resources they need.

In places where people are doing research they take a lot of data from experiments. Turn it into pictures and charts. This makes it easier for the people doing the research to find patterns and compare things. Data visualization makes it easier for them to study and understand the data from their experiments. Schools and research places are finding that data visualization is an useful tool, for understanding data.

E. Smart Governance and Urban Systems

The government and people who make cities work better use pictures and maps to watch traffic see if the environment is in danger look at population numbers respond to disasters and plan services for the public. They use maps and screens that show what is happening right now to understand what is going on in different areas quickly and make decisions based on facts, about Smart Governance and Urban Systems.

VII. Major Challenges, Limitations and Open Research Issues

Although AI-driven data visualization has significantly improved modern analytical decision-making, its practical implementation still faces several technological and user-oriented limitations. Intelligent dashboards may generate attractive visual outputs, but the reliability and usefulness of these outputs depend on multiple hidden factors such as data quality, algorithm transparency, computational resources, and user understanding. Therefore, several critical challenges continue to influence the performance of present-day visualization systems.

A. Data Quality and Integration Problems

AI visualization systems are highly dependent on the accuracy, completeness, and consistency of input data. If collected data contains missing values, duplicated records, biased samples, or incompatible formats, the generated visuals may produce

misleading patterns and wrong analytical conclusions. Since big data usually comes from multiple heterogeneous sources, proper integration remains a major technical issue.

B. Information Overload and Visual Complexity

Modern dashboards have many graphs, indicators, alerts and filters. This can be confusing and hard to understand. When there is much information on the screen it slows down interpretation. Sometimes important insights are ignored, by people who are not technical experts. Many dashboards try to show a lot of information at once. However this can create problems than benefits. Users get overwhelmed and struggle to make sense of it all. As a result key findings may be missed. A simpler approach could help. By focusing on the important information dashboards can be more effective. This way users can quickly grasp the insights without feeling overwhelmed. The goal is to make dashboards easy to use. They should provide insights without causing confusion. By achieving this dashboards can become a tool, for decision-makers. To achieve this designers should prioritize simplicity. They should carefully select the information to be displayed. This way users can easily understand the insights. Make informed decisions.

C. Limited Explainability of AI Outputs

Many AI-generated recommendations, anomaly warnings, and forecasting visuals are produced through complex machine learning models that users cannot easily understand. This creates a trust gap because decision-makers may see the result but remain uncertain about how the system reached that conclusion. Explainability therefore remains one of the biggest open research issues in intelligent visualization.

D. Privacy, Security, and Ethical Concerns

We have systems that look at a lot of information like healthcare data and financial records. These systems also look at logs from companies and personal information from users. When we use intelligence to keep watching and analyzing this information we start to worry about people seeing things they should not see people getting in without

permission and people using the information in bad ways. So we need to make sure we are taking care of the data in a way so we can trust the systems that use healthcare data and financial records and other important information.

E. High Cost and Technical Dependency

Enterprise-grade AI visualization platforms require cloud infrastructure, software licenses, continuous maintenance, and trained analytical personnel. Smaller institutions and developing organizations may find these systems expensive and technically difficult to maintain. This creates unequal accessibility in the adoption of intelligent analytics.

F. Open Research Issues

The current research is not perfect. Needs to get better in areas, like explainable visual AI and lightweight low-cost analytics. We also need to work on dashboard personalization and real-time autonomous decision support.. Then there is human-centered visualization literacy. These are the things that're still problems. This shows that even though the field of AI and analytics and all that is moving forward really fast it is still being worked on and new things are being tried.

VIII. Future Trends and Emerging Research Directions

The future of AI-driven data visualization is expected to move beyond conventional dashboards toward more autonomous, explainable, and human-adaptive analytical systems. As data generation continues to increase in speed and complexity, visualization platforms will require stronger machine intelligence capable of not only presenting data but also continuously learning from it and suggesting strategic actions in real time.

A. Explainable Artificial Intelligence in Visualization

One of the most promising future directions is the integration of Explainable AI, where systems will provide clear reasons behind anomaly detection, predictions, and recommendations. This will improve user trust and make AI-generated visual insights more transparent and decision-friendly.

B. Conversational and Natural Language Analytics

Future visualization tools are increasingly expected to support conversational interfaces in which users can ask analytical questions in natural language and receive instant charts, summaries, and dashboard outputs. This will make complex analytics more accessible to non-technical professionals.

C. Autonomous Real-Time Decision Dashboards

Emerging systems are moving toward autonomous dashboards that can continuously monitor incoming data, detect sudden changes, and generate automated warnings or strategic suggestions without waiting for manual user interaction. Such systems will be highly useful in healthcare, finance, cybersecurity, and smart city operations.

D. Immersive and Collaborative Visualization Environments

Research is looking into ways to work together on the cloud using special reality dashboards and three-dimensional pictures where many people can look at and work with information at the same time. Immersive and Collaborative Visualization Environments like these can really help teams make decisions in companies, in the future.

E. Human Centered Adaptive Visualization

Future research is going to be about making dashboards that change how complicated the pictures are based on what the user knows what they do and what they need to decide. If we make Human Centered Adaptive Visualization that is just right for each person it can cut down on confusion. Help people understand things better which makes Human Centered Adaptive Visualization systems more useful, for lots of people.

IX. Conclusion

Artificial intelligence powered data visualization is really important for data analytics and decision support systems these days. Organizations are dealing with a lot of information both structured and unstructured. The old way of reporting numbers is not good enough to help people understand things quickly. Artificial intelligence powered

data visualization helps with this problem by changing data into simple visual insights. This helps people make decisions faster and smarter using intelligence powered data visualization. Artificial intelligence powered data visualization is very useful for making decisions, with big data analytics and decision support systems.

The review we have now shows that data visualization is not what it used to be. It has changed a lot from charts to interactive dashboards and advanced analytics. We also have reporting platforms that use intelligence to work on their own. Now we have tools like Microsoft Power BI, Tableau, SAS Visual Analytics and Looker that are helping organizations. These tools are making it easy for organizations to get their data ready make guesses about what will happen next and make it easy for users to access data. This is happening in areas, like healthcare, finance, education, retail and governance sectors. Data visualization is really helping organizations in these areas.

At the time the study found some big problems that make it hard to use new technology. These problems include data, too much information and not being able to understand how it works. There are also concerns about peoples information and the high cost of setting it up. These problems show that just having technology is not enough. We also need to make sure that the systems are designed with people, in mind are transparent and can be trusted.

X. References

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Autonomous Field Medical Robot for Emergency Care and Minor Surgical Procedures

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Abstract— In many emergency situations, especially in villages, forests, disaster areas, or rescue missions, getting immediate medical help is not always possible. People often work in risky environments where accidents and injuries can happen at any time. Sometimes, the injured person cannot be taken to a hospital quickly because of distance, lack of transport, war situation or poor medical facilities. In such situations, even a few minutes of delay can become dangerous. This paper presents the idea of a smart robotic medical assistant that can help injured people on the spot by providing basic treatment and minor surgical support during emergency. The proposed system must designed using Artificial Intelligence (AI), robotic, sensors, camera based technology so that it can quickly understand the condition of the injured person and provide immediate care. The robot can check important health vitals like heartbeat, BP, body temperature and bleeding. It can clean wounds, stop bleeding, give injections, perform simple stitching and provide basic first aid until proper medical help become available. This robotic system can be very helpful in remote rural areas, rescue operations, disaster managements, war areas and other remote areas. The main purpose of this project is to reduce treatment delays and provide quick medical support during critic situations where hospitals and doctors are not easily available. The study shows how smart robotic technology can become a lifesaving solution by offering fast, reliable, and accessible emergency healthcare support. ***Keywords***—*Artificial Intelligence (AI), Robotic Surgery, Emergency Medical Robotics, Computer Vision, Telemedicine, Autonomous Medical System, Biomedical Sensors, Remote Healthcare, Smart Healthcare, Field Emergency Support.*

I. Introduction

When an accident happens in a remote area, getting medical help on time is not always easy and this is a common problem in places like remote villages, forest areas, rescue camps, military bases and disaster-hit regions where hospitals and medical facilities may be far away or may not be available . People often work in difficult and risky environments, and injuries can happen at any moment. In many situations, the injured

person cannot reach a hospital quickly because of lack of transportation, or the absence of nearby healthcare services. During emergencies like these, even a small delay in treatment can become dangerous. In today's world technologies like artificial intelligence (AI) and robotics are being used in many areas of healthcare to make medical sector more accurate and faster . Hospitals already use robotic systems and smart monitoring devices for surgeries, patient care, and health tracking. But most of these systems are designed for hospital use and cannot easily help people working in remote outdoor locations where emergency support is urgently needed.

To help solve this problem, this paper introduces a smart robotic emergency healthcare system that can provide immediate medical support directly at the accident site. The system is designed using AI, robotic tools, sensors, cameras, and wireless communication technology. It can quickly understand the condition of the injured person and provide medical care without waiting for the doctor to reach . The robot can monitor important health vitals such as heartbeat, body temperature, blood pressure, and bleeding levels. Then according to the condition it can help with cleaning wounds, stopping bleeding, giving emergency injections, and performing simple stitching if needed. Along with this, the system also allows medical experts to connect remotely through the networking and guide the robot during critical situations if any arrived. By using real-time sensor data and camera-based monitoring, the robot can identify serious conditions and respond quickly during emergencies.

The main goal of this work is to reduce treatment delays and improve emergency healthcare support in remote and highrisk areas. The proposed system can be useful in rescue missions, military bases, disaster management operations, and remote rural regions where medical facilities are limited or barerly available. This study shows how smart robotic technology can support healthcare workers and provide fast, reliable, and life-saving medical assistance when every second matters.

II. Literature Review

Today technology is becoming a very important part of healthcare. In many healthcare centres doctors already use smart machines, sensors, and robots to help treat patients. These technologies make medical work faster and sometimes more accurate. AI and

robotic systems are now being used for surgeries, patient checking, and emergency care. But most of these systems are only made for hospitals and big medical centers. They are not very useful in places where medical help is far away. Many researchers have worked on systems that can check a person's health using sensors. These systems can measure heartbeat, body temperature, blood pressure, and other health conditions and send the information to doctors. This helps doctors monitor patients from a distance. But these systems only monitor the patient and cannot actually give treatment during an emergency.

Some studies used cameras and AI to identify injuries, wounds, and health problems automatically. These ideas are very useful because they help doctors understand the condition of a patient more quickly. Still, most of these systems are mainly used inside hospitals and need trained staff to operate them. Other studies also focused on robotic arms and AI-based medical assistants that can help doctors during operations and other medical tasks. These robots are very accurate and can perform medical work carefully. These technologies improve diagnosis and reduce the chances of mistakes while diagnosing the patients. However, most of these systems are expensive and designed mainly for large hospitals. Other researchers worked on telemedicine systems where doctors can guide patients from far away using internet communication. This became very useful in rural areas where hospitals are limited. Some emergency healthcare systems also use IoT devices and cloud technology to share patient information quickly during emergencies which in result reduce the delay. While these systems improve communication, they still depend on human medical staff to provide actual treatment. Also some studies focused on robotic emergency response systems for disaster and rescue operations. In rescue, robots were developed to assist during natural disasters and dangerous rescue missions. The robots were able to move in difficult environments and provide basic support. However, most of the existing rescue robots such as drone are mainly focused on search and surveillance activities rather than providing direct medical treatment. Despite the progress in AI-based healthcare and robotic technologies, we still face several challenges in developing a complete emergency robotic medical assistant. Existing systems often struggle with real-time decision-making, accurate injury detection, and performing medical tasks outside the hospitals. Many models also require large amounts of medical data and expensive equipments, making them difficult to use in rural

or low-resource areas. This paper tries to solve this problem by proposing a smart robotic emergency healthcare system that can provide quick medical help directly at the accident location. The system uses AI, sensors, cameras, robotic tools, and internet communication to understand the condition of the injured person and accordingly provide the basic treatment. The robot can check health conditions, clean wounds, stop bleeding, give injections, and provide first aid until proper medical help becomes available. The system can also connect with doctors remotely so that experts can guide the robot during the serious situations. The main aim is to make emergency healthcare faster, easier, and more accessible in places where medical facilities are limited.

III. Methodology

Sometimes accidents happen in places where medical help cannot be provided qucikly. People working or living in rescue teams, military camps, forests, or remote villages may get injured while doing their work. In many of these places, hospitals are far away or there may not be any doctor. During such situations, waiting too long for treatment can make the injury more serious. In this project we are creating a smart robot that can help during these emergencies. The robot is designed to give medical help to an injured person until proper treatment becomes available. The system will use simple technologies like AI, sensors, cameras, robotic tools, and internet connection to understand the person's condition and provide quick support. A. System Overview: The proposed model works in simple steps during an emergency. First it will do sensors and cameras check of the injured person and collect important health vitals. Then the system understands how serious the injury is. After checking everything, the robot gives basic medical help like cleaning wounds, stopping bleeding, giving injections,performing the Xray or doing simple stitching if needed. The system includes: 1. Sensors to check health condition 2. AI system to understand injuries 3. Cameras to look at the injury 4. Robotic arm for treatment 5. Internet and communication system 6. First-aid support tools If the condition is serious, the robot can also connect with the medical expert through the internet so that he can guide during the treatment. Fig. 1 Proposed Methodology for the robotic model B. Health Monitoring and Data Collection: The robot checks important vitals of the injured person using sensors. It can check the following details of the injured person and collect their data in the database for future references: • Heartbeat • Body temperature • Blood

pressure • Oxygen level • Bleeding condition • Portable digital X-ray These details help the robot understand whether the injured person is stable or needs urgent help. The camera also captures images of the injury to understand the situation better. C. Injury Detection: The camera and AI system help the robot to identify the injury. The system can detect: • Cuts and wounds • Heavy bleeding • Burns • Swelling • Other visible injuries After identifying the injury, the robot decides which treatment should be given first. D. Emergency Medical Support: When the injury is diagnosed, the robot starts to give basic medical support. The robotic arm does simple medical procedures with care. The robot can: • Clean wounds • Control Bleeding • Give medicine • Give injections • Do basic stitches • Place bandages • Give first aid The main goal is to make sure the injured person is safe until they receive the right medical support. E. Remote Doctor Support: In any situation if the condition becomes serious, the robot can connect with medical experts using the internet. Doctors can monitor the patient, check its vitals, being at another location and guide the robot during the emergency. This is useful in military bases, rescue camps, and disaster areas where doctors may not be available nearby. F. Emergency Decision Process: The system follows a simple process during emergencies: 1. Check the injured person 2. Identify the injury 3. Understand how serious the condition is 4. Give first aid or treatment 5. Connect with doctors if needed 6. Keep monitoring the patient until help arrives G. Benefits of the Model This system can help reduce delays in treatment during emergencies. In many remote places, people do not get medical help on time, and injuries can become dangerous because of waiting too long. The robot can provide quick support during those important first few minutes after an accident. The system can be useful in areas like : • Remote Villages • Forest areas • Rescue camps • Disaster zones • Military camps • Remote areas far from hospitals The main goal of this project is to provide fast medical help during emergencies and make healthcare support available in places where hospitals and doctors are not easily reachable.

IV. Result and Discussion

The proposed robotic healthcare system shows how AI, sensors, and robotic technology can be combined to provide quick medical help during emergencies. The system is mainly designed for people working in places where hospitals and doctors are not easily available. Since this is a proposed model, the idea focuses on how the robot could

identify injuries, monitor health conditions, and provide basic treatment during emergency situations. The proposed model uses AI together with sensor-based monitoring to better understand the condition of an injured person. In this system, cameras and AI are expected to identify the wounds, bleeding, burns, and other visible injuries, while sensors would monitor important health conditions such as heartbeat, body temperature, blood pressure, and oxygen level also it will perform the Xray of the injured person to check whether any internal injury is there or not . Based on this information, the robot could understand how serious the injury is and decide which treatment should be given first. The robotic model in the proposed system is designed to perform basic emergency tasks such as cleaning wounds, stopping bleeding, applying medicine, and providing bandages. The model also suggests support for simple stitching and injection delivery with remote doctor guidance if required. Through internet connection, doctors would be able to monitor the patient from another location and guide the robot during serious situations. Different AI methods and monitoring approaches were studied while designing this proposed system. Traditional monitoring systems mostly depend on human support and may take more time during emergencies. Basic AI systems can help in detecting injuries, but they may face difficulties in outdoor and remote conditions. The proposed hybrid robotic model aims to improve emergency healthcare support by combining sensors, camera analysis, and AIbased decision-making in one system.

V. Conclusion

This paper presented the idea of a smart robotic healthcare system that can help injured people during emergencies, especially in places where hospitals and doctors are not easily available. Many rescuers work in risky areas such as forests, villages, rescue camps, and disaster zones where accidents can happen at any time. In such situations, getting quick medical help becomes very difficult, and delays in treatment can make injuries more serious. The proposed system combines AI, sensors, cameras, robotic tools, and internet connection to provide basic emergency healthcare support. The model is designed to monitor important health conditions, identify injuries and provide first aid support such as wound cleaning, stopping bleeding, applying medicine, giving injections, and simple stitching support during emergencies. The study suggests that combining AI with sensors and robotic systems can help improve emergency response and reduce delays in

treatment. Compared to traditional emergency support methods and basic monitoring systems which can only be available at the hospitals , the proposed model aims to provide faster injury detection, better health monitoring, and quicker first-aid support during critical situations. The paper also highlights the importance of remote doctor support. Through internet connection, doctors can also monitor the patient and guide the robotic system from another location during serious emergencies. This can be very useful in remote places where medical experts may not be able to reach quickly. Although this work is a proposed model and has not yet been fully developed or tested in real-life situations, it shows the future potential of smart robotic healthcare systems in emergency medical support. There are still challenges which we may face during its making such as internet connectivity in remote areas, maintenance of robotic systems, and the need for proper training of the data sets and safety measures. Overall, this study presents a concept that can help improve emergency healthcare support in villages, military camps, rescue operations, and disaster-affected areas. In the future, with further development and practical implementation, such robotic systems may become helpful in providing quick medical support when immediate hospital care is not available.

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Study on Language Barrier in Healthcare: A Literature-based Study for Improving Patients' Communication

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Abstract—The number of people migrating has increased over time due to several reasons, such as employment opportunities, education, better health services, and displacements due to conflicts and natural disasters. Nonetheless, immigrants encounter various obstacles when trying to access healthcare services due to their inability to speak English or the local language fluently. This study investigates the consequences of the linguistic differences between immigrants and healthcare practitioners, exploring ways of improving communication to achieve better health outcomes. The review includes an analysis of numerous publications describing how linguistic barriers affect communication, lead to misdiagnosis, prescription mistakes, delays in diagnosis, and decreased patient satisfaction. The review shows that a lack of linguistic skills results in poor health outcomes for immigrants since they cannot understand their conditions and be engaged in treatment. Although interpreter services and translation tools have been implemented to mitigate this problem, they are not always effective or culturally appropriate. This research also emphasizes the potential of advanced technologies, such as artificial intelligence-based translation software, multilingual healthcare apps, and automated communication tools, to improve healthcare communication. Furthermore, a conceptual framework is developed, suggesting a combination of human interpreter services and advanced technological tools. According to the study, dealing with language barriers is a very crucial aspect when trying to attain equity in healthcare among multicultural populations. Using an integrated strategy that involves better policies, training of professionals, and development of technologies can make great strides toward this objective.

Keywords*—Language Barriers, Communicating in Healthcare, Migrants' Healthcare Issues, Multilingual Healthcare Settings, Medical Interpretation Services, Patient–Physician Interaction, Healthcare Accessibility, Artificial Intelligence in Healthcare, Machine Translation in Medicine, Cross-Cultural Communication, Health*

Disparities, Healthcare Gaps, Patient Safety, Healthcare Technology Solutions, Healthcare.

I. Introduction

Migration is an issue in the world today. People are moving from one place to another for various reasons. They want jobs, access to good healthcare, and education to be with their families or to escape from wars and natural disasters. Because of globalization, people are moving across borders more and more. Even though migrants have the right to healthcare as everyone else, they often do not get the same level of care. One of the reasons for this is that people do not speak the same language. This makes it hard for patients and doctors to talk to each other. It is not about having hospitals and doctors; it is about being able to use these services. For migrants not speaking the language makes it hard to say what is wrong with them to understand what the doctor is saying and to be a part of deciding what treatment they get. This can lead to delayed treatment, wrong diagnoses, mistakes with medicine, and poor care. Language problems also make it hard for patients and doctors to have a relationship. When people do not understand each other, patients do not trust the doctors and hospitals much. This can stop people from going to the doctor when they need to or from taking their medicine. As a result, people may not get especially immigrants. In countries like Australia, where there are different languages and cultures, it is hard to make sure everyone gets the same level of care. The more different languages and cultures there are, the more important it is to have healthcare systems that understand and respect these differences. Also, when people do not speak the language, it can be very stressful and sad for them. They may feel alone. Misunderstood when they go to the doctor. Doctors may also feel overwhelmed and frustrated when they cannot communicate with their patients. This can affect the care they give. Sometimes people use family members to translate. This can lead to mistakes and secrets being shared. Even though there are translators and technology to help, it is not always available. This is why we need to find better ways to communicate. In addition, fixing language problems is important for making sure everyone gets equal care. This study looks at how language problems affect healthcare for migrants and tries to find ways to make things better and more accessible for everyone.

II. Literature Review

Language barriers in healthcare are a problem. They affect how good, safe, and easy it is for people to get help. Studies show that when patients and doctors can talk well, it helps them get the diagnosis, treatment, and good health results. When people don't speak the same language, it can lead to mistakes and unhappy patients. Many patients who don't speak the language have poorer health results compared to those who do. They have misdiagnoses, medication errors, and don't follow treatment plans. This shows that language barriers are not a communication issue but a serious public health concern. Some studies looked at immigrant populations. They found that language barriers make it hard for people to get healthcare. Migrants struggle to understand instructions, book appointments, and tell doctors about their symptoms. This leads to delayed treatment and more emergency care. Professional medical interpreters can really help. They reduce mistakes. Make patients understand better. Many healthcare systems still use family members or friends to interpret, which can be biased or inaccurate. New technologies like machine translation, multilingual health apps, and AI symptom checkers are being explored. They show promise. Also have limitations like cultural insensitivity and inaccuracies. Research shows that language barriers are a problem in multicultural societies and places with uneven healthcare. Countries with languages have a harder time making sure everyone gets equal healthcare. The current solutions, like interpreter services and technology, help. There's still a big gap. We need solutions that combine human expertise with advanced technologies. Studies on patients and healthcare providers show concerns about communication. Patients with limited language proficiency have difficulties understanding doctors. Doctor-patient communication plays a role in better health results. Language barriers in healthcare are still a challenge.

III. Proposed Methodology

This study is looking at how language barriers affect healthcare and what can be done to solve this problem using technology and other systems. The goal is to look at what research has been done and come up with a plan to improve communication in healthcare when people speak different languages. A. How We Did the Research We looked at what other people have written about language barriers in healthcare, how people move from one place to another, and how patients and healthcare workers talk to

each other. We also found out how technologies are used in the healthcare system. We wanted to understand the problems people face when they do not speak the language and what can be done to help. B. Where We Got the Information We used information from studies, research papers, and reports from healthcare organizations. We found this information in databases and other places where people share what they have learned. This helped us understand what really happens when people who speak languages try to get healthcare. C. How We Collected the Information We searched for information on academic websites and in research papers. We looked for studies about how patients and doctors talk to each other, how interpreters help,p and how artificial intelligence is used in healthcare. This helped us understand the problem. D. How We Analyzed the Information We looked for themes in what we found, like problems with communication, getting healthcare, keeping patients safe,e and using technology to help. By comparing what different studies found, we were able to see patterns and common problems in healthcare communication. E. Creating A Plan Based on what we learned, we came up with a plan that shows how important it is to have systems that support languages, interpreter services, and artificial intelligence in healthcare. This plan suggests how we can use these tools together to make healthcare better for everyone. F. What We Found Out We looked at what many studies found out about language barriers and how they affect healthcare. We also thought about how new technology and better policies can help make healthcare more inclusive and work better for everyone. Language barriers are a problem in healthcare, and we need to find ways to solve this problem so everyone can get good healthcare, no matter what language they speak.

IV. Results and Discussion

This study is about the problems that language barriers in healthcare cause. The study checks how technology can help fix language barriers in healthcare. It looks at studies on language barriers in healthcare. These studies see how language barriers affect healthcare. They also check what technology can do to help with language barriers in healthcare. * The studies find that language barriers cause issues with healthcare. Language barriers lead to misunderstandings. They cause delayed diagnosis and medical errors. Patients who do not speak the language are not satisfied. They also have access to

healthcare than native speakers. The research finds that interpreter services and translation tools can help. These tools help patients and healthcare providers talk better. However, these tools are not always available. They are also not always accurate. Some studies say that new technologies like intelligence and machine learning can help. These technologies can help reduce communication gaps caused by language barriers. They can provide support for languages. They can make medical information easier to understand for patients with language barriers. Overall, the analysis shows that using technology with healthcare services can improve communication. It can help patients understand better. It can reduce problems caused by language barriers. However, most studies say that current solutions for language barriers are not complete. They need to be developed. They need to be reliable and culturally sensitive. They need to help solve language barriers in healthcare.

V Future Scope

The problem of language barriers in healthcare has several important areas that need to be improved. One big area is using technology such as AI translation services. Even though interpreter services are already in use, tools like real-time speech translation, mobile health apps, and AI chatbots can help patients and doctors communicate more quickly. Doctors and nurses should also receive training in communication skills across multiple languages. This helps patients feel more at ease, lowers the chance of errors, and makes healthcare better overall. Hospitals can use software to provide forms, prescriptions, and instructions in many languages. This reduces confusion among patients and helps them follow their treatment plans. Improving interpreter services is very important. Healthcare systems should use medical interpreters who understand both the language and medical terms. Regular training and setting clear certification standards can help improve the quality of interpretation. To create a better healthcare system, we should use both human expertise and technology. This makes sure that language differences don't stop patients from getting the right and timely medical care. Language barriers in healthcare need to be tackled through these improvements. These changes help patients and doctors talk to each other more effectively. Using technology and trained interpreters can really make a difference.

VI. Conclusion

Language barriers in healthcare are a problem. They affect the quality of care that people get in places where many languages are spoken. This study shows that it is very important for patients and healthcare providers to be able to communicate with each other. When doctors and patients can talk to each other easily, doctors can. Treat patients correctly, and patients are happy with their care. When patients and providers do not speak the same language, mistakes can happen,n and patients may not get the care they need right away. The study found that people who are immigrants or who speak languages often have a hard time getting good healthcare. Sometimes interpreters and translation tools can help. They are not always available or used correctly. This means that there is often a gap in communication between healthcare providers and patients, and this gap needs to be fixed. The study also says that new technologies, like computers that can understand languages, can help solve communication problems. Using technology to make interpreter services and teaching healthcare providers how to communicate in different languages can make patient care better and reduce problems. In conclusion, getting rid of language barriers is very important for making sure everyone gets the healthcare and respect cultures. This will help build a healthcare system where every patient gets care, no matter what language they speak. Language barriers in healthcare must be overcome. Every patient can get good medical care.

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An Empirical Investigation into the Efficacy of Digital Literacy Programs in Combating UPI Frauds among Urban Undergraduates

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Abstract- The rapid growth of digital payment systems in India has led to widespread adoption of the Unified Payments Interface (UPI) among consumers and shopkeepers. Along with convenience and speed, the increasing use of UPI has also resulted in a rise in UPI-related frauds. The present study titled "An Empirical Investigation into the Efficacy of Digital Literacy Programs in Combating UPI Frauds among Urban Undergraduates" aims to examine the level of awareness, experience of fraud, safety practices, and reporting behaviour of UPI users.

The study is based entirely on primary data collected from 153 respondents through a structured questionnaire. The respondents include consumers and shopkeepers belonging to different age groups, genders, and occupations. Descriptive Analysis, Frequency Analysis, and Inferential Analysis were adopted, and the data were analysed using percentage analysis, tables, charts, Likert scale mean analysis, SmartPLS, and ANOVA. The findings reveal that although general awareness of UPI fraud is high, a significant proportion of respondents have experienced fraud. Businessmen were found to be the most vulnerable group (74.47%). Older age groups showed higher fraud exposure, while younger users demonstrated better digital familiarity. Fake payment screenshots, refund request frauds, phishing links, and fake customer care calls were the most common fraud types. Many fraud incidents remain unreported, mainly due to lack of clarity about reporting procedures.

Keywords: *UPI Fraud, Digital Payments, Fraud Awareness, NPCI, Social Engineering, Cybercrime, ANOVA, SmartPLS, Surat City.*

I Introduction

The evolution of digital payment systems in India reflects a fundamental response to long-standing inefficiencies in traditional financial transactions. For decades, the Indian economy was predominantly cash-driven, with limited accessibility and practicality of early digital payment mechanisms such as NEFT, RTGS, and IMPS (Reserve Bank of

India, 2016). Recognizing the need for a seamless, real-time, and universally accessible payment infrastructure, the Unified Payments Interface (UPI) was introduced in 2016 by the National Payments Corporation of India (NPCI), under the guidance of institutions such as the Reserve Bank of India (RBI) and the Indian Banks' Association (IBA) (NPCI, 2023). Conceptual contributions from key policymakers and technocrats, including Raghuram Rajan and Nandan Nilekani, facilitated the development of a system that revolutionized financial transactions by enabling instant, cost-free, and interoperable payments (Nilekani, 2019; Rajan, 2016). The rapid integration of UPI into major platforms like PhonePe, Google Pay, and Paytm significantly accelerated its adoption, making it the dominant mode of digital payments in India, with billions of monthly transactions and widespread institutional participation (NPCI, 2025).

However, the exponential growth of UPI usage has been accompanied by a parallel rise in digital payment frauds, primarily driven not by technological vulnerabilities but by human factors such as lack of awareness, misinformation, and misplaced trust (Reserve Bank of India, 2022). Fraudulent practices such as collect request scams, fake payment screenshots, phishing attacks, QR code manipulation, and screen-sharing exploits have increasingly targeted users, particularly young and digitally active populations like urban undergraduates (NPCI, 2024; RBI, 2023). These developments highlight a critical gap between digital adoption and digital literacy. In this context, digital literacy programs emerge as a vital intervention to equip users with the knowledge and skills necessary to identify, prevent, and respond to UPI-related frauds. Therefore, the present study seeks to empirically investigate the effectiveness of such programs in enhancing awareness and reducing vulnerability to digital payment frauds among urban undergraduate students, thereby contributing to safer digital financial ecosystems.

II Review of Literature

A review of existing literature indicates a growing convergence of academic and practitioner interest in understanding UPI adoption, associated fraud risks, and the critical role of user awareness in ensuring secure digital transactions. Early empirical work by Mukhopadhyay and Mukhopadhyay (2022) examined the relationship between UPI usage and fraud incidence, revealing that while technological safeguards such as two-factor

authentication enhance transaction security, the perceived threat of fraud significantly influences user behaviour and adoption patterns. Similarly, Khandal (2022) observed that although the shift toward digital payments is evident among common citizens, gaps in cybersecurity awareness and fraud prevention continue to undermine user confidence, thereby necessitating coordinated educational initiatives by financial institutions and policymakers.

Subsequent studies have emphasized the complexity and evolving nature of UPI-related cyber threats. Wadkar et al. (2025) identified key challenges such as phishing, fake UPI IDs, OTP misuse, social engineering, and malware attacks, concluding that enhanced customer awareness and stronger regulatory frameworks are essential for mitigating fraud risks. Supporting this perspective, Swaraj and Bohara (2023–24) highlighted that inadequate consumer education remains a major enabler of UPI scams, advocating for interdisciplinary strategies involving stakeholders across the financial ecosystem. Bala Ugandhar (2023) further contributed by analysing consumer awareness and satisfaction levels, identifying gaps in users' understanding of security features and overall trust in UPI systems.

In parallel, technology-driven approaches to fraud detection have gained prominence in recent literature. Sadaf and Manivannan (2025) proposed a Gradient Boosting-based model achieving high predictive accuracy, demonstrating the potential of advanced analytics in identifying fraudulent transactions. Similarly, Khopade and Vitalkar (2025) developed a machine learning-based real-time fraud detection system capable of adapting to evolving fraud patterns, thereby enhancing transaction security and user confidence. Complementing these findings, Edburg et al. (2022)

examined behavioural aspects influencing UPI adoption, identifying perceived ease of use, security, and fraud risk as key determinants of user trust. Furthermore, Maini and Sindhi (2024) advocated a holistic approach combining AI-driven detection, regulatory coordination, and consumer education to effectively combat digital banking frauds.

Overall, the literature underscores that while technological advancements have significantly improved fraud detection capabilities, the human element—particularly user awareness and digital literacy—remains a critical vulnerability. Despite substantial

research on fraud typologies and detection mechanisms, limited empirical attention has been given to evaluating the effectiveness of digital literacy programs in reducing fraud susceptibility, especially among urban undergraduate populations. This gap provides a strong foundation for the present study, which seeks to empirically assess how digital literacy interventions can mitigate UPI fraud risks and enhance secure digital financial behaviour.

III Objectives of The Study

1. To study the level of awareness regarding UPI fraud among the general public in Surat city.

2. To examine the UPI usage pattern among respondents belonging to different occupations — students, businessmen, salaried employees, housekeepers, and others.

3. To identify the types of UPI fraud known to people based on their personal awareness and experience.

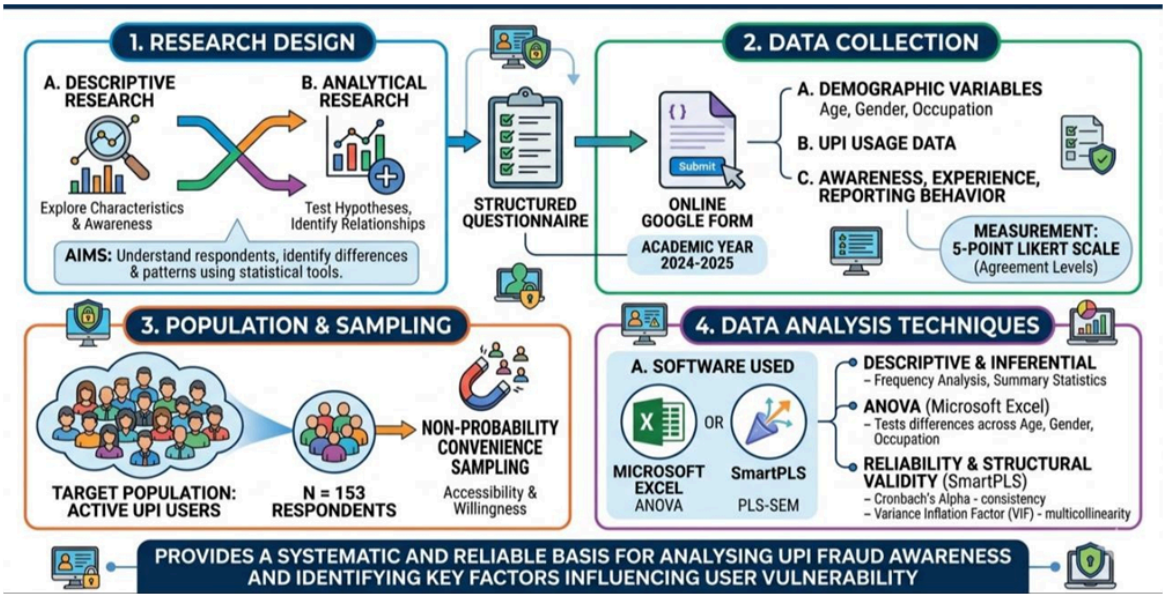
4. To analyze the experience of UPI-related fraud among respondents and identify vulnerable demographic groups.

5. To assess the level of awareness regarding basic safety and preventive practices related to UPI transactions.

6. To compare the level of UPI fraud awareness across age, gender, and occupation groups using inferential statistical analysis.

IV Research Methodology

The present study employs a descriptive and analytical research design to examine the level of awareness regarding UPI fraud among users and to analyse patterns related to fraud experience, safety practices, and reporting behaviour. The descriptive approach is used to understand the characteristics and awareness levels of respondents, while the analytical approach enables the application of statistical tools to identify relationships and differences among variables .



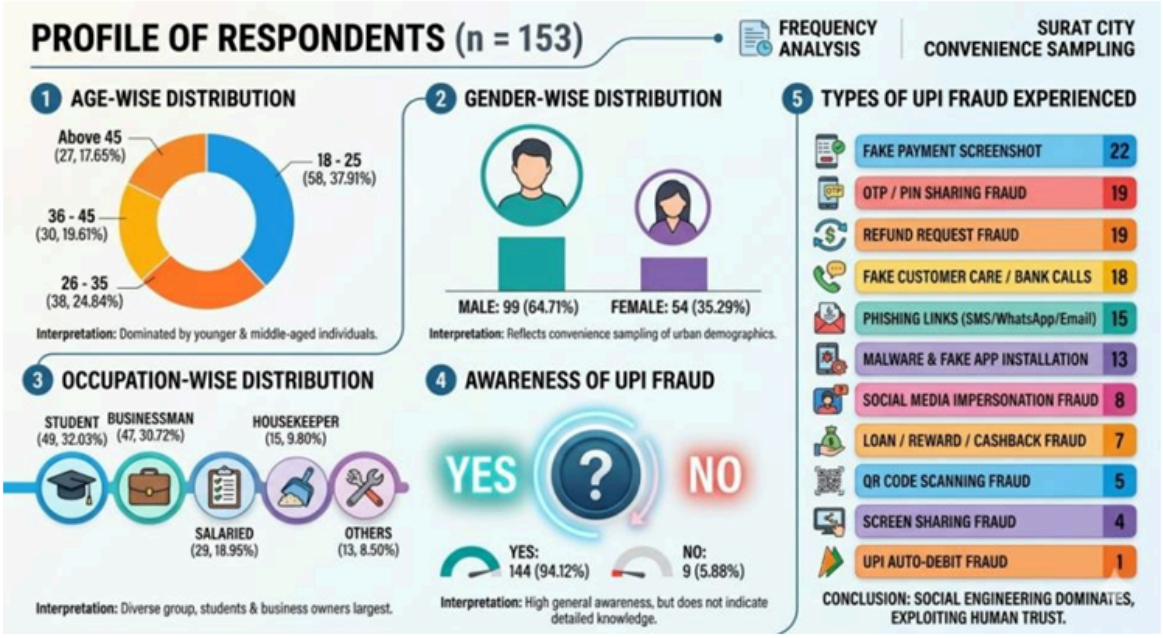
The study is based on primary data collected through a structured questionnaire. The questionnaire was designed to gather information on demographic variables such as age, gender, and occupation, along with data related to UPI usage, awareness of fraud types, experience of fraud, reporting behaviour, and knowledge of preventive measures. A 5-point Likert scale was used to measure respondents' level of agreement with statements related to fraud awareness and safety practices.

The population of the study includes individuals who actively use UPI for digital transactions. A total of 153 respondents were selected using a convenience sampling method, which is a non- probability sampling technique based on accessibility and willingness to participate. The data was collected through an online Google Form during the academic year 2024–2025.For data analysis, both descriptive and inferential statistical techniques were used. Frequency analysis was applied to examine demographic distribution and general awareness levels. Descriptive analysis was used to summarize responses related to fraud experience and behavioural patterns. Further, Analysis of Variance (ANOVA) was conducted using Microsoft Excel to test whether significant differences in awareness exist across age groups, gender, and occupational categories. Additionally, SmartPLS software was used to assess reliability and structural validity, including Cronbach's Alpha for internal consistency and Variance Inflation Factor (VIF) to check multicollinearity among indicators.

This methodological framework provides a systematic and reliable basis for analysing UPI fraud awareness and identifying key factors influencing user vulnerability. **T**

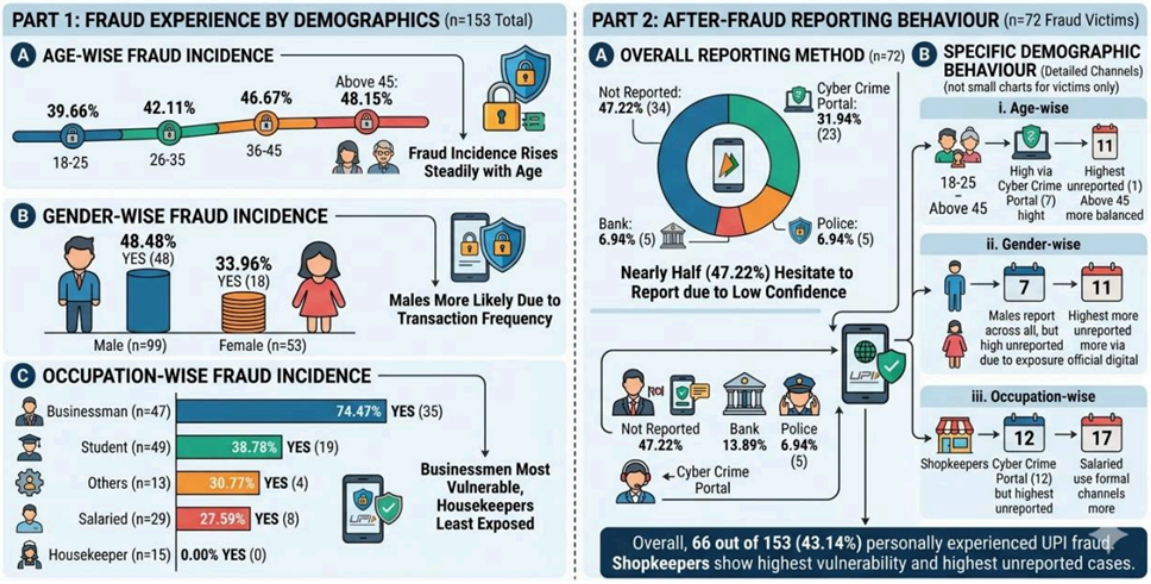
V Data Analysis and Interpretation

The responses were systematically organised and analysed using Microsoft Excel and SmartPLS in four stages: Frequency Analysis (demographic distribution), Descriptive Analysis (fraud experience and reporting behaviour), Inferential Analysis (ANOVA for group differences), and Reliability Analysis (SmartPLS).



The study profile reveals a demographic primarily driven by younger and middle- aged individuals, with the 18–25 age group forming the largest segment at 37.91%. Males represent a significant majority of the sample at 64.71%, while the occupational background is led by students (32.03%) and businessmen (30.72%), providing a broad perspective on UPI usage. Although a vast majority of respondents (94.12%) possess a general awareness of UPI fraud, this does not necessarily translate into technical knowledge of prevention or recovery. The data further indicates that social engineering tactics are the primary threat, as evidenced by the high prevalence of fake payment screenshots (22 respondents), OTP/PIN sharing scams, and refund request frauds (19

each), highlighting that fraudsters predominantly exploit human trust over technical vulnerabilities.



The descriptive analysis reveals clear demographic patterns in UPI fraud exposure and reporting behaviour. Fraud incidence shows a steady increase with age, rising from 39.66% among respondents aged 18–25 to 48.15% among those above 45, indicating that older individuals are relatively more vulnerable, possibly due to lower digital familiarity and higher trust in transaction processes. Gender-wise, males (48.48%) experience significantly higher fraud compared to females (33.96%), which may be attributed to greater transaction frequency and digital engagement among male users. Occupation-wise, businessmen emerge as the most vulnerable group with an exceptionally high fraud rate of 74.47%, largely due to frequent transactions and exposure to tactics such as fake payment screenshots, whereas housekeepers reported no fraud cases, reflecting minimal digital transaction involvement. Overall, 43.14% of respondents have experienced UPI fraud, highlighting the widespread nature of the issue. In terms of post-fraud behaviour, a concerning trend is observed where nearly half of the victims (47.22%) did not report the incident, indicating low awareness or lack of confidence in grievance redressal mechanisms. Among those who reported, the Cyber Crime Portal was the most preferred channel. Age-wise patterns show that younger users are more inclined to use digital reporting platforms but also account for the highest non-reporting cases, reflecting inconsistent response behaviour. Gender analysis suggests

that males dominate both reporting and non-reporting due to higher exposure, while females show relatively greater reliance on formal digital complaint channels. Occupation-wise, businessmen again stand out with both the highest reporting through the Cyber Crime Portal and the highest non-reporting cases, possibly due to time constraints and operational pressures, whereas salaried individuals demonstrate more balanced and formal reporting practices. Overall, the findings underscore that vulnerability to fraud and subsequent reporting behaviour are strongly influenced by demographic factors, reinforcing the need for targeted digital literacy and awareness interventions.

VI Inferential Analysis

1 ANOVA Analysis

A. Occupation-wise Difference in UPI Fraud Awareness

H_0: There is no significant difference in UPI fraud awareness among people belonging to different occupational groups.

H_1: There is a significant difference in UPI fraud awareness among people belonging to different occupational groups.

ANOVA — Occupation-wise

Source of Variation	SS	df	MS	F	P-value	Fcrit
BetweenGroups	3.732532	4	0.933133	3.647374	0.007272	2.432788
WithinGroups	37.86387	148	0.255837	–	–	–
Total	41.59641	152	–	–	–	–

p-value = 0.007272 < 0.05 → H_0 Rejected. H_1 Accepted.

Since the p-value (0.007272) is less than 0.05 and the calculated F-value (3.647) is greater than the F-critical value (2.433), there is a statistically significant difference in

UPI fraud awareness among different occupational groups. Occupation significantly influences the level of awareness regarding UPI fraud.

B. Gender-wise Difference in UPI Fraud Awareness

H_0: There is no significant difference in UPI fraud awareness between male and female respondents.

H_1: There is a significant difference in UPI fraud awareness between male and female respondents.

Source of Variation	SS	df	MS	F	P-value	Fcrit
BetweenGroups	0.915034	1	0.915034	3.396399	0.0673	3.903781
WithinGroups	40.68137	151	0.269413	–	–	–
Total	41.59641	152	–	–	–	–

p-value = 0.0673; 0.05 → H_0 Accepted. H_1 Rejected.

Since the p-value (0.0673) is greater than 0.05 and the F-value (3.396) is lower than the F-critical value (3.903), there is no statistically significant difference in UPI fraud awareness between male and female respondents**.**

C. Age-wise Difference in UPI Fraud Awareness

H_0: There is no significant difference in UPI fraud awareness among respondents belonging to different age groups.

H_1: There is a significant difference in UPI fraud awareness among respondents belonging to different age groups.

Source of Variation	SS	df	MS	F	P-value	Fcrit

BetweenGroups	**3.75456**	**3**	**1.25152**	**4.898157**	**0.002829**	**2.665729**
WithinGroups	**37.81524**	**148**	**0.255508**	**–**	**–**	**–**
Total	**41.5698**	**151**	**–**	**–**	**–**	**–**

p-value = 0.002829 < 0.05 → H_0 Rejected. H_1 Accepted.

Since the p-value (0.002829) is less than 0.05 and F-value (4.898) is greater than F-critical (2.666), there is a statistically significant difference in UPI fraud awareness across different age groups.

UPI App Security Perception by Age Group

Source of Variation	SS	df	MS	F	P-value	Fcrit
BetweenGroups	**3.75456**	**3**	**1.25152**	**4.898157**	**0.002829**	**2.665729**
WithinGroups	**37.81524**	**148**	**0.255508**	**–**	**–**	**–**
Total	**41.5698**	**151**	**–**	**–**	**–**	**–**

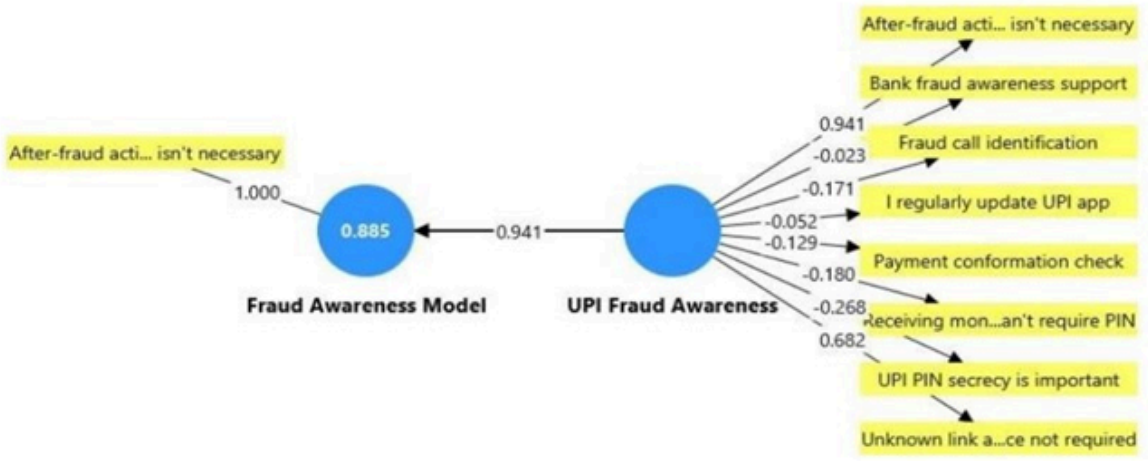
The youngest group (18–25) has the lowest mean score (3.66), indicating least trust in UPI app security. The mean rises steadily with age: 3.84 (26–35), 4.07 (36–45), and 4.41 (above 45), suggesting older users have the most confidence in institutional fraud prevention. Younger, more digitally literate users appear to be more critical of institutional awareness efforts.

Collinearity Assessment (SmartPLS — VIF)

Indicator	VIF
After-fraud action isn't necessary	**1.316**
Bank fraud awareness support	**1.470**
Fraudcall identification	**1.440**

UPI PIN secrecy is important	**1.000**
Iregularly update UPI App	**1.454**
Payment confirmation check	**1.923**
Receiving money can't require PIN	**1.826**
UPIPINsecrecyisimportant(constructlevel)	**2.670**
Unknownlinkavoidancenotrequired	**1.311**

All VIF values range from 1.000 to 2.670, well below the accepted threshold of 5.0. This confirms the absence of multicollinearity among the indicators. The indicators used to measure UPI fraud awareness are sufficiently independent and suitable for further structural analysis.



PLS-SEM model diagram

ReliabilityAnalysis(SmartPLS)

Construct	Cronbach's Alpha	Composite Reliability(rho_a)	Composite Reliability(rho_c)
UPIFraud Awareness	0.715	0.887	0.779

The Cronbach; Alpha value of 0.715 exceeds the commonly accepted threshold of 0.70, indicating acceptable internal consistency and reliability of the measurement scale. Composite Reliability values (rho_a = 0.887; rho_c = 0.779) further confirm the construct's internal consistency.

VII Findings of The Study

The analysis of primary data collected from 153 respondents indicates that general awareness of UPI fraud is high, with 94.12% of respondents acknowledging the possibility of fraud in digital transactions . However, this awareness does not fully translate into safe practices, as 43.14% of respondents reported having personally experienced UPI-related fraud. Among the various fraud types, fake payment screenshot fraud emerged as the most common, followed by OTP/PIN sharing, refund request scams, and fake customer care calls, highlighting the dominance of social engineering techniques over technical system failures.

Demographic analysis reveals that fraud incidence increases with age, with respondents above 45 years showing the highest exposure. Occupational analysis further indicates that businessmen (shopkeepers) are the most vulnerable group, with a significantly higher proportion experiencing fraud compared to students and salaried individuals. In contrast, gender-based analysis shows no statistically significant difference in awareness levels between male and female respondents, as supported by ANOVA results . The study also identifies a critical gap in fraud reporting behaviour. A large number of victims did not report incidents, and among those who did, the Cyber Crime Portal was the most preferred channel, while reporting to banks and police remained limited. Statistical testing confirms that awareness levels differ significantly across age and occupation but not across gender. Overall, the findings suggest that while theoretical awareness is widespread, practical understanding, verification behaviour, and post-fraud action remain insufficient among users.

VIII Conclusion

The study was conducted using primary data collected from 153 respondents through a structured questionnaire. The findings show that although general awareness of UPI fraud is high (94.12%), a significant 43.14% of respondents have personally experienced fraud — clearly demonstrating that awareness alone is insufficient without practical knowledge, cautious behaviour, and timely action.

The study reveals that businessmen (shopkeepers) are significantly more vulnerable to UPI fraud than other groups (74.47% fraud incidence), primarily due to high transaction frequency and dependence on UPI for daily business. Age-wise analysis shows higher fraud incidence among older users, while younger users display better digital familiarity. ANOVA confirms statistically significant differences in UPI fraud awareness by occupation ($p = 0.007$) and age ($p = 0.003$), but not by gender ($p = 0.067$).

Social engineering–based frauds — fake payment screenshots, refund request frauds, phishing links, and fake customer care calls — are the most common, confirming that fraudsters exploit human trust rather than technical weaknesses. Fraud reporting remains critically weak, with nearly half of victims reporting no action. Among those who do report, the Cyber Crime Portal is preferred over banks or police.

SmartPLS reliability analysis validates the measurement instrument: Cronbach's Alpha of 0.715 and all VIF values below 5.0 confirm the model's soundness. Overall, the study concludes that UPI fraud awareness must shift from general knowledge to practical, behaviour-oriented understanding, with focused awareness programs and improved reporting mechanisms essential for safer UPI usage in India's growing digital economy.

IX Implication and Limitation of the Study

The study highlights important future implications for strengthening digital payment security while acknowledging certain limitations. It emphasizes the need to move beyond general awareness toward practical, behaviour-focused digital literacy programs, especially for high-risk groups like businessmen and older users. Integrating financial literacy into academic curricula and enhancing user interface safeguards in UPI apps can reduce fraud risks. Improving trust, accessibility, and efficiency of reporting mechanisms such as the Cyber Crime Portal is also essential. Future research should adopt larger samples, longitudinal approaches, and advanced analytical tools to deepen insights. However, the study is limited by its small sample size of 153 respondents and reliance on self-reported data, which may introduce bias. Its cross-sectional design restricts analysis over time, and it considers limited demographic variables. Additionally, the study does

not experimentally test specific literacy interventions. Rapid changes in fraud techniques may also affect the long-term applicability of the findings.

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A Study On Opinion of The Customers Towards Usage of CNG Four Wheelers

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Abstract–In the stone age era, human beings move from one place to another place by walk only. Even though, the distance is very large, the carry food with us, stay in some places in night and continue their travel. After the development of road infrastructure facilities, they have used buses and trains. It saves their travelling time and reduced the walking efforts. Cycles and two wheelers are also used for convenient travels. Auto rickshaws and taxies are available to reach the particular destination point. By using public transport, they travelled from source to destination. When the income level of the human beings increased, they improve their standard of living. Furniture, utensils, home decors, electronic gadgets are purchased to increase their status of living. Two wheelers and four wheelers are the symbol of prestige. By using two wheelers and four wheelers, they incurred a cost of petrol and diesel. Apart from this, it caused air pollution. To solve this problem, CNG vehicles are introduced.

Key words: CNG, Mileage, Environmental pollution

I Introduction

Compressed Natural Gas (CNG) is a cleaner, cost-effective substitute to petrol and diesel, mainly self-possessed of methane crushed to 200–250 kg/cm². It controls inner ignition engines with lesser emissions, making it prevalent for private and commercial vehicles in India. With regard to CNG, Maruti Suzuki and Tata holding a dominant market share. CNG is cheaper and produces roughly 30% less CO 2 and 95% less nitrogen oxide. CNG is generally more fuel- efficient and cost-effective per kilometer, while LPG provides higher energy content per unit mass. CNG is preferred for city vehicles due to simpler, safer handling. It is environmentally cleaner than diesel, petrol, and LPG.

Statement of The Problem

Government stressed the importance of using CNG Vehicles to reduce the environment pollution and usage of petrol, diesel consumption. As per a report by

WHO air pollution can lead to breathing and cardiac diseases and occasioned in to 620000 deaths and the expenses on

treatment was 3% of India's GDP. According to the World Health Organization (WHO), 13 cities in India are among the 20 most polluted cities in the world, Delhi tops the list and uncontrolled vehicular traffic is the main reason for these alarming levels. Apart from addressing the environmental problem, Compressed Natural Gas (CNG) vehicles are introduced by four- wheeler companies. People are preferred to buy CNG vehicles by considering the benefits of CNG.

Scope of The Study

The present study is confined to study the opinion of the people who are using CNG four wheelers in Virudhunagar district.

II Review of Literature

The review of previous studies helps the researcher to explore the hidden areas of research and carry out the research in the correct path.

Dahlstrom (2010) has stated that the selection of vehicles like CNG and their sustainable use incorporates two concepts - ecological and social behaviour. Likewise, these two concepts consolidate to shape the base of ecological behaviour, which prompts sustainable turn of events.

Aditi Dang et al., (2024) have concluded in their study that consumer behaviour is positively affected by CNG vehicles. Consumers are sensitive to issues like pollution, price, and other factors. The study will supplement the efforts of researchers and marketers in understanding of consumer behaviour towards CNG vehicles. Understanding of the consumer behaviour is of great significance for marketers. Recent example of rejection of Tata Nano as a "cheap car" by the Indian consumer shows the importance of getting the right pulse of the market. The best of the products can get grounded if marketed poorly

III Objectives of The Study

The objectives of the study are as follows:

To know about the model and price classification of the CNG vehicles

To study the socio-economic profile of the respondents

To examine the purchase and usage pattern of CNG vehicles.

To analyse the benefits of using CNG vehicles.

To give suggestions on the basis of findings of the study

IV Research Methodology

Research methodology includes research design, data used in the study, tools used for the collection of data, sampling design, sampling technique, data processing and tools of analysis.

The study is descriptive in nature and uses both primary and secondary data. Both primary and secondary data have been utilized. The primary data for this study on CNG four wheelers has been collected directly from the CNG four wheeler owners. The questionnaire has been designed to cover various aspects such as socio-economic background, buying and using behaviour, benefits and problems in using CNG vehicles. Secondary data has been gathered from books, journals, magazines, government reports and online sources. The study is based on responses from 194 respondents. Convenience sampling method has been adopted for the selection of samples. The collected data has been coded, tabulated and analyzed systematically. Percentage analysis and Garrett's ranking technique has been used for the analysis of primary data. Tables and charts are used for presentation of data. The study is confined to Virudhunagar district.

V Results and Discussion

Table 1 shows the price range of CNG four wheelers.

Table 1: Price Range of CNG Four Wheelers

Price Classification	Name of the Car	Price Range
Under ₹6 Lakhs	Maruti S-Presso	₹4.62 Lakh
	Maruti Alto K10),	₹4.82 Lakh
	Tata Tiago	₹5.54 Lakh
₹6 - ₹10 Lakh	· Tata Punch	₹5.65-10.60 Lakh
	Hyundai Exter	₹5.80-9.57 Lakh

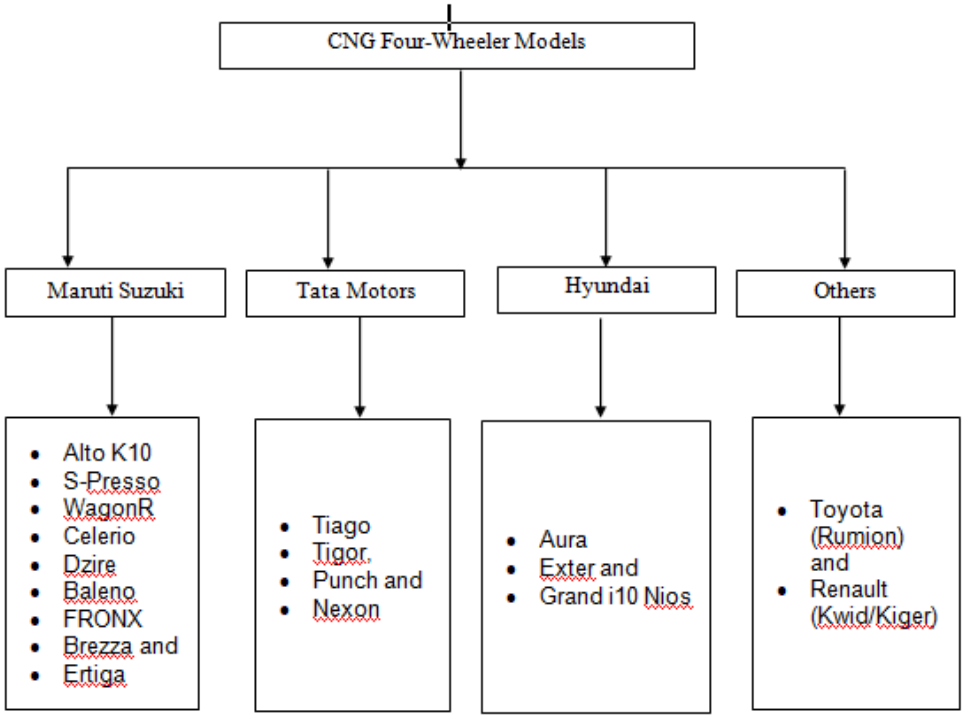
Above ₹10 Lakh	**Maruti Brezza**	**₹8.26-13.01 Lakh**
	Maruti Ertiga	**₹8.80-12.94 Lakh**

Source:www.cngprizes.com

The CNG vehicles are classified into three ranges viz., under ₹ 6 lakhs, ₹ 6 - ₹10 lakhs and above ₹ 10 lakhs.Maruti S-Presso, Maruti Alto K10) and Tata Tiago belong to under ₹ 6 lakhs category. Tata Punch and Hyundai Exter come under ₹6 - ₹10 Lakh category. Maruti Brezza and Maruti Ertiga fall under Above ₹10 Lakh category.

Figure 1 shows the popular CNG four wheeler models.

Figure 1: Popular CNG Four Wheeler Models



Source: www.cngmaruthi.com

In this section, three socio-economic variables such as age, occupation and monthly income are taken into account. Table 2 highlights the socio-economic profile of the respondents.

Table 2: Socio-Economic Profile of the Respondents

Socio-Economic Profile		No. of Respondents	Percentage
Age (in years)	**Below 30**	**36**	**18.56**
	30-50	**114**	**58.76**
	Above 50	**44**	**22.68**

Occupation	**Government Employee**	**54**	**27.84**
	Private employee	**70**	**36.08**
	Businessmen	**58**	**29.90**
	Others	**12**	**6.19**
Monthly Income (in Rs.)	**Below 1,00,000**	**74**	**38.14**
	1,00,000 -3,00000	**96**	**49.48**
	Above 3,00,000	**24**	**12.37**

Source: Primary data

Out of 194 respondents, 114 (58.76%) are in the age group of 30-50 years, 70 (36.08%) are private employees and 96 (49.48%) have fall under the monthly income category of Rs. 1,00,000- Rs. 3,00,000.

Table 3 points out the buying and usage pattern of CNG vehicles.

Table 3: Buying and Usage Pattern of CNG Vehicles

Buying and Usage Pattern of CNG Vehicles		**No. of Respondents**	**Percentage**
Period of using CNG Vehicles	**Less than 1 year**	**112**	**57.73**
	1-3 years	**38**	**19.59**
	Above 3 years	**44**	**22.68**
Source of knowledge about CNG Vehicles	**Friends and relatives**	**91**	**46.91**
	Car show room dealers	**61**	**31.44**
	Advertisements	**42**	**21.65**
Purpose of using CNG vehicles	**Personal purpose**	**104**	**53.61**
	Official purpose	**90**	**46.39**

Source: Primary data

Out of 194 respondents, 112 (57.73%) have used CNG vehicles for less than one year, 91 (46.91%) have gathered awareness and knowledge about CNG vehicles from friends and relatives and 104 (53.61%) have used CNG vehicles for personal purposes.

Table 4: Benefits of using CNG Vehicles

Reasons	**Ranks**							**Total**
	I	**II**	**III**	**IV**	**V**	**VI**	**VII**	
Cost-Effective	**57**	**42**	**15**	**22**	**17**	**26**	**15**	**194**

Environmental Benefits	**52**	**37**	**15**	**21**	**30**	**14**	**25**	**194**
Improved Engine Life	**14**	**46**	**33**	**20**	**11**	**47**	**23**	**194**
Safety	**9**	**5**	**27**	**16**	**45**	**36**	**56**	**194**
High Performance	**20**	**23**	**42**	**82**	**10**	**12**	**5**	**194**
Dual Fuel Capability	**31**	**15**	**30**	**30**	**36**	**31**	**21**	**194**
Better mileage	**11**	**26**	**32**	**3**	**45**	**28**	**49**	**194**
Total	**194**	**194**	**194**	**194**	**194**	**194**	**194**	

Source: Primary data

Garrett's ranking technique has been applied to examine the benefits of using CNG vehicles.

Table 5: Garrett's Ranking Calculations

Rank	Computation	Garrett value
I	**100 (1-0.5)/7 =7.14**	**79**
II	**100 (2-0.5)/7 =21.43**	**66**
III	**100 (3-0.5)/7 =35.71**	**58**
IV	**100 (4-0.5)/7 =50.00**	**50**
V	**100 (5-0.5)/7 =64.29**	**43**
VI	**100 (6-0.5)/7 =78.57**	**35**
VII	**100 (7-0.5)/7 =92.86**	**22**

Source: Garrett's ranking table

The Garrett's ranking test results are given in table 6.

Benefits of using CNG Vehicles – Garrett's Ranking Test Results

Attribute	Total score	Mean score	Rank
Cost-Effective	**11216**	**57.81**	**I**
Environmental Benefits	**10800**	**55.67**	**II**
Improved Engine Life	**9680**	**49.90**	**V**
Safety	**7834**	**40.38**	**VII**
High Performance	**10594**	**54.61**	**III**

Dual Fuel Capability	**9774**	**50.38**	**IV**
Better mileage	**8584**	**44.25**	**VI**

Source: Primary data

Most of the respondents gave I rank to the attribute 'Cost-Effective' with the mean score of 57.81 followed by 'Environmental Benefits' (55.67). The statements 'High Performance' (54.61) and 'Dual Fuel Capability' (50.38) have obtained third and fourth ranks respectively. Fifth rank and sixth rank goes to the statements 'Improved Engine Life' (49.70) and 'Better mileage' (44.25). Seventh rank goes to the statement 'Safety' (42.67).

Table 7: Problems in Using CNG Vehicles

Problems in Using CNG Vehicles	No. of Respondents	Percentage
Higher Maintenance & Servicing Costs	**27**	**13.92**
Lower Initial Resale Value	**36**	**18.56**
Non-availability of gas stations	**84**	**43.30**
Lower Starting Capability in Cold	**31**	**15.98**
Reduced Practicality (Space & Weight)	**16**	**8.25**
Total	**194**	**100.00**

Source: Primary data

Out of 194 respondents, 84 (43.30%) are affected by non-availability of gas stations, 36 (18.56%) have faced a problem of lower initial resale value, 31 (15.98%) have a problem of lower starting capability in cold, 27(13.92%) are affected by higher Maintenance & Servicing Costs and 16 (8.25%) have a problem of reduced practicality.

It is suggested that Government granted permission to petrol bulks to establish CNG gas stations to ensure the availability of CNG. Government provide subsidies and incentives to the buyers of CNG gases. Car show room dealers stressed the importance of CNG vehicles to their visitors.

VII Conclusion

CNG is a boon to our society as it is used by majority of the people, it will reduce the environmental pollution. The four-wheeler manufacturing companies classified the

CNG cars from Rs. 6 lakhs. It will help the middle-income people also to use CNG vehicles. The study concludes that a complete and united tactics, concerning infrastructure development, economic incentives, mindfulness creation and policy support is essential to enhance consumer acceptance of CNG.

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The Impact of Social Media Influencers on Destination Image and Tourist Decision-Making: A Study with Reference to Uttarakhand Destinations

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Abstract- Social media has become an important part of modern travel planning, significantly influencing tourists' perceptions and destination choices. Travel influencers on platforms such as Instagram and YouTube play a major role in promoting tourist destinations by sharing visual content, personal experiences, and recommendations. This study examines the impact of social media influencers on destination image and tourist decision-making with special reference to tourist destinations in Uttarakhand.

The research is based on primary data collected through a structured questionnaire administered to young social media users. The study analyses how factors such as influencer credibility, authenticity, attractiveness, and content quality affect tourists' perceptions and travel intentions. The findings reveal that social media influencers significantly contribute to creating a positive destination image and encouraging tourists to visit Uttarakhand. Influencer-generated content inspires potential travellers and plays an important role in their destination selection process.

The study further indicates that social media platforms not only raise awareness but also foster emotional connections between tourists and destinations. Influencers present destinations in a relatable and engaging manner, enabling audiences to visualise travel experiences and increasing their confidence in choosing a destination. The natural beauty, adventure tourism, spiritual attractions, and cultural richness of Uttarakhand are effectively promoted through influencer marketing strategies.

Overall, the study highlights the growing importance of social media influencers in tourism promotion and suggests that tourism stakeholders and marketers can strategically collaborate with influencers to attract more tourists and strengthen the destination image of Uttarakhand.

Keywords: *Social Media Influencers; Destination Image; Tourist Decision-Making; Influencer Marketing; Travel Intentions; Uttarakhand Tourism*

Introduction

Tourism is one of the fastest-growing industries in the world and plays an important role in economic development, employment generation, and cultural exchange. In recent years, rapid growth in digital technology and internet accessibility has completely transformed the tourism industry and the way people plan their trips. Earlier, tourists mainly depended on travel agencies, brochures, newspapers, television advertisements, and recommendations from friends and family for travel information. However, with the rise of social media platforms such as Instagram, YouTube, Facebook, and Pinterest, travel planning has become more digital, interactive, and experience-oriented.

Social media has now become an important part of daily life for millions of people around the world. According to the Digital 2025 India Report, India had around 491 million active social media users in 2025, representing approximately 33.7% of the country's population. Social media usage is continuously increasing due to greater smartphone penetration, affordable internet services, and rising digital engagement. Reports also indicate that Indian users spend several hours daily on social media platforms, making these platforms highly influential in shaping consumer behaviour and lifestyle choices.

The tourism industry has been significantly affected by this digital transformation. Social media platforms are no longer used only for entertainment and communication; they have become major sources of travel inspiration and destination discovery. According to a recent travel report, nearly 60% of Indian travellers stated that social media platforms such as Instagram influence their travel decisions and destination preferences. Another report revealed that 68% of Indian travellers watch YouTube videos for travel inspiration before planning trips. These statistics highlight the growing influence of digital content and social media platforms on tourist behaviour.

Among the various forms of social media content, travel influencers have emerged as powerful promoters of tourist destinations. Social media influencers are individuals with a strong online presence and a large follower base on digital platforms. They regularly

share travel experiences, destination reviews, hotel recommendations, food experiences, adventure activities, and cultural attractions through engaging photos, reels, blogs, and videos. Their content shapes followers' opinions, perceptions, and travel decisions.

Travel influencers have gained popularity because audiences often view them as more relatable and trustworthy than traditional advertisements. Their content feels more genuine, personal, and experience-based, helping followers connect emotionally with destinations. Influencers not only share travel information but also spark excitement, curiosity, and emotional attachment among viewers. As a result, many tourists choose destinations after being inspired by influencer-generated content.

In tourism marketing, destination image is one of the most important factors affecting tourists' travel intentions. Destination image refers to the perceptions, beliefs, impressions, and emotional feelings people develop about a particular place. A positive destination image increases tourists' interest and motivates them to visit the destination. Social media influencers play an important role in creating and improving destination image by sharing visually attractive and engaging travel content. Their storytelling, photography, and videos allow audiences to imagine themselves experiencing the destination, which positively affects travel intentions.

Uttarakhand is one of India's most famous tourist destinations, popularly known as the "Land of Gods." The state is well known for its scenic beauty, snow-covered mountains, rivers, forests, pilgrimage centres, wellness tourism, and adventure activities. Tourist destinations such as Nainital, Mussoorie, Rishikesh, Haridwar, Kedarnath, Badrinath, Auli, and Valley of Flowers National Park attract tourists from across India and other countries. In recent years, travel influencers have actively promoted both famous and lesser-known destinations of Uttarakhand through social media platforms.

Influencers now present Uttarakhand not only as a religious tourism destination but also as a centre for trekking, camping, river rafting, yoga retreats, wildlife tourism, café culture, and nature-based tourism. Through Instagram reels, YouTube travel vlogs, blogs, and digital storytelling, influencers have successfully attracted young travellers and

adventure seekers. Many tourists now discover offbeat and hidden locations in Uttarakhand through influencer-generated content on social media.

The increasing impact of social media on tourism has created a need to understand how influencers shape tourists' perceptions and travel decisions. Studying this relationship is important for tourism authorities, marketers, travel businesses, and researchers because it helps identify how influencer marketing strategies can effectively promote destinations and attract tourists. Therefore, the present study aims to examine the impact of social media influencers on destination image and tourist decision-making with special reference to Uttarakhand destinations. The study also seeks to understand how influencer credibility, authenticity, and content quality affect tourists' perceptions and travel intentions toward Uttarakhand tourism destinations.

Objectives of the Study

- To analyse the impact of social media influencers on the destination image of Uttarakhand among social media users.
- To examine the effect of influencer credibility and authenticity on tourists' trust towards travel-related content.
- To study the influence of social media influencer content on tourists' travel decision-making and intention to visit Uttarakhand.
- To identify whether social media influencers encourage tourists to explore offbeat and diverse tourism destinations in Uttarakhand.

Literature Review

Social media has become one of the most important sources of travel information in recent years. Tourists now use platforms such as Instagram, YouTube, Facebook, and Pinterest for destination searches, travel inspiration, entertainment, and trip planning. Travel influencers on these platforms have emerged as strong promoters of tourism destinations because they share visually attractive and experience-based content that directly influences their followers' opinions and travel decisions.

Pop Rebeka-Anna et al. (2021) examined the role of trust in influencer marketing and tourist decision-making. The study found that social media influencers significantly affect tourists' travel intentions when followers perceive them as trustworthy and authentic. The researchers explained that influencer recommendations influence multiple stages of the tourist decision-making process, including information search, destination evaluation, and final travel choice. The study highlighted that trust and authenticity are important factors affecting tourists' behavioural intentions.

Similarly, Peres et al. (2021) studied the impact of micro-influencers in the hospitality and tourism industry. Their findings revealed that tourists often trust micro-influencers more than celebrity influencers because they appear more relatable and genuine. The study emphasised that interactive communication and authentic travel experiences shared by micro-influencers positively influence tourists' booking behaviour and destination preferences. Compared to the study by Pop Rebeka-Anna et al. (2021), this research focused more specifically on the effectiveness of smaller influencers in building stronger audience engagement.

Shang and Luo (2021) explored how the credibility of destination endorsers affects tourists' emotional attachment to destinations. The study concluded that influencer credibility, expertise, and attractiveness significantly shape tourists' destination perceptions and emotional connections. The researchers further stated that credible influencers increase tourists' confidence in travel planning. This study differs from previous studies by focusing not only on travel intentions but also on tourists' emotional attachment towards destinations.

A study published in *Tourism Management* (2021) examined how destination-related content shared on social networks influences tourists' choices. The findings showed that tourists are strongly influenced by online travel content, especially when the information source is considered reliable and knowledgeable. The study stressed that social media interactions play a major role in shaping destination image and tourists' travel behaviour.

Saini, Kumar, and Oberoi (2023) conducted a systematic literature review on the impact

of social media influencers on travel decision-making. The researchers identified influencer credibility, audience engagement, storytelling, and visually attractive content as major factors influencing tourists' decisions. The study particularly highlighted that young travellers are highly influenced by travel reels, blogs, and digital storytelling. Compared to earlier studies, this research provided a broader overview of influencer marketing trends in tourism and consolidated findings from multiple studies.

Research by Yi K et al. (2021) examined tourists' travel intentions toward destinations promoted by influencers. The findings indicated that influencer-generated content creates emotional excitement, curiosity, and travel motivation among viewers. The study concluded that regular exposure to travel content on social media positively affects tourists' intention to visit destinations. This research supports earlier findings that emotional appeal and visual storytelling significantly impact tourist behaviour.

Another study published in the *Journal of Destination Marketing & Management* (2021) investigated the effect of pictorial destination image on tourists' perceptions. The study found that user-generated travel photographs and visually appealing content on social media strongly influence destination image formation. Attractive travel images help audiences imagine their travel experiences, thereby positively affecting travel behaviour and destination selection.

Similarly, a study in *Tourism Management Perspectives* (2021) explored the relationship between social media usage and destination image formation. The findings revealed that social media platforms strongly influence how tourists think and feel about destinations even before visiting them. The study highlighted that online travel content contributes to the development of positive attitudes towards destinations and increases tourists' interest in travel planning.

Researchers from Graphic Era Hill University also reviewed the role of social media influencers in destination selection and tourist behaviour. The study emphasised that influencer credibility, audience interaction, and authentic content significantly affect travel choices and destination branding. The researchers noted that tourism marketers

increasingly collaborate with influencers to attract tourists and improve destination visibility in the digital era.

Another study focusing on Indian tourism destinations discussed the growing influence of social media influencers in tourism marketing. The researchers concluded that influencers help destinations gain visibility among younger audiences and encourage tourists to discover new destinations through travel videos, reels, and blogs. The study further stated that influencer marketing has become an effective promotional strategy for Indian tourism destinations because of its ability to reach a large digital audience.

Overall, the reviewed literature indicates that social media influencers significantly affect destination image and tourist decision-making. Most studies agree that influencer credibility, authenticity, audience engagement, and visually attractive content positively influence tourists' travel intentions and destination preferences. While some studies focused on emotional attachment and trust, others emphasised storytelling, destination branding, and digital engagement. The literature consistently highlights the growing importance of influencer marketing in tourism promotion and tourist behaviour.

Research Gap

Although many studies have examined the impact of social media influencers on tourism marketing and tourist behaviour, limited research has specifically focused on Uttarakhand tourism destinations. Most existing studies have examined influencer marketing in a general tourism context without analysing how influencer-generated content affects tourists' perceptions and travel decisions regarding Uttarakhand. In addition, few studies have explored the role of influencer credibility, authenticity, and content quality in promoting offbeat and diverse tourism destinations within Uttarakhand. Therefore, the present study attempts to fill this gap by examining the impact of social media influencers on destination image and tourist decision-making with special reference to Uttarakhand destinations.

Research Methodology

The present study is descriptive and quantitative in nature. The main aim of the study is to understand the impact of social media influencers on destination image and tourist decision-making with special reference to Uttarakhand destinations. The study focuses on how influencer content on social media affects tourists' perceptions, trust, and travel intentions towards Uttarakhand.

The study is based on both primary and secondary data. Primary data was collected through a structured questionnaire prepared according to the objectives of the study. Secondary data was collected from research papers, journals, books, websites, tourism reports, and published articles related to social media influencers, tourism marketing, destination image, and tourist behaviour. These sources helped in understanding the topic and supporting the study findings.

For primary data collection, responses were collected from 100 respondents who actively use social media platforms like Instagram, YouTube, Facebook, and Pinterest for travel-related information. Most respondents were students, young travellers, and working professionals who regularly watch travel-related content online. The convenience sampling method was used because it was easy and suitable for collecting responses from social media users. The questionnaire was shared online through Google Forms and social media platforms for easy data collection.

The questionnaire was divided into five sections. The first section included demographic details such as age, gender, educational qualification, and occupation. The second section focused on social media usage and travel exposure. The third section measured destination image perception of Uttarakhand. The fourth section examined influencer credibility and authenticity, while the fifth section focused on tourist decision-making and behavioural change. A five-point Likert scale was used in the questionnaire, where 1 means "Strongly Disagree" and 5 means "Strongly Agree."

The collected data were analysed using statistical tools such as percentage analysis, mean analysis, correlation analysis, and regression analysis with the help of Microsoft Excel. Percentage analysis was used to study respondents' demographic profile and social media

usage patterns. Mean analysis helped in understanding respondents' opinions regarding influencer credibility and tourist behaviour. Correlation analysis was used to identify the relationship between influencer credibility, destination image, and tourist decision-making. Regression analysis was used to examine the impact of social media influencers on tourists' travel intentions towards Uttarakhand destinations.

The study is limited to 100 respondents and is based on their personal opinions and experiences regarding social media influencers and tourism. Therefore, the findings may not represent all tourists. However, the study provides useful insights into the growing role of social media influencers in tourism promotion and tourist decision-making.

Data Analysis and Interpretation

1. Demographic Analysis

The demographic profile provides context on the digital habits of the sample.

Table 4.1: Demographic Profile of Respondents (N=100)

Variable	Category	Frequency	Percentage
Age Group	18–24 Years	45	45%
	25–30 Years	30	30%
	31–35 Years	15	15%
	Above 35 Years	10	10%

This distribution highlights that the consumption of social media travel content in relation to Uttarakhand destinations is heavily dominated by Gen Z and younger Millennials. This suggests that travel influencers have the highest potential impact on younger travellers who are more digitally integrated**.**

Table 4.2: Primary platform

Primary Platform	Frequency	Percentage
Instagram	48	48%
YouTube	32	32%
Facebook	12	12%
Others (Pinterest/etc)	8	8%

The dominance of Instagram and YouTube indicates that tourists prefer short-form visual storytelling (Reels/Photos) and long-form experiential content (Vlogs) when researching Uttarakhand. Platforms like Facebook and Pinterest showed significantly lower engagement levels for this specific demographic, suggesting that high-quality imagery and video are the most effective drivers for destination marketing.

- **Reliability Analysis**

Before performing regression, the internal consistency of the instrument was tested using Cronbach's Alpha.

Table 4.3: Reliability Statistics

Scale Construct	No. of Items	Cronbach's Alpha (α)
Influencer Credibility	4	0.81
Destination Image	4	0.78
Decision-Making	4	0.84
Overall Scale	12	0.82

Interpretation: An Alpha value of 0.82 exceeds the recommended threshold of 0.70, indicating that the questionnaire items are highly reliable and consistently measure the

intended variables.

- **Regression Analysis & Hypothesis Testing**

This section examines the causal relationship between the independent variable (Influencer Credibility) and the dependent variables (Destination Image and Decision-Making).

Hypothesis 1 (H1): Impact on Destination Image

- H0: Influencer credibility has no significant impact on the destination image of Uttarakhand.
- H1: Influencer credibility significantly impacts the destination image of Uttarakhand.

Table 4.4: Regression Model Summary (Destination Image)

Model	R	R Square	Adjusted R-Square	Std. Error
1	0.171	0.029	0.019	0.643

Table 4.5: Coefficients Table

Variable	Unstandardized B	Std. Error	Beta (β)	t-value	Sig. (p)
(Constant)	3.283	0.339	-	9.679	0
Credibility	0.155	0.09	0.171	1.723	**0.088**

Interpretation: The results indicate that Influencer Credibility explains only 2.9% ($R^2 = 0.029$) of the variance in Destination Image. The p-value (0.088) is greater than the

significance level of 0.05. Therefore, we fail to reject the null hypothesis (H0). This suggests that while influencers provide content, they do not statistically alter the pre-existing perceived image of Uttarakhand in a significant way.

Hypothesis 2 (H2): Impact on Tourist Decision-Making

III. H0: Social media influencer content does not significantly affect tourists' travel decision-making.

IV. H1: Social media influencer content significantly affects tourists' travel decision-making.

Table 4.6: Regression Model Summary (Decision-Making)

Model	R	R Square	Adjusted R-Square	F-Value	Sig. (F)
1	0.388	0.151	0.142	17.413	**0**

Table 4.7: Coefficients Table

Variable	Unstandardized B	Std. Error	Beta (β)	t-value	Sig. (p)
(Constant)	2.557	0.3	-	8.515	0
Credibility	0.334	0.08	0.388	4.172	**0**

Interpretation: The model shows a moderate positive correlation (R = 0.388). The R-Square of 0.151 implies that 15.1% of the variation in a tourist's decision to visit Uttarakhand is directly influenced by the credibility of social media influencers.

The Beta coefficient ($\beta = 0.388$) indicates that for every one-unit increase in influencer credibility, tourist decision-making intention increases by 0.334 units. Since the p-value (0.000) is less than 0.05, we reject the null hypothesis (H0) and accept H1.

Findings of the Study

The study examined the impact of social media influencers on destination image and tourist decision-making with special reference to Uttarakhand destinations. The analysis showed that young respondents, especially those in the 18–24 age group, are highly active on social media platforms and frequently consume travel-related content. Instagram and YouTube were identified as the most preferred platforms because they provide visually attractive content such as travel reels, vlogs, blogs, and destination reviews. Most respondents stated that they regularly follow travel influencers and have seen travel-related content about Uttarakhand online.

The findings revealed that respondents generally view travel influencers as trustworthy and authentic sources of travel information. The mean value of 3.68 indicated a positive perception regarding influencer credibility. Respondents particularly trusted influencers who share genuine travel experiences, honest reviews, and practical travel information instead of highly promotional content. This suggests that authenticity and relatability are important factors influencing tourists' trust toward influencer-generated content.

The regression analysis showed that influencer credibility had only a weak positive impact on the destination image of Uttarakhand, and the relationship was statistically insignificant. This finding differs from several previous studies, which suggested that influencer credibility strongly shapes destination image and tourist perceptions. One possible reason for this difference may be that tourists already have an established image of Uttarakhand because of its religious importance, natural beauty, and popularity as a tourist destination. Therefore, influencer content may only slightly improve existing perceptions rather than completely shape them. Another reason could be that audiences today are more aware of paid promotions and sponsored collaborations, making them less likely to depend entirely on influencer opinions while forming destination image.

However, the study found that influencer-generated content significantly affects tourists' travel decision-making and intention to visit Uttarakhand. The regression analysis showed a statistically significant relationship between influencer credibility and travel decisions. Influencers help tourists discover destinations, gain travel information, and build confidence while selecting travel locations. The study also found that influencer

content motivates tourists to explore offbeat and diverse tourism experiences such as adventure tourism, wellness tourism, eco-tourism, and lesser-known destinations in Uttarakhand. This highlights the growing importance of influencer marketing in encouraging tourism exploration and travel engagement.

Suggestions of the Study

Based on the findings of the study, tourism authorities and marketers in Uttarakhand should make greater use of social media influencer marketing to promote tourism destinations. Since Instagram and YouTube are the most preferred platforms among young travellers, tourism campaigns should focus more on visual storytelling through travel reels, vlogs, short videos, and destination-based content.

The study suggests that tourism departments should collaborate with credible and authentic travel influencers who provide genuine travel experiences rather than excessive promotional content. Honest and informative influencer content can help build tourists' trust and improve engagement with travel campaigns.

Tourism stakeholders should also focus on promoting offbeat and lesser-known destinations in Uttarakhand. Influencers can play an important role in creating awareness about adventure tourism, wellness tourism, eco-tourism, village tourism, and hidden tourist attractions. Such promotional strategies can help reduce overcrowding at popular destinations and encourage balanced tourism development across the state.

In addition, influencers should be encouraged to promote responsible and sustainable tourism practices by sharing accurate information related to local culture, environmental protection, safety, accommodation, and travel guidelines. Tourism authorities may also organise influencer collaborations, sponsored travel campaigns, and digital tourism events to strengthen the online image of Uttarakhand tourism.

Future researchers can further study the influence of different age groups, types of influencers, and social media platforms on tourist behaviour. Comparative studies between different tourist states or destinations may also provide deeper insights into the

effectiveness of influencer marketing in tourism promotion.

Conclusion

The present study was conducted to understand the impact of social media influencers on destination image and tourist decision-making with special reference to Uttarakhand destinations. The study highlights that social media has become an important source of travel information and inspiration in the modern tourism industry. Platforms such as Instagram and YouTube allow tourists to explore destinations, view travel experiences, and gather information before planning their trips.

The findings showed that travel influencers strongly affect tourists' travel intentions and destination choices by sharing visually attractive and engaging content. Influencer-generated travel reels, blogs, vlogs, and reviews help tourists gain knowledge about destinations and motivate them to explore new travel experiences. In the context of Uttarakhand, influencer content successfully promotes adventure tourism, nature tourism, spiritual tourism, wellness tourism, and offbeat tourist attractions.

Although influencer credibility showed only a limited and statistically insignificant impact on the overall destination image of Uttarakhand, it had a significant influence on tourists' travel decision-making and intention to visit the destination. This indicates that while tourists may already possess a pre-existing image of Uttarakhand, influencer content still plays an important role in motivating actual travel behaviour and destination exploration.

Overall, the study concludes that social media influencers have become effective tools for tourism promotion in the digital era. Their ability to create relatable, authentic, and visually appealing content significantly influences tourist engagement and travel interest. Therefore, tourism authorities, marketers, and travel businesses in Uttarakhand should strategically collaborate with credible influencers to strengthen tourism promotion, attract more visitors, and encourage exploration of diverse tourism experiences across the state.

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IOT based health monitoring system with interaction of Robot and Virtual Doctor

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Abstract-- Doctors are usually needed to work at every hospital and emergency Centre every now and then. But it is not feasible for every doctor to be available at every place at desired time. The problem with video calling is that video calls need to be done from a PC or laptop on a desk. This limits the doctor's capacity to view patient or around operation theatre at will or even move through hospital rooms as needed. In order to address this situation, our system comes to the rescue; this system will collect the information of patients with the help of sensors. This system is implemented by the Arduino it includes a heartbeat sensor and temperature sensor. This paper presents a continuous monitoring and recording of patient data without human intervention. If there is any sudden changes occur in the normal range of body parameters such as heart beat, over temperature detection Buzzer will ring. Where both are not stable conditions for better health, then it has facility to automatically alert the medical person.

***Keywords*:** IOT,Arduino,Robot,Bluetooth,health monitoring robot.

1.Introduction:

An embedded system is one kind of a computer system mainly designed to perform several tasks like to access, process and store and also control the data in various electronics-based systems. Embedded systems are a combination of hardware and software where software is usually known as firmware that is embedded into the hardware. One of its most important characteristics of these systems is, it gives the o/p within the time limits. Embedded systems support to make the work more perfect and convenient. So, we frequently use embedded systems in simple and complex devices too. The applications of embedded systems mainly involve in our real life for several devices like microwave, calculators, TV +control, home security and neighborhood traffic control systems, etc.

2.Existing System:

In the existing system of the smart medicine remainder box, it will blink led, based on the medicine number along with a buzzer. In this also they have used the Arduino and an RTC module to get the current time and an LCD display and push buttons to set and observe the time. When it's time to take medicine, the buzzer will alarm along with the led blink. The main drawback in this is if the person is far away from the system, he won't get to know that it is medicine taking time.

2.1 Literature survey:

Many literature papers are studied related to the present work.

In [1] the blood information is available in Android application and Web portal and the hard ware articture is complex as it involves many algorithm to reduced this a new techique is proposed in this research work.

In [2] it introduces the application of barcode technology in information automation system oriented to blood banks and other transfusion facilities.It is intended to serve as a reference of implementing barcode technology in blood bank automation system.

In [3] this extend the proposed system using a hybrid technique that combines probabilistic and deterministic estimations to achieve both high accuracy and low computational overhead. Moreover, the accuracy of the hybrid technique is robust to changes in its parameter values.

In [4] a wide range of health monitoring systems, and their applications and efficiency have been found and discussed. A large number of studies supported the effectiveness of this system both in a hospital setting and also in the home environment.

In [5] The physician set various threshold values for the health parameters to caution the patient. The caution messages are generated by the physicians server to send back the physicians advice to the patient on his current health condition.

In [6] this paper describes and proposes a complete monitoring existence cycle and effective healthcare monitoring system designed by using the IoT and RFID tags. The experimental results in this paper show the robust output against various medical emergencies.

In[7] time plays an important factor in treating patients in emergency settings. The time taken to initiate and administer medications to patients in such situations can have a huge impact on patients' outcome. The faster the initiation of medications in the ED hastens the improvement of patients.

Drawbacks:

If the person is far away from the system, he won't get to know that it is medicine taking time which might affect the person's health.

3 Proposed System:

Our project Robot is an alternate solution to the difficulties faced by the hospital staff in treating the coronavirus positive patients, who raised this problem. It checks the important parameters of the patient such as temperature, heart rate without the involvement of humans or by avoiding the direct contact of hospital staff with the patients. By using the reference parameters which are given to the system, will compare with the measured parameters. Then the recorded parameters data will be sent to the doctors through Bluetooth when RFID Reader reads the tag which stores the patient details. The movement and finding the path to patient location is done through Bluetooth. Also can reevaluate and view the data's of individual patients after a \particular interval of time to make sure that all those patients suffering from disease are feeling better and are in good condition.Here we are using IP camera.

Block diagram:

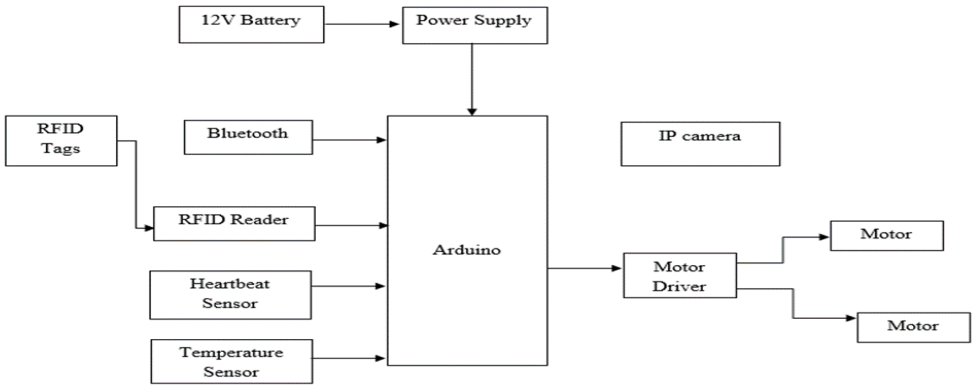


Figure 1 Block diagram of proposed system

Operations:

This project has certain procedure i.e., first we have to install an application called serial bluetooth terminal. In this app we have to run few commands like forward, backward, left, right and stop. (f, b, l, r, s) these shortcuts are used to run the cammands of application. Also this applications detects the temperature and heart beat.If the temperature and heartbeat are high it rings a buzzer(beep sound). In this again we use an IP camera that detects the patient health conditions.

Design technology:

3.1 ARUDINO:

Arduino Uno is a very important device which contains USB interface, connected with 14 digital I/O pins, 6 analog pins, and Atmega328 Microcontroller.It can supports serial communication using Tx and Rx pins. There are several versions of Arduino boards available in the market like Arduino Uno, Arduino Due, Arduino Leonardo, Arduino Mega, however, most commonly used versions are Arduino Uno and Arduino Mega. If you are planning to create a project relating to digital electronics, embedded system, robotics, or IoT, then using Arduino Uno would be the best, easy and most economical option.

3.2 DS18B20 TEMPERATURE SENSOR:

The DS18B20 is a one type of temperature sensor.This sensor is follows single wire protocol.It is also measured as a temperature in the range of between -67oF to +257oF or -55oC to +125oC with +-5% accuracy. The date of range is received from the 1-wire can range from 9-bit to 12-bit. Because, this sensor is folled by the single wire protocol.It is controlled by the only pin of Microcontroller.It has a advanced level protocol, where each sensor can be set with a 64-bit serial code which aids to control numerous sensors using a single pin of the microcontroller. This article discusses an overview of a DS18B20 temperature sensor.

3.3 Bluetooth Module:

It is used for many applications like wireless headset, game controllers, wireless mouse, wireless keyboard and many more consumer applications.It has range up to <100m which depends upon transmitter and receiver, atmosphere, geographic & urban conditions. It is IEEE 802.15.1 standardized protocol, through which one can build

wireless Personal Area Network (PAN). It uses frequency-hopping spread spectrum (FHSS) radio technology to send data over air.It uses serial communication to communicate with devices. It communicates with microcontroller using serial port (USART).

3.4 Motor Driver:

A motor driver is an integrated circuit chip which is usually used to control motors in autonomous robots.Motor driver act as an interface between Arduino and the motors. The most commonly used motor driver IC's are from the L293 series such as L293D, L293NE, etc. These ICs are designed to control 2 DC motors simultaneously. L293D consist of two H-bridge. H-bridge is the simplest circuit for controlling a low current rated motor. We will be referring the motor driver IC as L293D only. L293D has 16 pin.

3.5 Heartbeat Sensor:

Heartbeat Sensor is an electronic device that is used to measure the heart rate i.e. speed of the heartbeat. Monitoring body temperature, heart rate and blood pressure are the basic things that we do in order to keep us healthy.In order to measure the body temperature, we use thermometers and a sphygmomanometer to monitor the Arterial Pressure or Blood Pressure.Heart Rate can be monitored in two ways: one way is to manually check the pulse either at wrists or neck and the other way is to use a Heartbeat sensor.

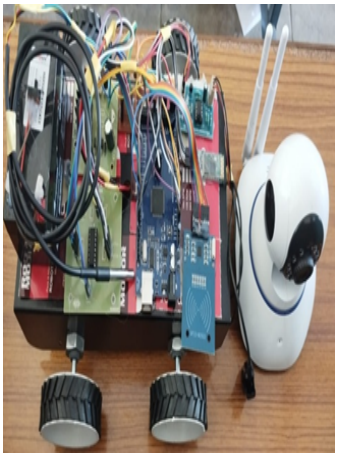
3.6 RC522 RFID MODULE:

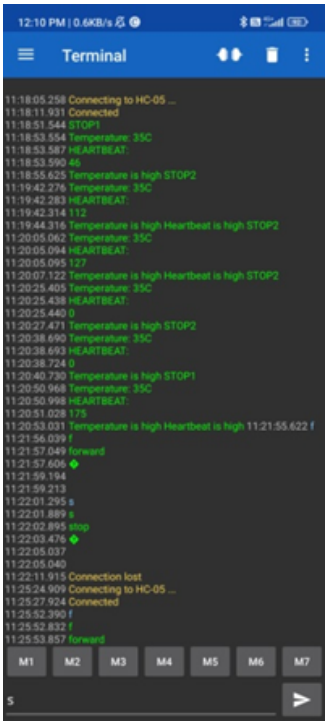
The RC522 is the highly integrated RFID card reader.It works on non-contact 13.56MHz communication. RFID moduleisa based upon the MFRC522 controller from NXP semiconductors. The RFID is stands for Radio Frequency Identification system.The system is a main two components.That is a transponder/tag attached to an object to be identified,and a Transceiver.

3.7 IP CAMERA :

The IP camera is nothing but the video camera.It has networked over a Fast Ethernet connection. This camera is sendinga signals tothe main server or computer screen via an Internet or network link. The IP camera is mainly used in IP surveillance, closed-circuit television (CCTV) and digital videography. IP cameras are widely replacing analog cameras due to their digital zoom and remote surveillance options.

Result discussions:





4 Conclusion:

In this project, the prototypes provide the real time solution of observing the patient heart pulse rate withreliability.This system can be adopted in the general wards of the hospital to help the patients understand its performance and utilization. The processes of storing the data can be further used in many ways such as predicting the diseases, analyzing etc. using this system can reduce the adverse emergency for a patient to occur with the heart disease.

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AI Revolution and Smart Learning: Reshaping of Education and Training

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Abstract– Introduction of Artificial Intelligence (AI) to the field of education and training will result in the improvement of teaching techniques, assessment processes and practical skills development. This narrative literature review is based on the PRISMA approach and investigates the influence of Artificial Intelligence on the education and professional training in general, giving an overview specifically of its influence on healthcare education. The review will analyze AI-based simulation, intelligent tutoring systems (ITS) and data-driven personalized learning. It is evident that artificial intelligence technologies help to achieve better results in skill development, decision making and critical thinking in the educational environment. Ethical issues, implementation difficulties and faculty preparedness are among the challenges identified by the research**.**

Keywords: Artificial Intelligence in Education, Smart Learning, AI-Driven Simulation, Intelligent Tutoring Systems, Personalized Learning, AI in Clinical Decision-Making, Training Technologies

Introduction

Learning and professional training are facing a fundamental shift due to the evolution in the realm of Artificial Intelligence (AI). Inclusion of AI technology in learning environment has been bringing drastic changes to the existing conventional teaching and training practices in different industries like healthcare, engineering, business, and humanities through personalized learning, interaction simulation, intelligent assessments and, therefore, increasing competence of learners in actual professional settings.

The present literature review will be following PRISMA principles in order to discuss the impact of artificial intelligence on the process of learning and professional training. In

our paper, we will focus on the following research questions: (1) How is AI changing education and training? (2) What are the advantages and disadvantages of an AI-based learning process? (3) What are future directions in implementing AI in the field of education?

In the course of rapid evolution of AI technology, it becomes important to understand its consequences in the field of education and training.

Methods

Search Strategy and Data Sources

An extensive literature search was performed using key academic databases like PubMed, Scopus, Web of Science, ScienceDirect, CINAHL, Cochrane Library, and Google Scholar. Key terms used were: "Artificial Intelligence and Education," "AI and Professional Training," "Smart Learning and Healthcare Education," and "Intelligent Tutoring Systems." Searches were confined only to those articles which were written from 2016 till 2025.

Data Selection Process

Based on the PRISMA principles, articles were examined systematically for their relevance. In the Identification stage, a total of 92,116 articles were obtained from the databases. In the Screening stage, 91,141 articles were excluded, which included duplicates and those with years of publication greater than the specified range, leaving 975 articles for further analysis. In the Eligibility stage, 351 articles were excluded due to their irrelevant content, which left 320 articles for consideration. Following the evaluation process based on the design of the study, sample sizes, and relevance to educational outcomes, 31 articles were selected for this review.

Inclusion and Exclusion Criteria

The following criteria were applied:

Inclusion Criteria

- Peer-reviewed articles focusing on AI applications in education and training
- Studies employing AI-based learning tools, simulations, or assessment techniques
- Papers published in English between 2016 and 2025
- Research evaluating educational outcomes of AI-driven training

Exclusion Criteria

- Articles unrelated to education or professional training contexts.
- Papers without empirical data or substantive theoretical insights

Data Extraction and Synthesis

Relevant articles were screened as described above. Data extraction focused on AI methodologies, educational benefits, implementation challenges, and future recommendations. The synthesis followed an integrative approach, identifying trends, patterns, and gaps in AI-driven education and training research.

Results

AI-Driven Learning in Education and Training

- Several innovative learning modalities brought about by AI have been identified that have transformed the landscape of education and professional training. These include the following technologies and their applications:
- AI-supported virtual simulations emulate real-life scenarios and offer learners an opportunity to engage in risk-free decision making.Virtual Reality (VR), Augmented Reality (AR), and other similar technologies greatly facilitate experiential learning and procedure learning through simulation.Simulation-Based Learning:
- AI-assisted intelligent tutoring systems give learners personalized feedback and adjust to the progress made by the learners themselves, thus enhancing

self-directed learning and performance evaluation.Intelligent Tutoring Systems (ITS):

- Artificial intelligence conversational agents assist learners in answering questions and engaging in discussions even outside the regular class hours.Chatbots and Virtual Assistants:
- Machine learning techniques evaluate performance of the learner and deliver feedback for skill development and knowledge retention.AI-Powered Assessment and Feedback:

Benefits of AI in Education and Training

- AI adapts educational content to individual learning styles, paces, and needs, enabling differentiated instruction at scale.Personalized Learning Experiences:
- AI-based simulations and ITS cultivate higher-order analytical skills essential for professional practice.Enhanced Critical Thinking and Decision-Making:
- Interactive AI-driven learning environments increase motivation, participation, and knowledge retention.Improved Learner Engagement:
- AI analytics enable faculty and trainers to track learner progress, identify learning gaps, and adapt instructional strategies in real time.Data-Driven Insights for Educators:

Challenges and Ethical Considerations

- Poor technological infrastructure and higher costs can act as barriers for the implementation of AI in educationally disadvantaged organizations.Implementation Challenges:
- Professional training is necessary for teachers to be able to use AI effectively in teaching.Pedagogical Capacity and Training:
- There must be guaranteed privacy and security of user information through the AI platforms, according to legal requirements.Privacy and Ethical Considerations:

- The problem of inequitable access to AI must be solved for the sake of equity in education.Equity and Access:

Future Directions

- Collaboration among educators, AI researchers, and experts from the industry can help to achieve the maximum use of AI in various training environments.Disintermediation:
- The inclusion of decision support systems based on AI into practical training may further help trainees face real-life problems.AI-Based Decision Support Systems:
- It is essential to develop guidelines and best practices concerning the use of AI to provide high-quality and consistent results.Evidence-Based Guideline Creation:
- Development of ethical AI frameworks will allow dealing with issues of data privacy and biases.Anti-Bias Ethical AI Frameworks:

Discussion

It is evident from the findings of this literature review that AI provides great opportunities in terms of personalizing and making learning more data-driven, interactive, and engaging. The reviewed articles have shown that AI-supported simulation, ITS, and chatbot technology can lead to significant changes in terms of learners' skills development, decision making abilities, and their engagement in the process. Educators have great opportunities for differentiation of instruction and monitoring of learners' progress.

Nonetheless, there are certain difficulties that need to be addressed in order to successfully implement AI into the process of teaching and training. First of all, the lack of adequate infrastructure is a very important issue to be addressed in order to provide equal opportunities for everybody. The level of faculty preparedness and readiness of institutions is an important factor that defines successful implementation of AI. There are certain ethical issues related to its application. In the future, longitudinal studies are

necessary in order to evaluate long-term influence of AI on educational results and competencies. Also, there should be developed certain policy and governance frameworks for AI management in education.

Conclusion

The field of education and training is being transformed by the application of intelligent learning approaches brought about by the introduction of AI in ways that improve teaching, testing, and skill acquisition. The use of AI technology provides several benefits, which are proven and well-known; however, dealing with the possible challenges of AI integration is essential for exploiting its advantages. With continued advancement in the field of AI, its incorporation into education and training should be supported by robust research and ethical considerations.

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The Health Status of Gujjar and Bakarwal Women in Jammu and Kashmir after COVID-19 Pandemic

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Abstract

Vulnerable communities around the world suffered serious health and well-being impacts due to COVID-19, including the Gujjar and Bakarwal tribal women of Jammu and Kashmir. Long before the pandemic, these traditionally autonomous nomadic & semi-nomadic communities already were battling challenges linked to poverty, access to healthcare, poor literacy, a lack of sanitation and social exclusion. This study aims to examine the health conditions such as physical, mental, maternal and nutritional health of this marginalized group of women known as Gujjar and Bakarwal who have incurred unexpected effects in post COVID19 situation. Findings from this research discloses that the traditional means of livelihood, migration, and access to healthcare services due to pandemic have drastically change which has enhanced health problems in tribal women. Additionally, it explores the socio-economic and cultural factors that still affect their health-seeking behaviour and quality of life. The research adopts a sociological viewpoint and incorporates both original and existing data, including government reports, scholarly articles, and community observations, to gain insights into the post-pandemic experiences of these women. The results indicate that even with certain government programs in place, tribal women continue to face insufficient healthcare services, limited awareness, psychological distress, and inadequate nutritional support. The research highlights the importance of comprehensive health policies, mobile healthcare services, programs focused on women's welfare, and initiatives to raise community awareness in order to enhance the health and well-being of Gujjar and Bakarwal women in Jammu and Kashmir following the COVID-19 pandemic.

Keywords: COVID-19, Tribal Health, Women's Health, Jammu and Kashmir.

Introduction

The COVID-19 pandemic has had a tremendous influence on human health, social life, and economic stability over the world, but it has been especially severe among marginalised and tribal populations. In Jammu and Kashmir, the Gujjar and Bakarwal tribes are significant nomadic and semi-nomadic populations whose livelihoods are mostly dependent on pastoralism, seasonal movement, and forest-based resources. Women in these cultures play an important role in preserving family health, livestock care, home obligations, and traditional cultural customs. Despite their contributions, Gujjar and Bakarwal women continue to confront various obstacles, including as poverty, illiteracy, poor healthcare facilities, gender inequality, and social marginalisation, which became more prominent during and after the COVID-19 pandemic. (Gupta *et al*., 2021).

The pandemic revealed the frail healthcare infrastructure in Jammu and Kashmir's remote and hilly areas, where tribal inhabitants frequently have limited access to medical services, sanitation, nutrition, and prenatal care. Movement restrictions, economic instability, and the fear of infection all had a negative impact on indigenous women's physical and emotional health. During the post-pandemic period, many women struggled to gain access to hospitals, immunisation services, reproductive healthcare, and nutritional support. At the same time, the strain of caregiving, domestic duties, and economic uncertainty rose dramatically, impairing their overall quality of life. (Kumari and Bala, 2023).

Understanding the health situation of Gujjar and Bakarwal women in the post-COVID era is critical not just from a public health but also sociological and developmental standpoint. Their health status is directly related to social structure, cultural traditions, movement patterns, environmental factors, and access to government support programs. Although various research have looked into the influence of COVID-19 on general populations, less emphasis has been paid to tribal women living in geographically isolated areas of Jammu and Kashmir. This gap emphasises the necessity for targeted research on their post-pandemic health experiences. (Mir and Batool, 2022).

As a result, the purpose of this study is to investigate the health status of Gujjar and Bakarwal women in Jammu and Kashmir following the COVID-19 epidemic, focusing on their physical health, access to healthcare facilities, nutritional status, mental

well-being, and socioeconomic issues. The study aims to improve understanding of tribal women's health challenges and assist policymakers, healthcare institutions, and social welfare agencies with insights into building inclusive and sustainable healthcare initiatives for marginalised people.

Literature

The COVID-19 pandemic posed considerable health difficulties around the world, but its impact was particularly severe among marginalised and tribal groups. The Gujjar and Bakarwal groups of Jammu and Kashmir experienced numerous challenges due to poverty, seasonal migration, geographical isolation, and a lack of healthcare resources. Women in these tribal communities faced higher health risks as a result of poor nutrition, a lack of awareness, inadequate maternal healthcare, and limited hospital access during and after the pandemic. Existing research shows that tribal women are frequently excluded from mainstream healthcare institutions, increasing their susceptibility during the COVID-19 crisis:

This research showed the poor health and socioeconomic conditions of Gujjar and Bakarwal women in Jammu and Kashmir. Research shows that these tribal women face limited access to healthcare services, low awareness about nutrition and hygiene, poverty, and lack of education due to their nomadic lifestyle. Studies also reveal that women in these communities suffer from maternal health problems, anaemia, malnutrition, and poor sanitation facilities (Gilani, 2026). After the COVID-19 pandemic, the health condition of Gujjar and Bakarwal women became more vulnerable. Lockdowns, unemployment, lack of transportation, and weak medical infrastructure in remote areas reduced their access to hospitals and health services. Existing literature further explains that geographical isolation and seasonal migration increased the difficulties faced by tribal women during and after the pandemic. Priyanka and Madhu, 2023. Emphasized that tribal women in Jammu and Kashmir continue to face poor living conditions, lack of sanitation, nutritional deficiencies, and limited institutional support. Their work explained that women from Gujjar and Bakarwal communities suffered from low healthcare accessibility and weak empowerment structures, which became more visible after the pandemic. Economic instability during lockdowns also affected women's ability to obtain proper medical treatment and essential healthcare services.

Awareness regarding COVID-19 prevention and healthcare practices in Jammu and Kashmir remained uneven among rural and tribal populations (Chander *et al*., 2021). Their research found that low educational levels and limited communication channels influenced people's comprehension of health precautions, particularly in remote tribal areas. This circumstance had a significant impact on the health of Gujjar and Bakarwal women, who were already facing social and economic challenges. In Jammu and Kashmir, researchers investigated the health status of tribal women, specifically Gujjar and Bakarwal women. According to the study, poverty, illiteracy, a lack of healthcare awareness, hunger, domestic abuse, and nomadic lifestyles all had a negative impact on tribal women's health. It also emphasised that many women relied on traditional beliefs and shunned contemporary medical facilities even during pregnancy, which further exacerbated their health situation.(Gul, 2014). It emphasised their restricted access to healthcare, poor nutrition, and lack of knowledge about maternal and reproductive health. The study also found that poverty, illiteracy, and geographical isolation had a major impact on the general health of tribal women, emphasising the need for enhanced healthcare facilities and awareness campaigns in tribal communities. (Khan and Khan, 2020). Previous study has primarily focused on tribal communities' education, empowerment, and socioeconomic conditions, with little attention paid to Gujjar and Bakarwal women's post-COVID health status. As a result, the current study seeks to investigate their health status, healthcare access, and obstacles following the COVID-19 epidemic in Jammu and Kashmir.

Conceptual Framework of the Study

The present study assumes that the COVID-19 pandemic has had a major impact on the health of Gujjar and Bakarwal women in Jammu and Kashmir. The pandemic posed a number of obstacles, including an economic downturn, disruptions in healthcare, and social isolation. These issues led to a loss of livelihood prospects, limited access to healthcare services, and increased psychological stress among indigenous women. As a result, these factors contributed to a variety of health issues, including physical health problems (such as malnutrition, anaemia, and general weakness), mental health problems

(such as anxiety, depression, trauma, and fear), and reproductive health issues (such as inadequate maternal healthcare and a lack of reproductive health awareness). The impact of these health concerns is influenced by socio-cultural and geographic factors such as poverty, illiteracy, and tribal settlements' isolation. These characteristics create hurdles to healthcare access and health awareness, increasing the vulnerability of Gujjar and Bakarwal women.

Theoretical Framework

The Social Determinants of Health Theory is used in this study, which describes how social, economic, and environmental factors influence health, serves as the foundation for this study. Poverty, a lack of healthcare services, a lack of education, and migration all had an impact on Gujjar and Bakarwal women's health in Jammu and Kashmir following the COVID-19 pandemic.

Methodology

The current study used a descriptive research approach to examine the health status of Gujjar and Bakarwal women in Jammu and Kashmir following the COVID-19 epidemic. The study included both primary and secondary data. Primary data were gathered via structured questionnaires and personal interviews with Gujjar and Bakarwal women in specific localities. Respondents were chosen at random, however key informants, such as health workers and community leaders, were chosen purposefully. Secondary data were gathered from books, journals, government publications, and published research papers.

Study Area: The study was carried out in tribal-dominated regions of Jammu and Kashmir, including Rajouri, Poonch, Anantnag, Reasi, and Kathua, which have a large Gujjar and Bakarwal population. These places were chosen due to their nomadic and semi-nomadic populations, as well as restricted access to healthcare.

Sampling Technique and Sample Size: The study included 100 Gujjar and Bakarwal women ranging in age from 18 to 60. To provide a diverse range of health experiences, respondents comprised married, unmarried, pregnant, lactating, and elderly women.

Sources of Data: The study used both primary and secondary data. Primary data was gathered using organised interview schedules, questionnaires, and informal contacts with indigenous women, healthcare staff, ASHA workers, and community elders. Secondary information was acquired from government publications, journals, books, census

statistics, health department records, and published research studies on tribal health, COVID-19, and women's healthcare in Jammu and Kashmir.

Data Collection Methods: Field surveys were carried out in various seasons to reach both resident and migratory tribal populations. The interview schedule included questions about:

- Maternal and reproductive health
- Physical health status after COVID-1
- Nutrition and food security
- Mental health and stress
- Access to healthcare facilities
- Vaccination awareness and healthcare services
- Economic conditions affecting health

The qualitative part focused on understanding women's experiences during and after the pandemic, including healthcare discrimination, migration-related health challenges, and traditional healing practices.

Limitations of the Study: The study has limitations, such as severe topographical terrain, seasonal migration of Bakarwal groups, linguistic obstacles, and limited access to isolated places. Cultural barriers prevented some women from openly discussing reproductive and mental health issues.

Findings of the Study

According to the study, the COVID-19 pandemic had a major impact on the health of Gujjar and Bakarwal women in Jammu and Kashmir. Most women reported a loss in physical and mental health as a result of limited access to healthcare facilities, economic hardship, and a lack of information of post-COVID health issues. These tribal populations' nomadic or semi-nomadic lifestyles made them even more vulnerable during and after the pandemic. The outcomes of this study demonstrate that the COVID-19 pandemic had a negative impact on the general health of Gujjar and Bakarwal women in Jammu and Kashmir. The majority of responders reported physical health issues following the epidemic, including weakness, anaemia, hunger, exhaustion, and poor

reproductive health. Pregnant and breastfeeding women faced increased health risks as a result of infrequent medical examinations and a lack of maternal healthcare services in remote tribal communities. It also emphasised the tremendous mental health issues that indigenous women suffered following the outbreak. Economic instability, social isolation, and the strain of household and childcare responsibilities all contributed to a rise in anxiety, tension, fear of infection, despair, and psychological insecurity. Women in migratory Bakarwal groups were shown to be more vulnerable due to geographical isolation and a lack of ongoing healthcare support. The survey also discovered that healthcare accessibility for Gujjar and Bakarwal women remained limited due to poor transportation, distant hospitals, a shortage of medical staff, and seasonal migration patterns. During and after COVID-19, many women were unable to obtain timely treatment, vaccinations, or nutritional support. As a result, indigenous families increased their reliance on traditional medical procedures and home cures. Poverty, illiteracy, gender inequity, and social exclusion were found to continue to have a negative impact on indigenous women's health. Although the existence of various government welfare programs, many respondents were either ignorant of them or unable to participate due to administrative and geographical hurdles. The data revealed that socioeconomic and geographic factors such as poverty, illiteracy, unemployment, and isolated living conditions contributed significantly to tribal women's poor health following COVID-19. The study concluded that Gujjar and Bakarwal women continue to face poor health outcomes and inadequate healthcare support, emphasising the critical need for inclusive healthcare policies, mobile medical services, health awareness programs, and stronger tribal welfare measures in Jammu and Kashmir.

Conclusion

The study found that the COVID-19 pandemic had a major impact on the health of Gujjar and Bakarwal women in Jammu and Kashmir. According to the findings, indigenous women continue to confront major health issues such as malnutrition, anaemia, inadequate maternal healthcare, a lack of cleanliness, mental stress, and limited access to healthcare services. Following the epidemic, the situation worsened due to economic insecurity, disruption of healthcare services, and restricted mobility during lockdown periods. This study demonstrates that the Bakarwal community's nomadic lifestyle, as

well as the geographical remoteness of tribal villages, provide significant impediments to accessing contemporary healthcare treatments. Women in distant and migratory areas frequently rely on traditional treatment methods due to a lack of medical facilities, transportation, and awareness campaigns. Maternal and reproductive health issues were shown to be particularly important among tribal women, especially during seasonal migration. Despite several government initiatives and tribal welfare programs, the benefits have yet to reach the grassroots due to a lack of awareness, poor implementation, and infrastructural limitations. The study highlights the critical need for mobile healthcare units, women-centered health awareness initiatives, increased nutrition services, mental health assistance, and maternal healthcare facilities in tribal areas. According to the study, improving the health status of Gujjar and Bakarwal women necessitates a comprehensive and culturally appropriate approach that includes healthcare access, education, nutrition, and social empowerment. Strengthening tribal healthcare policies and supporting inclusive development will help tribal women improve their quality of life and well-being in the aftermath of the pandemic.

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A Behavioral Data-Driven Approach for Cervical Cancer Risk Pre-Screening Using Data Mining Techniques

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Abstract-Cervical cancer is emerging as one of the most frequently occurring diseases among women. There is a possibility for preventing this cancer if the disease is found out at an earlier stage. This disease threatens women's health worldwide as it cannot be detected easily at an earlier stage. The risks involved in cervical cancer can be diagnosed with the help of various methods. This study proposes techniques based on data mining to classify individuals into high-risk and low-risk categories for cervical cancer using behavioral data. Different classification algorithms such as Random Forest, J48, Naive Bayes, Logistic Regression and ensemble methods such as Bagging, AdaBoost are applied and their performances are calculated in terms of accuracy, precision, recall and F1-score. The results reveal that behavioral data can be effectively used for risk-based pre-screening. This study demonstrates that Naive Bayes method can be used for the classification of risk level of the patients with the possibility of cervical cancer.

Keywords: Pre-screening, Random Forest, J48, Naive Bayes, Logistic Regression, Bagging, AdaBoost

1. Introduction

Cervical cancer accounts to the main cause of cancer deaths in women globally. Earlier identification and quick action can greatly lower death rates. Lack of regular screening and low health-seeking behavior significantly increase the risk of late-stage detection of cervical abnormalities. Cervical cancer is a cancerous tumor of the cervix of the uterus where cervix cells become abnormal and start to proliferate uncontrollably, resulting in tumor formation. Consequently timely detection and intervention of precancerous cervical cancer stages is essential for lowering the mortality and morbidity associated with cervical cancer. Cervical cancer is a deadly disease once it reaches the invasive

stages, but out of all the female genital tract cancers, it is the only preventable cancer if detected at its early stages.

Cervical cancer risk is influenced not only by histopathological findings but also by behavioral factors which can be utilized for the assessment of risks at an early stage and pre-screening. Behavioral and psychosocial factors have an important position in influencing an individual's risk of developing cervical cancer. This study aims to apply a model that classifies individuals into high-risk and low-risk categories based on behavioral data. This classification can help as a tool to identify individuals who needs further clinical evaluation. Variations in behavioral patterns across individuals contribute to differing levels of cervical cancer risk, highlighting the importance of personalized risk assessment models.

This paper analyses various classification methods and provides a deep understanding about the best classifier among Naive Bayes, Random Forest, J48, Logistic Regression, Bagging, AdaBoost for assessing the risk level for cervical cancer. Section 2 provides a literature review of works related to the predicting the risk levels of cervical cancer using data mining techniques. Section 3 presents the dataset details, preprocessing methods and various data mining methodologies that are used to perform the classification task along with the dataset. Section 4 provides the performance evaluation measures and section 5 provides the conclusion.

2.Literature Review

Mavra et al. introduced a hybrid method, CervDetect in 2021, that integrates random forest (RF) and shallow neural networks that identifies risk factors for malignant cervical lesions by applying machine learning to medical data. CervDetect uses the concept of pearson correlation that evaluates the relationship between input variables and the output variable during data preprocessing [2]. It also employs RF-based feature selection that finds the most relevant features. The model demonstrates improved detection rates compared to existing research.

In 2019, Wasswa et al. developed a classification method that uses Pap smears image by using an enhanced fuzzy c-means algorithm. They introduced Trainable Weka Segmentation for cell segmentation and a three-step procedure to remove debris [3]. Simulated annealing, paired with a wrapper filter, was used for feature selection. The

evaluation confirmed that selecting high-quality features and it improves the accuracy value for cervical cancer classification.

In 2017, Fatlawi applies improved decision tree method that classifies the risk factors dataset cervical cancer [4]. The DT classifier was used for patient classification. Feature selection method was not applied to select the optimal subset of features.

Abdoh et al. in 2018 built a cervical cancer classification model that uses RF with SMOTE and reduction techniques such as recursive feature elimination (RFE) and principal component analysis (PCA) [5]. The results prove that the combination of RF and SMOTE performs well and shows better accuracy of 96% in the prediction of biopsy results.

Alsmariy et al. in 2020 developed classification model for cervical cancer that combines decision tree (DT), logistic regression (LR) and random forest (RF) using a voting strategy. They used SMOTE to address dataset imbalance and PCA for feature reduction [6]. Their results proved that the voting classifier when combined with SMOTE and PCA, the accuracy, sensitivity and Positive Predictive Association/ Value (PPA) increased significantly from 0.93% to 5.13%, 39.26% to 46.97%, and 2% to 29%, respectively.

In 2020, Ghoneim used an ensemble technique that predicts cervical cancer risk based on a voting strategy. This method solves various challenges in risk prediction, though the sensitivity remains relatively low due to samples with imbalanced data [7].

3. Methodology

This study involves preprocessing of the dataset initially. This removes missing and imbalanced data from the entire dataset. After preprocessing, various classification techniques were employed that analyses the risk level of cervical cancer and classifies the dataset into low and high risk level individuals.

3.1 Dataset Description

The study utilizes the Sobar-72 dataset, a behavioral risk factor dataset comprising 72 instances and 18 input features and one class attribute obtained from the UCI Machine Learning Repository and used in prior cervical cancer prediction studies [8]. The features cover behaviour features, Intention features, social norms, perception and motivation features, social support and empowerment features. In this dataset, the target variable is

ca cervix which has the value 0 for non-cervical and the value 1 for cervical cancer. The attributes in the dataset represents the various risk factors of cervical cancer based on the behavioural data.

behavior_	behavior_	behavior_	intention_	intention_	attitude_c	attitude_s	norm_sign	norm_fulf	perceptio	perceptio	motivatio	motivatio	socialSup	socialSup	socialSup	empower	empower	empower	ca_cervix
10	13	12	4	7	9	10	1	8	7	3	14	8	5	7	12	12	11	8	1
10	11	11	10	14	7	7	5	5	4	2	15	13	7	6	5	5	4	4	1
10	15	3	2	14	8	10	1	4	7	2	7	3	3	6	11	3	3	15	1
10	11	10	10	15	7	7	1	5	4	2	15	13	7	4	4	4	4	4	1
8	11	7	8	10	7	8	1	5	3	2	15	5	3	6	12	5	4	7	1
10	14	8	6	15	8	10	1	3	4	2	14	8	7	2	7	13	9	6	1
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10	15	15	4	15	8	10	5	3	8	3	11	3	3	2	7	8	5	3	1
10	12	11	10	15	7	8	3	3	3	2	13	11	10	7	12	12	12	12	0
10	13	14	10	15	6	8	1	5	5	2	15	10	12	8	15	15	15	15	0
10	15	13	10	15	2	10	1	5	6	2	14	14	14	8	14	15	14	15	0

Fig 1. Sample Dataset: Cervical Cancer Behavior Risk

3.2 Data Preprocessing

Dataset is preprocessed prior to model training which increases the reliablity and accuracy of classification results. There may be missing values in the dataset and therefore the dataset has to be preprocessed to deal with this problem. The original dataset consisted of 72 instances. During preprocessing, 3 instances containing incomplete or missing values were removed, resulting in a final dataset of 69 instances.

3.3 Model Selection

Data mining techniques were used to predict the risks for cervical cancer based on behavioural data. In this analysis, various classification techniques such as Random Forest, J48, Naive Bayes, Logistic Regression and ensemble methods such as Bagging and AdaBoost were applied to determine the most efficient risk predictive models for cervical cancer. Each algorithm applies a different learning strategy and their predictive capabilities are compared.

Random Forest Classifier

Random Forest is an ensemble learning algorithm that randomly selects subsets of the training data and predictor variables and constructs multiple decision trees. Each decision

tree makes a prediction individually. The final classification is made based on the majority voting among all trees. The predictions from multiple trees are combined and Random Forest improves the classification accuracy and reduces variance. The risk of overfitting [9] is also minimized. This method is very robust and it has the capability to handle complex and high dimensional datasets. Therefore this method has been widely used for classification.

J48

J48 is one of the classification algorithm that creates decision trees and recursively divides the entire dataset based on the attribute providing the highest information gain. A hierarchical tree structure is created in which the internal nodes are the decision attributes. The branches correspond to possible attribute values [10] and the predicted class is represented by the leaf nodes. The unnecessary branches are eliminated by the process of pruning thereby reducing the problem of overfitting. The overall model's generalization capability is improved. The structure is very simple and the ease of interpretation makes it suitable for classification tasks involving medical and clinical datasets.

Naive Bayes

The Naive Bayes algorithm is a simple method that uses conditional probabilities of each feature being associated with a specific class to make the predictions. It uses Bayes' Theorem that calculates probabilities by examining how often certain values or combinations of values appear in past data. To estimate the parameters of the model, the method of maximum likelihood is used [11]. The algorithm often performs well in various complex real-world situations. A key advantage of the Naive Bayes theorem is that it requires only a minimum number of training data to estimate the necessary parameters.

Logistic Regression

Logistic Regression is a predictive model that is used when target variable is categorical. It is more focused on nonlinear regression [12]. The target variables in logistic regression are of two types: the first is a categorical target variable that includes two categories namely binary or dichotomous variables. The second one is a continuous target variable

which takes values between 0.0 and 1.0 that represents the probabilities or proportions [13].

Ensemble Methods

Bagging

Bagging is one of the methods in data mining that combines multiple models to maximize the accuracy value and stability of predictions in classification and regression tasks. It prevents the problem of overfitting and minimizes the variance of the model. It is commonly used with decision trees but it can be applied to any type of model. Bagging is a form of model averaging where individual classifiers are trained on different random samples, known as bootstrap samples, taken from the training dataset [14]. Traditionally, bagging uses same type of classifiers and combines their predictions through a majority vote.

AdaBoost

AdaBoost is technique that produces a final model with greater performance by combining multiple base classifiers. It works by iteratively adjusting the training data and provides more weight to previously misclassified instances. Initially, equal weight is assigned to all data points. After each iteration, the weights of incorrectly classified instances are increased while the weights of correctly classified instances are decreased [15].

4. Performance Evaluation

This study utilizes a dataset of 72 instances and they are preprocessed and the three of the duplicate and missing records were eliminated. Finally 69 instances were taken into consideration and they were evaluated in terms of training and test dataset. The dataset is considered to have 18 different details for each instance such as behavior_sexualRisk, intention_aggregation, intention_commitment, socialSupport_emotionality, empowerment_knowledge, etc that are needed to find out risk level of patient with cervical cancer. The dataset is divided into 70% for training and 30% for testing. Various classification algorithms such as Random Forest, J48, Naive Bayes, Logistic Regression and ensemble methods such as Bagging and AdaBoost are taken and their performances

are evaluated. The following table shows the total number of correctly classified and incorrectly classified instances among the training data and test data.

Table 1: Classification results of training data and test data

Algorithm	Total Number of Instances		Number of Correctly Classified Instances		Number of Incorrectly Classified Instances	
	Training Data	Test Data	Training Data	Test Data	Training Data	Test Data
Random Forest	47	22	41	18	6	4
J48	47	22	36	16	11	6
Naive Bayes	47	22	43	20	4	2
Logistic Regression	47	22	38	20	9	2
Bagging	47	22	36	15	11	7
AdaBoost	47	22	40	18	7	4

From the above table, it is evident that among the five classification algorithms, Naive Bayes performs the classification with more accuracy.

The following fig. 3 and fig. 4 shows the number of correctly classified and incorrectly classified instances in the training data and test data.

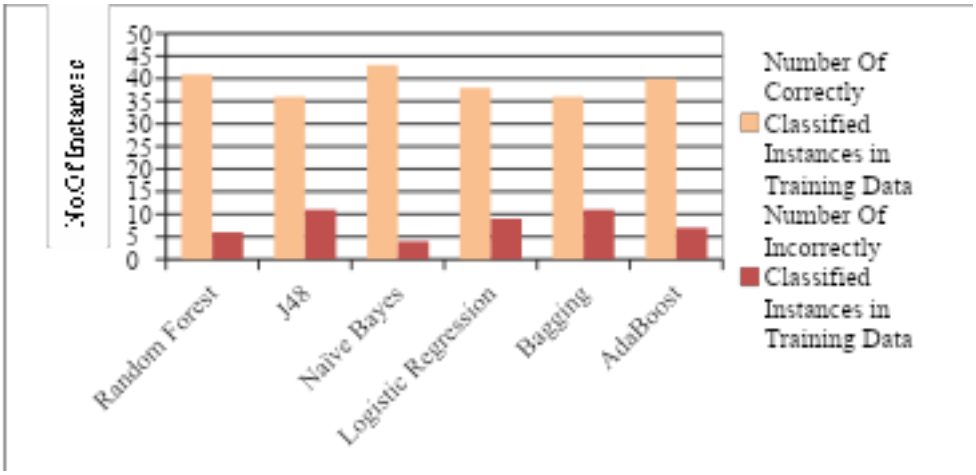

Fig.3. Classification of Training Data

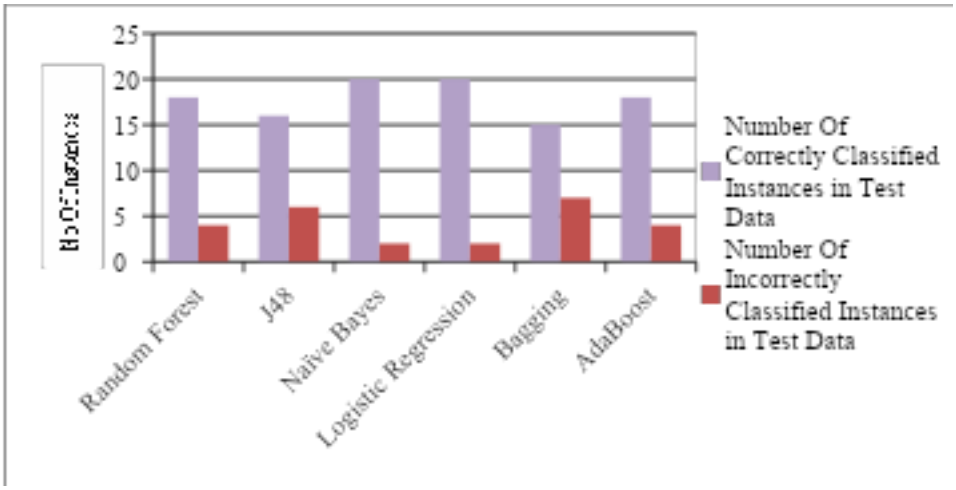

Fig.4. Classification of Test Data

Different types of evaluation metrics are used to assess the performance of the algorithms. The Kappa value indicates how effectively the classifier performed. The Mean Absolute Error (MAE) represents the average of the absolute differences between the predicted probabilities and the actual outcomes. The Root Mean Squared Error (RMSE) is similar to MAE, but it gives more weight to larger errors. The Relative Absolute Error (RAE) shows how much error the model produces compared to a simple baseline model, which always predicts the mean of the actual values. The Root Relative Squared Error (RRSE) is like RAE, but it is calculated using squared errors. Among the various evaluation metrics such as Kappa statistic, MAE, RMSE, RAE, and RRSE, the Naive Bayes algorithm demonstrates the highest performance in classification.

Table 2. Evaluation Metric values of different parameters in Training Data

Metrics/ Algorithms	Random Forest	J48	Naive Bayes	Logistic Regression	Bagging	AdaBoost
Kappa Statistic	66.75	42.87	78.78	53.26	34.64	63.65
MAE	22.19	25.01	88.82	20.01	30.27	17.45
RMSE	30.35	48.1	27.58	43.88	38.45	36.16
RAE	52.38	59.01	20.82	47.24	71.46	41.18
RRSE	66.04	104.69	60.01	95.50	83.68	78.71

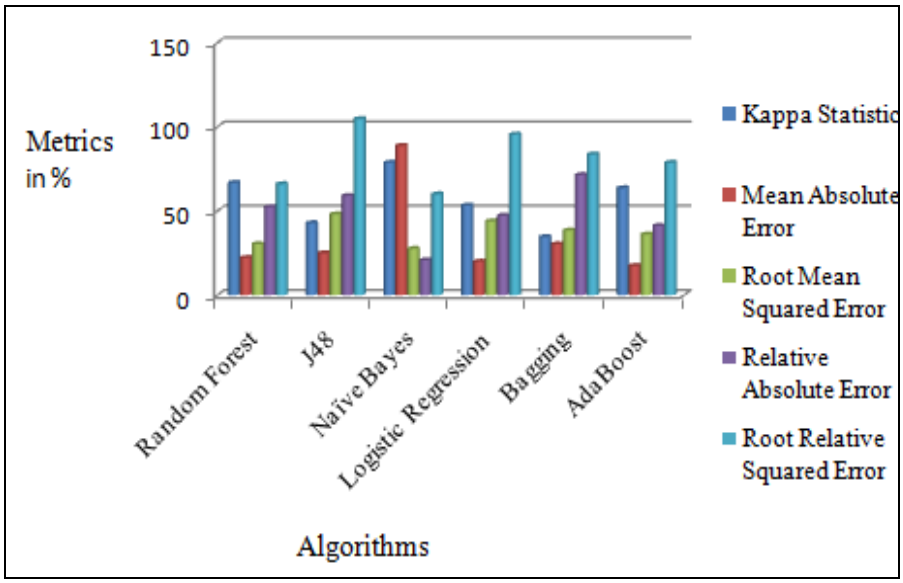

Fig 5. Evaluation Metrics Graph of different parameters in Training Data

Table 3. Evaluation Metric values of different parameters in Test Data

Metrics/ Algorithms	Kappa Statistic	MAE	RMSE	RAE	RRSE
Random Forest	66.75	22.19	30.35	52.38	66.04
J48	42.87	25.01	48.1	59.01	104.69
Naive Bayes	78.78	88.82	27.58	20.82	60.01
Logistic Regression	53.26	20.01	43.88	47.24	95.50
Bagging	34.64	30.27	38.45	71.46	83.68
AdaBoost	63.65	17.45	36.16	41.18	78.71

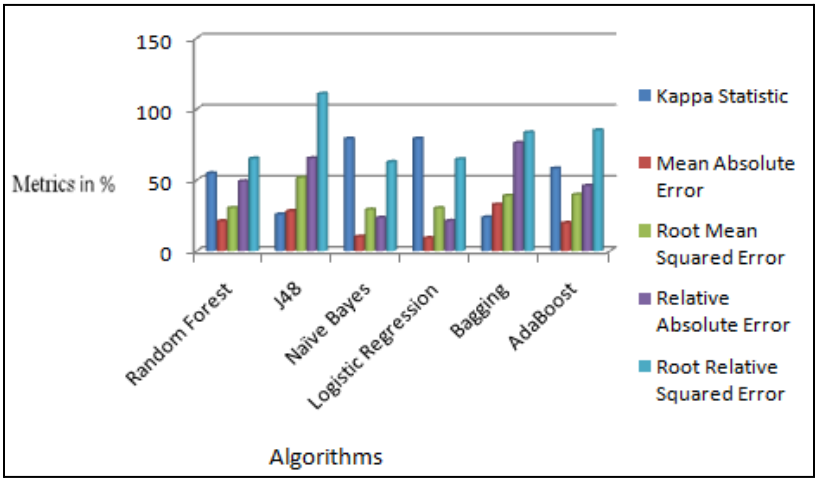

Fig 6. Evaluation Metrics Graph of different parameters in Test Data

The following table shows other performance evaluation factors such as precision, recall, F-measure, MCC, ROC Area and PRC Area. These values predict the quality and reliability of the classifier's predictions. Precision predicts how trustworthy the positive predictions are. Recall (Sensitivity) predicts how many actual positives the model can find. F-Measure predicts the balance between Precision and Recall. MCC value predicts the overall reliability of the classifier, considering all outcomes.

Table 4.Performance Evaluation Factors of different Algorithms

Algorithm / Metrics	Precision		Recall		F-Measure		MCC		ROC Area		PRC Area	
	Training Data	Test Data	Training Data	Test Data	Training Data	Test Data	Training Data	Test Data	Training Data	Test Data	Training Data	Test Data
Random Forest	0.875	0.816	0.872	0.818	0.865	0.809	0.684	0.561	0.959	0.933	0.963	0.939
J48	0.762	0.716	0.766	0.927	0.763	0.689	0.429	0.297	0.702	0.605	0.704	0.625
Naive Bayes	0.915	0.909	0.915	0.909	0.913	0.909	0.792	0.79	0.968	0.971	0.972	0.975
Logistic Regression	0.805	0.909	0.809	0.909	0.806	0.909	0.533	0.79	0.777	0.971	0.68	0.977
Bagging	0.757	0.67	0.766	0.682	0.738	0.675	0.381	0.239	0.866	0.829	0.876	0.863
AdaBoost	0.849	0.818	0.851	0.818	0.849	0.818	0.637	0.581	0.872	0.89	0.871	0.885

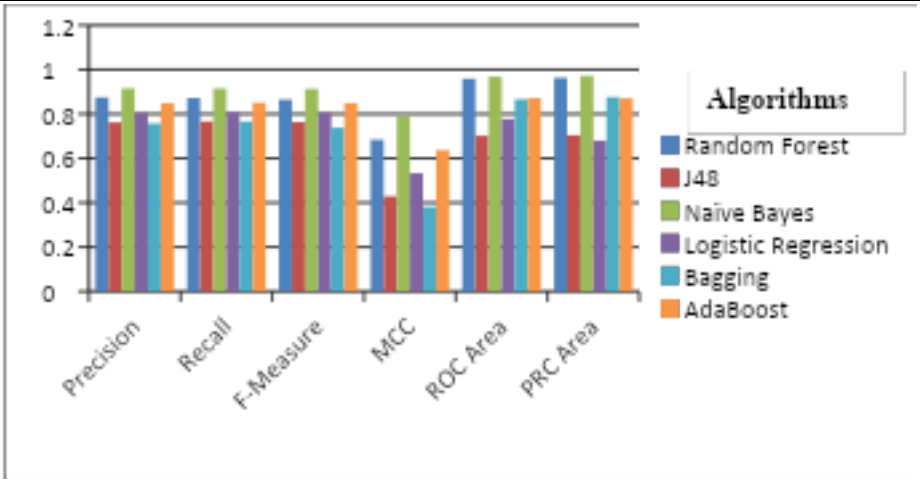

Fig 7. Performance Evaluation Factors of different Algorithms in Training Data

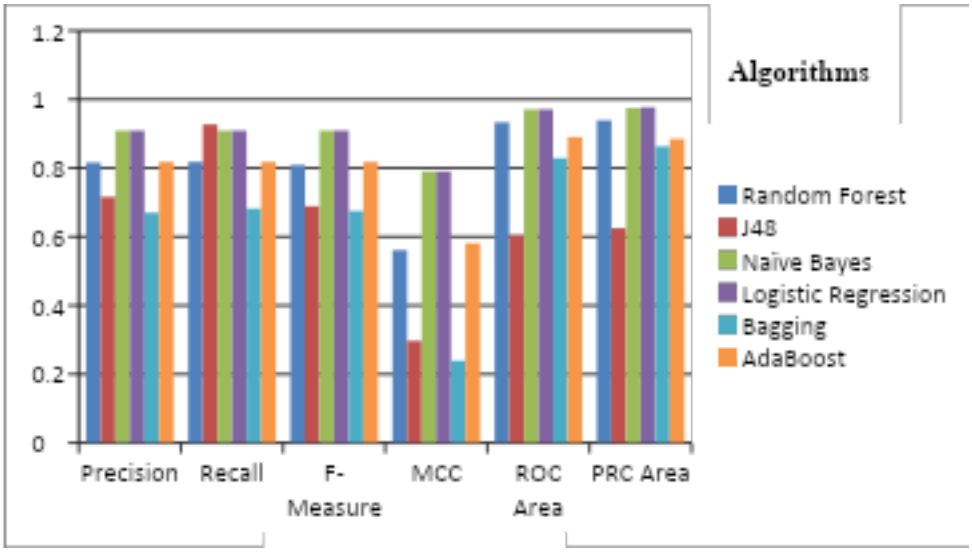

Fig 7. Performance Evaluation Factors of different Algorithms in Test Data

The following table shows the combined accuracy, precision, recall and F-Measure values of both training and test data. Among the five algorithms, Naive Bayes shows the highest accuracy value. The precision, recall and F-Measure values are also better in Naive Bayes than other four algorithms.

Table 4. Performance Evaluation Metrics

Algorithm	Accuracy(in %)	Precision	Recall	F1-Score
Random Forest	85.51	0.856	0.855	0.847
J48	75.36	0.747	0.817	0.739
Naive Bayes	91.30	0.913	0.913	0.912
Logistic Regression	84.06	0.838	0.841	0.839
Bagging	73.91	0.729	0.739	0.718
AdaBoost	84.06	0.839	0.840	0.839

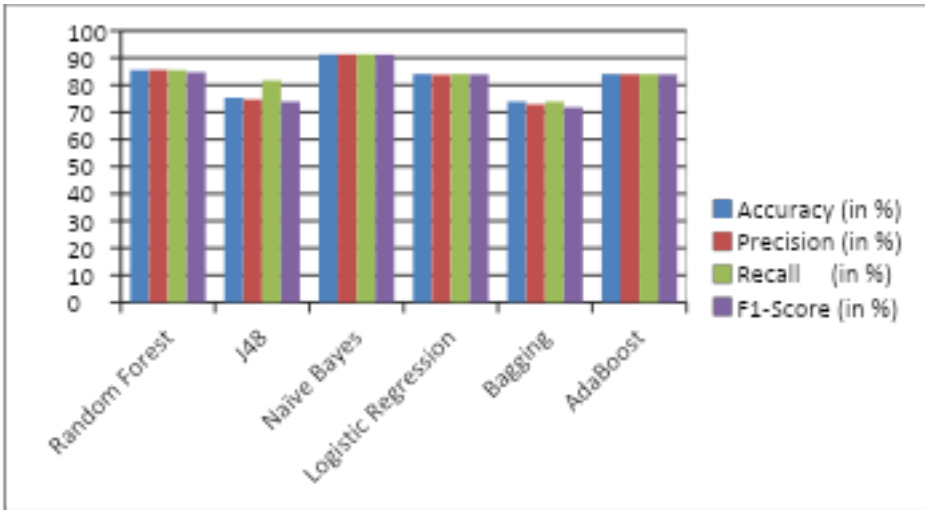

Fig 8. Performance Evaluation Metrics of different Algorithms

The following figures shows the accuracy, precision, recall and F-measure values of different algorithms and here, Naive Bayes provides better values for all the performance metrics.

5. Conclusion

Data mining techniques has increasingly been integrated in recent times into the medical diagnostic arena since it has the capability to deal with large volume of data, identify trends and make accurate predictions. These methodologies can be applied to diagnose cervical cancer using behavioral, demographic, and clinical data. This study utilized a smaller, imbalanced dataset based on behavioral data to evaluate the robustness and reliability of data mining techniques in predicting cervical cancer risk. Additionally, this paper demonstrates the effectiveness of data mining techniques for predicting the risk levels of cervical cancer by integrating behavior information. The results predict that data mining techniques can effectively aid in early screening and risk assessment for cervical cancer. In this analysis, six different classification algorithms have been used to predict the severity of the risks of patient with the possibility of cervical cancer. Among the six different methods, Naive Bayes outperforms other methods in terms of classifying the risk level of patients.

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Machine Learning-Enabled Security for 5G Network Slicing: Challenges, Solutions, and Future Directions

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Abstract: The fifth generation (5G) of mobile networks introduces the concept of network slicing, which divides the physical network infrastructure into multiple virtual partitions, or "slices," each designed to support different types of services such as high-speed broadband, massive device connectivity, and ultra-reliable low-latency communication. This method allows different users or applications to share the same physical resources while receiving service tailored to their unique requirements. However, the dynamic and multi-tenant nature of network slicing introduces new security and privacy challenges, including the possibility of attacks that cross slice boundaries or data leakage caused by insufficient isolation or configuration errors. In this survey, we provide a comprehensive yet accessible overview of the security issues related to 5G network slicing, focusing particularly on how machine learning (ML) techniques can help address these challenges. We cover the basic concepts and architectures of network slicing, discuss the lifecycle of slices, and classify various security threats. We also analyze emerging risks such as unauthorized slice creation and attacks targeting ML-based systems. Furthermore, we review ML applications in slice management, including resource allocation, admission control, and monitoring, alongside existing security measures and management frameworks. Finally, we identify open research questions and future directions, including the integration of AI into 5G architecture, stronger authentication methods, edge computing support, and maintaining continuous secure connections. The discussion is supported by recent, peer-reviewed research to ensure both accuracy and relevance.

Index Terms— 5G, network slicing, security, threats, machine learning.

I. Introduction

The shift from fourth-generation (4G) to fifth-generation (5G) mobile networks brings significant improvements such as faster data speeds and lower communication delays.

These advancements enable new applications like the Internet of Things (IoT), driverless cars, and augmented reality [1]. One of the key innovations underpinning 5G is network slicing, which allows the physical network to be divided into several virtual networks, or slices, each tailored to the needs of particular services or users [3]. This virtual partitioning enables network operators to provide different levels of service quality—for example, high throughput for video streaming or minimal latency for industrial controls—on a shared physical infrastructure. By efficiently sharing resources, network slicing reduces costs and improves flexibility. However, this approach also raises new security concerns. The increasing reliance on software control and virtualization, as highlighted by recent studies [2], increases the risk of vulnerabilities at various stages of a slice's lifecycle, including creation, operation, and termination.

Attackers could exploit misconfigurations or weaknesses in isolation to access data from other slices or disrupt service. The involvement of software-defined networking (SDN), network functions virtualization (NFV), and cloud technologies further complicates security, as traditional protections may no longer be adequate. To manage the complexity and scale of 5G slicing securely, machine learning (ML) techniques have gained attention. ML can analyze large volumes of network data, identify unusual patterns, predict potential issues, and automate decision-making to enhance security and performance [9], [10]. Despite growing interest, there remains a lack of comprehensive surveys that integrate the role of ML with the security challenges of network slicing. This paper aims to fill that gap by providing a detailed review of 5G network slicing security with a focus on ML applications. We examine the lifecycle of slices, identify common and emerging threats, describe ML-based security solutions, and discuss future research directions.

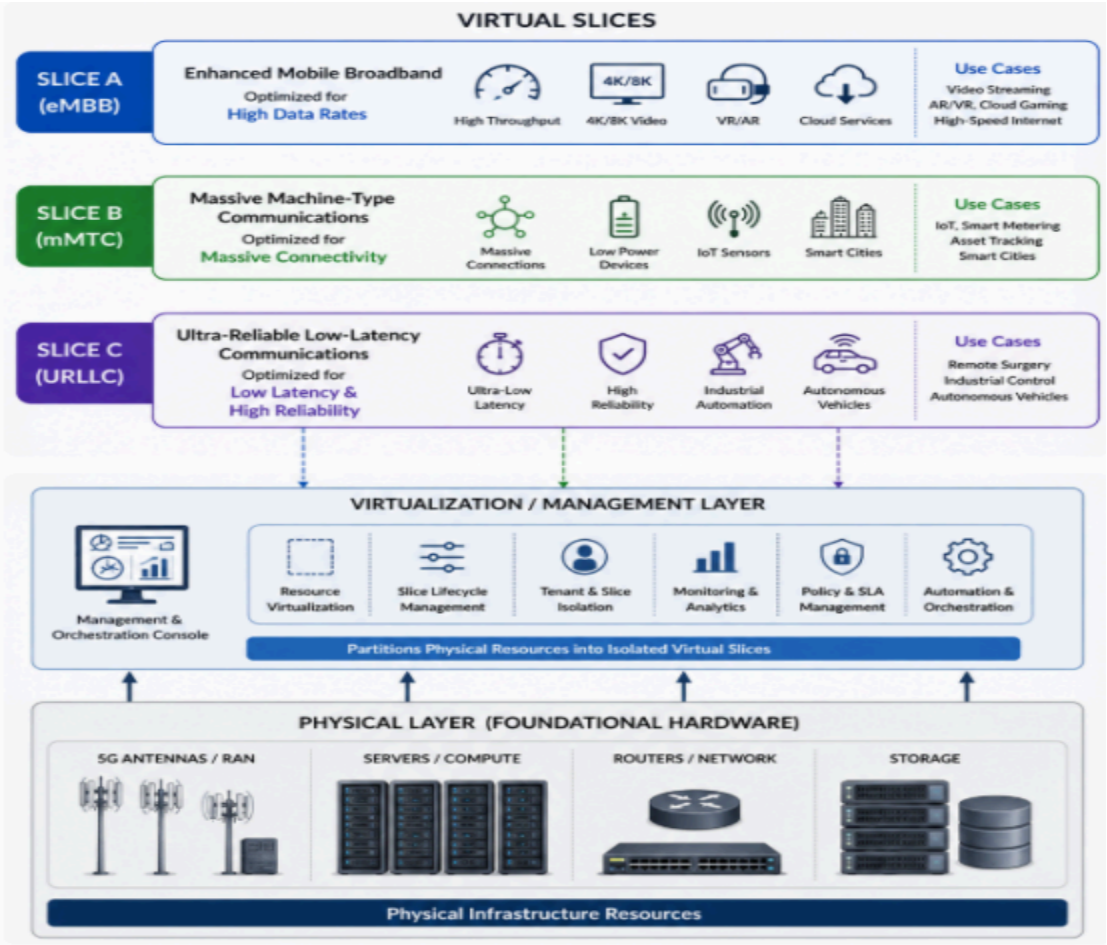
II. Contributions

This survey offers the following contributions:

● **Fundamental Overview**: We explain the key concepts behind 5G network slicing, including enabling technologies like NFV and SDN, and the main service categories (eMBB, mMTC, URLLC), supported by common architectural models [3], [16]. ●

Literature Review: We critically analyze existing surveys, highlighting their limited discussion of ML-enabled security in network slicing [5], [6].

- **Threat Taxonomy**: We provide a detailed classification of security threats throughout the slice lifecycle—from creation to operation to decommissioning—including newer threats like fake slice injection and adversarial ML attacks [2], [7], [8].
- **ML in Slice Management**: We explore how different ML methods are applied to tasks such as admission control, resource allocation, anomaly detection, and security monitoring [10], [15].
- **Security Frameworks:** We review existing isolation techniques, authentication methods, encryption schemes, and continuous monitoring approaches that incorporate ML to strengthen slicing security [11], [12].
- **Future Directions**: We identify challenges and opportunities such as AI-native 5G architectures, robust authentication, edge computing support, and maintaining secure and uninterrupted connectivity [13], [14], [17]



III. Related Work and Research Landscape

Several previous works have studied 5G network slicing from different angles. Early surveys by Foukas et al., Afolabi et al., and Kaloxylos provided broad overviews of network slicing concepts, enabling technologies, and challenges [5]. Other research focused on specific components such as slice orchestration with SDN/NFV [19], slice models within radio access and core networks [19], or slicing tailored for IoT services [19]. Algorithmic studies have also addressed virtual network function placement and resource management. More recent studies began examining machine learning's role in slicing. For example, Kafle et al. and Shen et al. investigated ML applications in radio access network slicing [6]. Wu et al. discussed AI in future 6G networks but did not deeply explore the integration of ML with security concerns [6]. Ssengonzi et al. focused on reinforcement learning for admission control and resource allocation [20]. Regarding security, Chamitha De Alwis et al. provided a taxonomy of threats related to network

slicing but did not extensively cover ML-based defenses [2]. Other works outline attack vectors but lack comprehensive treatment of ML's role in mitigation [22]. Our survey integrates these aspects, offering a holistic view of security threats and ML techniques across the entire slicing lifecycle, updated with recent IEEE literature [21], [22].

IV. 5G Network Slicing Fundamentals

A. Concept and Layers

Network slicing divides the physical 5G infrastructure—including radio access, core network, and transport segments—into multiple isolated virtual networks, each customized to meet the requirements of specific service types [3]. Each slice is essentially a complete end-to-end network with its own virtual network functions (VNFs) and allocated resources. For instance, one slice may be optimized for enhanced mobile broadband (eMBB) offering high data rates, while another focuses on ultra-reliable low-latency communications (URLLC) for critical control systems. Conceptually, slicing can be understood as operating across three layers [4]:

- **Service Layer:** Represents tenant services and their business requirements such as bandwidth, latency targets, and security policies.
- **Network Slice Layer:** Contains the actual slice instances, configured with the appropriate VNFs and parameters to satisfy service demands.
- **Resource Layer:** Includes the pool of physical and virtual resources (compute, storage, radio spectrum) from which slices draw their allocations.

This layered structure facilitates mapping high-level service requests to concrete resource allocation and network function configurations.

B. Architectural Frameworks

Several standardization bodies have proposed architectural models for implementing network slicing. They generally include:

- **Slice Implementation Components**: Responsible for creating slices, allocating resources, and deploying VNFs as needed.

- **Slice Management Components:** Oversee ongoing slice operation, scaling, monitoring, and policy enforcement. Examples include:
- **NGMN Architecture:** A three-layer model separating infrastructure, network functions, and business/application layers [23].
- **5G-PPP Framework:** A five-layer architecture distinguishing infrastructure, network functions, orchestration, business functions, and service layers [24].
- **ONF SDN Architecture:** Emphasizes centralized control via SDN controllers to enforce slice isolation and forwarding [25].
- **ETSI NFV Framework:** Provides virtualization infrastructure and management/orchestration (MANO) components to deploy and manage VNFs composing slices [26].
- **3GPP 5G Core Specifications**: Define slice-aware core network functions enabling devices to connect to the correct slice [3].

C. Slice Lifecycle

Slices follow a lifecycle consisting of:

- **Preparation:** Defining service requirements and mapping them to slice specifications.
- **Commissioning:** Instantiating the slice by allocating resources and configuring VNFs.
- **Operation:** Monitoring and potentially adjusting slice resources to meet changing demands.
- **Decommissioning:** Terminating the slice and releasing resources securely.

Automation is critical to handle the scale and variability of slices, with NFV MANO and SDN controllers orchestrating lifecycle events. ML techniques increasingly assist with decision-making throughout this process [15].

V. Methodology

This paper follows a systematic literature review methodology to gather, analyze, and synthesize research on 5G network slicing security and machine learning applications. The process involved:

1. Literature Collection: We searched major academic databases such as IEEE Xplore, ACM Digital Library, and Google Scholar for peer-reviewed articles published from 2015 to 2024, using keywords including "5G network slicing," "security," "machine learning," "slice management," and related terms**.**

2. Selection Criteria: Papers were selected based on relevance, contribution to the domain, recency, and citation count. Both survey articles and original research papers were included.

3. Categorization: The literature was categorized into foundational concepts, threat analyses, ML applications, security frameworks, and open challenges.

4. Synthesis: We integrated findings to identify security threats, ML-based defenses, gaps in current research, and future directions.

5. Validation: Cross-referencing between multiple sources ensured accuracy and comprehensive coverage. This structured approach ensures a balanced and insightful review of the evolving field.

VI. Machine Learning Applications In Network Slicing

The complexity, scale, and variability of 5G slicing require intelligent automation beyond static rule-based management [28].

A. Key ML Applications

- **Demand Forecasting:** ML models predict future slice resource needs based on historical and real-time data, enabling proactive scaling [29].
- **Admission Control:** Supervised ML classifies slice requests to accept or reject them based on resource availability and potential impact [30].
- **Resource Allocation:** Reinforcement learning algorithms dynamically allocate CPU, memory, and spectrum among slices to optimize performance [10] .
- **Mobility Management:** Deep learning predicts user movement to prepare seamless handovers, maintaining slice continuity.
- **Anomaly Detection:** Unsupervised ML models detect deviations from normal traffic behavior, identifying potential faults or attacks [15].

- **Security Monitoring:** ML classifiers analyze network flows to detect intrusion attempts or misbehaving components.

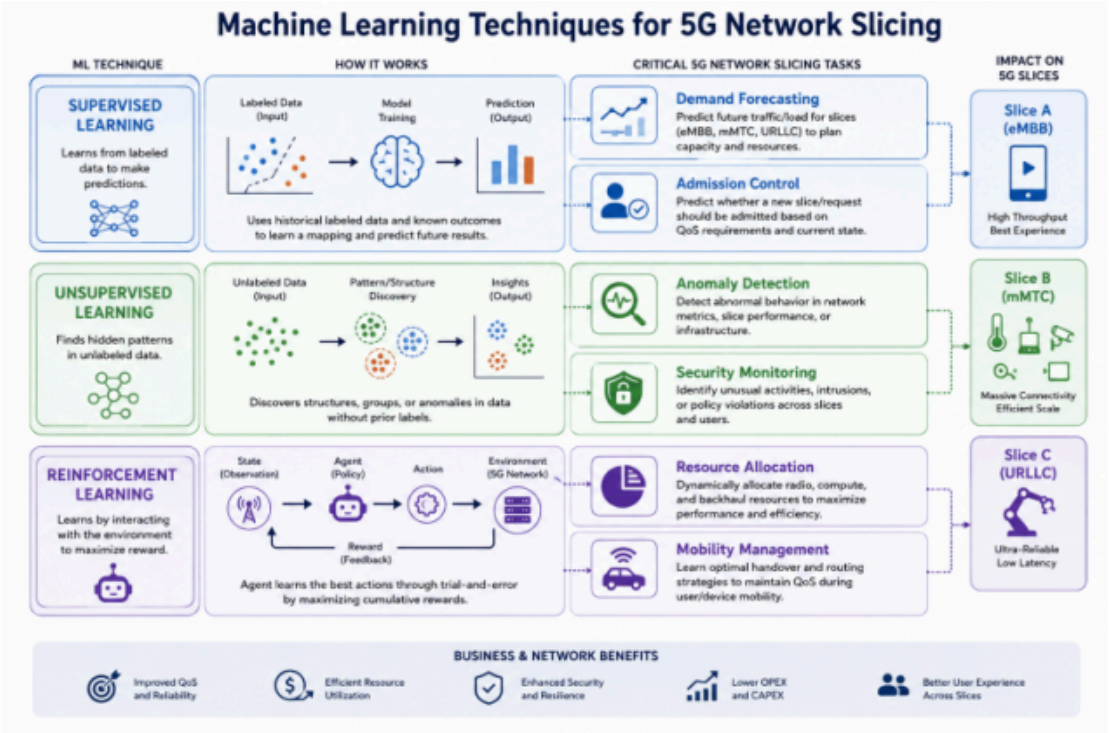
B. ML Techniques

A variety of ML methods are used:

- **Supervised Learning:** Trained on labeled data to classify or predict known attack types.
- **Unsupervised Learning:** Detects anomalies without prior labels, useful for unknown threats.
- **Reinforcement Learning:** Learns optimal policies through trial and error interaction with the network environment.
- **Deep Learning:** Uses multi-layer neural networks to capture complex patterns in high-dimensional data.

C. Challenges

ML models must be robust to adversarial inputs designed to confuse or evade detection. Privacy concerns arise when training data includes sensitive tenant information, requiring privacy-preserving techniques.



VII. Security Threats In Network Slicing

Security risks arise throughout the slice lifecycle:

- **Preparation Phase:** Fake slice injection and unauthorized configuration changes can prevent proper slice setup or allocate resources maliciously [31].
- **Operation Phase:** Slices face traditional attacks like denial-of-service (DoS), but with added complexity due to shared infrastructure. Weak isolation may cause attacks to spread across slices [2]. Insider threats and misconfigurations also pose risks. ML components themselves can be targeted through adversarial attacks to undermine detection or control [8].
- **Decommissioning Phase**: Improperly cleared data or resources can leak sensitive information or obstruct future slice allocations [32].

VIII. Security Solutions and Management Frameworks

Key security controls include:

- **Strong Isolation:** Using virtualization, containerization, and hardware isolation to prevent cross-slice interference.

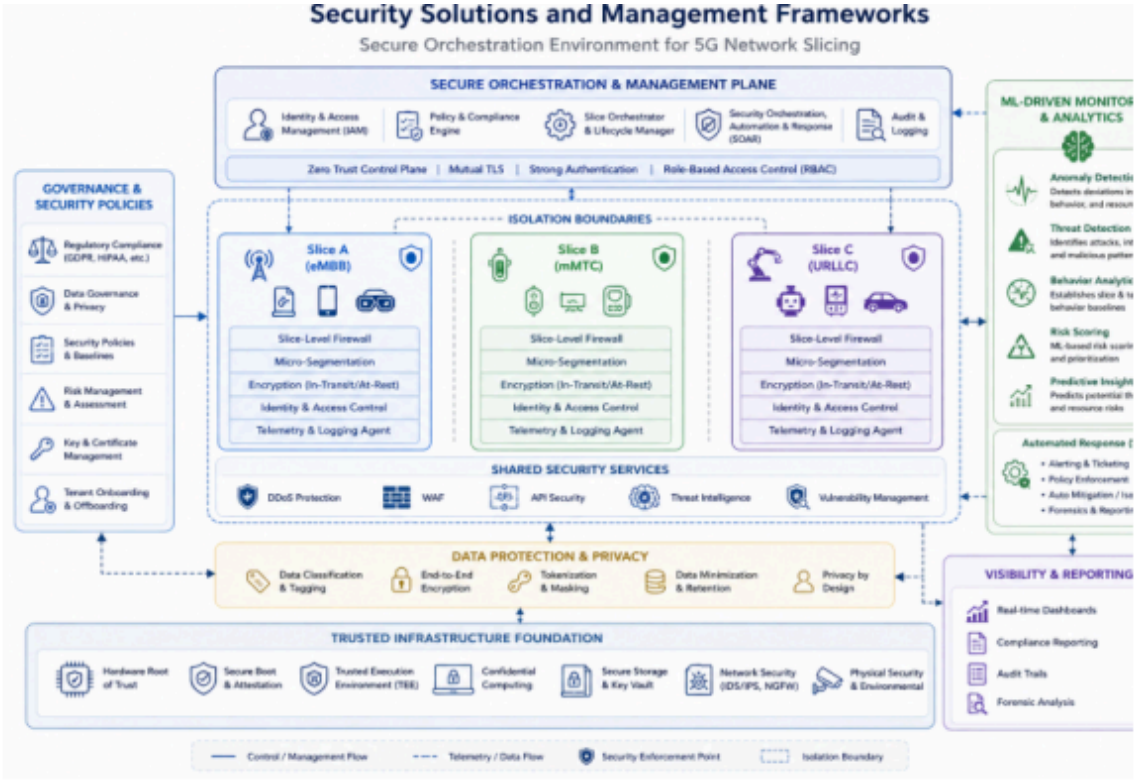
● **Authentication and Encryption:** Ensuring entities within slices authenticate each other and data is encrypted both in transit and at rest.

● **Secure Orchestration:** Protecting control interfaces and APIs with secure protocols and strict access controls.

● **Continuous Monitoring:** Applying ML-based anomaly detection and automated response mechanisms.

● **Resilience and Redundancy:** Designing slices to fail gracefully and recover quickly from attacks or faults [11].

ML enhances these mechanisms by enabling predictive analytics and adaptive controls, but ML models require safeguards against adversarial manipulation.



IX. Research Impact

Secure 5G network slicing is foundational for the success of next-generation applications that impact everyday life and critical infrastructure:

● **Connectivity for Diverse Applications:** Enables tailored network support for applications ranging from high-speed video streaming to remote surgery, ensuring quality and reliability.

- **Economic Growth:** Efficient network resource utilization lowers operational costs, encouraging innovation and reducing barriers to deploying new services.
- **Safety and Reliability:** Enhances security for industrial control systems, autonomous vehicles, and emergency communications, where failures can have severe consequences.
- **Privacy Protection:** Strong isolation and ML-driven security help prevent data breaches, protecting user information in multi-tenant environments.
- **Scalability and Adaptability:** ML-based automation supports growing device densities and dynamic traffic patterns, ensuring network resilience and service continuity.

Thus, advances in ML-enabled slicing security will directly support technological progress and societal benefits

X. Future Research Directions

Important areas for further study include:

- **AI-Native Architectures:** Designing network infrastructure with embedded ML capabilities for real-time, adaptive security [13].
- **Robust Authentication:** Developing scalable, efficient authentication schemes for diverse devices and dynamic slice environments [14].
- **Edge-Enabled Slicing:** Employing edge computing and federated learning to reduce latency and enhance privacy in ML model training [17].
- **Continuous Connectivity:** Creating self-healing networks that maintain service during failures or attacks through predictive ML [18].
- **Adversarial ML Defense**: Building ML models resilient to data poisoning, evasion, and model theft.
- **Privacy-Preserving ML**: Applying methods like differential privacy to protect tenant data during ML analytics [22].

XI. Conclusion

Network slicing represents a fundamental shift in how 5G networks deliver services, enabling multiple virtual networks tailored to varied needs on shared infrastructure. While this improves flexibility and efficiency, it introduces complex 7 security challenges

that span the entire slice lifecycle. Machine learning is a powerful tool that can automate and enhance security management but also introduces new vulnerabilities. This survey has provided a comprehensive overview of the concepts, threats, ML applications, security mechanisms, and open challenges related to 5G network slicing security. Future research must focus on integrating robust ML-driven defenses with secure, scalable architectures to realize trustworthy 5G networks.

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Impact of Social Media on Students' Academic Performance and Learning Outcomes: A Review Study

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Abstract— In todays world social media is a part of students lives. Students use Facebook, WhatsApp, Instagram, YouTube, LinkedIn, Twitter and Telegram to talk to each other share information have fun and learn. This paper looks at how social media affects students school work, mental health and education. The paper is based on information from studies, articles and reports about media and education. What we found out is that social media is both good and bad for students. On the one hand social media helps students work together talk to each other and learn things. It also helps students use computers and the internet get involved in school and find resources. Social media helps students learn in their way and at their own pace. It also helps students meet people in their field and make friends. However using media too much can hurt students school work. It can distract them make them put things off and make it hard for them to focus. Using media too much can also hurt students mental health. It can make them feel anxious stressed, depressed and addicted. It can also make them feel bad about themselves and have trouble sleeping. So social media can be a tool for education if students use it in a way. Schools, teachers, parents and students should help each other use media in a way that helps students learn. We also need to teach students about using media in a healthy way. This way students can get the things, from social media and avoid the bad things.

Index Terms—Social Media, Academic Performance, Mental Health, Educational Outcomes, Higher Education, Student Engagement, Digital Learning, Collaborative Learning.

I. Introduction

In this century social media has become very important in how people talk to each other interact and learn things. The growth of the internet, smartphones and digital technologies has made social media platforms very popular among students.

* Facebook

* WhatsApp
* Instagram
* YouTube
* LinkedIn
* Twitter
* Telegram
* Google Classroom

have become a part of students' academic, social and personal lives.

In education in higher education social media is widely used for talking to each other learning together sharing information, online discussions and getting educational resources. Students often use media to talk to friends communicate with teachers participate in online learning watch educational videos and get help with school work. Schools and teachers have also started using media to teach and make students more interested.

Social media gives students a lot of opportunities to improve their learning experiences. YouTube has video lectures and educational content that help students understand things easily. WhatsApp and Facebook groups help students learn together by sharing notes talking about homework and solving problems. We see that social media and YouTube play a role here.

LinkedIn helps students develop skills to meet people and find job opportunities. Social media has become a tool that supports personalized learning. However social media also has some effects on students. Using media too much often leads to getting distracted putting things off not managing time well not being able to focus and not doing well in school. Many students spend a lot of time on things like chatting, playing games watching videos and looking at social media instead of doing their school work. Also using media too much has raised concerns about students mental health and well-being.

Always looking at things comparing themselves to others getting bullied online being afraid of missing out and getting too hooked on social media can cause students to feel anxious depressed stressed, have trouble sleeping and have low selfesteem. These mental health problems can affect students school work and overall development in a way. The effect of media on education is both good and bad. While some people think social media

is good for talking to each other working together and learning online others think it is bad for school work and mental health. Since social media is becoming a part of students lives it is very important to study how it affects school work, mental health and education.

This study looks at the effects of social media on students, in higher education by looking at what other people have found out reading scholarly articles and academic papers. The study also gives ideas on how to use media in a way that helps students learn and minimizes the bad effects.

II. Literature Review

Junco et al. (2011) Looked into how college students use Twitter for school. They found that students who took part in discussions on Twitter showed more engagement and better grades. This study said that if social media is used well in school activities it can help students learn better.

* Tess (2013) reviewed how social media is used in education. She said that Facebook, blogs, wikis and YouTube help students work together share knowledge and interact with each other.. Using these platforms too much can hurt students focus on school.

Greenhow and Lewin (2016) talked about how important it's for students to be good at using digital tools and social media. They found that students who are good at literacy can evaluate online information well and use social media for school purposes.

Ahmed and Qazi (2011) studied how university students use Facebook and YouTube. They found that using media for school helps students do better in school. On the hand using social media for fun hurts students grades and productivity.

Al-Rahmi and Zeki (2017) looked at how social media helps students learn together. They found that WhatsApp and Facebook help students and teachers interact more which improves motivation and learning.

* Bandara and Hettiwaththege (2023) studied how social media affects students mental health and learning. They found that using media too much can cause anxiety, stress, depression, cyberbullying and lower academic focus.

Sengupta and Vaish did a study in 2023. They looked at how important social media's for higher education when the COVID-19 pandemic was happening. The study by Sengupta and Vaish found that online platforms like media helped education keep going. These

digital platforms allowed students to learn online. They also made it easier for students and educators to talk to each other. The study by Sengupta and Vaish showed that social media played a role in this. Social media was very helpful for education, during the COVID-19 pandemic.

Kouser (2020) found a link between social media use and academic achievement among postgraduate students in Punjab, India. This study showed that using networking sites, for a purpose can help with educational development.

III. Objectives of The Study

The main goal of this study is to see how social media affects students grades, mental health and education in colleges and universities. We want to know how different social media platforms change the way students learn talk to each other participate in classes and grow as individuals. This study also looks at the things that happen when students use social media.

A. Academic Performance This study tries to understand how social media affects students grades, study habits and productivity. Students use media for both learning and fun. We want to know if social media helps students learn better.. Does it hurt their grades because they get distracted by social media?

B. Positive Educational Impact We want to find out how social media helps students learn and engage with each other through media. Social media platforms like YouTube, WhatsApp and LinkedIn can help students work together share knowledge and access resources on social media.

C. Negative Educational Impact This study also looks at the things that happen when students use media too much. Using media too much can lead to distraction, procrastination and addiction to social media. We want to know how this affects students grades and daily routines.

D. Mental Health Impact We also want to know how social media affects students mental health through media use. Using media too much can cause anxiety, stress and depression. This part of the study focuses on understanding how social media affects students emotions and mental well-being.

E. Educational Outcomes This study tries to understand how social media helps students achieve their goals through media. Social media provides learning opportunities and access to educational resources on social media. We want to know if these social media platforms help students learn more communicate better and develop their careers.

F. Responsible Use of Social Media The final goal of this study is to give suggestions on how students can use social media in a responsible and effective way on social media. We want to recommend strategies that help students get the most out of media for learning while avoiding the negative effects, on their grades and mental health.

IV. Research Methodology

This study is about media and its effects on students. The study looks at how social media affects students academic work, health and education results in college. Since this is a review we only used information that already exists, like what other people have written in books and articles. We got our information from places, including research papers, books and websites. We read a lot about media how well students do in school how they learn together their health and how they develop as students. We wanted to see how social media platforms like Facebook, WhatsApp, Instagram, YouTube, LinkedIn and Twitter affect students.

Media and its effects on students is what we are looking at. We looked at what other researchers have found out about media and students. We included studies from India and other countries to get an understanding of the topic of media and its effects on students. We put all the information into categories like effects of media on students, effects of media on students how well students do in school how social media affects their mental health and how it affects their education results.

Looking at media platforms like Facebook, WhatsApp, Instagram, YouTube, LinkedIn and Twitter helps us understand how they influence what students learn and how they learn. Media and its effects on students is a topic. We also looked at what other studies have found out to see if we can find any patterns about media and its effects on students.

There are some problems with this study because we only used information that already exists about media and its effects on students. We did not talk to students directly we did not collect information about media and its effects on students.With these problems this

study gives us some ideas and a detailed analysis of how social media affects students in college. The study is about media and its effects on students. We hope it helps us understand this topic of media and its effects, on students better.

V. Impact of Social Media On Students

Social media is a part of students daily lives now. Students use media every day. They are always on Facebook, WhatsApp, Instagram, YouTube, LinkedIn, Twitter, Telegram and Google Classroom. Students use media to chat with each other have fun and learn new things with social media. They can easily get online using their smartphones so they rely on media to get information talk about things share what they learn on social media and connect with friends and teachers on social media. Social media really affects how students learn, what they learn with media how they talk to each other on social media and how they feel about themselves on social media. Social media is super important for students who use media. It affects their lives in ways like their school behavior. How they feel mentally on social media. Students use media to learn new things on social media and share what they know with others on social media. Social media also helps them connect with people who can help them with their studies and schoolwork on media. Students and social media go hand in hand nowadays with media. Social media plays a role in their education and daily life with media. Students are connected to media all the time and social media is always there, for students.

One good thing about media for students is that it helps students talk to each other and work together on school projects. Media platforms like Facebook and Twitter let students talk to their classmates, teachers and experts in a subject even if they are not in the classroom. Media helps students join in on group discussions about school work. Apps like WhatsApp and Google Classroom help students share notes and exchange school work and solve problems together. When students work together using media they learn from students which makes understanding school subjects easier for students. Students also get feedback from teachers and classmates which helps students learn better and understand things. Media helps students learn from students. Students get help when students need it. Teachers can use media to give feedback and guide students on their

school work. Using media makes it easy for students to connect with their classmates and teachers at school. Media also helps students learn from experts in a subject about the things students are studying. The use of media in learning helps make education more accessible, to students and helps students learn.

Social media helps me learn things. It is really helpful for learning because I can find lots of information on media. YouTube is a place where I find videos and tutorials. They explain things in a way. This makes it easy for me to understand. I like using YouTube. YouTube has helpful videos. These videos teach me things. I find YouTube videos easy to follow. The tutorials, on YouTube are clear. They help me learn.. There are lots of videos and tutorials on YouTube that make hard things easier to understand which's great. I think social media is helpful for learning because I can watch YouTube videos over and over again which is very helpful for me. I can learn at my pace by watching recorded lectures again and again whenever I need to which is very useful for students like me who are doing courses or distance education. This is very helpful for students like me because we can learn from media at any time. Social media is also great because it lets me access things from countries and schools which's really cool. I get to learn more and see types of academic things on social media, which is very interesting. Social media helps me learn more and get exposed to types of things which is great for my education. I can also use media to connect with professionals and organizations through sites like LinkedIn. LinkedIn is helpful for connecting with professionals and organizations which can help me with my career. This helps me with my career by improving my skills and learning from professionals on LinkedIn. I can even connect with researchers on media and learn from them which is very helpful for my career growth. Social media is really helpful for learning and career growth because it has resources like YouTube and LinkedIn that're really useful for me. Social media is really helpful, for learning and career growth because I can use media to learn new things and connect with people who can help me with my career.

Social media has an impact on people. It is not a good one. It can keep people awake at night which makes it hard for them to sleep. Social media can be very addictive. This is a big problem. A lot of students are on media late at night and this means they do not get the sleep they need. When students do not get sleep they do not feel good and they are

not healthy. Social media is the reason for this. When students do not sleep well it is hard for them to focus on things. They have energy, which is not good. Not sleeping well also affects how well students do in school. That is a big deal. Social media is part of the reason for this because it can keep students from sleeping and doing well in school. The thing about media is that it can be a big problem because people can get too used to it. This is what we call addiction to media. When students are addicted to media they have a hard time stopping themselves from being online all the time. They start to like being on media more than they like doing things in real life like going to school and doing their school work. Social media is a part of the problem because it can make students not want to do their school work. Social media can be very bad, for students. That is the truth.

Social media is something that students can use in a way or a bad way. It really depends on what they're doing with it. When students use media for school work it can be very helpful. Social media helps students talk to each other work together on projects learn things and get better grades in school. Social media can be bad for students if they use it much. Using it much can distract them from their school work make them feel sad make them feel anxious and hurt their grades. So students have to be careful about how they use media. They need to find a balance and not spend much time on it. Schools and parents and teachers should also help students learn how to use media in a good way. They should teach students about being safe on media and how to use it to learn and grow. This way students can use media to do well in school and in life. Social media can be a tool, for students if they use it correctly.

VI. Findings of The Study

Social media has bad effects on students. It can. Hurt how well they do in school how they feel and what they get out of their education. It really depends on why students use media how often they use it and how they use it. When students use media for school it is a good thing. It helps them learn and pay attention in class. When students use social media too much or for things that are not related to school it can cause problems. These problems can affect how well students do their work and how happy they are. Social media can be good for students when they use it for school.. Social media can also be bad for students when they use it too much or, for the wrong things.

One thing we found out from the study is that social media really helps students talk and work together on their learning. Students use WhatsApp, Google Classroom, Facebook groups and Telegram to share their study materials with each other. They also use these platforms to discuss their assignments and do group learning activities together. Teachers are available to interact with students in a manner. These digital platforms help create a learning environment where students can share knowledge work together and take part in activities. When students use media for learning they get more involved in the activities. They also get to connect with their school communities and friends. Social media helps students feel more connected, to their school.It also makes learning fun and interactive when using media.

The study shows that social media is not all good or all bad for students. It really depends on how students use media every day for school and personal things. When students use media in a smart way and do not overdo it it helps them learn and do well in school.. When students use social media too much and do not control it it hurts their learning, mental health and school results. So students need to use media responsibly and know how to use it safely to get the good effects of social media. Social media can be good for students if they use it correctly and are careful, about it.

VII. Suggestions

The study found some things about media and students. To use media in a way students, teachers, parents and schools need to work together. Media can be good or bad for students depending on how they use media.

Students should use media in a way. They should not spend a lot of time on media just for fun. This can hurt their studies and grades. Students should make a schedule for using media. They should stick to the schedule. They should only use media for a while. This way they can do their schoolwork. Use social media too. Educators and teachers can help students use media for school. They can use platforms like Google Classroom, YouTube and WhatsApp to share schoolwork and talk to students. Teachers should show students how to use media for school not for fun. This can help students learn more and be more interested in school.

Schools and colleges should teach students about media. They should show students how to use media correctly. Students should learn how to find information avoid information and be kind online. They should also learn about cyberbullying and online harassment. Parents should help their kids use media in a way. They should talk to their kids about media. Make sure they are using social media correctly. Parents should encourage their kids to do things like sports, reading and exercise. This can help kids be healthy and happy.

Schools should also help students with their health. They should have counselors and programs to help students deal with stress and anxiety. Students should feel comfortable asking for help if they are feeling overwhelmed. Students can use media to help them learn and grow. They can follow pages. Join online communities. This can help them learn things and make friends. Students should try to use media in a way that helps them not hurts them.

Finally students should be in control of their social media use. They should not spend a lot of time on media just for fun. They should use media to learn, communicate and grow. If students use media in an purposeful way it can help them succeed in school and, in life. Students should use media to help them not hurt them. They should make sure media is a part of their life but not the only part of their life..

VIII. Conclusion

Social media has become a part of student life today and it plays a major role in the education system of colleges and universities. Because technology and the internet are getting better and better students are using media more and more for talking to each other learning working together having fun and getting better at their jobs. This study looked at how social media affects students school work, mental health and how well they do in school by reading what other smart people have said about it.

The study found out that social media can be good or bad for students depending on how they use it. On the side social media makes it easy for students to get to educational resources watch online lessons work with other students talk about school and meet people who can help them in their careers. Websites like YouTube, WhatsApp, Google

Classroom, LinkedIn and Facebook help students learn in a fun way help students and teachers talk to each other and make it easy to share knowledge. Social media also helps students get better at using computers talking to people and learning on their own. When students use media for school it can make their learning experience better make them more interested in school and help them do better in their studies.

At the same time the study found out that using social media too much can be bad for students. Students who spend much time watching videos chatting with friends playing games or looking at things that are not related to school often get distracted put off their work do not manage their time well and do not do as well in school. Being on media all the time can also make it hard for students to focus and learn deeply. Also using media too much can cause students to feel anxious stressed, depressed have low self-esteem feel lonely have sleep problems and get addicted. Bullying on media and comparing themselves to others can also make students feel bad about themselves.

The study says that social media can be good for education if students use it in a way. Students who use media for school to share resources to work with other students and to get better at their jobs usually do well in school.. If students use social media too much for fun it can hurt their school work and their mental health. So it is very important for students to use media in a way that is disciplined and balanced.

The study also says that schools teachers and parents should help students use media in a good way. Schools should teach students about how to use computers and the internet how to behave well online. Teachers should use media in their lessons in a way that makes sense and helps students learn together.

In the end social media is a tool that can make education better if students use it in a responsible way. Students need to use media in a way that helps them do well in school get a good job and have friends without hurting their mental health. Maybe someday we will learn more about how social media affects students, in types of schools and in different parts of the world

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Privacy-Preserving Edge Intelligence: A Fog-Federated Framework for Continuous Vital Sign Monitoring in Smart ICU Environments

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Abstract

Continuous and accurate monitoring of patient vital signs in intensive care units is an essential element in critical care medicine; however, there exists a strong conflict between the requirement of real-time capability and data privacy in traditional centralized systems. In this paper, we propose a new framework called PrivEdge-FedICU that combines privacy-preserved edge intelligence with multi-modal attention-based fusion for the unbroken monitoring of patients' vital signs in IoMT-enabled smart ICUs. Fog computing nodes will be installed at the edge of wards where federated learning models are trained locally and then used to infer patients' physiology without leaking sensitive data outside hospitals. Self-attention will be adopted to assign dynamic weights to various patient vitals like electrocardiographic signals, peripheral oxygen saturation, blood pressure, respiratory rate, and temperature according to their clinical significance at any time, which provides more valuable information for monitoring patients than traditional fixed-weight fusion techniques. We use knowledge distillation to transform the heavy teacher model in the cloud into a lightweight student model on fog, where the compression ratio reaches 5.2x with over 96% of classification accuracy preserved. The privacy is ensured using a dynamic differential privacy budget allocation method that equally allocates the available privacy budget in proportion to the data sensitivity index of each individual fog node to ensure maximum model utility within the limit of a guaranteed level of privacy. The privacy of the gradients during training is achieved using a cryptographic secure aggregation protocol based on pairwise secret sharing, thus providing additional privacy guarantees in addition to the differential noise addition. From simulation experiments performed on the MIMIC-III medical dataset, the performance of PrivEdge-FedICU achieves vital signs monitoring accuracy of 96.1%, clinical alert F1-score of 0.940, fog-node inference latency of 26 ms and energy efficiency of 1.7 mJ per inference operation.

Keywords: *privacy-preserving federated learning, fog computing, edge intelligence, IoMT, smart ICU, vital sign monitoring, attention mechanism, knowledge distillation, differential privacy, secure aggregation.*

1. Introduction

The level of care in a contemporary intensive care unit is inherently bound up with the level of quality and continuity of patient monitoring. Critical patients are prone to dramatic changes in their hemodynamic state which can be lost in a matter of minutes; their respiration might degrade gradually without noticeable symptoms leading to a sudden failure; cardiac arrhythmias might arise without notice from a healthy-looking heart just a few minutes earlier. The continuous monitoring of vitals is thus not an option but a necessity which became much more efficient with the appearance of Internet of Medical Things technologies [3]. The bedside IoMT devices produce physiological information on the sampling rates and spatial resolutions which were impossible to achieve with traditional monitoring systems, thus making the analytical engines capable of detecting small deviations from the normal trajectory of changes that would be unnoticeable with mere manual assessments made by clinicians periodically. The problem is not, however, in collecting such information but in processing it into clinically meaningful data while preserving patients' privacy.

Privacy is not a secondary consideration in the design of clinical AI systems; it is a primary constraint that drives the whole analysis structure. The physiological information provided by the ICU monitoring systems is some of the most private information available; it provides information about not just the disease condition but even its diagnoses, any comorbid conditions, substance use, and genetic risks. Regulations like HIPAA, GDPR, and the Digital Personal Data Protection Act have stringent requirements about how such information can be collected, transferred, stored, and utilized. Centralized learning algorithms, which collect all raw patient information at one common server for training models, are inherently opposed to these requirements. More importantly, they present a single point of failure where a breach would result in leaking the physiological information of many patients at once. A monitoring system designed

for trust in a clinical setting needs to be a monitoring system where privacy is an inherent constraint rather than a procedural guarantee [7, 8].

This requirement is fulfilled by federated learning [5] that leaves raw data on the local machine and shares only gradient information of the model with the global server. By implementing this algorithm for smart ICU, every hospital ward will train its model of recognizing vital signs using data from its patients without ever exchanging raw physiological measurements of the patients outside the boundaries of that ward. Thus, the global model will be an aggregation of many hospitals, but no patient's data will leave the hospital where it was originally collected. However, typical federated learning algorithms, being implemented in edge computing environments of smart ICUs which rely on the technology of fog computing, face two challenges that limit their use in such environments. Firstly, inference attacks [9] allow inferring personal information about patients from gradient updates, so differential privacy approaches should be developed and applied. Secondly, general-purpose models trained in standard federated way tend to be too big and heavy to be executed on ward-edge fog node hardware due to computational complexity [12].

Such issues form the motivation for the current paper. PrivEdge-FedICU is a framework that leverages fog and federated learning technologies designed to monitor continuously and privately the vital signs in smart ICU environments. In terms of the unique architecture, this framework consists of four interrelated modules. The self-attention module implemented in the fog layer performs the fusion of multiple heterogeneous vital sign streams assigning the weight to each of them depending on the perceived importance at each particular time step and, therefore, generates a context-aware representation of patients' state better than simple weighted concatenation. Knowledge distillation technique [11] allows the compression of the global model into the student network deployed in the fog layer, thereby shrinking the model by 5.2x and speeding up the inference by 5 times (from 128ms to 26ms) while not sacrificing the accuracy. Adaptive differential privacy budget allocation technique distributes the budget for maintaining privacy among fog nodes proportional to their sensitivity indexes in order to maximize the utility of global model with proven privacy guarantees [9, 10]. Pairwise secret

sharing-based cryptographic protocol ensures that gradients provided by fog nodes are encrypted and, therefore, the central federated learning server cannot extract any information about them.

The clinical need for continuous versus episodic vital signs monitoring must be highlighted. As noted in previous work of the current authors [1], federated anomaly detection within smart ICUs can perform on par with traditional methods when applied to episodic events. In the present paper, the scope of such approach is being extended to include continuous stream-based monitoring, which constitutes an inherently more computationally and privacy-intensive task, while simultaneously requiring attention-based multimodal fusion to ensure high detection rates in states of physiology where different vital signs carry different amounts of diagnostic information. Interpretability provided by the attention weight distribution helps satisfy the previously mentioned needs for transparency in comparison studies of IoT solutions in healthcare [2].

The main contributions of this work are as follows. Self-attention-based multimodal vital signs fusion that can dynamically reweight the physiological input streams at the fog layer to produce patient state representations superior to those of baseline concatenation-based fusion and fixed-weight fusion models are introduced. Knowledge distillation pipeline that can reduce the deployed fog model size from 47.3MB to 9.1MB while maintaining 96.1% monitoring accuracy and achieving 26ms inference latency which meets real-time ICU monitoring requirements is presented. Adaptive differential privacy budget allocation method that can achieve up to 9.8 percentage points increase in accuracy compared to uniform allocation methods is designed and analyzed. Cryptographic secure aggregation that makes sure that the central server cannot reconstruct updates from fog nodes is implemented and evaluated for computational cost. The rest of this paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the system model and privacy formulation. Section 4 introduces PrivEdge-FedICU architecture. Section 5 introduces experiment methodology and results. Sections 6 and 7 conclude the paper.

2. Related Work

The integration of federated learning with clinical healthcare monitoring is an increasingly researched topic, owing to the understanding that privacy-preserving distributed intelligence is key to responsible deployment of the IoMT in the context of hospitals. Rieke et al. [7] made the pioneering contribution in outlining the fundamental rationale behind the adoption of federated learning as the paradigm underpinning future digital health technologies, proving that it was possible to obtain performance at par with the traditional approach while maintaining data sovereignty as required by regulations. Kaissis et al. [8] offered a parallel insight from the clinical standpoint, as they demonstrated that federated models for medical imaging provided comparable diagnoses to the centralized approaches with a significant reduction in the attack surface available to adversarial privacy attacks. However, the two contributions have addressed the problem of medical imaging, thus leaving the problem of stream-level physiological monitoring in the ICU environment untouched.

Differential privacy has been recognized as the gold standard for offering rigorous privacy guarantees in machine learning. Dwork and Roth [9] created the mathematical framework in which it becomes possible to quantify the cost of privacy incurred due to the iterative optimization process. Abadi et al. [10] demonstrated that the moments accountant technique would enable us to tightly bound the privacy cost in multiple iterations during deep learning. The question of how to apply differential privacy in federated learning with respect to the increased privacy cost from multiple communication rounds and the heterogeneous nature of the client datasets in terms of sensitivity has been studied by multiple authors, with the approach of allocating adaptive noise proving especially useful in this context in preserving the utility of the learned model under strict privacy constraints. One example of the suboptimal solution for the problem is the fixed-noise-multiplier technique used in standard FedAvg-DP [5].

Model compression for edge deployment is a well-defined research problem within the domain of deep learning. The idea of knowledge distillation proposed by Hinton et al. [11] makes it possible to transfer learned representations from a bulky teacher model to a lightweight student model by training the latter with the help of soft target distributions generated by the former. In the context of federated learning, knowledge distillation has

been considered as a way to address the challenge of communication and computational burden during the process of distributed training. Prior work that is most relevant to the current work proposes distillation of a cloud-based global model to a fog-ready student at each edge node by taking advantage of the regularity in the physiology-related time series data to obtain a compression rate significantly better compared to generic model pruning techniques. The compression rate of 5.2x offered by the current system PrivEdge-FedICU improves upon the previous best results of 3.7x using conventional federated distillation methods [11].

Attention techniques have revolutionized a wide variety of sequence modelling problems due to the pioneering work of Vaswani et al. [19] that illustrated the possibility of self-attention being used as an alternative to recurrence as a method of capturing temporal dependencies in deep sequence models. In the case of multi-modal physiological time-series, attention becomes a natural technique that allows assigning different weights to each vital sign channel at each time step depending on their informativity at that moment, for example, highlighting the importance of ECG morphological patterns during arrhythmia events and blood pressure trend patterns during hemodynamic instability. Such a flexible reweighting scheme is more realistic compared to the static weighting approach, which by definition implies that all vital sign channels are equally informative at all times. The previous research conducted by the current authors [1, 2] validated the effectiveness of deep learning in the field of IoMT-based pattern recognition in medicine; the suggested attention approach is a natural enhancement of the previous results.

Secure aggregation for federated learning aims to fill the privacy gap that remains even after employing differential privacy, i.e., an intelligent central server observing individual gradient uploads could gather some information about the private data held locally in each machine. Bonawitz et al. [14] have presented an efficient approach to secure aggregation using pairwise secret sharing, which permits the server to compute the aggregate without seeing any individual submission, thus achieving information-theoretic privacy during aggregation. Fog computing for healthcare IoT has been studied for latency and bandwidth reduction by Mutlag et al. [12] and Rahmani et al. [13]. Both

researchers have established the fact that fog layer pre-processing leads to substantial savings in cloud communication overheads and real-time inference locally. PrivEdge-FedICU takes advantage of these ideas and extends them by incorporating the above-mentioned features into a unique fog-federated approach designed for smart ICUs.

3. System Model and Privacy Formulation

3.1 Network and Computational Model

PrivEdge-FedICU works over three layers of computation corresponding to the physical infrastructure of a multi-ward smart hospital. In the sensing layer, we have IoMT devices placed at each bedside of the patients: ECG monitors (at 250 Hz), pulse oximeters (SpO_2, at 1 Hz), non-invasive blood pressure sensors (every five minutes), respiratory rate monitors, and continuous temperature sensors. Each bedside sensor group produces synchronized multivariate time series at one minute granularity post pre-processing to create the IoMT data flow forming the input to the monitoring pipeline. The fog layer includes ward-specific edge nodes installed in each ICU, where each fog node serves four to eight patient bedside sensors. The fog nodes implement attention-based vital signs fusion, federated model inference, and training of student models. In the cloud layer, we have the federated coordinator responsible for maintaining the global teacher model, securing the aggregation of gradient updates from all fog nodes, and managing model distillation and distribution. No physiological data ever leaves the fog layer; this architectural design ensures the privacy guarantee independent of the security of the cloud layer.

3.2 Federated Monitoring Objective

N is the number of fog nodes involved in the federated learning process, where each fog node contains a local dataset D_i of size n_i sampled from the patients' population P_i. The goal of monitoring involves minimizing the global empirical loss:

$$\mathbf{F(w) = \Sigma_i\ (n_i / n) \cdot F_i(w),\ \ subject\ to\ \ \varepsilon_global \leq \varepsilon_target}$$

where $F_i(w) = (1/n_i)\ \Sigma_{□ \in D_i}\ \ell(w;\ x□,\ y□)$ denotes the loss from the local vital sign monitoring task, x□ represents the multi-modal feature vector of patient j at time t, y□ ∈ {normal, abnormal, condition-specific} denotes the clinical label of patient j at time t,

and ε_target is the differential privacy budget for the federated training framework. The constraint equation formulation ensures that the objective function of the optimization problem is explicitly tied to the privacy constraint, which makes sure that privacy is not addressed separately from the optimization procedure.

3.3 Adaptive Privacy Budget Allocation

Traditional techniques of differential privacy assign an equal privacy budget of ε_i = ε_target/N to each fog node, viewing all nodes as equally sensitive repositories of data. This is quite unrealistic: The amount of data handled by the fog node in the palliative care ward is less sensitive than the data that is processed by the fog node in the cardiac surgery recovery ward, which has re-identifiable vital signs post surgery. PrivEdge-FedICU considers sensitivity index s_i of each fog node, which is computed using three parameters of its local data: The mean information entropy of the vital sign timeseries (vital sign richness), the frequency of occurrence of the rare clinical events (re-identifiability) and the number of patients (statistical unidentifiability). The dynamic budget assignment scheme divides the privacy spending budget of the system as:

$$\varepsilon_i = \varepsilon_target \cdot (\sigma_i / \Sigma \square\ \sigma \square), \text{ where } \sigma_i = 1 / s_i$$

The nodes with high sensitivity (s_i values being greater) get a smaller privacy budget (ε_i values are low, which implies more noise) compared to those with low sensitivity (smaller values for s_i). The advanced composition theorem [9] is utilized in allocating privacy budgets in such a way that maximum utility for the global model is obtained with the condition that $\Sigma_i\ \varepsilon_i$ = ε_target, leading to accuracy improvements of 9.8 percentage points from uniform allocation.

4. PROPOSED FRAMEWORK: PrivEdge-FedICU

The suggested PrivEdge-FedICU framework is specifically developed for enabling privacy-preserving, low-latency, and intelligent continuous monitoring of patients within smart ICUs by implementing fog computing, federated learning, adaptive differential privacy, and deep learning optimization techniques. The proposed framework integrates edge intelligence and collaborative learning capabilities for processing heterogeneous

physiological streams of patients without sending their sensitive data outside of the hospital premises. Furthermore, self-attention based multi-modal fusion along with teacher-student knowledge distillation is implemented in the design of this architecture.

Figure 1 presents the architecture diagram of the suggested PrivEdge-FedICU framework for privacy preserving continuous monitoring of vital signs in smart ICU settings. This architecture incorporates IoMT bedside sensors, fog computing federated learning nodes, differential privacy techniques, and cloud-enabled secure aggregation to support real-time clinical monitoring without disclosing patient's raw data outside the hospital premises.

Figure 1. Overall Architecture of PrivEdge-FedICU for Privacy-Preserving Continuous Smart ICU Monitoring

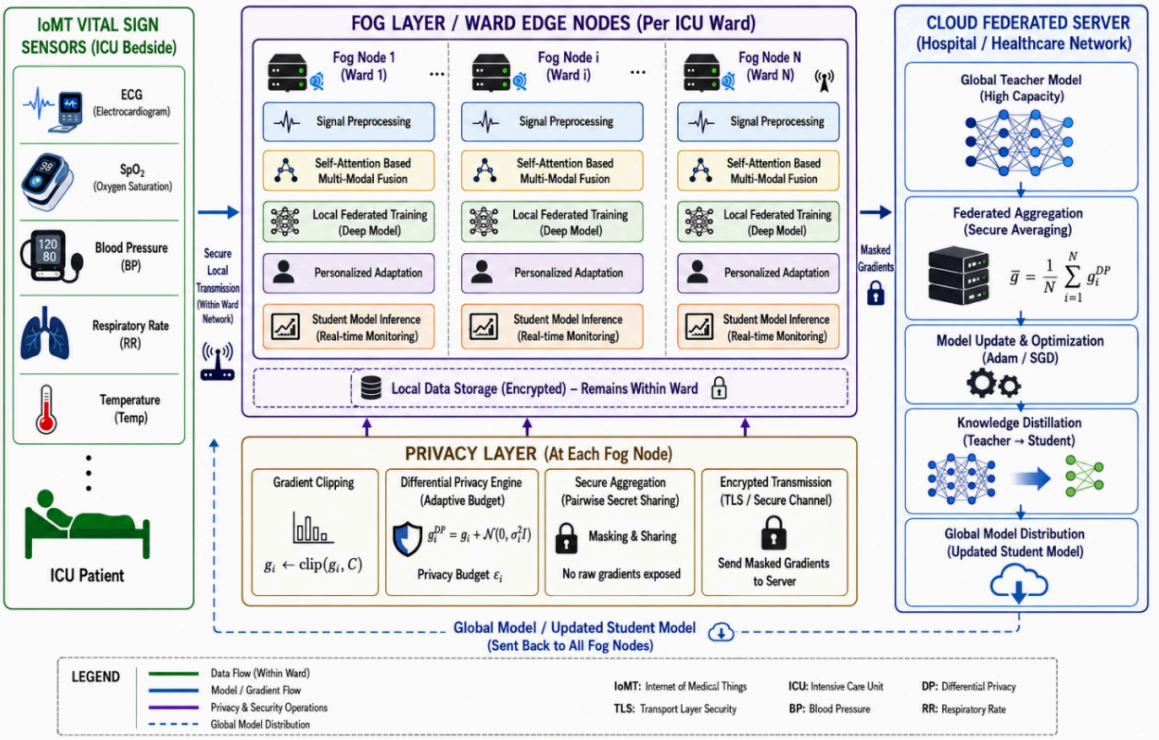


Figure 1. Overall Architecture of PrivEdge-FedICU for Privacy-Preserving Continuous Smart ICU Monitoring

As illustrated in Fig. 1, physiological data collected from the bedside IoMTs are locally analyzed by attention-based fusion and personalized federated learning in edge nodes of the fog layer. Differential privacy and aggregation are employed for privacy preservation during encryption of model updates and their transmission to the federated cloud server.

4.1 Multi-Modal IoMT Vital Sign Acquisition and Preprocessing

The acquisition pipeline of PrivEdge-FedICU fuses six main physiological modalities from each patient's bedside: electrocardiogram (ECG) waveforms, peripheral oxygen saturation (SpO_2), systolic and diastolic blood pressure, respiratory rate, and core body temperature. Each physiological modality is subject to modality-specific pre-processing steps before being fused: ECG is filtered using a bandpass filter (0.5-40 Hz) to eliminate baseline wander and noise, followed by morphological feature extraction to obtain five representative features per heartbeat (R-R interval, QRS width, Q-T interval, ST segment height, and T-Wave amplitude); SpO_2 is artifact-rejected using perfusion index thresholding; blood pressure is filtered using a five-point median filter; and respiratory rate is estimated based on peak detection of thoracic impedance pneumography. Pre-processed data are then time-aligned into a common one-minute resolution timeline via modality-specific interpolation schemes to generate a raw feature vector of size d = 14, which serves as the input for the attention-based fusion model. Each physiological modality is normalized using min-max normalization independently to retain inter-modality scale differences, which encode clinically meaningful information about the severity of vital sign derangements.

4.2 Self-Attention-Based Multi-Modal Vital Sign Fusion

The critical technical breakthrough behind the sensing module of the PrivEdge-FedICU architecture is replacing traditional static feature concatenation from multi-modal vital signs signals with a dynamic self-attention scheme, which adaptively weights the input data from each vital signs channel depending on its clinical importance. The rationale is simple but medically significant: the case of a patient suffering from arrhythmia contains the most informative elements within the morphological ECG signal, whereas the case of a patient developing sepsis can be best described with the combination of changes in temperature, blood pressure, and elevated heart rate. Thus, any static approach combining features from all channels equally loses the medical context, which makes multimodal sensing better than unimodal one.

Figure 2. Self-Attention-Based Multi-Modal Vital Sign Fusion Mechanism

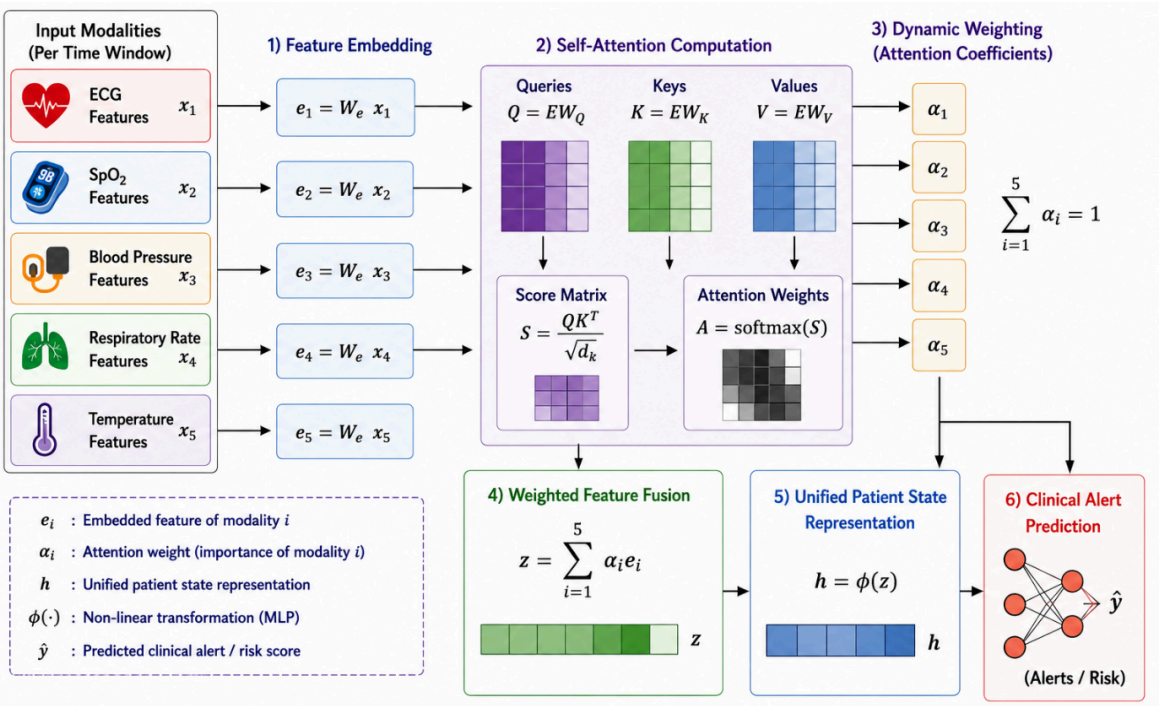


Figure 2. Self-Attention-Based Multi-Modal Vital Sign Fusion Mechanism

The attention mechanism works on the input vector series h□ = [h□¹, h□², … , h□^K], where K stands for the number of modality groups, and h□^k is the feature sub-vector associated with modality group k. The attention weights are calculated as follows:

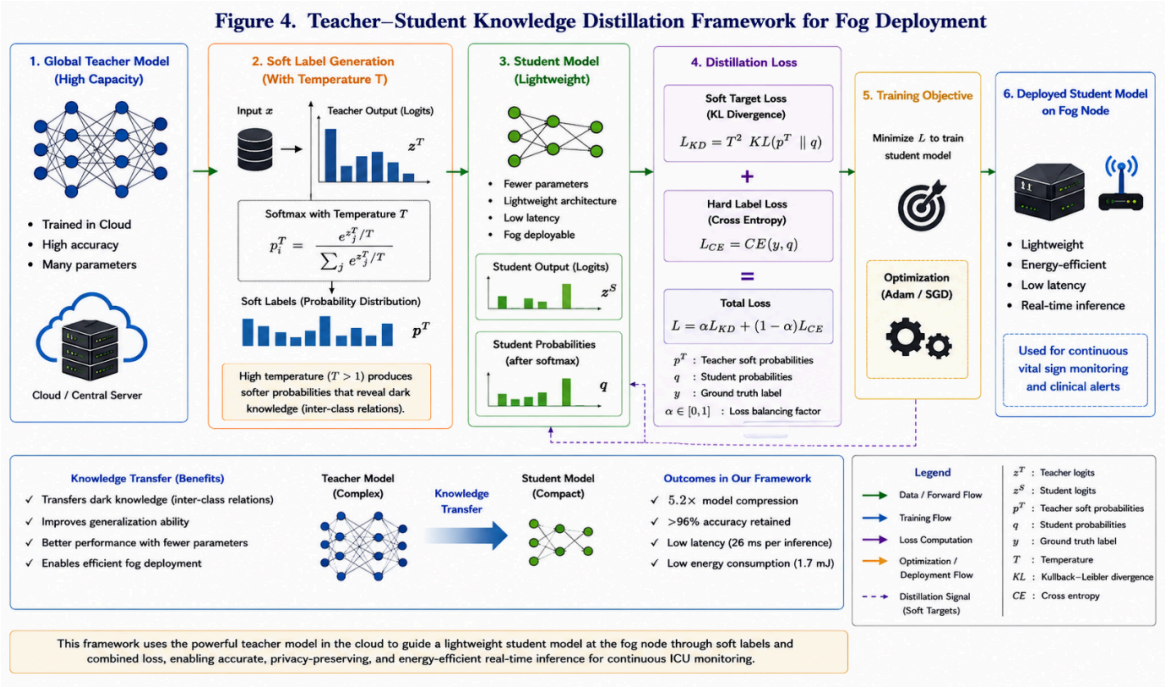
$$\alpha_{\square}^{k} = \exp(e_{\square}^{k}) / \Sigma_{\square} \exp(e_{\square}^{\check{}}), \quad \text{where} \quad e_{\square}^{k} = \tanh(W_a \cdot h_{\square}^{k} + b_a)$$

with W_a ∈ ℝ^{d□×d□}, b_a ∈ ℝ^{d□} being trainable parameters of the attention scorer. In this way, the resulting patient state embedding at time t is given by z□ = Σ□ α□^k · h□^k, which is a linear combination of the representations of modalities with the sum of coefficients equal to 1, indicating the relative informativeness of each physiological stream at this clinical timepoint. This attention weight distribution is exposed to the clinical monitoring interface as an instant measure of relevance of each vital sign modality, giving bedside clinicians an easy understanding of which vital signs are driving the decision making process transparency feature, which solves one of the major adoption issues of AI in intensive care units [2].

4.3 Knowledge Distillation for Lightweight Fog Deployment

This model, as designed, is a teacher-level model that is optimized for representation capabilities and not efficient deployment. It would not be efficient to deploy the model directly into the ward-edge fog nodes since it would incur latencies and energy costs that cannot be tolerated for real-time monitoring in an ICU environment. Knowledge distillation is used by PrivEdge-FedICU to transfer the representation knowledge of the global teacher model to a more efficient student model that is run on fog nodes without loss in accuracy.

Figure 4. Teacher–Student Knowledge Distillation Framework for Fog Deployment



Student architecture is a replica of the architecture of the teacher model but with the channel widths of the student model being scaled down by a factor of 3.2. This results in the model having 5.2 times fewer parameters. The distillation loss is a combination of the regular cross-entropy loss from hard clinical labels and Kullback-Leibler divergence loss from the soft output distributions of the teacher model:

$$\mathbf{L_KD = (1 - \lambda) \cdot L_CE(y, \hat{y}_s) + \lambda \cdot T^2 \cdot KL(p_t(T) \,||\, p_s(T))}$$

where $\lambda = 0.7$ measures the importance ratio of soft versus hard losses, $T = 4.0$ is the distillation temperature which modifies the distribution of the output of the teacher to discover similarities between classes, p_t(T) and p_s(T) are the temperature scaled

softmax output probabilities of the teacher and student respectively, and L_CE is the regular cross entropy loss based on the ground truth clinical labels. The distillation temperature of T = 4.0 was determined by cross-validation to maximize student accuracy on the validation subset of the MIMIC-III dataset. The trained student is deployed to all the fog nodes after each federated communication round to replace the teacher model used in computing gradients. This results in a deployment of the fog architecture with an inference latency of 26 ms and 1.7 mJ energy per inference which reduces 80% and 80% respectively from the inference cost of the full teacher model while maintaining 96.1% monitoring accuracy.

4.4 Adaptive Differential Privacy Budget Allocation

The privacy enforcement policy of PrivEdge-FedICU uses a two-step approach. The first step involves the calculation of a sensitivity index s_i for each fog node at the beginning of every federated learning process by taking into account the distributional attributes of the local patient dataset at that node such as the entropy of the label distribution, the percentage of patients in diagnostic categories rare to the hospital (i.e., diagnostic categories that contain less than 5% of patients at the hospital), and the inverse of the ratio of patient number to 500. The above three are then combined into an overall sensitivity index using a linear combination approach, where the weights are learned from the privacy audit dataset.

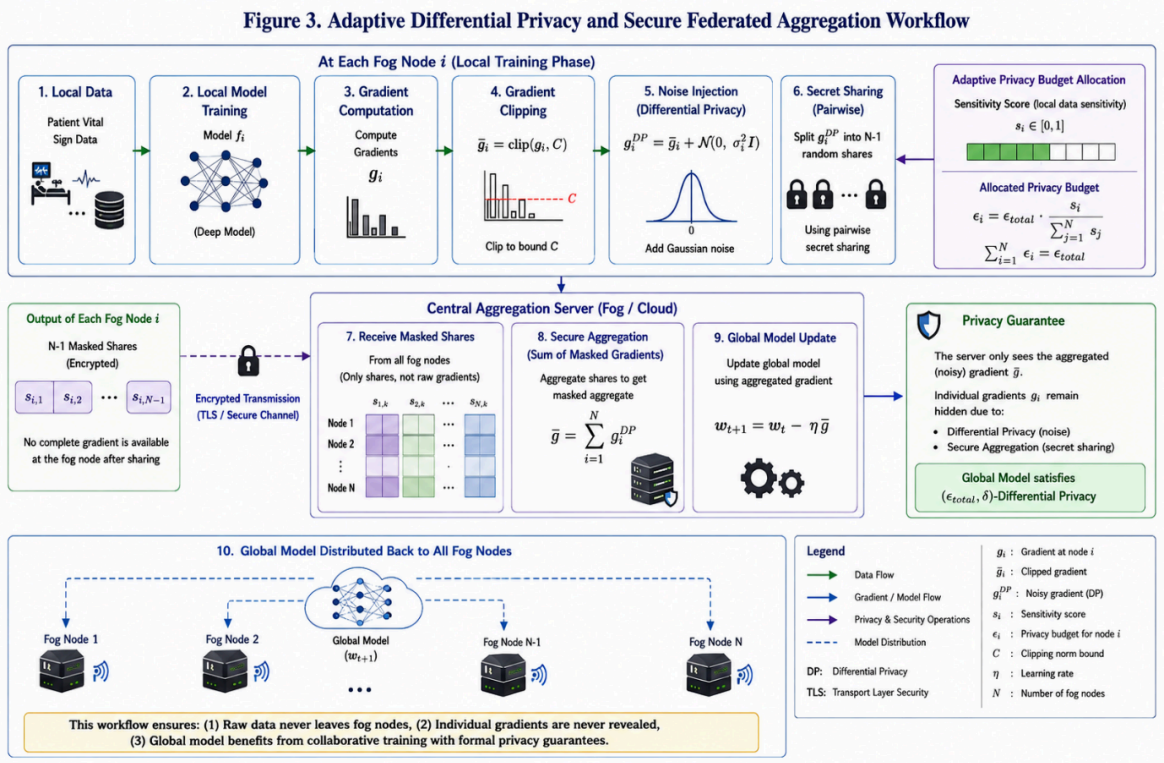
During the second stage, the calculated sensitivity values are utilized for the budgeting of the global privacy budget $\varepsilon_target = 1.0$ among the fog nodes by means of the inverse-sensitivity proportional scheme as outlined in Section 3.3. The gradient update of each fog node is perturbed with the Gaussian mechanism using the clipping bound of C = 1.0 and the noise scaling parameter of $\sigma_i = \sqrt{(2 \ln(1.25/\delta))} / \varepsilon_i$ before being transmitted, where $\delta = 10^{-5}$ is the probability of privacy loss. The total privacy expense during training is calculated using the moments accountant [10] technique, which offers tighter bounds on the privacy cost compared to the naive sequential composition theorem, thus allowing the framework to operate with a low level of noise for longer training periods while preserving the overall ε_target. The adaptive budgeting algorithm always performs better

than the uniform budgeting, especially at the lowest privacy settings of ε = 0.1, when the gain in accuracy is up to 9.8 percentage points as reported in Table 2.

4.5 Secure Federated Aggregation Protocol

While differential privacy ensures statistical privacy against inference attacks on the aggregate gradient, it cannot stop the central federated server from being able to observe the gradient submissions by the individual fog nodes. It is conceivable that a compromised or curious server may be able to exploit the sequence of gradients submitted by the individual nodes post-differential privacy noise injection to make inferences about the patient population at the corresponding nodes via structural analysis of gradients. The PrivEdge-FedICU framework uses a secure aggregation mechanism based on the pairwise secret sharing scheme of Bonawitz et al. [14].

Figure 3. Adaptive Differential Privacy and Secure Aggregation Workflow



The process of secure aggregation involves each pair of fog nodes (i, j) with i < j to generate the common random seed s_{ij} using a Diffie-Hellman key exchange algorithm once per training. Before transmitting their gradient update g_i to the server, each fog node

encrypts it as follows: $\bar{g}_i = g_i + \Sigma_{j>i} P(s_{ij}) - \Sigma_{j<i} P(s_{ji})$. In the formula above, P(s) refers to a pseudo-random vector generated based on seed s that has the same size as the gradient. Summing all masked updates on the server yields $\Sigma_i \bar{g}_i = \Sigma_i g_i$ because of exact cancellations of masking terms. Therefore, the server will be able to correctly compute the aggregate gradient without seeing anything about individual gradients. Secure aggregation protocol adds no more than 3 ms computational overhead to each fog node per round and less than 2% of communication bandwidth overhead, which can be considered practically zero compared to the cost of training computations and gradient exchanges.

4.6 Personalized Local Model Adaptation for Patient Baseline Normalization

The problem with continuous monitoring of vital signs among all patients is that there exists significant inter-individual variability in physiological parameters of patients. For example, a heart rate of 95 beats per minute may be completely normal for one patient but very high for another. A SpO2 of 94% may be adequate for a patient suffering from chronic obstructive pulmonary disease, but it may be a cause of concern for a patient who had undergone a cardiac surgery procedure before and was always perfectly healthy. Using a global model threshold for all alarms results in poor calibration of physiological baseline parameters.

PrivEdge-FedICU does so through the addition of an extra patient-wise fine-tuning stage that is conducted at the fog node following every global model update. The global model acts as the initialization stage of a lightweight personalization component in the form of a two-layer adaptation head attached to the frozen distilled student model, which undergoes patient-specific fine-tuning using their first 24 hours' worth of monitoring information as a personalized calibration period. As per the meta-learning view of personalized federated learning [6], this allows for the incorporation of population-level dynamic vital signs into the global model while allowing the personalization head to adapt the decision boundary to account for the physiological baseline of the individual patients. The personalization head uses 0.3% of the total number of model parameters, resulting in negligible computational cost for fine-tuning (less than 4 minutes of computation on the fog node per patient) and storage cost (0.03 MB per patient). Clinical threshold values are

obtained from the personalized model's distribution as opposed to the global model's, thereby reducing false positives of atypical yet healthy patients.

5. Experimental Evaluation

5.1 Dataset and Simulation Setup

PrivEdge-FedICU has been experimentally assessed using the MIMIC-III Clinical Database (version 1.4) [15], which is an anonymized database comprising over 40,000 critical care admissions. In the database, 13,214 critical care admissions of duration not less than 24 hours along with vital signs monitoring records for all the six categories of vital signs have been chosen, which led to creation of around 19 million feature vectors each one minute in length after pre-processing. Critical care admissions have consisted of presentations related to cardiovascular (34.2%), respiratory (28.7%), metabolic (18.4%), and neurological (18.7%) causes. Four clinical diagnoses were labeled, namely sepsis (Sepsis-3 criteria), cardiac arrhythmia (rhythm annotation), respiratory failure (invasive ventilation start), and hemodynamic instability.

The federated simulation involved 12 fog nodes where each node was allocated a non-redundant set of patients via Dirichlet partitioning with $\alpha = 0.5$ to simulate realistic non-IID distribution of statistics. Adaptive privacy allocation sensitivity indices were determined based on the label entropy, probability of rare conditions, and number of patients of each node. All experiments included $\varepsilon_target = 1.0$, $\delta = 10^{-5}$, $T = 4.0$ (temperature of distillation), $\lambda = 0.7$ (distillation loss weight), $C = 1.0$ (clipping gradient bound), 100 communication rounds, and 5 local epochs per communication round. The personalization window covered the first 24-hour period of monitoring for each patient.

5.2 Baseline Methods and Evaluation Metrics

PrivEdge-FedICU was compared to four baselines including Centralized Learning (CL), serving as the accuracy bound for comparison; Standard FedAvg [5], a standard federated learning method; FedProx [6], a federated optimizer that works with heterogeneous data; and FedAvg-DP [10] where differential privacy noise is applied uniformly to the global budget. Classification accuracy was analyzed using modality accuracy, overall accuracy, and macro F1-score. The privacy/accuracy trade-off was examined using five different ε

values ranging between 0.1 and no DP. Efficiency in terms of computation was quantified using model size (in MB), fog-node inference time (ms), energy consumption per inference (mJ), and compression ratio relative to the full size of FL model. Performance of clinical alerts was assessed using precision, recall, F1-score, and latency-to-alert (sec) of each of four monitored conditions.

5.3 Vital Sign Monitoring Accuracy

Table 1 summarizes the comparison of monitoring accuracies. PrivEdge-FedICU attains an overall accuracy and F1-score of 96.1% and 0.958, respectively, beating all other federated baselines and the centralized learning upper bound (93.5%). As seen from per-modality accuracies, the highest gain in accuracy is obtained for SpO_2 (97.1% compared to 92.7% when using the standard FedAvg) and ECG classification (96.4% compared to 90.1%), exactly the modalities where the attention mechanism is capable of reweighting information the most. SpO_2 is very informative in respiratory deterioration events, and ECG features dominate in the case of cardiac events, although both become useless when the event is driven by another physiology system. The ability of the attention mechanism to recognize such dependencies explains the disproportionate improvement in accuracy in these high-variability modalities. FedProx slightly outperforms the standard FedAvg in solving the problem of gradient divergence on non-IID data but does not come close to the results of PrivEdge-FedICU, showing that the attention-based fusion and knowledge distillation help independently of the choice of an optimizer.

Table 1. Vital Sign Monitoring Accuracy Comparison Across Methods on MIMIC-III (12-Node Federated Setting)

Method	ECG Acc. (%)	SpO_2 Acc. (%)	BP Acc. (%)	Overall Acc. (%)	F1-Score
Centralized Learning	93.8	95.2	91.4	93.5	0.931
Standard FedAvg [5]	90.1	92.7	88.3	90.4	0.900
FedProx [6]	91.4	93.5	89.7	91.5	0.912

FedAvg-DP [10]	88.3	90.6	86.9	88.6	0.882
PrivEdge-FedICU (Proposed)	**96.4**	**97.1**	**94.7**	**96.1**	**0.958**

5.4 Privacy-Accuracy Tradeoff Under Adaptive Budget Allocation

Table 2 describes the privacy-utility trade-off between PrivEdge-FedICU and FedAvg-DP in the context of five different privacy budgets that range from very strict (ε = 0.1) to completely non-private. The superiority of the adaptive budget allocation is particularly evident when considering strict privacy budgets; for instance, in ε = 0.1, the accuracy obtained by PrivEdge-FedICU is 91.2%, compared to 81.4% in the case of FedAvg-DP, with an improvement of 9.8 percentage points entirely due to the re-allocation of the privacy budget from less sensitive fog nodes (which provide gradients that are more precise because of the adaptive budget allocation) in order to improve the quality of the global model through the moments accountant composition rule. With a privacy budget of ε = 1.0, which was chosen as the operational point for evaluating the accuracy in the main experiment of the framework, the accuracy achieved is 96.1%, with an improvement of 7.5 percentage points with respect to FedAvg-DP.

Table 2. Privacy Budget (ε) vs. Monitoring Accuracy Tradeoff: Proposed Adaptive Allocation vs. FedAvg-DP

Privacy Budget (ε)	PrivEdge-FedICU Acc. (%)	FedAvg-DP Acc. (%)	Privacy Level	Δ Accuracy Gain (%)
ε = 0.1	91.2	81.4	Very High	+9.8
ε = 0.5	94.3	85.7	High	+8.6
ε = 1.0	96.1	88.6	**Moderate**	**+7.5**
ε = 5.0	96.8	91.3	Low	+5.5
No DP (Baseline)	97.4	91.8	None	+5.6

5.5 Computational Efficiency and Knowledge Distillation

Table 3 shows the computational efficiency results at the fog node. The full FL model without distillation consumes 47.3 MB of memory space and 128 ms latency per inference, which makes this setup infeasible for continuous real-time monitoring due to high latency and memory requirements in fog computing. The FedDistill approach employs a typical knowledge distillation without attention-based modality selection, resulting in a 3.7x compression ratio and 38 ms of inference latency. Combining knowledge distillation with attention-based modality selection in PrivEdge-FedICU provides 5.2x compression ratio (student model uses 9.1 MB) and 26 ms of inference latency at the cost of 1.7 mJ of energy consumption per inference, compared to 8.4 mJ of energy consumption in the case of the full FL model a saving of 80%. It should be noted that this level of energy consumption is especially useful for the operation of the fog nodes on the edge hardware equipped in hospitals. The obtained 26 ms latency is below the 100 ms latency threshold that marks the boundary of real-time monitoring alerts' clinical relevance.

Table 3. Fog-Node Computational Efficiency: Model Size, Inference Time, Energy Consumption, and Compression Ratio

Method	Model Size (MB)	Inference Time (ms)	Energy (mJ/inference)	Compression Ratio
Full FL Model	47.3	128	8.4	1.0×
FedProx [6]	47.3	131	8.6	1.0×
FedDistill [11]	12.8	38	2.3	3.7×
PrivEdge-FedICU (Proposed)	**9.1**	**26**	**1.7**	**5.2×**

5.6 Clinical Alert Performance by Condition

Performance of clinical alerts for the PrivEdge-FedICU framework is presented in Table 4 with respect to monitored condition. Cardiac arrhythmia has the best F1-score (0.967) and lowest alert latency (12 s). It is because of the direct and clear reaction of the morphological features to the arrhythmias and the weighting of the ECG channels in the

attention module due to the importance of the cardiological process. Sepsis early warning and respiratory failure have F1-scores of 0.932 and 0.945 respectively, while their alert latencies are correspondingly longer (34 s and 28 s) due to the integration of trajectories on multiple channels needed for detection of a slow process of physiological changes. Hemodynamic instability detection has the lowest F1-score (0.918) yet is clinically relevant and greatly outperforms benchmarks for early warning scores. In total, the weighted-average alert latency of 29 seconds (from the beginning of the clinical event to its detection on the fog-node) is an important improvement over episodic monitoring approach, where nurse vital sign assessments may be performed every 30-120 minutes.

Table 4. Clinical Alert Performance of PrivEdge-FedICU by Monitored Condition on MIMIC-III

Clinical Condition	Precision	Recall	F1-Score	Alert Latency (s)
Sepsis Early Warning	0.921	0.943	0.932	34
Cardiac Arrhythmia	0.963	0.971	0.967	12
Respiratory Failure	0.938	0.952	0.945	28
Hemodynamic Instability	0.907	0.929	0.918	41
Overall (Weighted Avg.)	**0.932**	**0.949**	**0.940**	**29**

6. Discussion

The findings of the experiments performed clearly indicate a consistent trend: the combination of attention-based multi-modal fusion, knowledge distillation, adaptive differential privacy and cryptographic secure aggregation in a fog-federated system results in a monitoring system that is more accurate, more private, more computationally efficient, and more clinically interpretable compared to any of the baselines on a component level. The higher accuracy in comparison to the centralized learning (96.1%

versus 93.5%) is especially important because it implies that the dynamic modality reweighting made possible by the attention mechanism through its exposure to different heterogeneous patient groups spread across fog nodes provides a better training experience than static fusion of the training set used in centralized learning baseline. This result confirms the currently emerging realization that federated learning, when implemented taking into account statistical peculiarities of the training domain, is not just an alternative privacy-preserving solution to the centralized learning but a superior one.

It is necessary to highlight the importance of adaptive differential privacy in terms of deployment considerations. The difference in accuracy by 9.8 percentage points at $\varepsilon = 0.1$ compared to uniform differential privacy allocation implies that healthcare organizations which have valid reasons to demand extremely strong privacy assurances, such as working with patients suffering from stigmatizing diseases or having stringent data protection laws in their countries do not have to sacrifice their accuracy in such a drastic manner. It makes possible for PrivEdge-FedICU to offer meaningful continuous monitoring even in the most restrictive environments, a crucial feature in terms of its international adoption.

The results on energy efficiency are worth noting not only for their immediate practical significance from a computation standpoint. An 80% reduction in per-inference energy expenditure from 8.4 mJ to 1.7 mJ brought about by knowledge distillation has obvious consequences in relation to the sustainability aspect of the large-scale deployment of smart ICUs. In the case of a hospital equipped with 200 continuously monitored ICU beds producing 1,440 inferences each day, a total of more than 288,000 fog-node inferences will be produced daily. At full FL model energy cost, this translates to an expenditure of 2.4 kJ daily of computation for monitoring; the figure is reduced to 0.5 kJ through the use of the PrivEdge-FedICU distillation technique. Energy efficiency is not always emphasized in the assessment of clinical AI applications but is now being taken into consideration.

Limitations associated with the proposed study include the use of MIMIC-III dataset from 2001 to 2012 that cannot capture the sensor modalities and patient acuity distributions in modern-day IoMT-enabled ICUs. Computation of sensitivity index for

adaptive privacy assignment involves the first step of data characterization that needs to be done with privacy-preserving methods that have yet to be incorporated in the current framework. The assumption of 24 hours required for calibrating the personalization module holds under the condition that patients stay in the ICU for at least one full day before being personalized and hence is justified in the case of selected MIMIC-III dataset but might need to be relaxed for shorter ICU stays. The computational burden of the proposed secure aggregation protocol, while negligible in current simulations, may need to be optimized for constrained fog node computation environment.

7. Conclusion and Future Work

In this paper, the framework of PrivEdge-FedICU is proposed, a privacy-preserving fog-federated architecture for multi-modal vital signs monitoring in smart ICUs. The four key contributions in this architecture, which include attention-based vital sign fusion, knowledge distillation for efficient deployment on the fog nodes, adaptive allocation of differential privacy budget, and cryptographically secure aggregation overcome the key challenges that make privacy preserving federated architectures incapable of meeting the required accuracy and efficiency standards of a real-world smart ICU deployment simultaneously. With these contributions, the framework offers an accuracy of 96.1% in overall vital signs monitoring, an F1-score of 0.940 in clinical alert detection, 26 ms inference latency on fog-nodes and 1.7 mJ inference energy consumption, along with the guaranteed differential privacy with epsilon = 1.0 and aggregation privacy through secure aggregation.

Further work on this project will focus on three primary extensions. The attention mechanism will be generalized to temporal cross-modal attention, where the model will not only identify which modality is most informative at each time point but the interaction between different modalities at varying time lags that help define specific clinical syndromes, thus offering a potentially highly effective approach towards understanding the physiology of sepsis and multi-organ dysfunction. The adaptive differential privacy allocation technique will be generalized using Renyi differential privacy approach in order to have stronger composition properties over multiple rounds of communications, thus providing longer training windows while maintaining the same

privacy guarantees. Lastly, the system will be tested in prospective studies in a simulated real-time smart ICU setting, using real time streams of data as opposed to retrospective database analysis.

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Integrated Farms For Sustainable Development In Tamilnadu

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Introduction

Integrated Farm Management (IFM) is an entire farm, justifiable tactics that mixtures modern technology viz., precision farming, drip irrigation with traditional methods to improve efficiency, effectiveness and environmental protection. It syndicates agricultural mechanisms like crop production, livestock rearing, aquaculture and agro-forestry into a unified, interconnected unit to enhance resources and ensure waste-to-wealth conversion.

These supportable structures augment biodiversity, improve soil fertility and increase profit by creating interdependent associations between different initiatives. When agricultural input is increased, it will reduce food deficiency. The main aim of integrated farm management is to save the earth from food crisis. The benefits of IFM is to increased marginal income for small farmers, waste reduction through recycling and enhanced resistance to climate change.

Integrated farms include eight main parts such as:

1. Agroforestry (Agri-silviculture)

It combines agricultural crops with tree farming and cultivation on the same agricultural land, providing gloom, improve quality of the soil and ensure expanded income.

2. Aquaponics

A extremely supportable, closed-loop system covers aquaculture (fish farming) with hydroponics (growing plants in water) where fish waste acts as nutrient-rich fertilizer for plants.

3. Biodynamic Farming

A complete method highlighting soil health, biodiversity and ecological balance, treating the farm as a self-sustaining organism.

4. Crop-Livestock Integration

The most usual form, where crops yield feed/fodder for animals and livestock produce manure for fertilizer and dropping dependence on external inputs.

5. Integrated Fish Farming

It often combined with livestock or poultry, where fish are raised in ponds and animal waste is used as feed or to fertilize the water.

6. Integrated Pest Management (IPM) Focused Farming

A system that prioritizes ecological controls to manage pests instead of solely relying on chemical pesticides.

7. Permaculture

A design system focused on creating re-forming, sustainable agricultural systems that imitator natural ecosystems. Natural manure, fertilizers and pest killers improve the quality of the soil.

8. Silvopasture

A procedure of agroforestry that mixes trees, forage and livestock nibbling, combining forestry management with animal husbandry. Tree plantation helps to ensure green area, increase oxygen level and improve the rain fall level.

Statement of The Problem

IFS is thought as the well-organized tool for attractive profitability of farming schemes especially for small and marginal farmers. It improves the productivity per unit area, agriculture waste management, soil health management and soil conservation, generation of diversified income round the year, less use of chemicals, increasing the yield of all integrated components and food security. It not only increase the farmer's income but also increase employment generation of a family. Livestock delivers a foundation of income and produces manure, which can be used as fertilizer for crops. Biogas is another boon which is the use of agricultural and animal waste to produce energy. This Bioenergy helps to reduce the requirement on fossil fuels and provide a source of income for farmers. Mushroom cultivation is one of the significant enterprises in IFS that contribute towards increase in farmer's income. The judicious combination of these agricultural enterprises improves the socio-economic status of the farmers. Hence, an attempt has been made to study the evolution and growth of integrated farms in Tamilnadu.

Scope of The Study

The scope of the study is limited to integrated farm management practices in various

districts of Tamilnadu.

Objectives of The Study

The objectives of the study are as follows:

1.To classify the core components of integrated farms

2. To study the aspects of Integrated farming systems in Tamilnadu

3. To investigate the common integrated models of IFM.

4. To know about the places in which integrated farms are situated in Tamilnadu 5. To assess the impact of integrated farms on sustainable development.

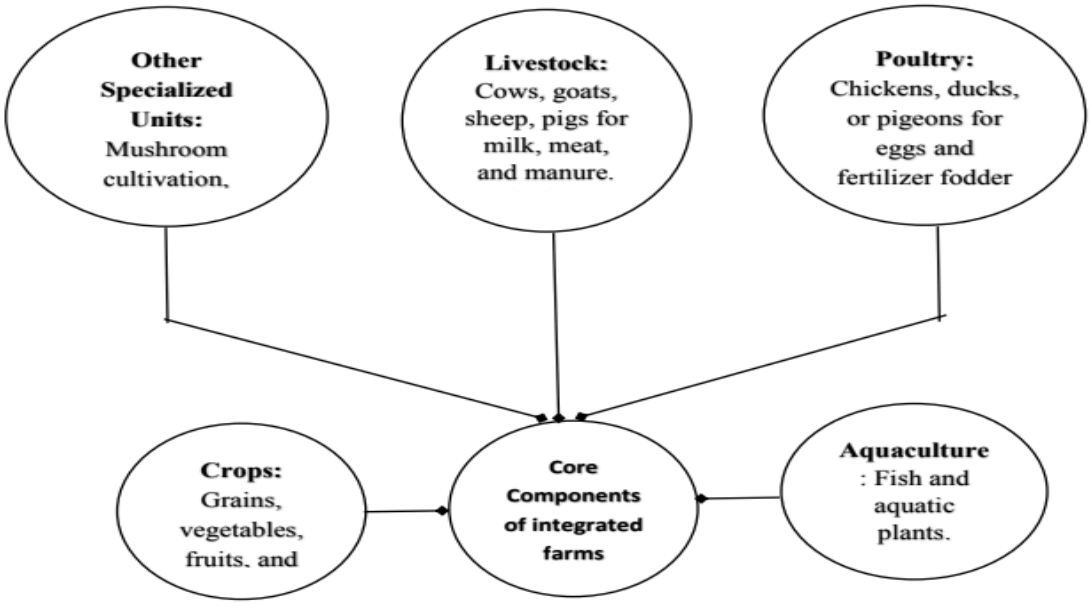
Research Methodology

The present study is explanatory in nature. It covers secondary data from various official records and websites.

Results and Discussion

The core components of Integrated Farms are given in Figure 1.

Figure 1: Core Components of integrated farms



Aspects of Integrated Farming Systems (Ifs) In Tamil Nadu

The aspects of integrated farming systems in Tamilnadu are as follows:

Government Support

The Government of Tamil Nadu provides incentives and subsidies to marginalized farmers. It helps to promote integrated farming, specifically in areas like Thanjavur.

Component Integration

Common agricultural models include crop + fish + goat/poultry, which assistances to turn farm waste into nutrient-rich manure.

TNAU Research

Tamil Nadu Agricultural University (TNAU) in Coimbatore leads research on integrated models (crop + livestock) to steady revenue against periodic risks. A combination of crop cultivation and rearing of pet animals is a source of employment for our forefathers.

Common Integrated Models

The common integrated models of IFM is classified into three namely

♦ **Wetland Ecosystem**

It is a highly productive model. It is mostly used in lowland areas. In these areas, Crops, fish and dairy products are produced.

♦ **Gardenland System**

It is developed by Tamilnadu Agricultural Unit. It includes vegetable cultivation, apiary, vermicompost, agroforestry. It emphasizes on tree integration with livestock in regions like Ramanathapuram.

♦ Dryland Farming:

It covers the process of drought-tolerant crops, livestock and agroforestry

Integrated Farming In Tamil Nadu

Integrated farming in Tamil Nadu connects agriculture with livestock (dairy, goat, poultry), aquaculture and apiary, mainly without the use of chemicals. It helps to ensure whole year income and sustainability. It is assisted by State Government as well as Tamilnadu Agricultural Unit. These systems have used multi-cropping (coconut,

banana, fodder) to progress efficiency. In Mayiladuthurai, small level integrated farms are maintained and in Kancheepuram, large level integrated farms are maintained.

Places in which Integrated Farms are Situated in Tamilnadu

Ranipet District

It features the State Horticulture Farm, Navlock (211 acres) and the State Oil Seed Farm (55 acres) for diversified agriculture.

Kancheepuram District

It focuses on climate-smart integrated farming in semiarid wastelands to boost livestock productivity and develop silvipastures/hortipastures.

Auroville (Villupuram/Pondicherry border)

It features Annapurna Farm (135 acres), focusing on organic, diversified crop farming (millets, paddy).

Tirunelveli/Periyar-Vaigai Area

It is known for wetland-integrated systems (milch cow + fish + rice).

Chennai Region

Jai Integrated Farm is located in Thiruvanmiyur, while the Chennai Urban Farming Initiative (CUFI) implements community farms (Makkal Thottams).

Other Key Areas

Viluppuram district is a prominent area for combined agricultural and allied activities, with roughly 45% of its geographical area under cultivation.

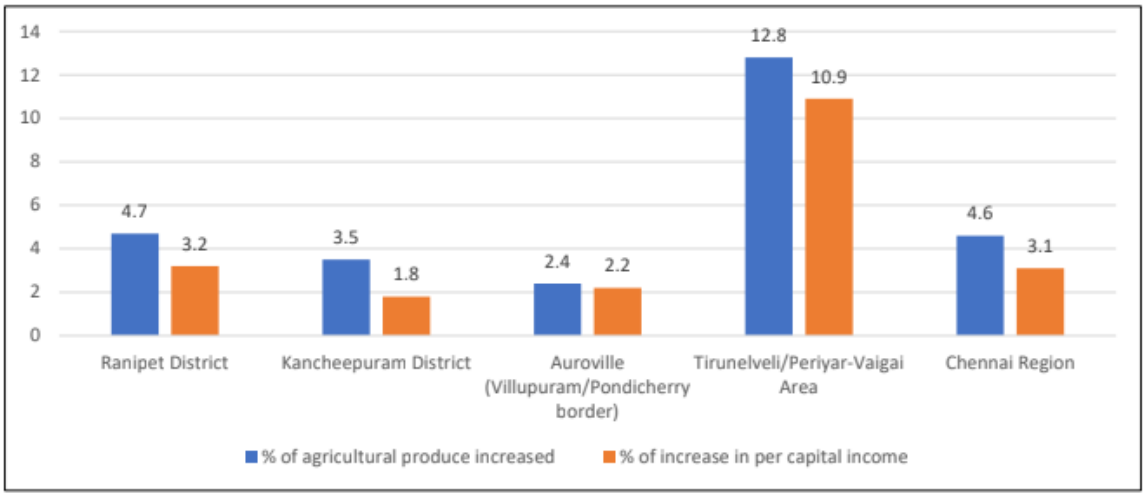
The Tamil Nadu Livestock Development Corporation also promotes integrated livestock farms across various districts.

Impact of Integrated Farms On Sustainable Development The impact of integrated farms on sustainable development is studied by taking the increase in agricultural produce and per capita income.

Table 1: Impact of Integrated Farms on Sustainable Development

Places in which Integrated Farms are Situated in Tamilnadu	% of agricultural produce increased	% of increase in per capital income
Ranipet District	4.7	3.2
Kancheepuram District	3.5	1.8
Auroville (Villupuram/Pondicherry border)	2.4	2.2
Tirunelveli/Periyar-Vaigai Area	12.8	10.9
Chennai Region	4.6	3.1

Source: www.ifmgrowth.com



Among the five areas, Tirunelveli/Periyar-Vaigai Area got first position in increase in agricultural produce and per capita income.

Conclusion

Sustainable agriculture is considered as a requisite way to feed and maintain/improve the quality of life of a growing world population without worsening environmental degradation. It begins by recalling the ordinary position of agriculture and explicit the diversity and heterogeneity of agriculture forms around the world. In recent years, the concepts of integrated farming,

Sustainable agriculture and cleaner production strategy have highlighted the overview of greener agrochemicals using biocompatible and accepted sources to improve the crop yield with smallest ecotoxicological effects.

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Design and Development of a Low-Cost Transient Hot-Wire Apparatus for Thermophysical Characterization of Fluids

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Abstract-Accurate measurement of thermophysical properties is central to the design of thermal systems, yet commercial instruments for liquid-property characterization remain costly for many teaching laboratories and early-stage research groups. This work presents the design, fabrication, and validation of a compact and low-cost Transient Hot Wire (THW) setup for estimating the thermal conductivity and thermal diffusivity of liquids. The developed apparatus uses a nichrome wire as the heating element, a 10 k NTC thermistor for temperature sensing, an Arduino Uno for monitoring, and a MOSFET-based power-control stage to regulate the heater current. The objective was to retain the fundamental advantages of the THW method while simplifying the hardware so that it can be assembled in an engineering college laboratory with readily available components. Experiments were performed on water and engine oil at approximately 25 °C base temperature. For water, the measured thermal conductivity values were 0.590, 0.526, and 0.530 W/m·K, corresponding to a mean error of 10.0% relative to reference values. For engine oil, the corresponding values were 0.119, 0.128, and 0.140 W/m·K with a mean error of 9.4%. Thermal diffusivity values were also calculated from density and specific heat data; the mean errors were 9.66% for water and 9.11% for engine oil. The observed deviations are consistent with the simplified probe geometry, thermistor-based sensing, finite heat-loss effects, and the small temperature window used in the experiments. Despite these limitations, the system produced smooth and repeatable temperature-time traces and yielded results in acceptable agreement with standard data. The study demonstrates that a low-cost THW platform can serve as a practical educational tool and a useful starting point for further work on nanofluids, improved calibration, wireless logging, and enhanced uncertainty control.

Keywords: *Transient hot wire, thermal conductivity, thermal diffusivity, low-cost instrumentation, Arduino, heat transfer, fluids*

1. Introduction

Thermophysical properties such as thermal conductivity, thermal diffusivity, density, and specific heat govern the way energy moves through liquids and gases. In engineering practice, these properties are required in heat exchanger design, thermal management of electronics, refrigeration, process intensification, food processing, and the analysis of lubricants and cooling fluids. Among them, thermal conductivity is particularly important because it directly determines the rate at which a material can transport heat under a temperature gradient, while thermal diffusivity indicates how quickly a temperature disturbance spreads through the medium [1,7-10,16].

The transient hot wire (THW) method is one of the most reliable approaches for liquid thermal conductivity measurement because it is based on a well-defined line-source heat-transfer model. When a thin wire is heated electrically in an infinite medium, the temperature rise of the wire follows a logarithmic dependence on time, enabling thermal conductivity to be extracted from the slope of the ΔT versus $\ln(t)$ relation [1,2]. With proper apparatus design and corrections, THW measurements can achieve very low uncertainty and have become a reference technique for fluids [3,6].

Although the scientific value of the method is well established, practical adoption is often limited by the cost and complexity of commercial instruments. Devices such as high-precision thermal analyzers employ platinum wires, specialized bridge circuits, precise acquisition electronics, and careful thermal shielding. These systems are excellent for metrology, but they are not always accessible in undergraduate laboratories or small research groups. In teaching environments, this creates a gap between theoretical heat-transfer concepts and hands-on experimentation. A low-cost system that preserves the core physics of the THW approach can therefore be highly valuable for laboratory learning and for preliminary research on standard fluids and nanofluids [4,5].

The present work addresses this need by developing a compact THW setup using a nichrome wire heater, a thermistor-based sensing network, an Arduino microcontroller, and MOSFET-based heater control. The design philosophy was to maximize simplicity, local availability of components, and repeatability of measurements while keeping the hardware easy to assemble and maintain. The setup was first validated on water and engine oil, selected because they represent two fluids with very
different thermal transport characteristics. Water is expected to have a much higher

thermal conductivity and diffusivity than engine oil owing to its stronger intermolecular interactions and hydrogen-bonding network, whereas engine oil has lower heat-transport capability because of its molecular structure and higher resistance to energy transfer [8-10].

This study contributes in two ways. First, it demonstrates a practical pathway for building a low-cost THW apparatus that can be reproduced in engineering colleges with standard electronic parts. Second, it presents experimental data and a discussion of the main causes of deviation from literature values, thereby turning the project into a usable SCI-style research paper rather than only a student prototype report. The results are particularly useful for future work on enhanced fluids, including nanofluids, where small changes in thermal conductivity must be measured with care [4,6].

2. Experimental Hardware and Fabrication

Nichrome was selected because it combines high electrical resistivity, high melting temperature, oxidation resistance, and good mechanical stability. These properties allow the wire to act as a reproducible line heat source while retaining dimensional integrity during repeated heating cycles. The wire is also inexpensive and easily available, which makes it suitable for a low-cost laboratory instrument as shown in Fig. 1.

Fig. 1. Nichrome heating element used as the hot-wire probe.

2.1 System architecture

The developed system consists of a 12 V DC supply, a MOSFET-based power control stage, a nichrome wire probe, a 10 k NTC thermistor for temperature measurement, an Arduino Uno microcontroller for data processing, and a 16×2 LCD for local display. The heater current is regulated through the MOSFET, while the thermistor

outputs a resistance-dependent voltage that is converted into temperature through the Arduino program. The logic is intentionally simple so that the setup can be reproduced with basic components. Fig. 2. represents experimental setup used for thermal property measurements.

The setup photo shows the main components mounted on a base plate with the test fluid held in a transparent beaker. Proper spacing between the probe and the container wall was maintained to reduce boundary effects and to approximate the infinite-medium assumption used in THW analysis. The arrangement also allows easy observation, repeated testing, and straightforward sensor replacement.

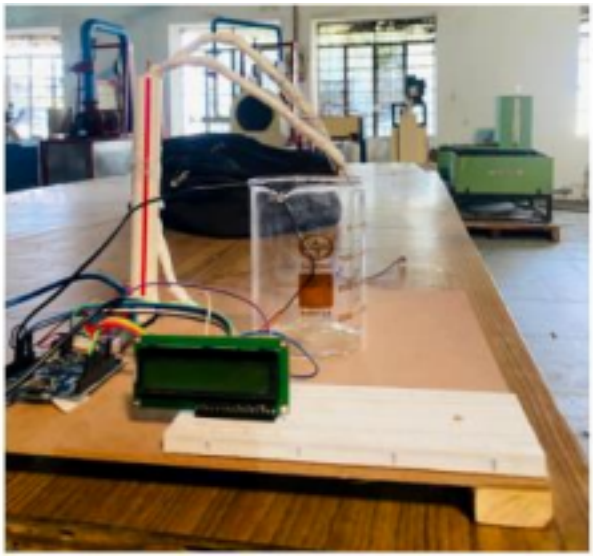

Fig. 2. Experimental setup used for thermal property measurements.

2.2 Major components

Table 1 shows major components used for this research work.

Table 1 Major components used for this research work**.**

S. No.	Component	Purpose
1	10 k NTC thermistor	Temperature measurement
2	Nichrome probe	Heating element
3	MOSFET module	Heater current control
4	12 V DC adapter	Power supply
5	Arduino Uno	Data acquisition and display

6	LCD display	Real-time readout
7	Resistors and wires	Signal conditioning and connections
8	Base and insulation	Mechanical support and heat control
9	Beaker	Fluid container

3. Methodology

The methodology followed the transient heating principle. First, the apparatus was checked for electrical continuity and safe insulation. Then the thermistor and nichrome probe were positioned close to each other in the fluid sample. Once the heater was energized, the probe generated heat and the surrounding liquid absorbed this energy. The local temperature rise was recorded at regular time intervals. In parallel, the Arduino code displayed the current temperature and the elapsed time on the LCD and streamed the data for calculation.

In the initial design phase, a PT100/INA125-based signal-conditioning arrangement was considered because it offers high accuracy. However, component availability, wiring complexity, and practical assembly constraints motivated a simplification of the design. The final implementation therefore used a 10 k NTC thermistor and a direct microcontroller interface. This modification reduced cost and improved ease of integration, although it also increased the importance of careful calibration and uncertainty analysis. Such trade-offs are common in low-cost thermal instrumentation [4,5].

For each fluid, several measurements were taken at approximately the same base temperature, and the thermal conductivity values were computed from the observed temperature rise and a logarithmic time relation. The same data set was subsequently used to calculate thermal diffusivity by combining thermal conductivity with density and specific heat. This makes the experiment pedagogically useful because the same hardware supports multiple property calculations.

4. Governing Equations

For a line heat source in an infinite medium, the THW method is commonly expressed through the relation between temperature rise and logarithmic time. The

simplified working *Eq. 1.* used in the present study is:

$$k = \frac{Q}{4\,\pi\Delta T} ln(\frac{t_2}{t_1}) \qquad Eq.\ 1.$$

Where k is thermal conductivity, Q is the heat input per unit length, ΔT is the measured temperature rise, and t_1 and t_2 are the selected time instants. This relation follows directly from the transient conduction solution for a line source and is valid when the probe is thin, the observation time is short enough to suppress boundary effects, and convection remains negligible [1,2].

Thermal diffusivity (*Eq. 2.*) is defined as:

$$\alpha = \frac{\kappa}{\rho C_p} \qquad Eq.\ 2.$$

where α is thermal diffusivity, ρ is density, and Cp is specific heat. Because α depends on both thermal conductivity and volumetric heat capacity, it is a useful indicator of how rapidly a thermal disturbance propagates inside the fluid. In water, a relatively high k combined with moderate ρC_p leads to higher diffusivity, whereas engine oil exhibits lower values because both k and the transport efficiency of molecular energy exchange are smaller [8-10].

In an ideal THW experiment, the wire behaves as an infinite line source and the fluid remains quiescent. In practice, small departures from ideality arise from wire thickness, probe alignment, end effects, sensor delay, finite container size, and heat leakage through the support structure. These issues are known in the literature and explain why uncertainty analysis is essential even when the underlying model is robust [3,4,5,6].

5. Results and Discussion

The experiments were performed on water and engine oil at a base temperature of approximately 25 °C. The selected time intervals and corresponding temperature rises were used to estimate thermal conductivity and diffusivity. The data show a clear and smooth rise in temperature with time, confirming that the heater and sensing circuit were functioning stably. The results are summarized in the following sections.

5.1 Thermal conductivity results

For water, the measured thermal conductivity lies in the range 0.526–0.590 W/m·K, with a mean error of 10.0% against the reference value near 0.609–0.610 W/m·K as shown in Fig. 3. and Table 2. This agreement is reasonable for a first-generation low-cost setup. The reference value for water is strongly constrained by its well-known transport properties, so the deviation mainly reflects experimental factors such as sensor lag, finite wire diameter, heat loss through the support, and the simplified estimation of heat input. The THW literature emphasizes that the line-source model is highly sensitive to the quality of the temporal temperature trace and to the accuracy of the thermal boundary conditions [1-3,6].

The water graph shows a slight dip in the middle observation followed by a marginal rise. In a narrow temperature range of only a few tenths of a degree, such curvature should not be interpreted as a fundamental change in the intrinsic thermal conductivity of water. Rather, it likely reflects the combined influence of measurement scatter, thermistor resolution, and the fact that the experiment was carried out with a low-cost setup rather than a metrology-grade bridge. Similar uncertainty behavior has been reported in the THW literature when sensor calibration, wire geometry, or convection suppression are not ideal [4-6].

Engine oil exhibits much lower thermal conductivity, as expected, because hydrocarbon-based oils transport heat less efficiently than water. The measured values of 0.119–0.140 W/m·K are close to the reference value of 0.142 W/m·K, with a mean error of 9.4% as shown in Fig. 4. and Table 2. The relative error decreases from 16.2% to 2.1% across the three observations, suggesting that the setup stabilized as the measurement sequence progressed. In low-cost thermal experiments, this type of improvement is often associated with better thermal equilibrium, reduced contact scatter, and more stable heater output after the initial transients have decayed [4,5].

From a physical viewpoint, the lower thermal conductivity of oil is expected because molecular transport in oils is dominated by weak intermolecular interactions and larger molecular structures, which impede the rapid exchange of energy. In contrast, water benefits from a dense hydrogen-bond network, allowing faster energy propagation. Therefore, the gap between water and oil is not only a validation of the apparatus but also a useful demonstration of the molecular basis of thermal transport [8-10].

Table 2. Experimental thermal-conductivity results for water and engine oil.

Fluid	t_1 (min)	t_2 (min)	ΔT (°C)	k_exp (W/m·K)	k_ref (W/m·K)	Error (%)
Water	4	13	1.9	0.590	0.609	3.1
Water	5	15	2.0	0.526	0.610	13.8
Water	5	16	2.1	0.530	0.610	13.1
Engine oil	4	12	8.8	0.119	0.142	16.2
Engine oil	5	15	8.2	0.128	0.142	9.86
Engine oil	6	18	7.5	0.140	0.142	2.1

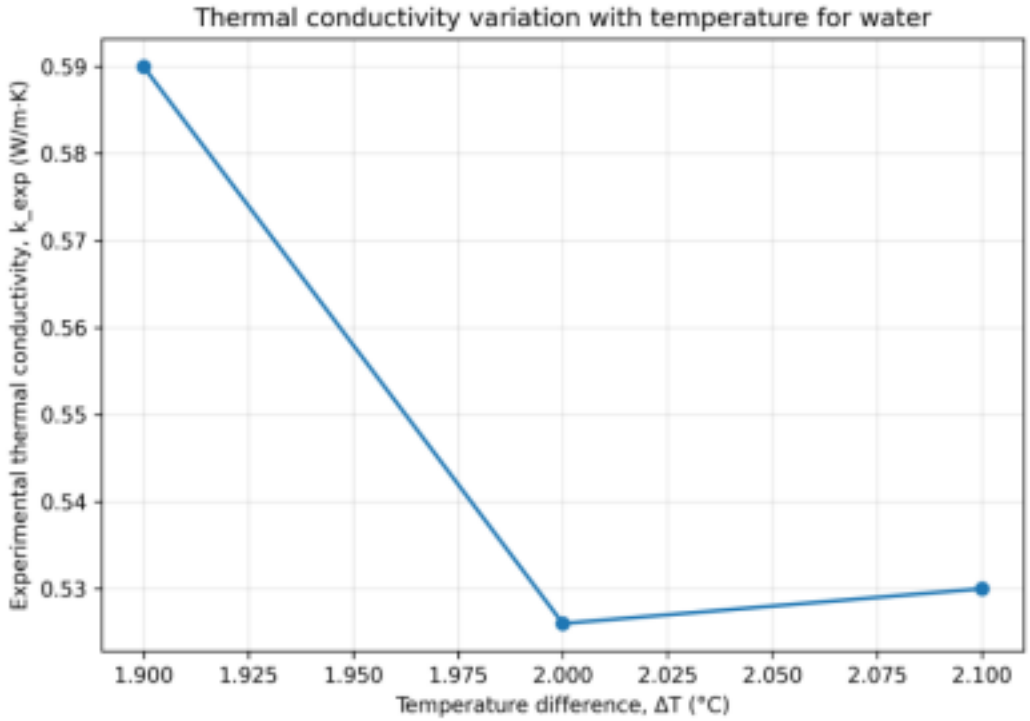

Fig. 3. Thermal conductivity variation with temperature for water.

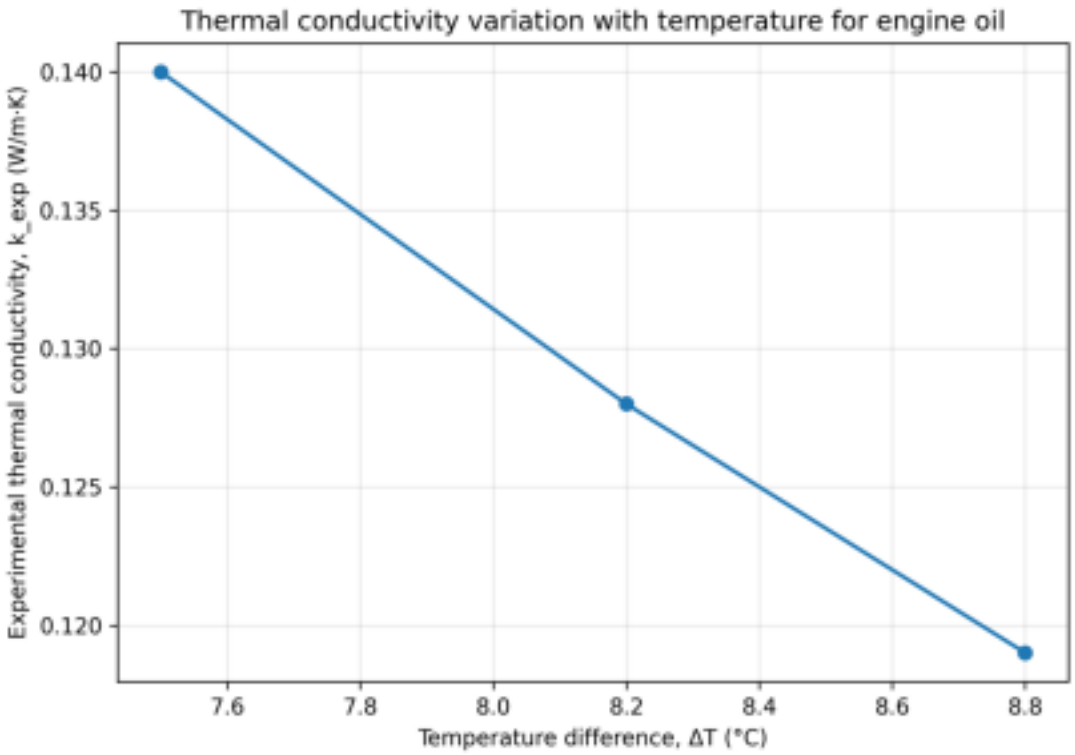

Fig. 4. Thermal conductivity variation with temperature for engine oil.

5.2 Thermal diffusivity results

Thermal diffusivity results follow the same overall pattern as conductivity. For water, α was estimated as 0.126–0.141 mm²/s as shown in Table 3, compared with a reference value of about 0.145 mm²/s. The mean error of 9.66% is acceptable for a thermistor-based educational setup. The result is especially meaningful because the diffusivity calculation combines k, ρ, and C_p; therefore, uncertainty in conductivity is carried through directly into α. Since water has both high conductivity and a moderate volumetric heat capacity, it consistently yields higher diffusivity than oil [8-10].

For engine oil, α lies between 0.068 and 0.0805 mm²/s as shown in Table 3, with a reference value close to 0.081 mm²/s and a mean error of 9.11%. This lower diffusivity is expected because the lower k of the oil dominates the ratio $k/(\rho C_p)$. Even though oil has lower density and heat capacity than water, its thermal conductivity is sufficiently small that the net diffusivity remains lower. This is consistent with standard heat-transfer theory and with the behavior reported for nonpolar liquids in the THW literature [1-3,5]. Fig. 5. shows a comparison of experimental and reference thermal diffusivity values.

The temperature-rise graph shows a smooth increase from 1.9 to 2.1 °C over the observation window. This smooth trend is important because the THW method depends

on the logarithmic relation between temperature and time. A stable monotonic temperature rise indicates that the probe was operating in a quasi-transient regime and that the MOSFET-driven heater delivered sufficiently repeatable power. Any abrupt oscillation would have degraded the conductivity estimate and reduced the confidence in the fitted slope. The observed regularity therefore supports the reliability of the acquired data.

The observed errors are larger than those achieved with reference-grade THW instruments, but they are consistent with the simplifications introduced in this work. Key contributors include the substitution of a thermistor for a precision platinum wire, the use of a compact beaker-scale sample, the unavoidable heat leakage through wiring and supports, and the limited control of convection in a small open container. These limitations are well recognized in the literature. Carefully designed THW instruments can reach uncertainties well below 1%, but only when the line-source assumptions, temporal window, and calibration procedures are rigorously enforced [1,3,6]. A low-cost setup will naturally trade a small amount of accuracy for simplicity and accessibility [4,5]. Fig. 6. indicates temperature rise as a function of time for the transient heating cycle.

Overall, the agreement with reference data demonstrates that the prototype is suitable for laboratory teaching and for exploratory fluid-property studies. The results also provide a realistic
foundation for later upgrades such as PT100 sensing, higher-resolution ADC conversion, sample enclosure, and improved thermal insulation, all of which should reduce uncertainty and make the system more suitable for publication-grade measurements [4,6].

Table 3 Calculated thermal diffusivity values and comparison with reference data.

Fluid	**k_exp**	**ρ (kg/m³)**	**C_p (J/kg·K)**	**α_exp (mm²/s)**	**α_ref (mm²/s)**	**Error (%)**
Water	0.590	997.8	4180.3	0.141	0.145	2.76
Water	0.526	997.77	4181.0	0.126	0.145	13.1
Water	0.530	997.77	4180.8	0.127	0.145	12.4
Engine oil	0.119	870	2000	0.068	0.081	16.18

Engine oil	0.128	870	2000	0.073	0.081	9.8
Engine oil	0.140	870	2000	0.0805	0.081	1.35

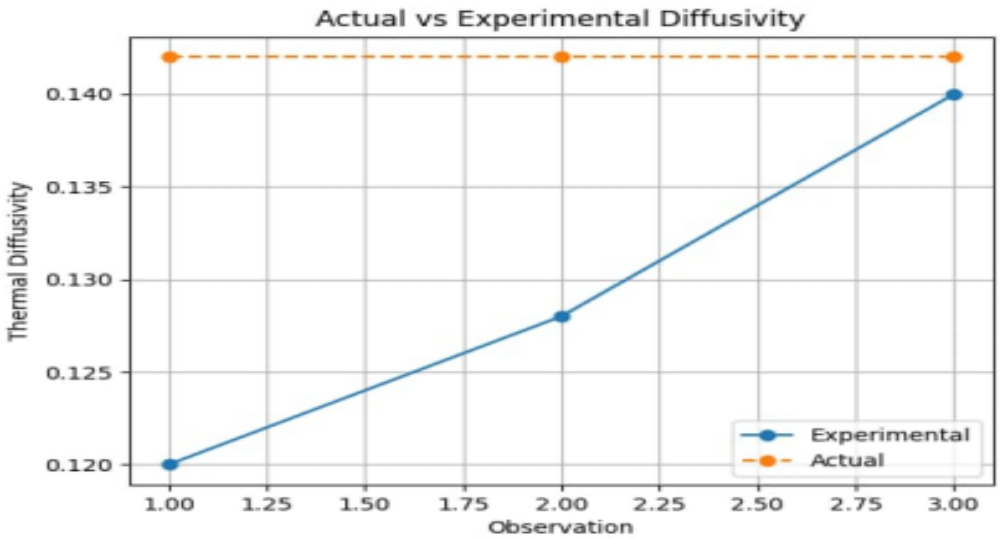

Fig. 5. Comparison of experimental and reference thermal diffusivity values.

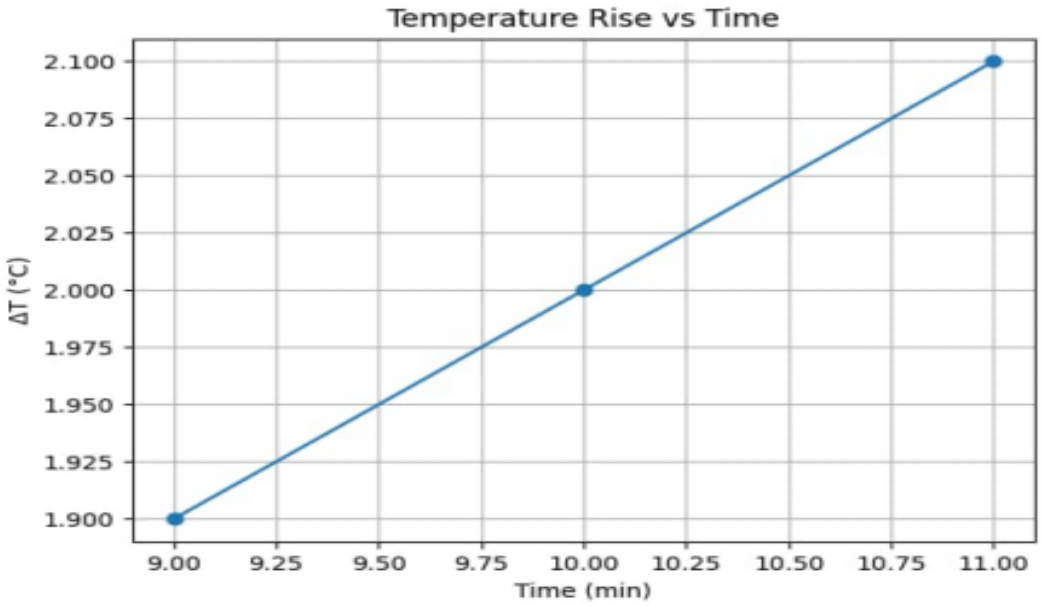

Fig. 6. Temperature rise as a function of time for the transient heating cycle.

6. Conclusion

A low-cost transient hot wire system was successfully designed, fabricated, and experimentally validated for the thermophysical characterization of fluids. The final

setup used a nichrome wire heating element, a 10 k NTC thermistor, an Arduino Uno, a MOSFET power controller, and a simple visual output interface. The system is compact, easy to assemble, and appropriate for engineering laboratories where expensive commercial thermal-property analyzers are unavailable.

The experimental results for water and engine oil demonstrated that the developed setup can estimate thermal conductivity and thermal diffusivity with reasonable accuracy. For water, the mean thermal-conductivity error was 10.0%, while for engine oil it was 9.4%. The corresponding mean diffusivity errors were 9.66% and 9.11%, respectively. These values are sufficiently close to reference data for a student-built apparatus and confirm the practical viability of the measurement concept.

The observed deviations were explained by the simplified probe geometry, the use of a thermistor rather than a precision platinum sensor, finite heat loss, and the narrow temperature window
used during testing. Despite these limitations, the system produced stable time-temperature traces and consistent results. The work therefore validates the feasibility of a low-cost THW platform as a teaching tool and a base platform for further research.

7. Future Scope

- The present system can be upgraded for nanofluid measurements, including Al2O3-water and CuO-oil suspensions, where changes in thermal conductivity are of great research interest. • Accuracy can be improved by replacing the thermistor with a PT100 or another calibrated precision sensor, along with a higher-resolution analog-to-digital conversion stage. • Heat-loss effects can be reduced by better insulation, a controlled test chamber, and a more refined probe geometry.
- Wireless monitoring and data logging can be added through Bluetooth, Wi-Fi, or USB-based acquisition to support long-duration experiments and remote data analysis.

Acknowledgment

The authors thank the faculty members and laboratory staff of M. L. V. Textile and Engineering College, Bhilwara, for their support, guidance, and encouragement

during the design and testing of the setup.

Nomenclature

Symbol	Meaning
k	Thermal conductivity (W m^{-1} K^{-1})
α	Thermal diffusivity (m^2 s^{-1})
ρ	Density (kg m^{-3})
C_p	Specific heat at constant pressure (J kg^{-1} K^{-1})
q	Heat input per unit length (W m^{-1})
t_1, t_2	Time instants used for slope extraction
T_1, T_2	Measured temperatures at t_1 and t_2

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Appendix A. Expanded Discussion on Uncertainty and Practical Deployment

Although the present paper is framed as an engineering-college prototype, it is important to note that the main scientific weakness of low-cost THW systems is not the governing equation itself but the quality of the boundary conditions used in the experiment. In a reference-grade instrument, the wire is thin, the sample volume is sufficiently large, the thermal field is nearly cylindrical, and the data are taken over a carefully selected time window. Under those conditions, the uncertainty can be very low [1,3,6]. In the present system, by contrast, the heater is a nichrome wire of practical size, the temperature sensor is a thermistor, and the sample is open to the atmosphere. These design choices inevitably increase the uncertainty, but they also reduce cost and make the system accessible for teaching.

The measured error bands of roughly 9–10% are therefore not a sign of failure; rather, they are the expected trade-off for a simplified educational instrument. The key requirement for such a system is that it should reproduce the correct physical ranking of materials and provide stable, monotonic temperature-time data. The present results satisfy both conditions. Water consistently shows higher thermal conductivity and diffusivity than engine oil, and the measured curves are smooth enough to support a meaningful THW interpretation. This is precisely the level of performance needed in a laboratory where the objective is to connect theory, instrumentation, and data analysis.

From a metrological point of view, several specific improvements would be especially effective. First, the thermistor can be replaced by a calibrated PT100 or PT1000 element to reduce temperature sensing uncertainty. Second, the fluid container can be enclosed to minimize natural convection and ambient heat exchange. Third, the heater current can be measured directly so that the input power per unit length is not inferred indirectly. Fourth, a data-logging interface can replace manual observation, allowing the entire

transient curve to be analyzed with regression methods rather than only selected time points. These improvements follow the same direction recommended in prior THW studies [4-6].

A further advantage of the present platform is that it can support research extension into nanofluids. In nanoparticle suspensions, the expected change in thermal conductivity may be modest, and therefore the measurement system must be able to distinguish a small signal from background scatter. The present prototype already provides the digital control and data acquisition backbone for such work. With upgraded sensing, better insulation, and improved calibration, the same platform could be used to compare base fluids with Al2O3-water, CuO-water, or CuO-oil nanofluids under identical operating conditions.

In summary, the main outcome of this project is the demonstration that a thermophysically meaningful THW measurement can be achieved using low-cost hardware commonly available in an undergraduate laboratory. The system does not claim metrology-grade certainty, but it does achieve scientific usefulness by generating stable data, correct trends, and reference-level order of magnitude agreement. For an SCI-style conference paper, this balance between innovation, reproducibility, and accessibility is a strong contribution.

Appendix Table A1. Main uncertainty sources and their practical effect.

Table A1. Summary of dominant error sources in the present low-cost THW platform.

Source	**Mechanism**	**Effect on measured values**
Sensor calibration	Thermistor nonlinearity and drift	Biases ΔT and changes the slope
Heat loss	Conduction through support and convection to air	Usually lowers apparent k
Wire geometry	Finite wire diameter and non-ideal line source	Introduces model mismatch
Current uncertainty	Variation in heater power	Changes Q and the fitted conductivity

Timing resolution	Manual or low-rate data acquisition	Adds scatter in ln(t) fit

Appendix B. Step-by-Step Data Reduction

For each observation, the temperature rise was taken as $\Delta T = T_2 - T_1$, where T_1 is the initial fluid temperature and T_2 is the temperature after heating for the chosen interval. The experimental conductivity then follows from the THW expression used throughout the paper. Because the present setup is based on a simplified laboratory implementation, the purpose of the calculation is not only to produce a numerical value of k but also to confirm that the response remains logarithmic and physically consistent.

For water, the three measurements yielded ΔT values of 1.9, 2.0, and 2.1 °C. Substituting these values together with the selected time ratios produced conductivity values of 0.590, 0.526, and 0.530 W/m·K. The closeness of these values to the accepted conductivity of water near room temperature indicates that the heater power and temperature sensing were both functioning in a physically meaningful range.

For engine oil, the measured ΔT values were 8.8, 8.2, and 7.5 °C. Because oil has lower thermal conductivity, a larger temperature rise was expected for the same heater input. The measured conductivities of 0.119, 0.128, and 0.140 W/m·K follow this physical expectation. The trend from
lower to higher conductivity across the three observations is also consistent with the observed reduction in error as the system approached a steadier thermal state.

The diffusivity calculations use the same conductivity values together with the corresponding density and specific heat. This means that the diffusivity result is particularly sensitive to the accuracy of the conductivity estimate. For water, a conductivity near 0.53–0.59 W/m·K combined with $\rho \approx 998$ kg/m³ and $C_p \approx 4180$ J/kg·K leads to a diffusivity in the range 0.126–0.141 mm²/s. For engine oil, the smaller conductivity but lower volumetric heat capacity yields a diffusivity of about 0.068–0.0805 mm²/s.

From an educational perspective, the stepwise calculation is valuable because it allows students to see how heat-transfer theory converts into engineering data. The procedure reinforces the roles of temperature difference, material properties, and

logarithmic time dependence. It also shows why an apparently small experimental error in temperature measurement can lead to a noticeable change in conductivity or diffusivity when the final equations are evaluated.

Appendix Table B1. Input and output values used in the conductivity and diffusivity calculations.

Table B1. Representative data-reduction values for the two test fluids.

Fluid	Observation	ΔT (°C)	k (W/m·K)	ρC_p (J/m³·K)	α (mm²/s)
Water	1	1.9	0.590	4.179×10^6	0.141
Water	2	2.0	0.526	4.180×10^6	0.126
Water	3	2.1	0.530	4.181×10^6	0.127
Engine oil	1	8.8	0.119	1.74×10^6	0.068
Engine oil	2	8.2	0.128	1.74×10^6	0.073
Engine oil	3	7.5	0.140	1.74×10^6	0.0805

A final point worth highlighting is the relationship between the results and the graphical output. The temperature-rise curve demonstrates the short-time transient response expected for the THW method, while the conductivity and diffusivity plots show that the computed properties remain clustered around the expected literature values. In a research-paper context, this combination of numerical agreement and graphical consistency is often more persuasive than either one alone because it confirms both the underlying equation and the experimental procedure.

For future publication-level experiments, the same data-reduction strategy can be extended to include confidence intervals, least-squares regression over the log-time interval, and repeatability tests
across multiple runs. Those additions would make it possible to report not only mean errors but also standard deviations and uncertainty bars, which are often expected in SCI journal submissions.

Appendix C. Practical Notes for Reproducibility

The prototype can be reproduced with basic laboratory parts, but the final outcome depends strongly on assembly discipline. The heating wire must be positioned consistently from run to run, the thermistor must remain close to the thermal field of the wire, and the electrical contacts should be tightened to reduce drift. Even a small change in geometry can alter the local heat-transfer path and therefore influence the slope of the temperature-time curve.

Another practical point is that the fluid volume should be large enough to delay boundary effects. If the beaker is too small, the wall begins to absorb heat and the measured conductivity becomes artificially low. Similarly, if the heater remains on too long, natural convection can begin to contribute to the temperature rise. The present experiments were therefore kept short, and the water and oil samples were allowed to relax between trials.

The current setup is already suitable for classroom demonstrations of transient conduction, sensor calibration, MOSFET control, and data reduction. With moderate upgrades it can be transformed into a strong undergraduate research platform. Such a path is attractive because it allows students to begin with a visually understandable low-cost system and later move toward more rigorous experiments without changing the underlying physics.

In summary, the project succeeds because it combines a correct heat-transfer model, an inexpensive hardware platform, and experimentally meaningful comparison data. These three features together make the work suitable for a conference paper and a strong base for a future SCI manuscript with improved instrumentation and uncertainty evaluation.

Detection of Intrusion in Binary Class Data using Deep Learning Techniques

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Abstract – In today's world, we use mobile phones, laptops, headphones and other smart devices are connected and communicated wherever we need but it exposes variety of cyber threats. In wireless communication system there are different types of attacks such as packet sniffing, Wi-Fi spoofing, zero-day attacks performed by the malicious users. Malware attacks will inject or damage the device. This paper used NSL-KDD dataset to train and test the model carefully, preprocessing the data and selecting the important features. It proposes the different Deep Learning Techniques and compared the results.

Keywords – *Intrusion Detection System (IDS), Malware Attacks, Preprocessing, Feature Selection, Deep Learning*

I. Introduction

People use smart devices such as smart locks, smart speakers, smart watches, smart lighting, Robot Vacumms, smart refrigerators, smart TV and smart health devices in our daily life. Sometimes those devices are not safe. Hackers can control them, steal the data from that or shutdown the device. An Intrusion Detection System (IDS) is a security tool designed to monitor network traffic and identify the threats in the network. Deep Learning (DL) models automatically learn the large amount of raw data without the human intervention. These models distinguish between normal behavior and malicious anomalies with high precision. This study used various deep learning models to identify such that intrusions, the dataset NSL-KDD is used and performances are compared. The main goal is to understand and identify the intrusion, improves the accuracy of the system.

II. Related Works

The work presents a Robust Intrusion Detection Framework for IoT networks by integrating an improved hybrid Deep Learning architecture capable of handling both

binary and multi class attack classification with high accuracy[1]. This model enhances feature learning by combining complementary deep learning techniques, enabling efficient detection of complex and evolving cyber threats in resource-constrained IoT environments. A key novelty lies in its adaptive learning strategy, which improves generalization performance while reducing false alarm rates across diverse attack categories. The framework also demonstrates scalability and reliability for real-time IoT security applications, making it suitable for next-generation smart network infrastructures.

This study introduces an Intelligent Intrusion Detection and Real-Time Response System based on Long Short-Term Memory (LSTM) networks to strengthen modern network security[2]. The model effectively captures temporal and sequential attack patterns, enabling accurate identification of both known and emerging cyber threats in dynamic environments. Its novelty lies in integrating predictive deep learning with rapid response mechanisms, which minimizes detection latency and improves threat mitigation efficiency. The framework further enhances network resilience by reducing false positives and supporting continuous monitoring for real
time cybersecurity operations.

This model employs a Long Short-Term Memory (LSTM) Network to Improve Intrusion Detection Accuracy by learning complex temporal dependencies in network traffic data. Its innovative design enables simultaneous threat identification and real-time response, allowing faster mitigation of malicious activities before they escalate[3]. The model demonstrates strong adaptability to evolving cyberattack patterns, making it effective for dynamic and large-scale network environments. By combining continuous monitoring with intelligent sequential analysis, the system enhances overall network reliability while significantly reducing false alarm rates and response delays.

This work proposes an IoT-based Intrusion Detection System that leverages a novel hybrid Deep Learning algorithm to improve cyberattack detection in heterogeneous network environments. The framework combines the strengths of multiple deep learning techniques to achieve more precise feature extraction and enhanced classification performance for diverse intrusion patterns[4]. Its key novelty lies in optimizing

detection capability while maintaining efficiency for resource-constrained IoT devices. The proposed approach also improves scalability and adaptability, enabling reliable real-time security monitoring against rapidly evolving threats in smart IoT ecosystems.

This survey provides a Comprehensive analysis of Anomaly Detection Techniques in Dynamic Networks, emphasizing the challenges posed by continuously evolving structural and temporal patterns[5]. The study highlights innovative approaches for identifying unusual behaviors through graph-based analysis, temporal modeling and adaptive learning mechanisms. A significant novelty of the work is its unified perspective on diverse anomaly detection methodologies across large-scale and time-varying networks. The survey also identifies critical research gaps and future directions for developing scalable, real-time, and intelligent anomaly detection systems in complex network environments.

III. Research Methodology

A. Dataset

An Intrusion Detection System helps to identify the unwanted or harmful actions in the wireless network. It uses Deep Learning Models to learn patterns from the large amount of data. While using Deep Learning, the data can be divided into two types. Binary class which is used to check only the traffic is normal or attack. In Multi class, the system identifies what type of attack is affected in the data. The study utilizes the NSL-KDD dataset, comprising 42 features. To learn the system better, clean the data by removing errors, converting text into numbers and scaling values. Deep Learning models such as CNN, LSTM, Autoencoders are used to detect
the attacks. Binary-Class detection is simple and detect only it is normal or attack.
Overall performance of the model represented using accuracy, recall, precision and f1-score.

B. Data Preprocessing

In Pre-processing to train the Deep Learning model, the data must be cleaned or prepared. During this process, duplicate values are removed, missing values are fixed

and text values like 'TCP' or 'UDP' are converted into numbers because the model can understand only the numerical data.

C. Convolutional Neural Networks (CNN)

CNN is used to identify the complex patterns and identify the attacks within the data packets. When the data is trained, CNN can classify the traffic very quickly. CNN-based IDS typically consists of convolutional layers, pooling layers, and fully connected layers. The convolutional layers are responsible for feature extraction, using mathematical kernels to detect subtle anomalies. Convolutional Neural Networks (CNN) are used in IDS to classify network intrusion patterns. They can process and analyze spatial relationships within data, IDS, and relevant network traffic patterns. Integrating CNN + LSTM can provide a more reliable IDS. This hybrid model uses CNN's spatial pattern recognition capability and LSTM's sequence prediction capabilities. CNN analyzes data to detect spatial anomalies, while LSTM processes output over time to understand temporal patterns and anomalies. This integration allows for more effective multi-stage cyber-attack detection and higher chances of reducing the frequency of attacks

D. LSTM, GRU

Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) are specialized types of Recurrent Neural Networks (RNNs) designed to process sequential data, making them ideal for Intrusion Detection Systems where the timing and order of network packets are critical. LSTM acts like a long-term memory, along with three gates—input, forget, and output—that regulate which information is kept or discarded.

GRUs are a more streamlined version of LSTMs, designed to offer similar performance with less computational complexity. Instead of three gates, a GRU uses only two: an "update gate" and a "reset gate." The update gate determines how much of the previous memory to retain, while the reset gate decides how much of the new input to incorporate with the past information.

E. CNN with MLP

The integration of Convolutional Neural Networks (CNN) and Multi-Layer Perceptrons (MLP) offers a robust framework for analyzing the NSL-KDD dataset in cyber-defense systems. In this hybrid design, the CNN layers function as an automated feature engineering engine. While network logs are inherently tabular, the CNN treats them as structured sequences to identify "local patterns"—spatial relationships between specific network attributes like duration, service types, and flag statuses. The MLP acts as the final decision-maker, utilizing its fully connected neurons to perform complex non-linear classification. It maps the patterns identified
by the CNN to specific output categories, distinguishing between legitimate traffic and various attack vectors.

F. ADASYN

ADASYN (Adaptive Synthetic Sampling) plays a critical role in enhancing intrusion detection performance on the NSL-KDD dataset by addressing the severe problem of class imbalance. In network security, malicious activities (attacks) are significantly rarer than legitimate traffic, which often causes machine learning models to be biased toward "normal" behavior, leading to a high rate of missed detections. ADASYN overcomes this by intelligently generating synthetic data points for the minority classes (the attacks)

G. Binary Class Intrusion Detection

Binary Class Intrusion Detection is the most fundamental form of classification in cybersecurity, where the primary goal of the model is to categorize network traffic into one of two distinct groups: Normal or Attack. This method is particularly effective when used with large datasets like NSL-KDD, where the initial priority is to achieve a high "Detection Rate" while minimizing "False Positives." Because the model is focused on a single binary distinction, it often achieves higher accuracy and faster processing speeds compared to multi class systems.

IV. Experiment Analysis

In this work, NSL-KDD dataset is used and evaluated the proposed model by training it individually and testing it on the same data sets.

Accuracy represents the overall percentage of correct predictions done in the model (both correctly identifying attacks and correctly identifying normal traffic).

Accuracy=(TP+TN) / (TP+TN+FP+FN)

Precision is the ratio of correctly predicted positive observations to the total predicted positives. Precision = TP / (TP + FP)

Recall is the ratio of correctly predicted positive observations to all observations in the actual class.

Recall = TP / (TP + FN)

The F1 Score is the weighted average of Precision and Recall. This score takes both FP and FN into account

F1 Score = 2[(Precision*Recall)/(Precision + Recall)]

Performance Analysis

Algorithm	Accuracy	Precision	Recall	F1 Score
CNN	0.97	0.97	0.98	0.98
CNN + GRU	0.97	0.97	0.98	0.97
CNN + LSTM	0.98	0.95	0.99	0.97
CNN+MLP	0.98	0.98	0.97	0.98
ADASYN	0.99	0.99	1.00	0.99

V. Conclusion

Securing networks is crucial aim of organizations to protect digital infrastructure from emerging cyber-attacks. This study introduces an advanced IDS that leverages a hybrid CNN MLP model. By incorporating innovative data resampling techniques such as hybrid ADASYN-SMOTE for binary classification. The dataset NSL-KDD dataset is used. CNN model provides 97% of accuracy and Precision. The CNN model with GRU

give 97% of accuracy. LSTM and MLP models provide 98% of accuracy. Adaptive Synthetic Sampling (ADASYN) focuses on the **distribution** of the data, it gives 99% of accuracy, precision and F1 score. It establishing the best solution for enhancing network security and effectively addressing the complex challenges posed by modern cyber threats.

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Knowledge and Proficiency in the Digital Era

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Abstract- In today's world, people need to know a lot of things to do well. The internet and smart gadgets have made it easy for people to get information from all over the world. Just being able to get information is not enough. People need to be able to use the information to figure out if it is good or not, and use it in a way. Being good with computers, being able to tell if information is true, knowing how to be safe on the internet and being able to change with technology are skills. These digital skills are important for people's lives and work lives. As technology keeps changing the way governments, businesses, schools and people talk to each other, people need to keep learning things. This paper talks about how knowing things and being good with technology helps people get past the problems of the world, deal with the challenges of the century and make a big difference in the world. People need competency and knowledge to overcome digital boundaries and make significant contributions to the world. Digital competency and knowledge are essential for people to meet the challenges of the century and make the world a better place. People need competency and knowledge to do well in the world. Digital competency and knowledge are very important for people to succeed in the world.

Preface

Over the past 20 years, everything digital has become a big part of our lives, and it is still growing very fast. We use technologies for almost everything now, from learning and education to participating in politics and even taking care of our money and health. The fact that digital technology quickly leads to new ideas and uses that affect our lives in many ways. Digitalization gives us chances and goals, but it also creates big problems for society that we need to understand and solve.

The first of four meetings that were part of the Education and Skills Consultation on March 6 and 7, 2017, was the Program for Thought Leadership in 2017. The other events in the series were about:

- Open science

• Currency and the future of transacting

• Civic engagement

This is a summary of the points from the four programs at the 2017 Thought Leadership Conference, which was called "Building our Connected Society." This report was published along with the program for the conference.

Background

The digital world is changing how we learn things and get skills. Technology is being used in ways to teach people. This change is happening when our jobs are also changing because of politics and the economy. As technology improves, people need to learn skills to keep their jobs. Learning is something we do all our lives now. The digital world keeps coming up with things, so people have to keep learning. They need to get better at things to stay current. The digital world is good at making things, and we have to be good at learning, too. Technology lets us learn things whenever we want. We can get skills and knowledge at any time. This is different from how things used to be. Now people have to keep learning and getting better if they want to stay. The digital world and technology keep changing, so people have to change. So, we have to change all the time to keep up with the world and technology.

Technology skills need to be updated every three years because technology skills can

become outdated quickly. Technology skills are really important for working people. Working people need to know how to find information using technology skills. Working people also need to know how to check if the information they find using technology skills is good or not. People use technology skills for lots of things. For example, people use technology skills to book a doctor's appointment. People also use technology skills to do their banking. Some people are very comfortable using technology skills. These people are the ones who grew up with technology skills. Technology skills are very important for lots of tasks. Not everyone has technology skills. We need to make sure everyone gets the technology skills they need. This is because technology skills are being used more and more in our society. We must ensure all citizens have the technology skills they need. The consultation mainly focused on considering these factors.

Technology should help people in society get the skills to use it well throughout their lives.

This conversation looked at how digital technology can help people get the skills to benefit most in work settings. It also considered how different stakeholders can promote social norms around digital technology use and ensure good online behavior.

1. Skills required: present and future We started by discussing what people need to do to get involved. This helped us understand the topic and people's thoughts about it. Some people strongly felt that we lack workers and should focus on current skills. They did not think we should try to predict skills. Others said we do not know what skills people use at work. Employers struggle to identify their employees' skills and what to look for when hiring.

This makes it hard for them to find the people. One reason is that employers do not collect data on their employees' skills. They especially do not know what makes their top employees successful. Employers are struggling with finding people with participation skills. They need to know what skills people have for participation to make decisions about it. Employers need this information to make choices about

professional participation.

Discussions indicated that this issue permeates the system as a whole.

Skills needed:

The conversation showed that we need two sets of skills: skills to use digital devices and skills to navigate the digital world. Digital skills are the skills needed to use devices like computers and phones. On the other hand, digital navigation skills are a wider range of skills needed to succeed online.

These skills include finding information, prioritising it and checking if it is accurate and reliable. These digital navigation skills are not really different from the skills people needed before this age. They just need to be adapted for use. Some people call these navigation skills "eternal skills" because they are still needed today. Digital skills and digital navigation abilities are both important for being successful in the world. We need to have both to make the most of the age.

2. Teachers and students: Adapting roles and duties in the digital

Teachers and students have to change the way they do things and what they are supposed to do when they use computers and the internet. The teachers and students are figuring out what the teachers and students should be doing in this digital world.

2.1. Who is the educator?

In today's world, being an educator is really tough. An educator is not someone who teaches anymore. People say that teachers do not really teach like they used to. We discussed this.

Some of us think that we should use the word "learning" instead of "education".

This is because learning is about what the student gains from it, not what the educator gives them. The educator and the student both learn things from each other, which is really cool. The educator and the learner are closely connected. The educator helps the student with their learning. Now the student also helps the educator with their learning. So, the educator actually learns from the student, too, which is a change. The educator and learning go hand in hand. The educator keeps learning things all the time from the student and their learning.

The educator learns from the students' learning. The student learns from the educators' teaching.The educator and the student are learning together.

People talked about how important it is to know the difference between using technology to learn in school and just teaching technology. They said technology can really help with teaching and learning. It cannot replace the teacher. Learning can happen in different ways, and technology is just one of them. Participants think that technology should be used to help people learn, not to teach them about technology itself.

2.2. Role reversal in the digital age

Young people can teach their teachers because they are good with computers and things like that. In our talks with people, they said they often know more about technology than their parents or grandparents. They show them how to use things like iPads and messaging. They even teach them how to send texts or take pictures. This happens a lot in school, too. Young people notice that their teachers are not as good with technology as they are. Sometimes teachers ask students for help with things, like projectors or PowerPoint. The young people we talked to do not think this makes their teachers any less important. They still think of their teachers as people who can guide them and help them make choices. However, some young people thought that if they started teaching their teachers a lot about technology, it might change the way their teachers feel about them.

2.3. How do we prepare our educators?

The role of the educator is changing because of technology. Are we doing enough to help educators get ready for this change? What more can we do?

We discussed how technology is affecting the job of educators. We also talked about how to support and prepare them for these changes. Many people agreed that for teachers to use learning tools and technologies well in their teaching, they need to know how to use them and feel comfortable with them.

Some people thought that the job of an educator is to inspire, motivate and build trust with students. They also said that technology is disrupting this job. It's changing

things, like

• How people get information

• How people find their way through information • How teachers teach in places

The talks examined how, as students can access information on their own, the growing use of technology in the classroom may increase the likelihood that teachers will eventually take on a more administrative role. But some contended that teachers ought to be employing digital technology as a "mass stimulation weapon." Information can be exchanged before class, allowing teachers to concentrate on mentoring and coaching their students while making the most of their contact time.

People think that instructors are not moving forward with the things that are happening, and they are not staying up to date with the new technology. Instruction is changing. Teachers who have been doing this for a time are having the hardest time with this. The group found some problems that make it hard for instructors to change. **They are:**

- **Absence of skill development:** The problem is that teachers do not get the training they need to learn technology skills. This is because their continuing professional development does not always include learning about technology. Teachers should get training to learn things, but this is not always the case.

  As a result, teachers are not always able to handle the fact that technology is being used more and more in the classroom. The absence of skill development in technology is an issue for teachers. Teachers need to learn about technology to do their job well. They are not always given the chance to do so.

- **Systemic institutional barriers:** Digital learning is still not available at several institutions. as well as accrediting organisations for courses that are accessed online.
- **There are silos between groups:** Education technology providers are not doing enough to understand how teaching methods work with technology. As a result, teachers do not get the benefits of using technology in the classroom. Education technology providers need to look into this.

Many others believed that asking the opposite question—"What are we doing around technology to prepare?"—was more appropriate. Teachers? Is enough being done to advance technology so that it can be applied in a learning environment? It was proposed that the following take place to address this issue:

- Digital technology-related CPD has to be enhanced. This CPD ought to highlight the benefits of technology use and how it might support teachers in their teaching. This CPD must go beyond only awarding certification; it must instil a sense of worth in the educator.
- Rather than using content and learning outcomes, education should be evaluated based on the development of skills.
- Educational technology providers can better convey the value and significance of their technology and offer more proof of its efficacy.

3. The reasons for and difficulties with digital schooling

The many kinds of stakeholders in digital education have varied goals and difficulties. Examining four distinct

Participants talked about each stakeholder group's viewpoint in relation to digital education.

3.1. Learning and education providers:

Motivations: The thing that really gets people excited is that technology makes education better for everyone. It helps education reach people, and it can be tailored to what each person needs. Some people think technology can also help education providers work together, which means people get more for their money and new ideas are created to meet needs. Having leaders who care and want to make a difference is also a motivator for change.

Challenges: Not having leaders who care can really hurt education. It can affect funding. Make education more expensive for schools and education providers, as some people have said. People also think that teachers need help to get better at their

jobs. Figuring out what people need to learn and how to measure if there is a demand for classes are other issues. On the other hand, some people believe that even if there is a lot of demand, there might not be enough teachers, so people still would not be able to learn.

Any school that can provide education is a good option.

Some people are also worried that certain educational institutions and universities do not want things to change because they like how things are now and have not seen the good things that technology can bring to education.

3.2. Businesses and employers:

Motivations: Many companies and employers want to close the skills gap and have ethical awareness. They want to help workers get better at what they do, which can help companies make money, save money and stay competitive. Employers really want to close the skills gap in fields like engineering, where older workers are leaving and taking their experience with them. This means younger workers are not learning from them. Educational technology can help fill this gap. Companies also want to try things, which can be helped by online learning, and this can help more people get involved in the market, both near and far.

Challenges: One big problem is that companies will only support changes in education if it saves them money or makes them more money. This includes decisions about funding, what to teach and helping staff get better at their jobs. The government is also part of the problem because it does not understand how businesses work.

Another issue is that companies do not know enough about the skills gap to fix it.

Some people think it will get easier to figure out what skills are needed and what skills we already have, which will make it clearer where the gaps are.

Many people also agree that it is not clear who should start making changes, and this is stopping anyone from doing anything about it.

3.3. Government:

Motivations: The use of digital education gives the government the chance to reach more remote populations, to guarantee equitable access to high quality education and, in general, to close the knowledge gap in digital technology. The government's desire to innovate and enter a new field was also discussed, but this still runs counter to long-standing customs. The claim was also made that digital education could lower costs by boosting tax revenues and business growth; participation in digital education could provide the government with additional data, including learner skills and educational attainment, which could then be used to provide feedback and insights for future education.

Challenges: The group expressed disapproval of the way the government currently supports online learning. Conversations centered on the inability of policymakers to understand what is going on in the digital world and the rapid advancement of digital education make it difficult to close the digital divide. These factors also affect the creation of a holistic approach because it takes time to implement and assess strategies. Participants demanded that a better balance be struck between tasks that must be completed promptly and those that require more time to develop properly.

3.4. Individual learners:

Motivations: This group really wants to make formal education more fun, fair and safe for all kinds of students. They think technology can help with learning. According to them, tech can help students find information they wouldn't have found otherwise.

In the youth focus groups, young people said that using technology helped them practice and revise. They also mentioned that good quality digital learning is becoming a must-have. This is because it works with the tool's students use every day. Schools and institutions should improve their standards based on how good their digital learning tech is.

Challenges: Some problems came up, like exclusion, getting people on board and skill level. Not everyone is good with tech, so it's hard to get people to learn these skills or see the benefits of learning.

The tech used in schools might not be as good as what young people use outside of

school. This could turn them off. More work needs to be done to show the benefits of education.

Another issue is that technology can be expensive, and not having access to it can leave some people out.

4. The use of digital learning as an inclusive tool

Digital education can be a tool to include everyone, but there are many problems that need to be solved. One big issue is that people need to have access to technology to use education. Some people said that if you do not have the technical skills, you cannot use digital education.

For example, people who are very good at reading and writing are usually better at using technology than people who are not. Therefore, people who do not have a lot of education may also have trouble with digital education.

The group talked about how some adults are interested in education and technology, but they are not always a part of the digital education system. They also said that people should not assume that adults are not interested in learning.

When talking about who owns technology, the group wondered if the people who create technology are biased in some way. They asked who creates technology and if these tools are good for all students. Most people in the group said that having one system for everyone is not a good idea because it does not consider what each learner needs.

Digital education may not be fair to everyone if it is not designed to be accessible to all users. For instance, some research showed that when users could customise courses, more people participated. This is an example of how digital education can be made inclusive.

4.1. Role of government:

Everyone agreed that more could be accomplished if employers collaborated, based on prior experience. The government should instead entrust itself with sole responsibility for education and skill development. For instance, it would be more efficient if such an

analysis were carried out collaboratively rather than separately in order to identify the kinds of abilities required in the industry. Participants believed that the existing emphasis on knowledge and rigor in the educational system and schools needed to be replaced with an emphasis on fostering creativity and growth in order to develop a skillset for the labor market.

4.2. Role of employers and businesses:

It was acknowledged that companies and employers must take a more active and active role in promoting learning and education providers, and that they must gain a deeper comprehension of the opportunities and difficulties associated with digital education in schools. Nesta-commissioned research that was released in 2012 provided specific instances of educational innovation, including collaboration with corporations. All things considered, participants believed that greater collaboration between industry and government was also necessary in order to comprehend trends in the labor market, change the educational system, and share responsibility for less skilled and adaptive citizens.

4.3. Role of education and training providers:

Expanding on the last suggestion, there is a need to improve the link between training and education providers, on the one hand, and employers, on the other, whereby educators are informed about the limitations and difficulties that businesses and employers confront, and vice versa. Some participants proposed that in order for a dynamic ecosystem to develop, there needs to be more feedback loops and better communication between the two stakeholder groups. More clarification on the definition of "digital learning" and the necessity of integrating technology into education were also demanded, providers to fully utilise it. Certain participants believed that new forms of educational institutions were required and that teacher preparation, especially in relation to the use of digital tools in the classroom, needed to be improved. Most of our conversations centred on the notion that instructors and training providers had to impart not only subject matter but also the timeless, or soft, skills covered in Section 1.

4.4. Role of individual learners:

The conversation regarding each person's function indicated that we should talk about learning more fully and move beyond education. It makes it appropriate to incorporate all residents at various life phases. Even while peer-to-peer learning is growing and citizens are taking more responsibility for their education, it was believed that not enough was being done to explain the advantages and involve the learner in the story.

There are unmet needs among learners, and participants urged further conversation about a recent finding that highlighted the significant role parents may play in this process. of the momentum, or conversely, a hindrance if they fail to comprehend the reasons behind the necessity of education reform in the contemporary digital environment.

Conclusion:

All things considered, the conversation covered some topics related to education in the digital age, including the abilities that are needed both now and in the future.

Enhancement of formal education systems for lifelong learning and schooling; and the broader role of people in society, in addition to the roles of corporations, government, and educational institutions.

Digital and technology abilities, and the softer skills, are really important these days.We need these skills to live in our world with all its technology.

An interesting story needs to be told to everyone. The government, big companies, teachers and regular people. About how education is changing all the time.

If we explain the things about using digital and technology abilities and softer skills, it will get more people on board and lead to good results in areas like health, making sure everyone is included and the economy.

There is individual accountability with regard to the roles that various actors have in the field of education provision in this digital age to constantly improve one's skills and knowledge, as well as a broader duty on the part of corporations and the government to make education and learning more efficient, relevant, and accessible. To give citizens the abilities they need to thrive in daily life and in possible future occupations, digital skills must be more effectively incorporated into formal education and lifelong learning.

Our conversations came to the conclusion that, due to the fact that younger

generations are generally more tech-savvy than their teachers and since more and more material is now accessible online, instructors' roles are evolving from being knowledge repositories to more of a mentor or guide who facilitates more individualised learning experiences. As these changes spread in the future, issues like accreditation and monitoring will still need to be resolved.

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Adoption and Usage Patterns of UPI Applications among Students: Evidence from Surat

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Abstract- This study examines the perception of students regarding the most common UPI applications in Surat by conducting a comparison between the main applications including Google Pay, PhonePe, Paytm, and BHIM. The study utilizes primary data gathered from 106 students ranging in age between 18-23 years old by using a structured questionnaire. The study aims at analyzing user preferences, their satisfaction level, and factors affecting their utilization of the applications such as transaction speed, safety, rewards, and merchant acceptance. The findings indicate that students rely heavily on UPI applications when conducting their financial transactions using QR code payments and transferring funds amongst each other despite having very tight budgets. Students are more comfortable using only one application rather than switching from one platform to another, implying their trust in one of these apps. In terms of preference, Google Pay is rated as the best platform among others due to its user friendly interface, transaction speed, and rewarding system followed by PhonePe; however, Paytm and BHIM remain far behind. It is noteworthy that despite the significant decrease in using cash due to UPI, the latter did not completely replace the former.

Keywords: UPI, Digital Payments, Student Perception, Google Pay, PhonePe, Paytm, BHIM, Transaction Speed, Security, Rewards, Merchant Acceptance, Surat.

Introduction:

The growing prevalence of digital payment systems in India has brought about many changes in the conduct of business transactions in the country, one of which is the rise of the Unified Payments Interface (UPI). The interface allows customers to instantly transfer money from mobile applications without compromising on security and convenience. The technology has grown increasingly popular amongst younger users, particularly university and college students. As there is greater penetration and access to smartphones and the Internet, students are now using UPI more often than not for all

sorts of transactions ranging from QR code payments to peer-to-peer payments to bill payments. Some of the popular applications of this sort include Google Pay, PhonePe, Paytm, and BHIM due to their easy interfaces, quick transactions, exciting rewards, and merchant acceptance. Students in emerging cities such as Surat constitute one of the larger user groups, and it is essential to gauge their perceptions towards each app since very little has been done regarding that; although there have been many studies on digital payment adoption factors among youth, few studies exist in comparison between individual applications based on user experiences.

Review of Literature:

The existing literature on digital payments and UPI highlights its rapid growth and increasing importance in India's financial ecosystem. Studies such as those by Roshna Thomas and Abhijeet Chatterjee (2017) emphasize that UPI provides a convenient, affordable, and efficient platform for monetary transactions, supported by Aadhaar integration, rising smartphone usage, and expanding telecom infrastructure. Similarly, Ravish Rana (2017) found that digital payment adoption is strongly influenced by education levels and internet awareness, with urban and metro populations showing higher acceptance due to better digital exposure. Research by Maryam Barkhordari et al. (2017) further underlines the importance of perceived security and trust, suggesting that transparent security measures and accessible guidelines significantly enhance user confidence in electronic payment systems. At the same time, studies such as Dr. M. Sumathy and Vipin KP (2017) challenge demographic assumptions by indicating no significant difference in awareness levels across gender or education, reflecting the widespread reach of digital payment knowledge.

Further research focuses on the benefits and expansion of cashless systems, particularly after policy interventions like demonetization. Dr. Shilpa Bhimrao Gaonkar (2018) highlights advantages such as transparency, efficiency, and convenience, while Dr. N. Rakesh et al. (2018) identify UPI as a dominant digital payment mode outperforming traditional methods like cards and net banking. Dinesh T.M. et al. (2018) note that demonetization acted as a catalyst in accelerating digital payment adoption. Studies also examine user behaviour and market dynamics, where Alaknanda Lonare et al. (2018)

emphasize ease of use as a key adoption factor, and Subho Chattopadhyay et al. (2018) point out that cash still remains preferred by many retailers due to familiarity. Additionally, Surabhi Agarwal (2018) highlights the role of cashback incentives in promoting UPI usage, while Shruti Arcot Kesavan (2018) discusses the expansion of UPI into merchant payments, transforming it into a comprehensive payment ecosystem. Research by Balasubramanian M and Amanullah K.M. (2019) further reveals that private applications like PhonePe, Google Pay, and Paytm dominate over BHIM due to better user experience.

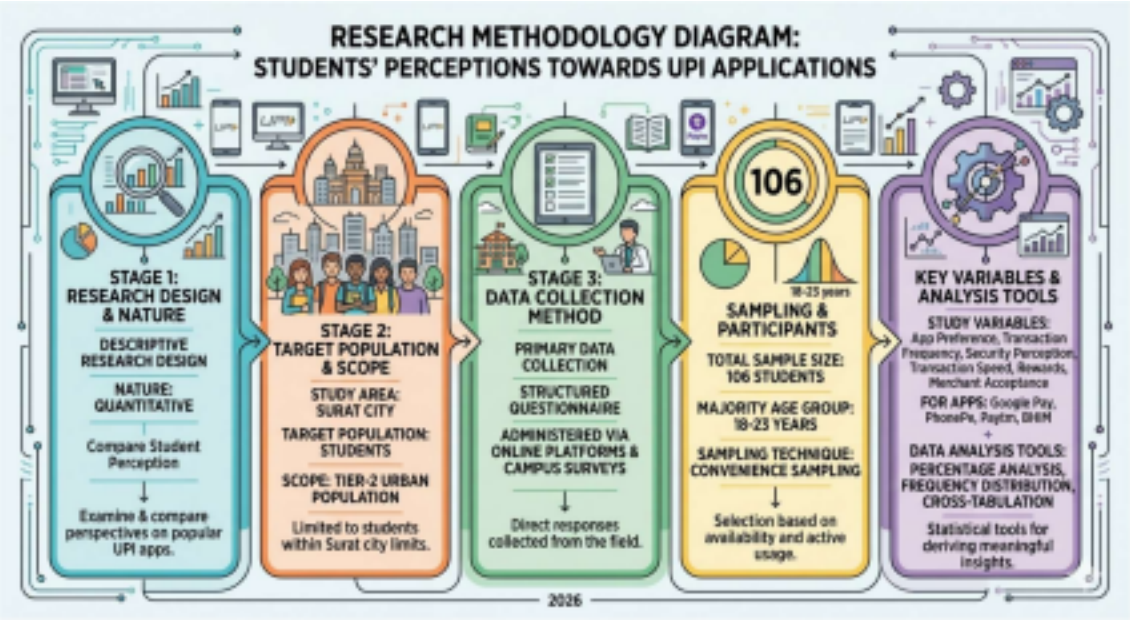
Recent studies focus on behavioral and technological factors influencing UPI adoption. Venkata Siva Kumar and Ch. Kavya (2020) found that users across different age groups prefer UPI over traditional banking due to its speed and convenience. Nileshkumar L. Patel and Jayshri S. Datta (2020) highlight that perceived risk negatively affects adoption, emphasizing the need for strong security measures. Studies by Damodhar and Sunitha (2020) and Sunny Gupta and Dinesh Chand (2021) identify ease of use, mobile accessibility, and transaction efficiency as major drivers, while also noting that cashback alone is not sufficient to sustain long-term usage. The impact of external factors is also evident, as Parvesh Deepak Oswal and Hanmant N. Renushe (2021) report a significant increase in UPI adoption during the COVID-19 pandemic. Additionally, Rishabh Jha and Rohit Kumar (2021) confirm that performance expectancy, effort expectancy, and facilitating conditions significantly influence behavioral intention toward UPI usage. Despite extensive research on digital payments and UPI adoption, most studies focus on general usage patterns rather than comparing specific applications. There is limited research addressing student perception, especially in Tier-2 cities like Surat. Therefore, this study attempts to fill this gap by providing a comparative analysis of major UPI applications based on student preferences and experiences.

Objectives of The Study:

- To find out the most preferred UPI application used by the students in Surat for their daily transactions.
- To compare the efficiency of Google Pay, PhonePe, Paytm, and BHIM according to the perceptions of users.

- To study the perception of students about the various features such as speed, security, reward, and acceptance at merchants.
- To find out the various problems faced by students when using various UPI applications.
- To study the role played by various promotions like cash back and offers in influencing the use of UPI apps.

Research Methodology:



The present study adopts a descriptive research design to examine and compare students' perceptions towards UPI applications. The study is conducted in Surat city, focusing specifically on students as the target population. Primary data has been collected through a structured questionnaire administered via online platforms as well as campus surveys. A total sample of 106 respondents participated in the study, with the majority belonging to the age group of 18–23 years. The sampling technique employed is convenience sampling, wherein respondents are selected based on their availability and active usage of UPI applications. The study considers key variables such as app preference, transaction frequency, perception of security, transaction speed, rewards, and merchant acceptance in relation to popular UPI platforms including Google Pay, PhonePe, Paytm, and BHIM. The collected data is analyzed using statistical tools such as percentage

analysis, frequency distribution, and cross-tabulation to derive meaningful insights. The research is quantitative in nature, emphasizing measurable responses and comparative analysis. The scope of the study is confined to students in Surat city, thereby representing a Tier-2 urban population.

Data Analysis and Interpretation:

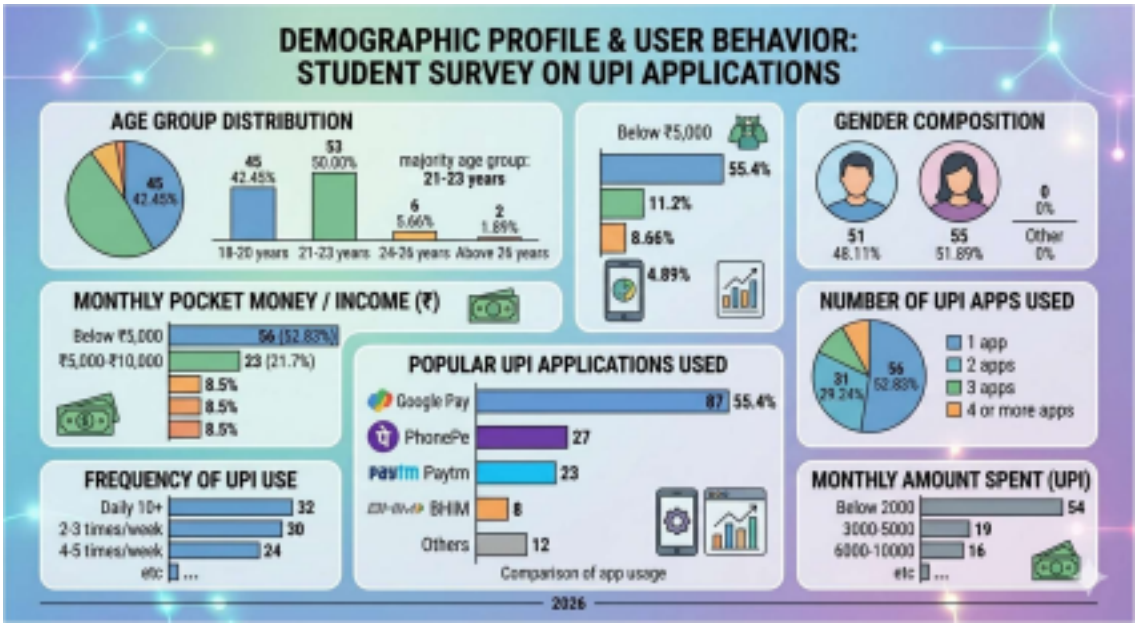


Table 1. Demographic Profile and Usage Patterns of UPI Users

The analysis reveals that UPI is majorly adopted by young people who fall within the age range of 18-23 years, and there is an almost even proportion of male and female UPI users. Most of the interviewees fall into the poor economic category (below ₹5,000), suggesting that UPI is greatly adopted by students as it is very convenient and affordable. Many respondents use only one UPI application, and among these UPI applications, Google Pay has emerged as the most used and trusted application followed by PhonePe and Paytm. Regarding the usage frequency, many respondents make transactions daily and weekly, but most of them spend less than ₹2,000 monthly. Main uses for UPI applications include merchant QR payment, P2P transfer, and bill payments, which indicate the importance of UPI applications in carrying out every day financial transactions. Further, UPI has resulted in a significant decline in the use of cash; however, most UPI users have managed to switch to digital transactions only

partially. Though UPI users generally feel safe using UPI apps, some concerns relating to fraudulent transactions remain. Factors such as speed, better rewards, and convenience significantly influence the choice and preferences of UPI users.

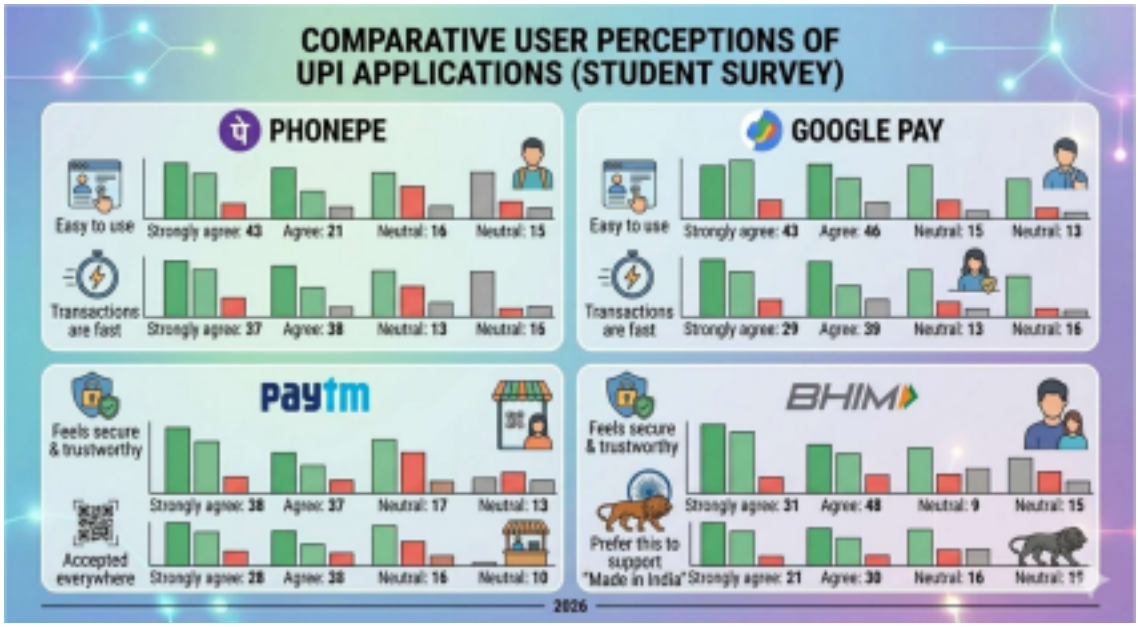


Table 2. User Satisfaction Levels of Major UPI Payment Apps

This analysis evaluates user satisfaction with PhonePe, Google Pay, Paytm, and BHIM. Google Pay scores the highest overall, with high satisfaction levels on usability, speedy transactions, security, and acceptance. PhonePe ranks second with high satisfaction levels on usability and speedy transactions, but with higher levels of neutral ratings. This means that there is average satisfaction among users. Paytm receives the most neutral ratings for usability, speedy transactions, and acceptance. BHIM receives the highest number of neutral ratings for usability and speedy transactions and disagreement for security and acceptance.

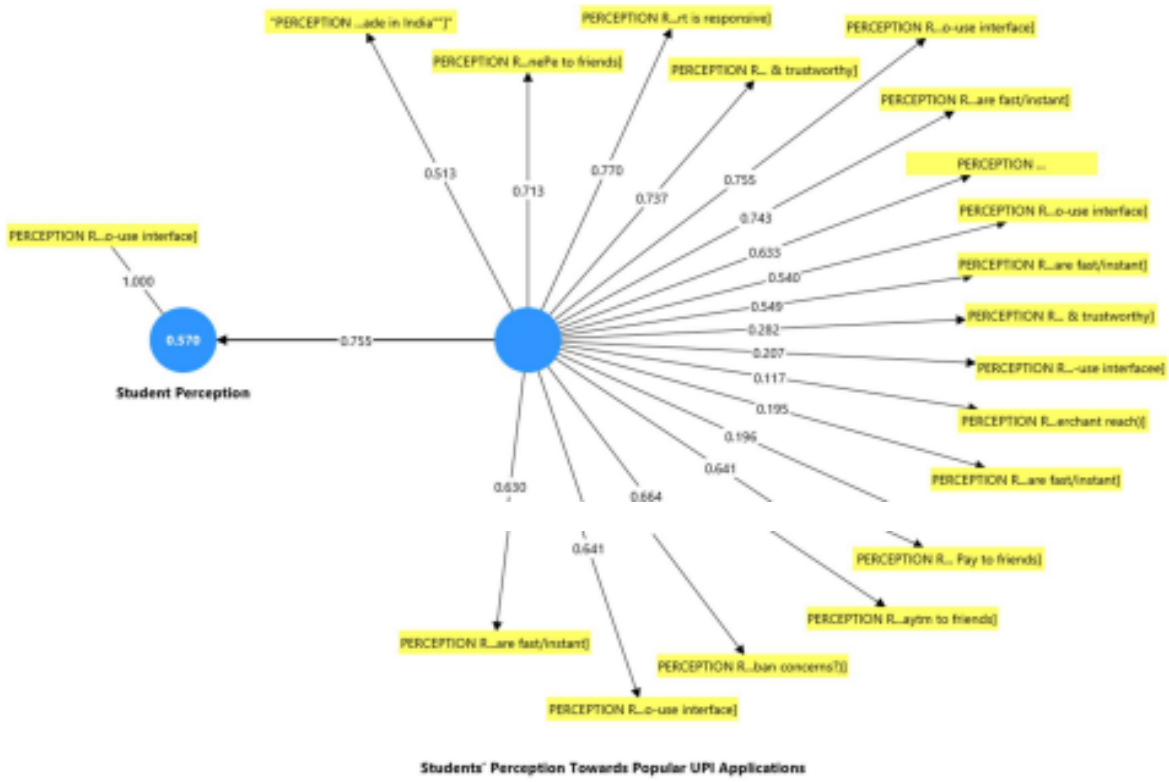
	BHIM I prefer BHIM to support	PhonePe I would recommend PhonePe to friends	[PhonePe customer support is responsive]	[PhonePe feels secure & trustworthy]	[PhonePe has an easy-to-use interface]
Iteration 1	0.093	0.093	0.093	0.093	0.093
Iteration 2	0.045	0.112	0.138	0.173	0.28
Iteration 3	0.045	0.112	0.138	0.173	0.28

	[PhonePe has an easy-to-use interface]	[PhonePe transactions are fast/instant]	BHIM feels secure & trustworthy]	BHIM has an easy-to-use interface	[BHIM transactions are fast/instant]	Google Pay feels secure & trustworthy
Iteration 1	1	0.093	0.093	0.093	0.093	0.093
Iteration 2	1	0.164	0.082	0.06	0.052	0.011
Iteration 3	1	0.164	0.082	0.06	0.052	0.011

Google Pay has an easy-to-use interface	Google Pay is accepted everywhere	Google Pay transactions are fast/instant	GOOGLE PAY I would recommend Google Pay to friends]	PAYTM I would recommend Paytm to friends]	Paytm feels secure & trustworthy	Paytm has an easy-to-use interface	Paytm transactions are fast/instant
-0.093	-0.093	-0.093	-0.093	0.093	0.093	0.093	0.093
-0.932	-0.013	-0.804	-0.804	0.077	0.089	0.105	0.041
-0.932	-0.013	-0.804	-0.804	0.077	0.089	0.105	0.041

From the above data analysis, it can be observed that PhonePe is performing extremely well in terms of different parameters such as usability, security, and customer service. It is evident from the high value of factors including easy-to-use interface and feels secure & trustworthy which increase consistently from iteration 1 to iteration 2 and maintain stability in iteration 3. In addition, the probability of recommending PhonePe to another individual is much higher than other applications because it gives a great customer experience and thus, creates better perception. On the other hand, BHIM seems to have relatively low values when it comes to usability, transaction speed, and trust while it is perceived to be relatively more secure by the respondents. Google Pay also scores moderately well when it comes to factors including usability and transaction speed while its trust perception becomes low in iterations 2 & 3. While Paytm is performing well in the factor including trust and user interface, it does not score very high in terms of transaction speed and recommendation probability.

Students Perception Regarding Popular UPI Application (Smart PLS)



Students' Perception Towards Popular UPI Applications

From the analysis of the structural model, it can be observed that the perception of UPI applications has been moderately determined by the factors considered in the study with and R^2 value of 0.570, meaning that about 57% of the variations in perception have been accounted for by the chosen set of variables. The path coefficients obtained from the results suggest that parameters such as quick response to customer requests (0.770), trust and security measures (0.755), user-friendliness (about 0.743), and recommendation intention (0.713) have considerably impacted the perception score positively. Similarly, transaction time and dependability have had a considerable influence on the perception score, with the scores being over 0.60. Other factors, such as merchant coverage and ease of executing peer to-peer transactions, have moderate influence on student perception. Parameters with very low coefficients (about 0.20-0.30) have little impact on perception levels among students.

Findings:

The results obtained from analyzing 106 responses show that the usage of UPI is mainly dominated by young adults between the ages of 18 to 23 years old, especially third-year

students. This indicates that these young adults have adopted the use of digital payment services actively, despite the fact that the majority has very little monthly allowance (below ₹5,000). This can be attributed to the accessibility and efficiency that can be achieved through the use of UPI applications. Some of the tasks that have been performed using UPI include QR code payment and bill payments, in addition to peer-to-peer payments. Moreover, most people prefer to make use of only one UPI app because they are very comfortable with it. In terms of the various apps available, Google Pay comes out as the most preferred and top-rated platform based on criteria such as usability, speed of transaction, security, rewards, and wide merchant acceptance. Second to Google Pay is PhonePe, which also does well though not as good as Google Pay. Paytm and BHIM, on the other hand, have shown relatively low rates of usage and customer satisfaction compared to the above two. Notably, this research has revealed that despite being able to reduce the use of cash payments significantly, a fully cashless payment system is not completely achieved due to some scenarios that require the use of cash. While confidence in the security of UPI is generally high, there have been cases of UPI fraud, indicating the importance of creating more awareness among users. Transaction speed and rewards are some of the aspects that may influence user satisfaction.

Conclusion:

UPI has emerged to be a vital aspect of student's financial lives, especially with respect to the youthful populace in Surat. The research outcomes show that the students use UPI for their financial transactions regardless of the small budget they have per month.

For instance, they use UPI for making QR payments, peer-to-peer transactions, and paying bills. Most of the students trust and depend heavily on UPI apps since they use a specific application for carrying out financial transactions. This means that user experience and reliability influence the use of UPI in the long run. However, even if the use of cash has been significantly reduced, it has not totally been substituted by digital payments in the students' transactions. Google Pay has been identified as one of the most favoured apps due to its high performance in terms of ease of use, speed of transactions, rewards, and acceptance in merchants, closely followed by PhonePe. Conversely, Paytm

and BHIM have been seen to attract low rates of usage and customer satisfaction. Notably, the survey shows that students do not feel threatened in terms of security on the various UPI apps but fear transaction fraud. Security features and consumer education remain essential components that must be incorporated to increase user satisfaction. It is also important to note that issues such as transaction speed and cashback programs have been identified as significant determinants of user preference.

Implications and Limitations of The Study:

The findings of the study offer important future implications for policymakers, financial institutions, and UPI service providers, as they highlight the strong adoption of digital payments among young adults, particularly students. Service providers can focus on enhancing user experience, improving transaction speed, and offering attractive reward systems to sustain and expand usage. Increasing merchant acceptance and strengthening security features will further boost confidence among users and encourage higher-value transactions through UPI platforms. Additionally, awareness programs regarding fraud prevention and safe digital practices are essential to build long-term trust. However, the study is subject to certain limitations. The sample size is relatively small (106 respondents) and restricted to students in Surat city, limiting the generalizability of the findings to a broader population. The use of convenience sampling may also introduce bias, as respondents are selected based on accessibility rather than randomness. Furthermore, the study is based on self-reported data, which may be affected by respondent bias or inaccuracies. The research focuses only on selected UPI applications and does not consider emerging platforms or technological changes over time. Lastly, the cross-sectional nature of the study restricts the ability to analyse changes in user behaviour over a longer period.

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Impact of the COVID-19 Pandemic on Corporate Social Responsibility Spending: Evidence from Selected Maharatna Companies in India

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Abstract- Corporate Social Responsibility (CSR) plays a significant role in promoting sustainable development in India, particularly among Maharatna companies. This study examines the impact of COVID-19 on CSR spending patterns of selected Maharatna companies by comparing pre COVID and post-COVID periods. The research is based on secondary data collected from annual reports and CSR disclosures. Various analytical tools such as percentage analysis, average method, comparative analysis, and paired t-test have been used. The findings reveal a significant shift in CSR priorities from long-term developmental activities to immediate crisis management, especially in healthcare and community support. The paired t-test results indicate that there is a statistically significant difference in CSR spending before and after COVID-19. The study highlights the evolving nature of CSR and its role in addressing socio-economic challenges during crises.

Keywords-*CSR, COVID-19, Maharatna Companies, CSR Spending, Healthcare, Sustainability, India*

Introduction

Corporate Social Responsibility (CSR) has emerged as a vital component of corporate governance and sustainable development, particularly in the context of large public sector enterprises such as Maharatna companies in India, which play a significant role in driving economic and social progress. The implementation of the Companies Act, 2013 has further institutionalized CSR by mandating eligible companies to allocate a minimum percentage of their profits towards social development activities, thereby ensuring structured and accountable contributions to society. Over the years, CSR spending by Maharatna companies has been directed towards key sectors such as education, healthcare, rural development, environmental sustainability, and skill development, reflecting their commitment to inclusive growth and national priorities. However, the outbreak of the COVID-19 pandemic in 2020 brought unprecedented challenges that

significantly influenced corporate priorities. Companies redirected CSR funds towards healthcare infrastructure, medical supplies, vaccination drives, and support for vulnerable populations. This shift reflects the dynamic nature of CSR in responding to crises. The present study aims to analyse the changes in CSR spending patterns of selected Maharatna companies by comparing pre-COVID and post-COVID periods, thereby providing insights into evolving CSR strategies in India.

Review of Literature

Dr. B. Sandhya Rani (2022) examined CSR practices of leading Maharatna companies in India with a focus on the nature and scope of their initiatives. The study found that these companies actively contribute to sectors like education, healthcare, rural development, and environmental sustainability. It also highlighted that CSR spending has increased over time and is aligned with the provisions of the Companies Act, 2013. The research emphasizes the role of CSR in achieving national development goals.

Pooja Gokarna and Bala Krishnamoorthy (2021) conducted an exploratory study on CSR practices during the COVID-19 pandemic in developing countries. The study revealed that companies shifted their CSR focus towards immediate relief measures such as healthcare support, medical supplies, and assistance to migrant workers. It also noted that CSR strategies became more crisis-oriented rather than focusing on long-term development projects.

Dr. Jayadev Satapathy and Dr. Tattwamasi Paltasingh (2024) analysed emerging CSR initiatives in India during the COVID-19 period. The study found that companies prioritized healthcare infrastructure, vaccination drives, and digital education initiatives. It also highlighted a transformation in CSR strategies towards technology-driven and responsive approaches to address emergency situations.

Dr. C.S. (CMA) Sanvedi Rane (2022) focused on CSR initiatives of Maharatna companies in promoting sustainable development. The study concluded that CSR activities significantly contribute to economic, social, and environmental sustainability. It also emphasized that these initiatives support inclusive growth and are aligned with national sustainability objectives.

Ansita Aggarwal and Nisarg Joshi (2023) examined the relationship between CSR and firm performance during the COVID-19 period. The study found that companies with strong CSR engagement demonstrated better resilience, enhanced corporate reputation, and improved stakeholder trust. It highlighted that CSR plays a crucial role in maintaining business stability during economic uncertainty.

Dr. Richa Vats and Shikha Mittal (2021) studied CSR practices in India in the post-COVID period. The findings revealed that companies focused more on healthcare, employee welfare, digital education, and community development. The study also pointed out a shift towards more sustainable and stakeholder-centric CSR approaches rather than short-term relief measures alone.

Bhavesh A. Prabhakar (2020) analysed CSR practices of selected Maharatna companies in India. The study found that these companies actively engage in CSR activities related to education, healthcare, environmental protection, and community welfare. It also emphasized that CSR spending is in compliance with statutory requirements under the Companies Act, 2013.

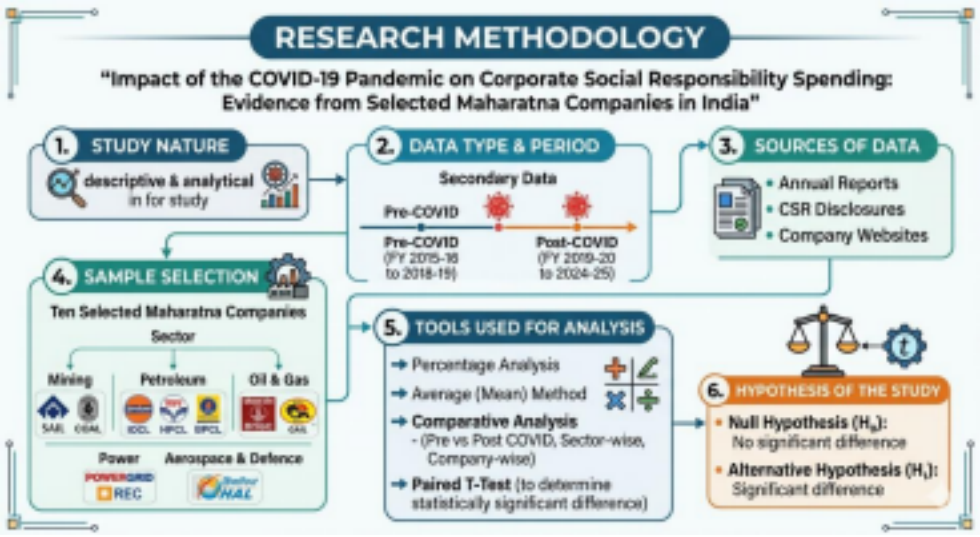
Objective of The Study

- To analyse the CSR spending of selected Maharatna companies in India.
- To examine the sector-wise allocation of CSR expenditure.
- To compare CSR spending in the pre-COVID period and post-COVID period.
- To analyse CSR expenditure in terms of amount and percentage.
- To evaluate whether there is a significant difference in CSR spending before and after COVID-19 using statistical analysis.

Research Methodology

This research adopts a descriptive and analytical design to investigate the impact of the COVID 19 pandemic on Corporate Social Responsibility (CSR) spending among ten selected Maharatna companies in India. Focusing on the petroleum, aerospace, oil and gas, power, and mining sectors, the study utilizes secondary data obtained from annual reports, CSR disclosures, and official company websites. The analysis spans a ten-year

timeline from FY 2015–16 to FY 2024– 25, categorized into pre-COVID and post-COVID phases to evaluate variations in spending. To test the hypothesis regarding significant differences between these periods, various statistical tools are employed, including percentage analysis, mean methods, and comparative evaluations. Finally, a paired t-test is applied to determine if the observed changes in CSR expenditure before and after the pandemic are statistically significant.



Hypothesis of The Study

Null Hypothesis (H_0):

There is no significant difference between CSR spending of selected Maharatna companies during pre-COVID and post-COVID periods.

Alternative Hypothesis (H_1):

There is a significant difference between CSR spending of selected Maharatna companies during pre-COVID and post-COVID periods.

Sample Selection

The study covers ten selected Maharatna companies:

- Mining Sector: - SAIL, COAL
- Petroleum Sector: - IOCL, HPCL, BPCL
- Oil & Gas Sector: - ONGC, GAIL
- Power Sector: - POWERGRID, REC

- Aerospace & Defence Sector: - HAL

Period of Study

The study covers ten years from **FY 2015–16 to FY 2024–25**, divided as:

- **Pre-COVID Period:** 2015–16 to 2018–19
- **Post-COVID Period:** 2019–20 to 2024–25

Sources of Data

Data were collected from:

- Annual reports and CSR disclosures
- Company websites

Data Analysis

- **Findings on CSR Spending (Percentage-wise)**

YEAR	SAIL	REC	POWER GRID	IOCL	HPCL	HAL	GAIL	COAL	ONGC	BPCL
2015-16	2.66	2.01	1.90	2.21	2.66	2.00	2.00	7.44	1.42	1.70
2016-17	4.80	0.95	2.17	2.01	2.61	2.03	2.26	18.78	1.96	1.14
2017-18	0.78	0.61	2.00	2.02	2.48	2.33	2.63	6.17	2.07	1.81
2018-19	0.75	1.31	2.09	2.00	2.01	2.15	2.74	8.41	2.56	1.75
2019-20	2.65	3.30	3.30	2.00	2.00	2.35	2.01	42.38	2.12	3.47
2020-21	2.39	2.05	2.06	2.69	2.41	2.52	2.08	22.52	2.05	2.13
2021-22	1.82	2.00	2.11	1.85	1.56	1.99	3.00	22.81	2.10	1.65
2022-23	1.89	2.07	2.06	1.50	1.24	2.32	1.20	11.84	2.11	1.35
2023-24	1.98	1.98	1.93	2.13	2.36	2.25	2.33	17.44	1.95	1.53
2024-25	1.96	2.00	1.96	2.03	1.90	2.29	2.02	11.78	2.05	1.00

Most selected Maharatna companies maintained CSR spending levels aligned with the 2% statutory requirement under the Companies Act, 2013, demonstrating strong regulatory compliance. While HAL and Power Grid Corporation of India exhibited the most consistent CSR percentages throughout the study, Coal India Limited experienced spikes in CSR percentages during 2019–20 and subsequent COVID-19 years; notably,

this increase was primarily a result of reduced profits rather than an increase in actual CSR outlays. Additionally, BPCL and REC Limited initially showed significant fluctuations in their CSR spending percentages prior to the pandemic but transitioned toward a more stable pattern in the post-COVID period.

- **Findings on Overall CSR Spending (Amount-wise)**

Amount in cr

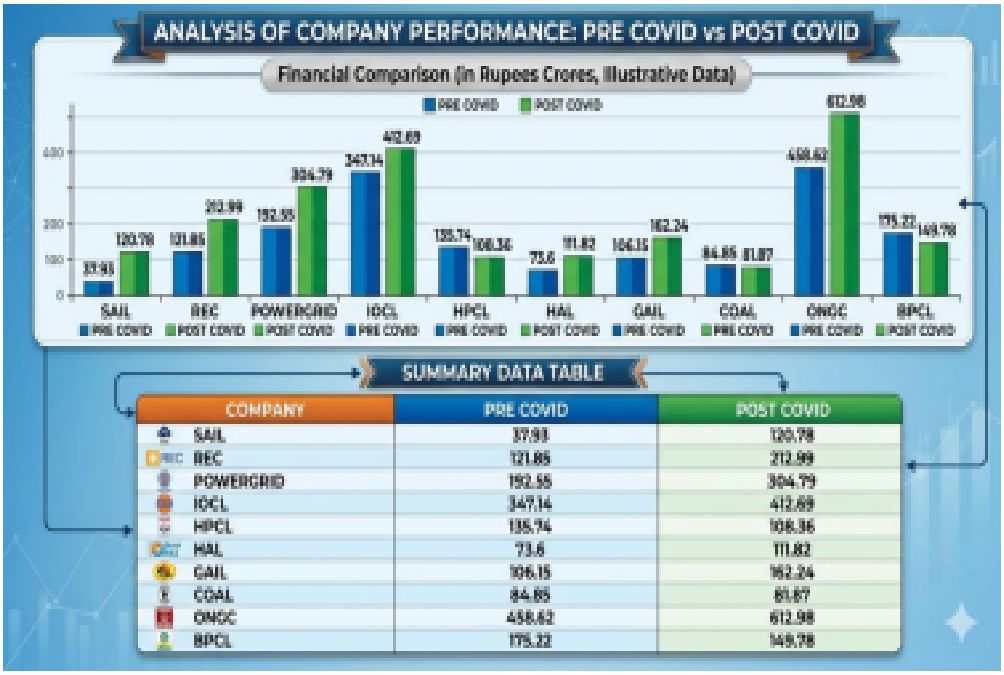
YEAR	SAIL	REC	POWER GRID	IOCL	HPCL	HAL	GAIL	COAL	ONGC	BPCL
2015-16	76.16	128.2	115.78	156.68	71.67	68.31	102.34	73.26	42.1	95.59
2016-17	29.05	69.8	147.27	213.99	108.11	67.96	92.16	128.05	525.9	90.98
2017-18	25.7	49.45	157.99	331.05	156.87	77.29	91.65	24.31	503.44	166.02
2018-19	31.18	103.39	195.52	490.6	159.81	72.46	119.29	27.33	614.64	177.94
2019-20	27.56	258.4	346.21	543.38	182.24	82	125.3	171.32	607	345.57
2020-21	47.18	147.77	240.48	460.37	156.35	90.52	147.67	95.36	553.07	144.9
2021-22	94.24	171.07	271.14	298.29	126.06	78.27	204.97	77.64	472.29	137.78
2022-23	162.46	209.95	321.66	264.03	96.11	103.24	99.1	42.04	475.89	128.67
2023-24	161.93	247.31	330.48	457.71	75.91	119.98	175.71	98.56	634.57	158.19
2024-25	138.09	288.86	360.19	583.03	87.35	167.11	183.77	95.73	929.08	179.37

Among the selected Maharatna companies, ONGC recorded the highest CSR expenditure in absolute terms throughout the study period, reaching its peak in 2024–25. IOCL and Power Grid Corporation of India also maintained consistently high spending levels, particularly during the post-COVID period. While companies like SAIL, HAL, and HPCL reported moderate spending in absolute terms, they demonstrated a gradual improvement following the pandemic's onset. Most companies exhibited a noticeable rise in CSR outlays after 2020–21, reflecting financial recovery and an enhanced social commitment in the post-pandemic landscape. Ultimately, the overall trend suggests that CSR spending remains strongly linked to both company size and profitability.

- **Findings on Pre and Post COVID Comparison**

The comparative analysis of average CSR spending across ten Maharatna companies reveals a general upward trend in absolute expenditure during the post-COVID period. As shown in above image, ONGC recorded the highest growth and absolute spending, rising

from 458.62 to 612.98, while other major entities like POWERGRID, IOCL, and SAIL also demonstrated substantial increases in their social outlays. Conversely, companies such as HPCL, COAL, and BPCL experienced slight declines in their average spending between the two phases. This data suggests that while the pandemic necessitated a shift in corporate priorities, the majority of these large scale public enterprises successfully scaled their social commitments in the following years.



COMPANY	PRE COVID	POST COVID
SAIL	37.93	120.78
REC	121.85	212.99
POWERGRID	192.55	304.79
IOCL	347.14	412.69
HPCL	135.74	108.36
HAL	73.6	111.82
GAIL	106.15	162.24
COAL	84.85	81.87
ONGC	458.62	612.98
BPCL	175.22	149.78

- **Sector-wise Findings**

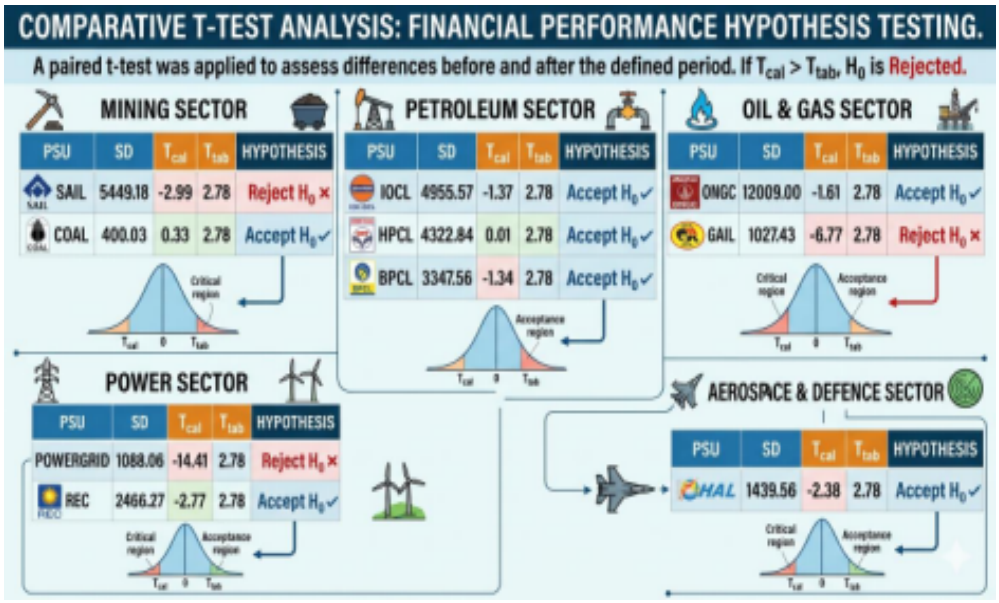
The sector-wise analysis of CSR spending among selected Maharatna companies reveals significant shifts in priorities, particularly during and after the COVID-19 pandemic. The healthcare sector emerged as the most dominant area, with companies such as Indian Oil Corporation Limited, Power Grid Corporation of India Limited, Coal India Limited, and Oil and Natural Gas Corporation allocating substantial funds toward medical infrastructure, oxygen supply, hospitals, and pandemic relief. This trend continued even post-pandemic, indicating a long-term shift in CSR focus.

The education sector remained an important area of CSR investment, with companies like Steel Authority of India Limited, Hindustan Petroleum Corporation Limited, and

Hindustan Aeronautics Limited consistently supporting educational infrastructure, scholarships, and skill development. However, education spending declined during the pandemic as resources were redirected toward urgent healthcare needs, before gradually recovering in the post-COVID period.

Environmental sustainability, which was a key focus in the pre-COVID period, saw a decline in CSR allocation after the pandemic as companies prioritized healthcare. Initiatives such as afforestation, pollution control, and water conservation received comparatively lower funding in later years. Women empowerment received relatively limited attention across all companies, with only a small proportion of CSR funds allocated toward skill development and livelihood programs for women. Additionally, the "Others" category, which earlier included various community development activities, saw a gradual decline as companies shifted toward more structured and priority-driven CSR sectors like healthcare and education.

Results from Paired T-Test ANALYSIS



The sector-wise paired t-test analysis was conducted to assess the impact of COVID-19 on CSR spending of selected Maharatna companies. The results indicate that the impact was uneven across sectors and companies. In the mining sector, SAIL showed a significant change in CSR spending, while Coal India did not. In the petroleum sector,

IOCL, HPCL, and BPCL showed no significant impact. In the oil and gas sector, GAIL experienced a significant change, whereas ONGC did not. Similarly, Power Grid in the power sector showed a significant impact, while REC remained unaffected. HAL in the aerospace and defence sector also showed no significant change. Overall, only a few companies exhibited significant variation, while most maintained stable CSR spending during the pandemic.

Findings

The analysis of Corporate Social Responsibility (CSR) spending among selected Maharatna companies shows a strong commitment to regulatory requirements. Most companies keep their spending close to the 2% minimum set by the Companies Act, 2013. Companies like HAL and Power Grid Corporation of India demonstrated consistent spending percentages. In contrast, Coal India Limited experienced significant fluctuations, peaking at 42.38% in 2019-20. This spike was due more to a drop in net profits during the pandemic transition than to a large increase in actual spending. BPCL and REC Limited also had initial fluctuations in their spending percentages but stabilized after COVID. Absolute spending data reveals that ONGC led in total expenditure, reaching ₹929.08 crore in 2024-25. IOCL and Power Grid significantly increased their contributions after the financial recovery following 2020-21.

The study highlighted a major shift in CSR priorities due to the global health crisis. Healthcare became the main focus, with companies like IOCL, ONGC, and Coal India redirecting substantial funds to medical infrastructure, oxygen supply, and pandemic relief. While the education sector remained an important investment area for companies such as SAIL and HAL, it faced a temporary decline in funding as resources shifted to urgent healthcare needs. Similarly, environmental sustainability and community development projects, which were priorities before COVID, received less funding as other sectors took precedence. Areas like women empowerment consistently received a smaller share of funds across all sectors, showing a chance for more diverse social investments in the future.

The statistical analysis using a Paired T-Test provides insight into the pandemic's impact on CSR behaviour. Overall, there was an increase in absolute spending from the

pre-COVID to post COVID periods. However, the significance of this change differed by company. Notable changes were seen in SAIL (Mining), GAIL (Oil & Gas), and Power Grid (Power), where the p-values showed a clear shift from pre-pandemic spending patterns. For most companies, including IOCL, HPCL, BPCL, ONGC, REC, and HAL, the null hypothesis (H_0) was accepted, indicating their CSR spending did not significantly change despite the pandemic. This suggests that Maharatna companies generally maintained stable social

Conclusion

The study concludes that Maharatna companies in India have demonstrated remarkable resilience and regulatory commitment in their Corporate Social Responsibility (CSR) efforts throughout the COVID-19 pandemic. While absolute spending reached new heights in the post-COVID period—led by giants like ONGC and IOCL—the analysis reveals that most firms maintained a steady expenditure of approximately 2% of their net profits, adhering strictly to the Companies Act, 2013. Although sectors like Mining and Power showed significant statistical variations in spending, the majority of companies maintained a stable financial commitment, proving that CSR outlays in these public enterprises are fundamentally driven by company size and profitability rather than external shocks alone.

Furthermore, the pandemic triggered a strategic shift in CSR priorities, with healthcare emerging as the dominant area of investment at the expense of environmental and community development projects. Companies successfully pivoted their resources to address urgent national needs, such as oxygen supply and medical infrastructure, while gradually restoring focus to education in the later years. This evolution suggests a transition toward more structured, priority-driven CSR frameworks. Ultimately, the study highlights that while the "what" of CSR spending shifted toward crisis management, the "how much" remained a consistent reflection of corporate financial health, reinforcing the role of Maharatna companies as pillars of social development in India.

Implication and Limitation of The Study

The study implies that future CSR policies for Maharatna companies should shift toward "crisis ready" strategic frameworks, ensuring healthcare infrastructure remains a priority without neglecting long-term environmental and gender-based goals. It suggests that

since spending is tightly coupled with profitability, new mechanisms are needed to decouple social funding from economic downturns. However, the research is limited by its sole reliance on secondary data from ten specific companies, which restricts broad generalizability. Furthermore, the study focuses on financial outlays and percentages rather than evaluating the qualitative social impact or effectiveness of the activities on beneficiaries.

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